

Exploring Foeto-Placental Serotonin Regulation within a Rodent Maternal Immune Activation Model

by

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Abstract

Maternal infections during pregnancy are associated with increased incidence of many neurodevelopmental and neuropsychiatric disorders (NDD/NPDs) in offspring, including autism spectrum disorder, schizophrenia spectrum disorders, and major depressive disorder. While it is generally understood that maternal infections lead to an inflammatory response that disrupts neurodevelopment, it remains unclear which developmental processes or signalling pathways underpin the association between maternal immune activation (MIA) and NDD/NPD outcomes. Emerging evidence suggests that dysregulation of serotonin availability within the foetoplacental unit may be involved in the MIA response. Changes to serotonin production, degradation, and transport have been demonstrated in the foetoplacental unit following MIA, although the direction and extent of these changes vary across gestational ages and tissues.

This thesis expands upon existing work by exploring the effects of MIA in mid-gestation on the serotonergic system, through administration of the viral mimetic polyinosinic:polycytidylic acid (poly I:C) or saline vehicle to pregnant Sprague-Dawley rats on embryonic day (E)12.5. C-reactive protein in maternal serum, increased litter loss, and altered placental growth all confirm the successful induction of MIA in poly I:C-exposed offspring. Although amniotic fluid serotonin concentrations are not affected by MIA at the time points assessed in this thesis (E14.5 and E18.5), mRNA expression of key serotonin-associated genes are significantly altered in poly I:C-exposed brain and placental tissue. Particularly notable is the E14.5 foetal brain, where treatment*sex interactions are observed in mRNA expression, including in serotonin transporter (*Slc6a4*) expression. Extensive analysis of existing literature and gene structure further revealed that investigating DNA methylation of *Slc6a4* is a promising, yet underexplored avenue of MIA research, particularly in rodent models.

Overall, this thesis provides evidence to suggest that mid-gestational MIA influences serotonin regulation in the foetoplacental unit, although the changes observed are relatively less extensive than models where MIA is induced earlier or later in gestation. These findings also demonstrate the importance of considering translatability, inter-litter variability, and offspring sex when assessing prenatal development. In addition, this thesis provides valuable methodological developments and preliminary analyses for future, more novel avenues for investigating serotonergic changes following MIA.

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Abbreviations

5-HIAA	5-hydroxyindole acetic acid
5-HT	Serotonin
5-HTP	5-hydroxytryptophan
5-HTTLPR	5-HT transporter linked polymorphic region
AARIVE	Animal Research: Reporting of In Vivo Experiments
ACN	Acetonitrile
ActB	Beta-actin
ADHD	Attention deficit hyperactivity disorder
AF	Amniotic fluid
AIC	Akaike information criterion
ANOVA	Analysis of variance
ASD	Autism spectrum disorder
B2m	Beta-2 microglobulin
Bdnf	Brain-derived neurotrophic factor
BzCl	Benzoyl chloride
CBCL	Child Behaviour Checklist
CBD	Cannabidiol
cDNA	Complementary DNA
CRP	C-reactive protein
CV	Coefficient of variance
Ddc	Dopa decarboxylase
ddPCR; RT-ddPCR	Digital droplet polymerase chain reaction; reverse transcription
DSM-5	Diagnostic and statistical manual of mental disorders, 5 th edition
dsRNA	Double-stranded RNA
E	Embryonic day
ELISA	Enzyme-linked immunosorbent assay
ESI	Electrospray ionisation
FC	Frontal cortex
fPLA	Foetally-derived placenta
Gapdh	Glyceraldehyde 3-phosphate dehydrogenase
GD	Gestational day
GDM	Gestational diabetes mellitus
gDNA	Genomic DNA

HPLC	High performance liquid chromatography
Hprt	Hypoxanthine phosphoribosyltransferase 1
i.p.	Intraperitoneal
i.v.	Intravenous
IDO	Indolamine 2,3-dioxygenase
IFN-	Interferon
IL-	Interleukin
IRFs	interferon regulatory factors
IUGR	Intrauterine growth restriction
KA	Kynurenic acid
Kdm5d	Lysine (K)-specific demethylase 5D
KYN	Kynurenine
LC-MS	Liquid chromatography-mass spectrometry
LMM	Linear mixed model
LPS	Lipopolysaccharide
MaoA	Monoamine oxidase A
MDD	Major depressive disorder
Mecp2	Methyl-CpG-binding protein 2
MIA	Maternal Immune Activation
mRNA	Messenger ribonucleic acid
MsRE	Methylation sensitive restriction enzyme
m/z	Mass to charge ratio
NDD	Neurodevelopmental disorder
NeuroD1	Neurogenic differentiation 1
NPD	Neuropsychiatric disorder
Oct3	Organic cation transporter 3
PAMP	Pathogen-associated molecular pattern
pBR	Posterior brain region
PBS	Phosphate buffered saline
PCR, qPCR, RT-PCR	Polymerase chain reaction, quantitative, reverse transcription
PI:C, Poly I:C	Polyinosinic:polycytidylic acid
PND	Postnatal day
Ppia	Peptidylprolyl isomerase A
PTB	Preterm birth
QA	Quinolinic acid

RDoC	Research Domain Criteria
RT	Reverse transcription
s.c.	Subcutaneous
SAL	Saline
SCZ	Schizophrenia
Sert	Serotonin reuptake transporter
Slc6a4	Solute carrier family 6 member 4, gene encoding SERT
s:n	Signal to noise ratio
SNRI	Serotonin-Norepinephrine Reuptake Inhibitors
SR	Scavenger receptor
SSRI	Selective Serotonin Reuptake Inhibitors
STin2	Serotonin transporter intronic enhancer
Tbp	TATA box binding protein
TLR-3	Toll-like receptor 3
TNF α	Tumour necrosis factor alpha
Tph-	Tryptophan hydroxylase
TRIF/NF-kB	TIR domain-containing adaptor protein inducing interferon beta/Nuclear factor kappa-light-chain-enhancer of activated B cells
TSS	Transcription start site
USV	Ultrasonic vocalisation
UTI	Urinary tract infection
VNTR	Variable number tandem repeat
Ywhaz	Tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein, zeta

Chapter 1:

General Introduction

1.1. Psychological Disorders and Prenatal Development

The prenatal period in human development is complex and dynamic. It is well-known that environmental factors may have significant effects on foetal development during pregnancy: poor nutrition and maternal alcohol use, for example, lead to long-lasting consequences for offspring such as poorer cognition, language development, motor coordination, and foetal alcohol spectrum disorders [1, 2]. Further environmental factors including (but not limited to) maternal smoking, antidepressant use, stress, infections, air pollution, extreme temperatures, and pre-term birth have been associated with increased risk of neurodevelopmental and neuropsychiatric conditions in offspring [3-8].

Neurodevelopmental disorders (NDDs), as defined by the *Diagnostic and Statistical Manual of Mental Disorders* (5th ed, Text Revision; DSM-5-TR), are a group of disorders that present during early childhood with origins in the prenatal developmental period [9]. Hallmarks of these disorders include delayed achievement of developmental milestones in areas such as language acquisition or motor skills, and the development of atypical behaviours such as repetitive or disruptive verbal and motor activity [9]. Underlying differences in brain connectivity and/or neuronal signalling pathways are often observed in these disorders and thought to significantly contribute to the aetiology and variability of disorder symptoms [10, 11]. While it is widely understood that these changes must occur or be primed for during the prenatal period, influenced by both genetic and environmental factors, specific causal mechanisms for many NDD symptoms and diagnoses remain elusive [11].

In contrast to NDDs, adolescence or adulthood is the typical age of onset for neuropsychiatric disorders (NPDs). These diagnoses and symptoms range broadly, usually anchored in dysregulated emotion regulation, reward processing, and/or trauma, and are less likely to be associated with learning or developmental delays than NDDs [9]. Many NPDs are commonly pharmacologically treated by antipsychotics, antidepressants, and anti-anxiety medications upon diagnosis. While NPDs have historically thought to be driven by postnatal triggers such as excessive stress or physical/emotional trauma, increasing evidence suggests NPDs have a neurodevelopmental component, and an individual may be primed during the prenatal period for developing an NPD later in life [4].

This introduction will first introduce a few key examples of NDD/NPDs that have been associated with altered prenatal development, and then focus on exploring one key environmental risk factor implicated in these disorders: the role of maternal immune activation (MIA) during pregnancy [12].

1.3.1. Autism Spectrum Disorder

Autism Spectrum Disorder (ASD) is classified as an NDD within the DSM-5-TR [9]. Individuals diagnosed with ASD experience difficulty with social interaction (e.g. challenges engaging with eye contact, understanding or reciprocating non-verbal and verbal social cues, adjusting behaviour in social environments, or managing relationships) and display at least two types of restricted, repetitive behaviour (e.g. repetitive motor movements known as self-stimulatory behaviour or 'stims', echolalia, fixated and intense interests, inflexible reliance on routine/sameness, and atypical responses to sensory stimuli). While many of these behaviours may be observed to an extent in non-autistic individuals, an ASD diagnosis requires traits to be significantly affecting the quality of life of the individual. Pharmaceuticals are not typically prescribed to manage the core symptoms of ASD, with behavioural and psychotherapies (e.g. speech and occupational therapies, cognitive behavioural therapy) being the most useful tools available for supporting individuals with ASD [13, 14].

Importantly, ASD is recognised as a spectrum – while difficulty with social and repetitive behaviours are diagnostic requirements, the presentation of these challenges varies widely across individuals. For example, one individual might be classified as 'requiring support' and be primarily affected by intense/unregulated responses to sensory stimuli and difficulty recognising and reciprocating social cues, another may be considered 'requiring very substantial support' and experience significant learning, cognitive, and speaking difficulties. This variability is further exemplified by the range of additional conditions that may accompany an ASD diagnosis such as intellectual impairment, language impairment, catatonia, mental disorder comorbidities, and genetic or medical comorbidities [9]. This last category includes monogenetic disorders with a high prevalence of ASD such as Rett Syndrome (originally estimated at ~78% with comorbid ASD, more recently 17-58%) [15, 16] and

Fragile X Syndrome (estimated 15-60% with comorbid ASD) [17, 18], as well as conditions with less-defined aetiology such as epilepsy (5-37% with comorbid ASD) [19-21].

Dysregulation of a number of broad pathways have been implicated in ASD. Many of these are shared with the common comorbidities of ASD; for example, mutations in the *FMR1* gene underlying Fragile X Syndrome lead to altered dendritic growth/morphology and altered synaptic plasticity [22, 23]. These changes are also reflected in ASD [24-26], producing complex patterns of hypo- and hyperconnectivity that are associated with altered learning, memory, emotion regulation, and speech and motor skills [27-29]. In Rett Syndrome and epilepsy, altered excitability of neural networks and neurotransmitter signalling are the core mechanisms of interest – changes to noradrenergic, dopaminergic, serotonergic, and glutamatergic/GABAergic signalling are observed across these diagnoses [30-33]. ASD exhibits excitability changes in many of these systems alongside changes in connectivity, primarily in serotonergic signalling and the balance between glutamatergic and GABAergic signalling [34, 35].

ASD is typically diagnosed in early childhood, with diagnoses before the age of eight years old being 3-4x more common in males than females [36]. Late diagnoses of ASD are also possible; lower-support needs children for example may develop coping strategies (masking the difficulty experienced in social or overstimulating environments) and remain undiagnosed until adulthood [37], or females in particular may receive mis-diagnoses such as anxiety or personality disorders before finally receiving a late diagnosis of ASD [38, 39]. While diagnostic biases influence the rate at which symptoms are recognised in females and likely contribute to the reduced diagnosis rate in females, there is also evidence to suggest a biological basis for female resilience/male susceptibility to ASD. Genetic predisposing factors are often found in a higher load in females with ASD and their families compared to males with ASD, suggesting the threshold required for females to develop ASD symptoms is higher than males [40]. Male offspring may also be more susceptible to developing ASD after prenatal exposure to environmental risk factors such as maternal selective serotonin reuptake inhibitors (SSRI) use [41], with supporting rodent models demonstrating sex-specific consequences of SSRI exposure [42, 43]. While the

mechanisms underlying these sex differences are unclear, the ratios and activity of sex hormones during development are suspected to play an important role [44, 45].

1.3.2. Schizophrenia

Schizophrenia (SCZ) is an NPD specified along the schizophrenia spectrum of psychotic disorders, characterised by both negative and positive symptomology [9]. Negative symptoms are considered to be typical traits or behaviours absent from the affected individual – periods of avolition (a lack of motivation and engagement in activities), asociality (lacking engagement in or opportunity for social interaction), anhedonia (diminished experience of pleasure from positive stimuli), and overall diminished emotional expression may be observed. Positive symptoms of SCZ include delusions (fixed beliefs that will not change based on evidence, often involving a feeling of immediate danger or receiving the approval/attention of others), hallucinations, disorganised/unintelligible thinking and speech, and abnormal motor behaviour (encompassing excessive, emotional activity to catatonia). Symptomatic periods of at least 6 months (with at least one month of positive symptoms) are considered acute episodes for diagnosis, with symptom onset typically occurring in adulthood [9].

SCZ and similar disorders are treated through individual combinations of antipsychotic medications and psychotherapies. First generation antipsychotics such as haloperidol act as dopamine D2 receptor antagonists, preventing the elevated nigrostriatal dopaminergic signalling that is considered to be a driver of the positive symptoms of SCZ [46]. Dopaminergic hyper and hypo-excitability across the brain plays numerous roles in the symptomology of SCZ and has been a major research focus, however additional neurotransmitters such as glutamate and serotonin (5-HT) are receiving renewed attention as potential therapeutic targets [47-49]. Current clinical trials for a new wave of antipsychotics targeting multiple neurotransmitter systems are showing promise; Ulotaront for example targets receptors involved in both dopaminergic and serotonergic signalling, improving positive SCZ symptoms similarly to D2 antagonists but with a significantly higher improvement in negative symptoms [46, 50]. 5-HT signalling imbalances have been particularly associated with negative symptomology, but growing evidence suggests a role for 5-HT in the onset of SCZ. This includes 5-

HT-regulated aberrant dendritic spine growth and synaptic pruning that may prime an individual for SCZ as early as the prenatal period, as well as 5-HT-mediated changes in neurotransmitter release that are associated with acute episodes [51-54].

There are a number of striking similarities between ASD and SCZ, in spite of the difference in age of onset. For example, sex differences are also observed in SCZ – females are approximately 1.5x less likely to be diagnosed with SCZ, and are more likely to be diagnosed later in life [55]. Later diagnoses and worsening symptoms occur around the menopause period (when female sex hormone levels are decreasing), again supporting a hypothesised protective effect of female sex hormones [56]. As mentioned above, both disorders are also associated with differences in synaptic connectivity and neurotransmitter imbalances that may be primed for during development [57]. This is reflected in symptomology, where symptoms between disorders are distinct but occur in overlapping areas such as atypical responses to stimuli and motor behaviour, emotionality, and sociality [9]. Other disorders marked by distinct differences in emotionality have increased comorbidity with both ASD and SCZ, including anxiety (~40% in both ASD and SCZ) and depressive (wide ranging estimates, up to 50% in both ASD and SCZ) disorders [58-61]. Some debate exists regarding the measurable comorbidity between SCZ and distinct depressive disorders, as depressive symptoms significantly overlap with the negative symptoms of SCZ [9, 60].

1.3.3. Major Depressive Disorder

Major Depressive Disorder (MDD) is a neuropsychiatric mood disorder with a growing prevalence in New Zealand [62, 63]. Key criteria of MDD include chronic difficulty with sleep (either hypersomnia or insomnia), fatigue, indecisiveness/difficulty thinking, significant weight/appetite fluctuation, reduced interest/enjoyment of typically pleasing activity (similar to the avolition/anhedonia characteristic of SCZ, although distinct in the efficacy of pharmacological treatments [64]), and excessive negative emotion (e.g. feelings of worthlessness, guilt, sadness, hopelessness, and/or thoughts of death). These feelings and behaviours must also persist for a minimum of 2 weeks (often much longer), significantly affect daily functioning, and be carefully weighed against

immediate life events that may naturally trigger such thought/behaviour (e.g. grief, sudden financial or medical jeopardy) [9].

Psychotherapies are often effective in the treatment of MDD and other depressive disorders, particularly when co-administered with antidepressant medications such as SSRIs [65]. SSRIs target the serotonergic system by binding to the serotonin reuptake transporter (SERT), increasing the duration of serotonergic signals at the synapse by preventing 5-HT reabsorption [66]. After long-term administration, SSRIs are further thought to increase 5-HT signalling and increase neuronal and synaptic plasticity, improving the disrupted functional connectivity observed in executive functioning and emotion regulation centres of the brain [67, 68]. Just as in ASD and SCZ, altered neural connectivity from disrupted prenatal or childhood development may drive susceptibility to MDD – genetic variance in serotonergic genes (e.g. Tryptophan hydroxylase 2/*TPH2*, Monoamine oxidase A/*MAOA*, and the SERT-encoding *SLC6A4*), maternal depression, and gestational SSRI use are all associated with changes in neural connectivity and MDD outcomes [69-72]. While 5-HT is the most common system associated with MDD, other factors are known to play a role in symptomology; alternative medications such as serotonin and norepinephrine reuptake inhibitors (SNRIs) for example also target the dysregulated noradrenergic signalling observed in MDD, which is particularly associated with avolition and fatigue symptomology [73]. Neuroinflammation is also observed in MDD, which impacts processes such as synaptic pruning and plasticity, and may contribute to treatment-resistant MDD [74, 75].

In contrast to ASD and SCZ, the age of MDD onset ranges broadly. Symptoms may be exhibited in childhood, however diagnosis is typically from adolescence and most common in early adulthood [76]. While sex differences are observed in MDD, these also differ from ASD and SCZ – rather than experiencing the hypothesised protective effect, females are approximately twice as likely to be diagnosed with MDD and experience more severe symptoms [77]. The diagnostic disparity extends across the broad range in age of onset, being particularly prominent at 10-15 years and during perimenopause [77-79]. Notably, this also aligns with the beginning and end of the female reproductive period – the cycling of sex hormones during the menstrual cycle, particularly the regular decrease in oestrogen, is thought to influence acute susceptibility to environmental stimuli triggering MDD symptomology [79]. SCZ onset

in perimenopausal females somewhat mirrors this effect (sex hormone reduction aligning with disorder onset), however the developmental effect of sex is less clearly defined in MDD. In contrast to the female protective effect of SCZ and ASD, sex hormones are thought to modulate the effects of developmental risk factors in MDD and lead to distinct areas of vulnerability rather than mediating overall risk [80, 81].

1.3.4. The Multiple Hit Hypothesis

It is widely accepted that multiple risk factors (rather than one specific causal factor) play a role in the aetiology of NDD/NPDs by disrupting typical development [12, 82, 83]. This is referred to as the 'two hit' or 'multiple hit' hypothesis of disorder incidence, with each 'hit' referring to any disorder-associated genetic or environmental risk factor. Multiple hits may occur/confer risk during the critical prenatal/neonatal periods of neurodevelopment, leading to the early emergence of disorder symptomology and childhood diagnoses such as ASD [12]. In other cases, this could consist of at least one hit *in utero* that alters foetal neurodevelopment and increases susceptibility to secondary hits later in life [12]. This second version is more aligned to the onset of NPD symptomology in adulthood; factors such as maternal infections or inherited genetic predisposition may influence prenatal development and increase SCZ risk for example, with secondary hits such as significant life stress or perimenopause triggering symptom onset [56, 84-86].

While many prenatal environmental risk factors have been associated with NDDs and NPDs, there is still a lack of understanding in how many of these risk factors directly increase the likelihood of disorder outcomes. This is particularly of interest when a risk factor is associated with multiple different NDD and NPD diagnoses, suggesting the risk factor can have a significant developmental impact but the individual outcomes may vary significantly. As NDD/NPD diagnoses significantly affect quality of life, and diagnosis rates of disorders such as ASD and MDD are rising both in New Zealand and globally, exploring how common risk factors contribute to various disorder incidence and symptomology is growing increasingly important [62, 87, 88].

1.1.4.1. *Maternal Infections During Pregnancy*

One prenatal environmental risk factor shared by numerous NDDs and NPDs is the presence of a maternal infection during pregnancy. The majority of early research focused on the risk of offspring developing ASD or cognitive delays following maternal infections that were directly transmitted to the developing foetus, such as congenital rubella and cytomegalovirus [89, 90]. Subsequent large birth cohort studies investigated a wider scope of infections and diagnoses, identifying associations with disorder diagnoses as well as earlier/general markers of NDD/NPD risk (Table 1.1).

Associations between maternal infections and ASD incidence are the most common in current literature, however the specific details of those associations vary widely (Table 1.1). Risk does not appear to be conferred from a specific type of infection or trimester of infection, however viral infections are more commonly associated with early pregnancy and bacterial with late pregnancy [91-93]. Incidence of SCZ has most commonly been associated with influenza outbreaks and early pregnancy infections, while offspring MDD diagnoses do not show an association with any particular timing or type of infection. The association between offspring MDD diagnosis and maternal infection is rarely studied compared to both ASD and SCZ, however this is partially due to a shift in research techniques; rather than focusing on specific diagnoses, NDD/NPD risk in children and adolescents is now more commonly assessed through non-diagnostic tests and questionnaires (Table 1.1). Depressive symptomology [94, 95] and internalising behaviours [96, 97] in adolescents have been associated with maternal infection during early pregnancy in recent years, strongly suggesting a trajectory towards MDD and NPDs with similar symptomology that are typically diagnosed in adulthood. These non-diagnostic, early screening approaches have also allowed preliminary research into the effects of the COVID-19 pandemic, where maternal SARS-CoV-2 infection has been repeatedly associated with developmental delays in offspring [98-100]. These studies suggest a further increased risk of ASD is likely, which was unable to be directly explored in such young cohorts but should be explored in future work.

	Increased Risk: Specific Diagnoses			Increased Risk: NDD/NPD Indicators	
	ASD	SCZ	MDD	Developmental, Social, or Cognitive Delays	Emotion and/or Behaviour Dysregulation
General Infection	Al-Haddad, 2019 [101] Brown, 2008 [102] ^f Hall, 2021 [103] Hope, 2024 [104] ^a Lee, 2015 [92] Lin, 2025 [105] ^{d,e} Zerbo, 2015 [106] ^{b,e,g}	Debost, 2016 [107] ^f Nielsen, 2011 [108] ^e	Al-Haddad, 2019 [101]	Green, 2018 [109] Kwok, 2022 [110] ^c O'Connor, 2022 [111] ^{a,b} Suleri, 2023 [96] ^d	Lipner, 2022 [95] ^b Murphy, 2017 [94] ^b O'Connor, 2022 [111] ^{a,b} Patel, 2021 [97] ^g Suleri, 2023 [96]
Viral Infection	Atladóttir, 2010 [91] ^{a,e} Lee, 2015 [92] ^{a,c}				
Influenza	Atladóttir, 2012 [112]	Byrne, 2007 [113] Brown, 2004 [114] ^a Mednick, 1988 [115] ^b Takei, 1993 [116] ^f	Cannon, 1996 [117]	Ellman, 2009 [118] ^{c,d}	
Covid-19				Edlow, 2022 [98] ^e Hill, 2025 [99] ^{c,e} Fajardo Martinez, 2023 [100]	

Bacterial Infection/ Antibiotics	Atladóttir, 2010 [91] ^{b,e} Atladóttir, 2012 [112] Lee, 2015 [92] ^{b,c} Fang, 2015 [93] ^{c,e} Zerbo, 2015 [106] ^{e,g}	Clarke, 2009 [119] ^{d,e}	Lavebratt, 2019 [120] ^g		
Genito- urinary Infections	Al-Haddad, 2019 [101] Fang, 2015 [93] ^{c,e}	Babulas, 2006 [121] ^a	Al-Haddad, 2019 [101]	Langridge, 2013 [122]	
Fever	Atladóttir, 2012 [112] Brucato, 2017 [123] ^c Croen, 2019 [124] ^b Hornig, 2018 [125] ^{b,g} Zerbo, 2013 [126]			Fajardo Martinez, 2023 [100] Zerbo, 2013 [126]	

Table 1.1: Thirty-six studies finding a significant association between prenatal infections and offspring NDD/NPD risk. Included studies assess risk through either direct diagnosis prevalence or utilise validated measures such as the Child Behaviour Checklist (CBCL) or the Social and Communication Disorders Checklist (SCDC) to identify atypical behaviour/developmental trajectory. Associations were not reported to be trimester-specific or severity-specific unless annotated accordingly. a = Strongest in or specific to the first-trimester; b = Strongest in or specific to the second trimester; c = Strongest in or specific to the third trimester; d = only significant in combination with an additional risk factor; e = significant when limited to severe illness e.g. requiring hospitalisation; f = significant sex effects observed; g = risk increases with severity and/or >1 infection during pregnancy.

While diagnosis of a maternal infection is associated with NDD and NPD outcomes, the mechanism(s) underlying this association are poorly understood. Fever itself has been associated with ASD risk repeatedly, emphasising inflammation as an important mechanism (Table 1.1). Some birth cohort studies have directly explored inflammatory markers in maternal serum; Canetta et al. for example found SCZ risk in offspring in a Finnish cohort increased 28% per every 1 mg/mL increase in maternal C-reactive protein (CRP) [127]. Increases in maternal tumour necrosis factor alpha (TNF α) and interleukins 1 β , 6, and 8 (IL-1 β , IL-6, IL-8) have also been associated with NDD/NPD risk in large cohorts, however findings for any specific cytokine are often inconsistent [128-132]. Just as with maternal infection overall, the association between inflammatory markers and NDD/NPD outcomes are likely impacted by variability in the timing and severity of the inflammatory response, and further mediated by offspring sex or heritable risk factors within the multiple hit model.

Maternal infections and inflammation have not only been associated with specific NDD/NPD diagnoses in offspring, but a limited number of studies further explore underlying changes in brain structure and function. Increased maternal CRP for example has been associated with decreased cerebellar volume in offspring and reduced sensory gating, which in turn are associated with motor activity, sensory sensitivity, and auditory processing differences across NDD/NPDs [133, 134]. Maternal IL-6 has also been associated with reduced cognitive scores on offspring and increased relative volume in the pars triangularis, which is involved in language processing within Broca's area [135]. Interestingly, other studies have suggested the relationship between maternal inflammation and neurodevelopment may be advantageous in some conditions; Spann et al. demonstrated that increased maternal CRP and IL-6 concentrations were associated not only with connectivity changes in offspring, but also improved cognitive testing in toddlers [136]. These studies overall support the role of maternal infection as a driving factor in altered neurodevelopment, however further exploration of the mechanisms underpinning this relationship are limited by the non-invasive techniques available and the challenges in accounting for/isolating a wide variety of confounding factors. Animal models have become a useful and flexible tool for managing these challenges, furthering understanding of the direct developmental impact of maternal inflammation.

1.2. The Maternal Immune Activation (MIA) Rodent Model

To model the impact of maternal infections during pregnancy, a number of approaches have been used to induce maternal immune activation (MIA) during the gestational period in animal models. This includes the use of live viruses and bacterial cultures, and the more commonly used non-infectious immunogenic agents lipopolysaccharide (LPS) and polyinosinic:polycytidylic acid (poly I:C) [12]. LPS is a protein found within the bacterial cell wall, mimicking a bacterial infection, while poly I:C is a double-stranded RNA (dsRNA) analogue that mimics RNA present during many viral infections. While general infection/inflammation (rather than specific infection types) is most commonly associated with NDD/NPD outcomes as described previously, virus-like infection models are of particular interest due to recent viral outbreaks such as the global COVID-19 pandemic beginning in 2020 and the emerging 2024-2025 dengue epidemic across countries in the Americas and the Pacific Islands [137, 138].

The validity of poly I:C-induced MIA in modelling NDD/NPDs has been explored extensively. This includes the ability of the model to reflect disorder phenotypes/behaviours (face validity), reflect the underlying biological mechanisms of those phenotypes (construct validity), and reflect/predict the phenotypic response to pharmaceutical treatment options (predictive validity). As the ability to control the developmental environment and directly access developing tissues are major advantages of using rodent models to explore MIA, the comparability between human and rodent gestation must also be considered when utilising and interpreting poly I:C models.

1.2.1. Poly I:C -induced MIA Model Validity

1.2.1.1. Offspring Outcomes: Face and Predictive Validity

The poly I:C -induced MIA model has demonstrated face validity through a range of behavioural outcomes in rodent offspring that reflect NDD/NPD phenotypes. While rodents have reduced social and cognitive complexity than humans and cannot specifically replicate NDD/NPD symptomology, rodents exhibit a range of developmental milestones (such as altered eye opening and achievement of righting

reflexes [139]) and social and affective behaviours that may be measured as proxies for human outcomes. While classical research approaches associated rodent behaviours with specific human symptoms/diagnoses to show face validity, recent approaches such as the Research Domain Criteria (RDoC) framework instead use a generalised structure of overlapping behavioural domains that are implicated across human diagnoses [140]. This approach better reflects the ability of animal models to reflect areas of human behaviour rather than complex symptomology, as well as reflecting the symptom and biological overlap observed across NDD/NPD diagnoses (see 1.1).

The sensorimotor systems domain of the RDoC framework is particularly relevant to ASD and SCZ, where repetitive and/or atypical motor activity is commonly observed. Juvenile rodents exposed to poly I:C-induced MIA have displayed altered climbing/rearing/locomotor activity and increased repetitive behaviours such as marble burying [139, 141]. Adult offspring have also displayed increased marble burying, as well as increased self grooming, impaired gait on ladder apparatus, and impaired pre-pulse inhibition that reflect altered responsivity to stimuli [142-146]. Non-steroidal anti-inflammatory drugs such as Aceclofenac, which have potential as adjuvant treatments to reduce inflammation in SCZ and MDD, have been shown to improve pre-pulse inhibition deficits in MIA offspring as well as impaired social activity [147].

Systems for social processes are also implicated in MIA models, as well as being highly relevant across ASD, SCZ, and MDD. This includes altered identification and/or production of social behaviours, as well as the overall level of direct engagement in social situations. Interestingly, behavioural tests exploring social processes often overlap with the positive valence systems RDoC domain, as social interaction in rodents is inherently rewarding. The positive valence systems domain encompasses the processing, learning, and valuation of positive stimuli/rewards; in humans, the negative symptoms of SCZ and many MDD symptoms fall within this domain such as a loss of motivation and reduced enjoyment of typically rewarding stimuli.

Atypical social communication is observed early in rodent development, at approximately postnatal day (PND) 10 where MIA-exposed pups typically emit fewer ultrasonic vocalisations (USVs) during maternal separation [143, 148]. Reduced USV

calling and play behaviour is also observed during juvenile social play, potentially reflecting both social and positive valence domains [141]. In adulthood, reduced time spent in social interaction is commonly observed; this includes reduced sniffing and direct interaction with novel partners, typically paired with decreased preference for partners when also provided a novel object [143, 146, 147, 149, 150]. Alongside Aceclofenac mentioned previously, administration of cannabidiol (CBD) and the SSRI vortioxetine have been demonstrated to ameliorate the reduction in social behaviours exhibited by MIA-exposed offspring [146, 149]. These findings reflect outcomes in human subjects, where CBD and vortioxetine have been used to improve social and positive valence symptoms in MDD and other NPDs (with additional efficacy in treating cognitive symptoms) [151, 152].

The RDoC domain of cognitive systems is focused on the regulation of attention, decision making, and memory. Memory processes in particular are relevant to rodent models, where spatial and working memory paradigms are used to assess learning, memory recall, and cognitive flexibility. For example, MIA-exposed offspring have displayed reduced latent inhibition (the typical delay in associating a familiar, previously irrelevant stimuli with a new conditioning response) and decreased discrimination indices when faced with a novel and familiar object, suggesting poorer recall of familiar stimuli and/or impaired screening of unimportant stimuli [147, 149, 153]. Adults have also displayed slower reversal learning (when a conditioned stimuli no longer produces the expected response), indicating reduced cognitive flexibility in changing environments [150, 154]. These outcomes reflect common traits presented in NDD/NPDs, including the altered perception of stimuli experienced during acute SCZ episodes and the cognitive rigidity and preference for familiar stimuli experienced in ASD [9].

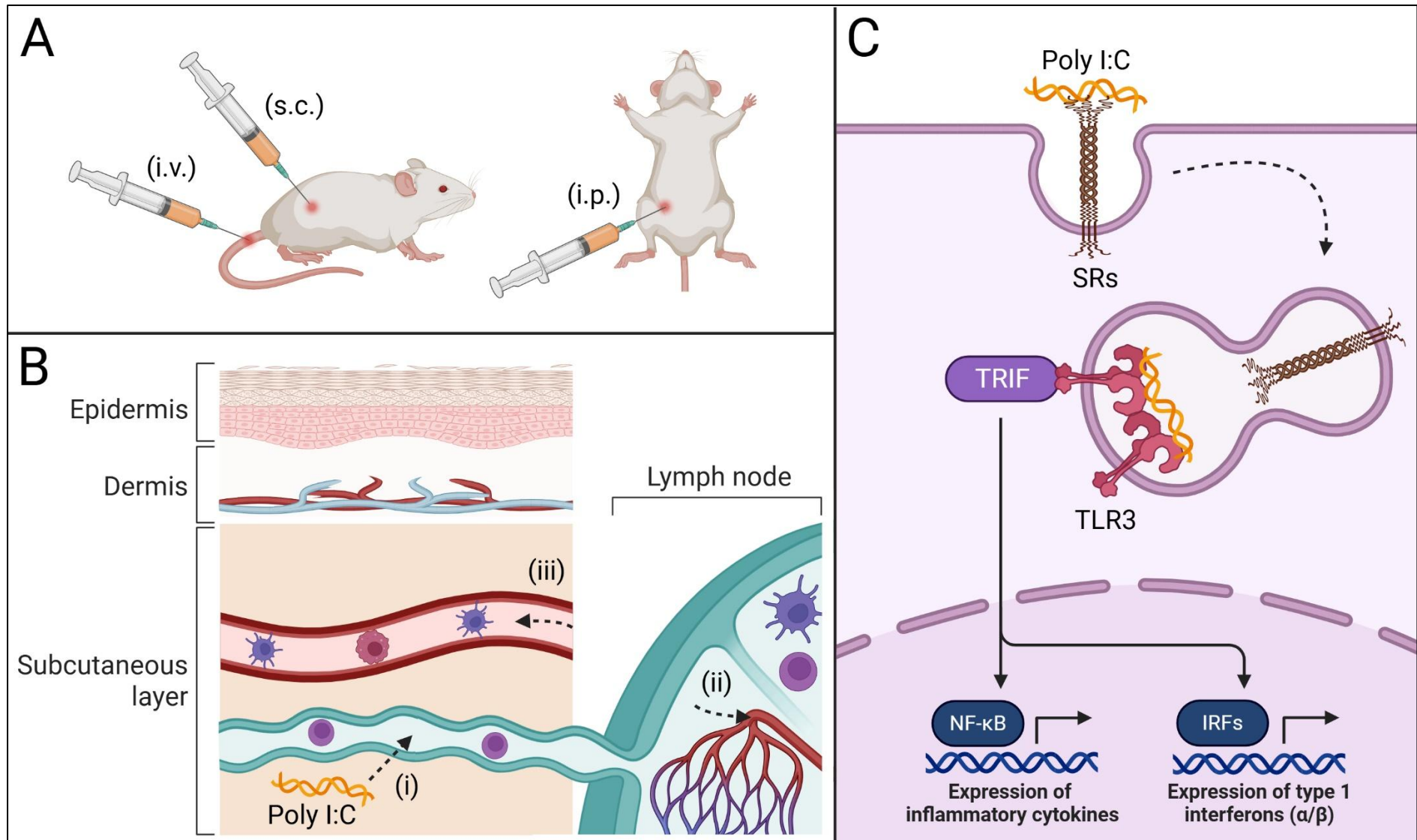
It should be noted that not all publications replicate the specific MIA-associated behavioural alterations reported above, with suggestions that variation in model establishment, MIA cytokine responses, additional risk factor modelling, or mediating factors within the laboratory environment may influence offspring outcomes [155, 156]. Offspring sex has also been repeatedly reported as a mediating factor in behavioural outcomes, with comparative studies finding sex-specific differences in measures across RDoC domains including USVs, gait, and reversal learning following MIA [139, 142, 148, 154]. This variation in behavioural outcomes additionally reflects the

heterogeneity of NDD/NPD presentation, particularly the varied phenotypes and sex-specific diagnostic differences that may be observed within a single diagnosis such as ASD. Rodent MIA models are allowing deeper exploration of the factors that contribute to behavioural outcomes following MIA, including outlining the specific inflammatory markers implicated in the MIA response and demonstrating which developmental systems may be vulnerable to MIA dysregulation at different stages of development.

1.2.1.2. Poly I:C-induced Immune Responses: Construct Validity

Poly I:C-induced MIA models are typically established with a single injection to a pregnant dam on a known day of gestation, leading to a short-lived maternal immune response that is apparent by approximately three hours post-injection and abates within 24 hours [157]. This reflects an approximate 0.5-2 week time frame in human gestation, relative to the 21 day rat gestational period. Intravenous (i.v.) or intraperitoneal (i.p.) injection are the most common route of poly I:C administration, although subcutaneous (s.c.) injection models are also reported [158] (Figure 1.1). This preference is likely due to early studies comparing administration routes finding inconsistent fever and sickness responses following subcutaneous injection, however studies finding no observable response to poly I:C were carried out at significantly lower doses (<1 mg/kg) than currently used to induce immune responses within the MIA literature (typically 4-20 mg/kg) [155, 159-162].

While i.v. administration directly supplies poly I:C into the maternal bloodstream, poly I:C administration via i.p. or s.c. injection is unlikely to be readily taken up by blood capillaries due to its large size (an 100bp segment, considered to be in the low molecular weight range for poly I:C, being ~60 kDA) [163-165]. Poly I:C by these routes must first enter lymphatic circulation and may subsequently reach the maternal bloodstream through capillaries within the lymph node (Figure 1.1B). Numerous immune cells along this route may recognise poly I:C as a viral pathogen-associated molecular pattern (PAMP), including T cells, dendritic cells, and macrophages [166]. Recognition initially occurs through scavenger receptors (SRs) on the outer cell membrane that detect a wide range of PAMPs, leading to poly I:C being endocytosed and released into endolysosomes that express toll-like receptor 3 (TLR3) [167-169]. TLR3 specifically recognises dsRNA and binds poly I:C, subsequently activating



signalling cascades that lead to cytokine expression that have been thoroughly reviewed in existing literature (Figure 1.1C) [169-171]. Interestingly, poly I:C has been found to recruit additional TLR3 units in a length dependent manner, leading to an amplified signalling cascade and additional cytokine expression with higher molecular weight segments [172-174].

As previously discussed, a range of inflammatory markers have been implicated in offspring NDD/NPD risk including IL-1 β , IL-6, IL-8, CRP, and TNF α (see 1.1.4.1). Increases of IL-1 β , IL-6, and TNF α in maternal serum/plasma in particular have been reflected in poly I:C-induced MIA models [157]. As reviewed by Solek et al, increased circulating cytokines are often observed between 2-6 hours post-injection, with factors including dosage, poly I:C molecular weight, administration route, and animal strain likely to affect the timing and magnitude of the response [155, 157, 175].

Figure 1.1: Overview of the absorption and recognition of poly I:C in rats. A) Illustration of the commonly used administration routes; intravenous (i.v.), subcutaneous (s.c.) and intraperitoneal (i.p.). B) The likely absorption route of poly I:C from the subcutaneous tissue layer. Injected poly I:C is taken up by lymphatic vessels (i), transferred into maternal circulation through blood capillaries within the lymph node (ii), and travels through the maternal bloodstream (iii). Immune cells such as T cells, dendritic cells, and macrophages within lymphatic and blood circulation (shown in shades of purple/red) may take up and respond to poly I:C through TLR3 signalling. C) Poly I:C is detected by scavenger receptors (SRs) and endocytosed. This endosome merges with an endolysosome containing TLR3, which binds poly I:C in a length-dependent manner and initiates primarily the TIR domain-containing adaptor protein inducing interferon beta/Nuclear factor kappa-light-chain-enhancer of activated B cells (TRIF/NF- κ B) signalling pathway to activate expression of proinflammatory cytokines. TLR3 also specifically upregulates interferon expression through activation of interferon regulatory factors (IRFs). Diagram not to scale. Created in <https://BioRender.com>.

Of particular note is IL-6, which has received significant attention in the literature and is considered to be a major driver of the rodent MIA response [176]. Animal models of acute maternal IL-6 administration, interruption of IL-6 signalling by use of anti-IL-6 antibodies or IL-6 knockout mice, and chronic IL-6 upregulation in offspring have all demonstrated the importance of IL-6 signalling in MIA-associated outcomes [176, 177]. This may be partially due to the role of IL-6 in the downstream production of other inflammatory markers such as CRP and IL-17 α through the maturation of specific naive T cells into Th17 cells [178, 179]. IL-17 α increases that have been discovered in rodent MIA models have been complemented by human studies highlighting changes in the regulation of IL-17 α and IL17a receptors in ASD [180, 181].

MIA models have also allowed exploration of inflammation within foetal tissues such as the amniotic fluid, placenta, and most commonly the foetal brain. IL-1 β , IL-6, IL-17 α , and TNF α are among the most commonly investigated inflammatory markers increased in foetal tissues [158, 182]. Conflicting findings are apparent across studies however, with a meta-analysis by Hameete et al. finding a significant increase only in foetal brain IL-6 following poly I:C [158]. Evidence is also divided regarding the origin of these markers. Dahlgren et al. for example utilised radiolabelled ^{125}I -IL-6 administered to a pregnant dam between E11-13 and found accumulation of ^{125}I -IL-6 in fetuses and amniotic fluid, illustrating cytokine transfer from maternal circulation *in vivo* [183]. Wu et al by contrast found that poly I:C on E12.5 increased mRNA and protein expression of IL-6 within the foetal brain, and this change was lost in mice lacking IL-6 receptor- α at the foetal-maternal interface, suggesting maternal IL-6 does not enter foetal circulation and instead triggers new inflammatory processes within foetal tissues [184]. Interestingly, TLR3 is also expressed in trophoblast cells within both rodents and humans, which could potentially result in direct poly I:C-induced stimulation of placental TLR3 signalling at the foetal-maternal interface [185-187].

1.2.2. Comparative Gestational Development

Even though rat models are well established in biological research, translatability is an ongoing challenge across non-human research that should be carefully considered. This is particularly important when utilising a gestational rat model, as the tissues within the foeto-placental unit are undergoing rapid and complex development. Understanding the gestational similarities and differences between humans and rodents is therefore critical for drawing informed hypotheses and highlighting reasonable implications that may be relevant for human development.

The overall foetoplacental unit of both humans and rats share a similar mammalian structure – the developing foetus is contained within an individual amniotic sac, surrounded by amniotic fluid, and connected to an individual placenta by the umbilical cord (Figure 1.2). Human twins may be an exception to this rule, for instance monozygotic twins that have individual amniotic sacs but share a placenta. Dizygotic twins however concurrently develop within individual foetoplacental units, which reflects the multiple individual foetoplacental units found along the rodent uterine horns during gestation of a litter (Figure 1.2A). These distinct units support the individual development of each pup, and are subject to individual genotype or sex-specific developmental differences that vary across pups [188]. Pups within a litter however may affect each other; testosterone administered to one foetoplacental unit has been shown to travel across neighbouring units in the uterine horn, and position relative to male littermates has even been implicated in moderating brain TNF α concentration and postnatal pre-pulse inhibition changes following MIA [189, 190]. Additionally, all pups share the same maternal bloodstream for support and are therefore subject to the same maternal health conditions. As a result, rodent pups within a litter have a mixture of similarities and differences that require careful statistical balancing in MIA models due to a lack of biological independence [155, 191]. Human maternal infection research does not have this particular challenge, as study cohorts are typically heavily comprised of or restricted to singleton pregnancies that each have independent gestational environments [108, 118].

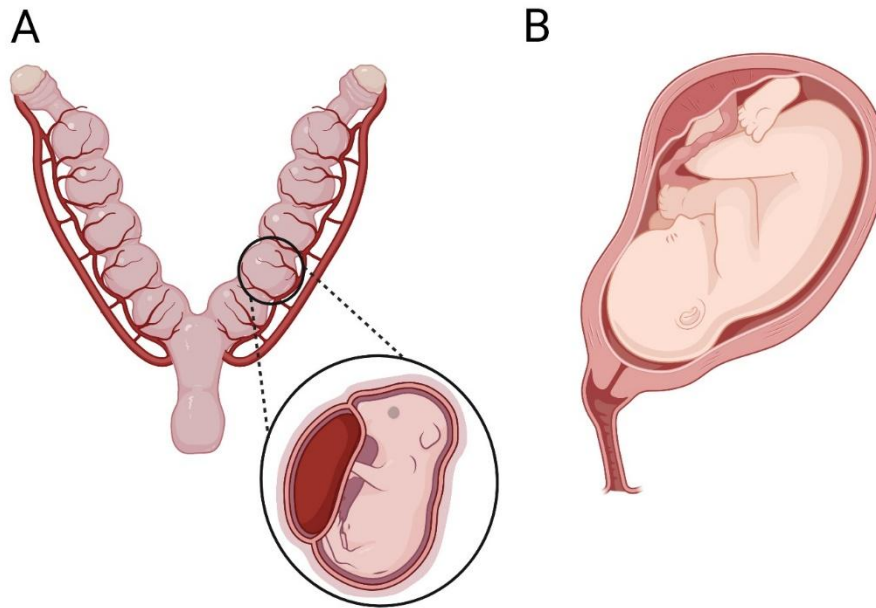
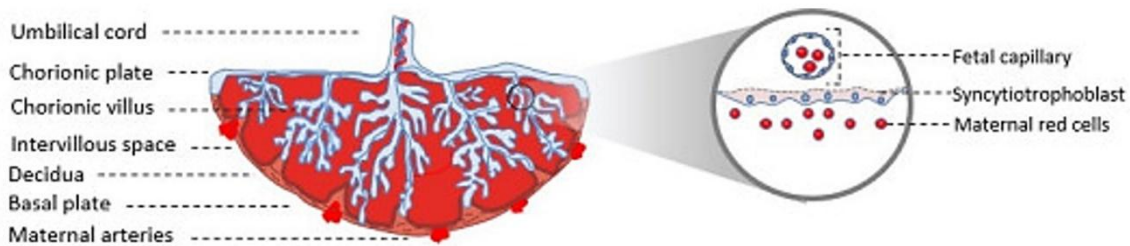


Figure 1.2: General view of the rodent (A) and human (B) uterus during pregnancy. Created in <https://BioRender.com>.

1.2.2.1. Placental Development

The placenta is one of the most complex and rapidly changing organs in both the human and rodent body. It is formed from both maternal and foetal cell lineages, acting as the interface that controls transport of essential resources for foetal development. Both humans and rats have a haemochorial placenta, with only a thin cell layer separating maternal and foetal circulation (Figure 1.3) [192]. Notably, the human placenta is haemomonochorial (containing one cell layer between circulatory systems) while the rat placenta is haemotrichorial (containing three layers). These layers form the basis of the human chorionic villi and rat labyrinth zone, which both act as the primary sites of foetal-maternal transfer and are surrounded in maternal blood (Figure 1.3). The junctional zone (also known as the basal zone) in rodents is considered analogous to the basal plate in humans, both implicated in the production of placental hormones and glycogen storage, and borders the maternally-derived decidua as the site of implantation.

A) Human Placenta



B) Rat and Mouse Placenta

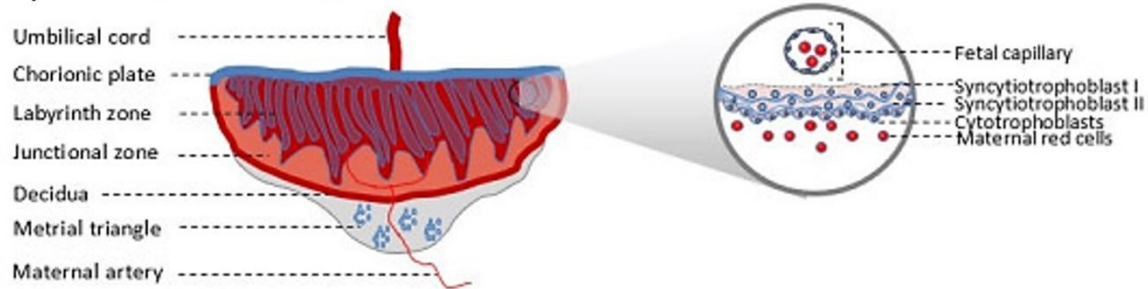


Figure 1.3: Structural comparison of the human (A) and rodent (B) haemochorial placenta. Adapted from Aguilera et al. 2022 under the Creative Commons Attribution License [193].

In the context of the MIA model, the foetally-derived labyrinth zone and syncytiotrophoblast/cytotrophoblast layers are a crucial site where maternal inflammatory cytokines may either damage the foetal-maternal interface or induce signalling changes that alter foeto-placental development. Although the rat placenta has a more complex barrier at the foetal-maternal interface with additional cell layers than the human placenta, a range of studies have suggested similar responses to maternal inflammation across humans and rats. This includes altered glucose transport, as well as changes to amino acid transport systems that may have implications for processes such as foetoplacental 5-HT synthesis [182].

Importantly for MIA models, rat placental tissue development over the course of gestation is fairly well-characterised (Figure 1.4) [192]. This allows MIA models to incorporate a fuller understanding of developmental stage in the analysis of morphological and functional changes in the placenta. For example, Kowash found a significant decrease in overall placental weight at E16 following poly I:C administration on E15, that was accompanied by structural and transporter expression changes (but

not volume changes) in the labyrinth zone. This aligns with poly I:C administration occurring after the peak proliferative period of the labyrinth zone, suggesting more subtle changes to the foetal-maternal interface than overall labyrinth area after significant expansion has already occurred [194]. Monteiro et al. by contrast administered poly I:C at E13.5 during high proliferation, and found decreased labyrinth volume [195]. While not investigated in MIA models as of yet, sex differences over gestational stage have also been identified, where relatively decreased volume of the labyrinth zone was identified in females at E15 that equalised with male relative labyrinth volume at E20 [188].

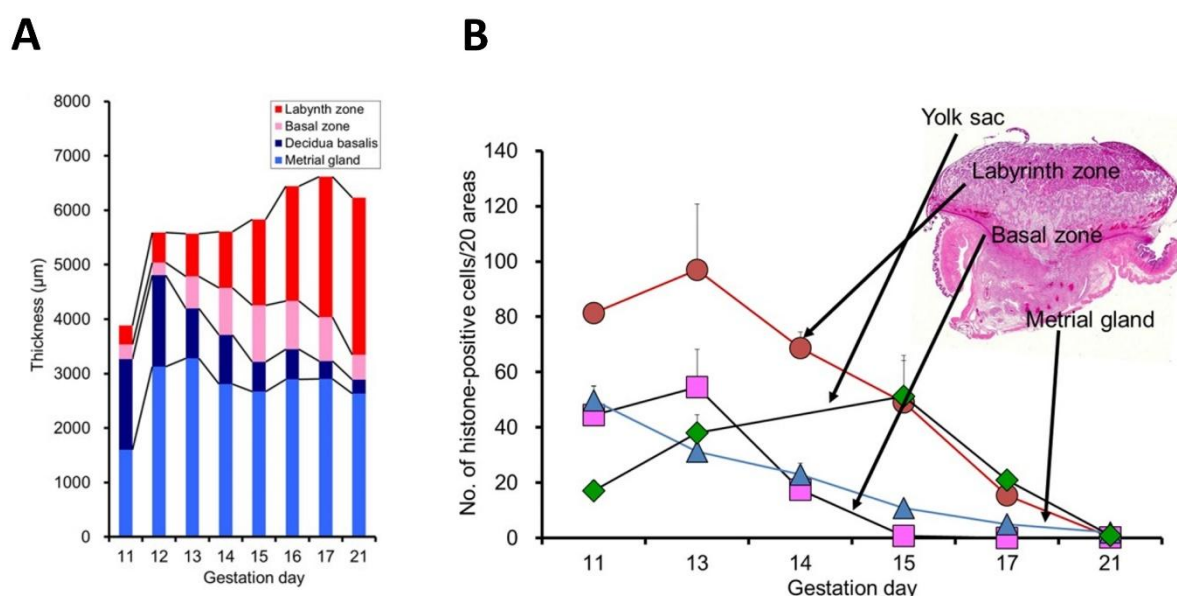


Figure 1.4: Development of the rat placenta across gestation. A) Thickness of each placental layer from gestational/embryonic day 11-21 in Wistar Han rats. B) Cell proliferation activity within each placental layer during pregnancy. Error bars represent standard deviation. Adapted from Furukawa et al. 2018 under the Creative Commons Attribution License [192].

Significant ethical limitations prevent free and direct analysis of the human placenta by contrast, limiting our understanding of its overall developmental timeline past the initial implantation and expansion stage in the first few weeks of gestation [196]. Current work is limited to studies with access to full term placentas, with first or second trimester tissue typically sourced from excess medical sampling or elective

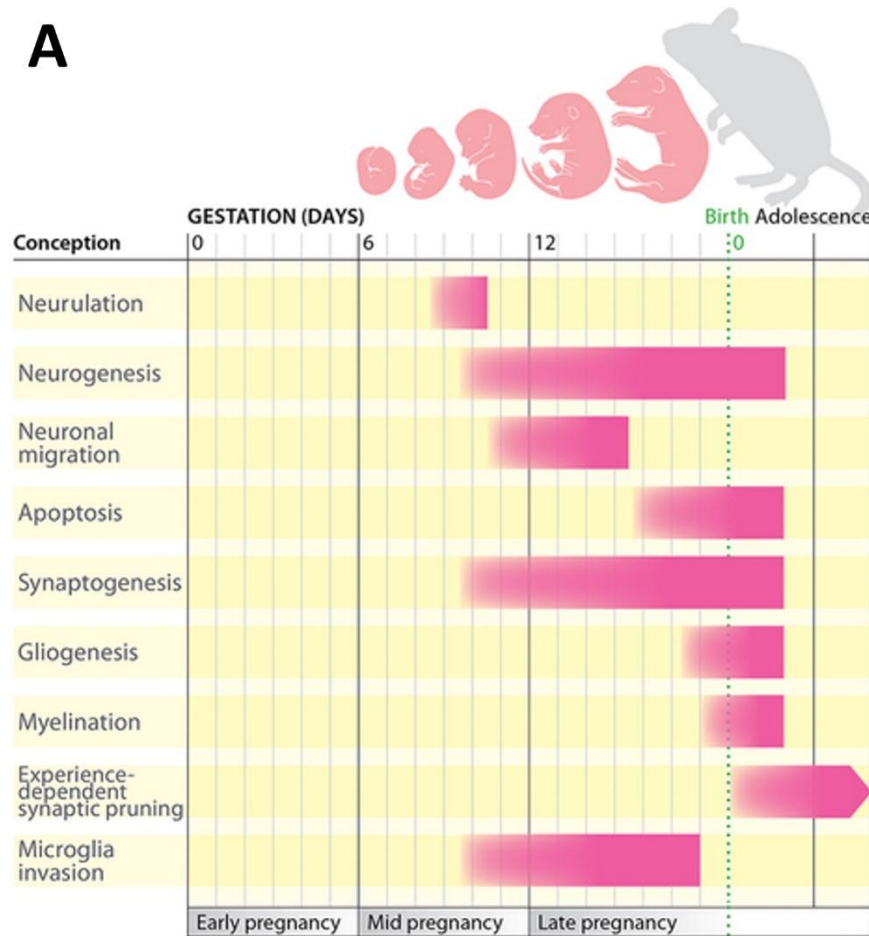
terminations; this provides limited global developmental information however allows useful windows of insight into specific tissue layers mid-gestation [197, 198]. Chorionic villi in the human placenta for example are established early in placental development across the first trimester of pregnancy [198], corresponding with the rat labyrinth zone having high proliferation at mid-gestation with the establishment of the rat placenta (Figure 1.4). It should be noted that prior to E11, the rat foetus utilises the yolk sac as a transient placenta, which is not observed in human development [192]. Based on this understanding, MIA models investigating development from E11 most closely resemble the early human foetal-maternal interface.

1.2.2.2. Neurodevelopment

Large comparative analyses using developmental milestones have been instrumental in providing researchers with resources to compare neurodevelopment across model organisms [199]. Using the Translating Time model developed from the comparative analyses from Findlay, Darlington, and colleagues for example, rat prenatal neurodevelopment between E12-21 can be roughly aligned with weeks 5-17 of human neurodevelopment [200]. As displayed in Figure 1.5, this period is characterised by critical neurogenesis and neuronal migration processes in both rodents and humans, patterning developing brain regions with crucial and interconnected neurotransmitter systems [201, 202]. Both synaptogenesis and microglial invasion processes begin relatively earlier in rodent development than in humans (likely associated with accommodating a shorter gestational window), however extend to similar end points relative to birth [201]. Species differences in prenatal synaptogenesis and microglial development are particularly relevant to MIA models that often assess altered synaptic connectivity or foetal neuroinflammation, and may limit the translatability of some findings.

Similarly to human placental development, direct evidence of human neurodevelopment is limited [199, 202]. Early foetal development is the most well characterised prenatal period, exemplified with resources such as the Carnegie Stages that illustrate the first 8 weeks of human morphological development [199]. Understanding of the general neurodevelopmental processes after this period (demonstrated in Figure 1.5) was historically estimated from conserved processes in model organisms such as rodents and primates, typically supported by histological

A



B

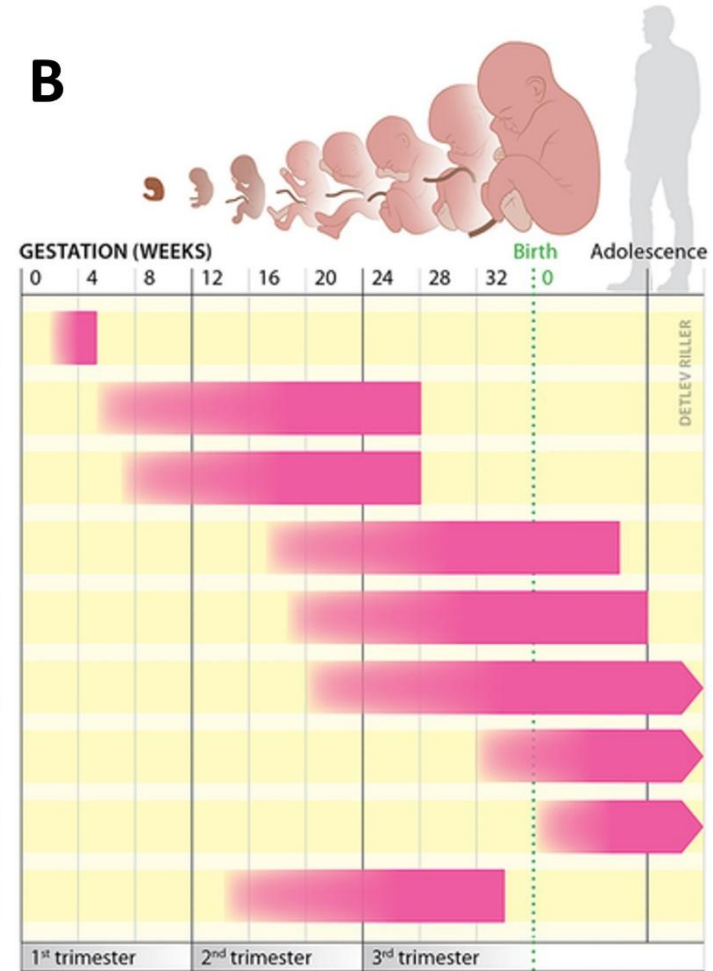


Figure 1.5: Comparative timing of neurodevelopmental processes across rodent (A) and human (B) gestation. Adapted from Schepanski et al. 2018 under the Creative Commons Attribution License [201]. While mouse gestation is approximately two days shorter than rat gestation (19 and 21 days respectively) and is depicted in A, rodent fetuses past mouse E10/ rat E12 follow a highly similar neurodevelopmental pattern [200].

analyses using a limited number of human foetal tissue samples [199, 203, 204]. This has been further improved by relatively recent advances in neuroimaging techniques that allow less-invasive assessment of structural development within the brain, however much of our understanding of human foetal brain development is reliant on small-scale studies prior to the 1990's [204, 205].

It should also be noted that specific brain regions develop at different rates across prenatal and postnatal life, contributing to the extended neurogenesis and synaptogenesis window across both species [199]. Regional development should therefore be specifically considered when relevant for evaluating the utility and findings from an MIA model. As an example, peak synaptogenesis in the somatosensory cortex begins before birth in humans, however this is primarily a postnatal process in rodents which are born at an earlier stage of development with closed eyelids and pinned ears [199, 206]. In contrast, the development of the 5-HT-producing raphe nuclei occurs at a similar stage in human and rodent development [207]. Serotonergic neurons have been identified in the developing human medulla (the location of caudal raphe nuclei) as early as week 7 of human gestation, while extension of serotonergic neurons into the cerebrum from the rostral raphe nuclei in the midbrain/pons (the location of the dorsal raphe nucleus) is apparent from approximately week 13 [204, 208]. As mentioned previously, this period generally aligns with neurodevelopment in the second half of rodent gestation [200]. In rats, the birth of serotonergic neurons and their extension from the dorsal raphe nucleus is evident in this analogous period, typically demonstrated between E12-17 [209, 210]. Based on these comparable developmental timelines, findings from MIA models implicating changes in prenatal serotonergic activity may have important relevance for human neurodevelopment [200, 207].

1.3. MIA and Serotonergic System Interactions

In addition to MIA, the serotonergic system is directly implicated in many NDD/NPD related processes. As would therefore be expected, serotonergic dysregulation is commonly observed in NDD/NPDs. For example, serotonergic signalling is implicated in sensory-motor integration and overall motor activity; hyperserotonemia in ASD is associated with increased stereotyped behaviours in humans, which is supported by increased repetitive behaviours in mouse models with hyperserotonemia from ASD-linked mutations in the SERT-encoding gene *Slc6a4* [211-213]. 5-HT also plays a role in regulating mood and stress responses, with altered activity through 5-HT receptors including the 5-HT_{1A}, 5HT_{1B}, 5-HT_{2C}, and 5-HT₄ receptors being implicated in MDD [214]. Antidepressant treatment has been demonstrated to modulate signalling and expression of these receptors to influence the regulation of fear and anxiety responses [215-217]. Serotonergic activity even extends to executive function, impulse control, and working memory processes through modulation of synaptic plasticity in the prefrontal cortex and corticolimbic pathways [218-220]. While serotonergic signalling is implicated across postnatal processes and symptoms, the importance of serotonergic signalling also extends to the prenatal period, where 5-HT has additional roles in regulating foetal development. This includes influencing nutrient availability, blood circulation, and the morphology and connectivity of the developing foetal brain.

Given the role of 5-HT in NDD/NPDs and neurodevelopment in general, it is possible that specific genetic or environmental changes in serotonergic signalling may combine with MIA in a multiple hit model to drive NDD/NPD outcomes in offspring. Previous work in our laboratory explored this possibility by assessing pup communication and juvenile hippocampal expression of synaptic plasticity related genes in the Wistar SERT knockout rat model following poly I:C-induced MIA (see Appendix A for drafted publication) [221]. MIA-SERT genotype interactions were observed in both tests, with MIA resulting in reduced call complexity in heterozygous males and reduced plasticity-associated mRNA expression in wildtype offspring. Genetic reduction of SERT expression was not simply associated with increased susceptibility to MIA, suggesting differences in SERT expression may contribute to the variability in MIA and NDD-associated outcomes.

Rather than focusing on the addition of separate MIA and 5-HT related risk factors, it is also possible that MIA leads to changes to serotonergic signalling. Firstly, there is extensive evidence implicating 5-HT in inflammatory and anti-inflammatory processes; immune cells throughout the body and brain express SERT and a range of 5-HT receptors, resulting in 5-HT signalling processes that influence activation states and subsequent activity [222]. Inflammation may also influence serotonergic activity in neuronal cells, particularly through regulation of SERT expression and activity leading to region-specific 5-HT signalling increases [222, 223]. MIA models have reflected these findings, demonstrating changes such as altered epigenetic SERT regulation and 5-HT abundance in the mouse hippocampus following gestational poly I:C exposure [224-226]. Most published research examining the relationship between gestational MIA and serotonergic signalling however has focused on postnatal offspring/tissues, with limited research investigating relatively acute interactions during the prenatal period. The existing prenatal research indicates that MIA can induce acute dysregulation of 5-HT signalling and production, however the extent of these changes across gestational ages and tissues remains largely unknown [227].

1.3.1. Gestational Changes to Serotonergic Activity

1.3.1.1. Nutrition and Overall Foetal Growth

The relationship between 5-HT and MIA is implicated in a number of pregnancy complications related to the health and development of the foetal brain. Risk of a number of conditions such as preeclampsia, gestational diabetes mellitus (GDM), intrauterine growth restriction (IUGR), and preterm birth (PTB) is increased following a maternal infection and/or associated with maternal inflammation [228-232]. Similarly to the overall NDD/NPD risk following maternal infections, no specific infection is required for increased risk; Yefet for example demonstrated GDM risk is increased following UTIs, bacterial infections, and Covid-19 [233]. In turn, existence of these pregnancy complications can confer an increased risk for NDD/NPD outcomes [234, 235]. The mechanisms implicated in these conditions involve impaired maternal metabolic support or immune tolerance that are mediated by changes to placental function/stress, often driven or accompanied by changes in the serotonergic system.

Changes to placental SERT are common among these conditions. PTB for instance is associated with increased placental *SERT* expression, with the strongest association demonstrated in pregnancies with increased inflammatory markers such as IL-6 in the amniotic fluid and maternal CRP [236]. The use of SSRIs during pregnancy also increases risk of PTB, however whether risk may be conferred through maternal or foetoplacental SERT activity is unclear [237]. In GDM and IUGR placental *SERT* expression is typically decreased, with work in human GDM placentas also revealing epigenetic downregulation through increased DNA methylation at the promoter of the *SLC6A4* gene that encodes *SERT* [198, 238, 239].

Maternal hyperserotonaemia is also commonly reported in GDM and preeclampsia. In GDM, 5-HT has an inhibitory effect on insulin and glucagon release, leading to significant challenges regulating blood glucose and therefore supplying consistent and adequate glucose to the developing foetus [240]. Cultured human trophoblasts treated with poly I:C have demonstrated impaired glucose uptake and insulin signalling, suggesting dysregulated glucose management is common to both GDM and placentas directly exposed to MIA [241]. Hyperserotonaemia in preeclampsia on the other hand is typically associated with maternal hypertension and increased vasoconstriction [242]. The placenta in preeclampsia exhibits an increased concentration of 5-HT as well as increased sensitivity to 5-HT mediated vasoconstriction, contributing to poor placental circulation and risk of hypoxia [242, 243]. In contrast to the heavier birthweights seen in GDM, preeclampsia is often linked to IUGR and a reduced foetal weight. MIA models often observe temporary maternal weight loss, as well as foetal or placental weight decreases, following poly I:C administration, further emphasising IUGR as a potential comparison of interest [244]. Interestingly, IUGR exhibits decreased placental *SERT* expression similarly to GDM, however placental 5-HT production and concentration in IUGR is increased [198]. Work by Ahmadzadeh has also demonstrated IUGR has significant effects on the developing foetal brainstem and the serotonergic raphe nuclei, with reduced 5-HT and increased inflammation in the medullary raphe nuclei [245]. While this study only examined the raphe nuclei that extend down toward the spinal cord, this may encourage future exploration of the relationship between IUGR and inflammation of the rostral raphe nuclei that innervate the brain.

1.3.1.2. Serotonergic Neuron Development

As mentioned in 1.2.2.2, serotonergic neurons have been detected in the developing dorsal raphe as early as E12 [209, 210]. Axons begin to innervate the foetal rat brain over the following days, identifiable in the cortex at ~E16-18 [210]. Direct research assessing the role of MIA in serotonergic neuron development is limited, however Ohkawara in 2015 identified an 11% increase in serotonergic neurons at E15 following poly I:C administration in pregnant rats at approximately E9.5 [246]. This increase was observed in serotonergic neurons originating in the rostral raphe nuclei specifically, including the dorsal raphe nucleus, which produce serotonergic neurons that extend from the midbrain to innervate regions across the cerebrum. Ohkawara did not assess whether the increased serotonergic neurons increased innervation of any specific brain regions. Their follow-up study in 2025 however found that poly I:C-exposed offspring were more susceptible to secondary postnatal immune challenge by LPS, and identified 1,700 differentially expressed transcripts in the midbrain/brainstem that demonstrate MIA-induced changes to immune system-related gene expression which persist postnatally [247]. An altered immune environment in the midbrain may have important implications for serotonergic activity, for example through proinflammatory cytokine upregulation (or anti-inflammatory downregulation) of *SERT* expression and activity [222, 248].

Additional factors associated with both MIA and serotonergic signalling provide further complexity to this relationship. A key example of this is brain-derived neurotrophic factor (BDNF), which promotes development and maturation of serotonergic neurons [249]. BDNF alters expression of key 5-HT receptors on maturing neurons such as the 5-HT_{2A} receptor, with BDNF knockout mouse models showing significantly reduced 5-HT_{2A} receptor expression in the dorsal raphe and prefrontal cortex and increased 5-HT_{2A} in the hippocampus [250, 251]. Other serotonergic targets show altered activity rather than expression BDNF-related changes, with both SERT and the inhibitory autoreceptor 5-HT_{1A} receptor showing reduced activity in BDNF heterozygote knockout mice [252-254]. Evidence of serotonergic dysregulation impacting BDNF is also available; increased hippocampal BDNF is associated with serotonergic hyperinnervation of the hippocampus in rodents with genetically reduced

brain serotonin production, while decreased BDNF in the hippocampus and prefrontal cortex is found in rodents with genetically reduced *SERT* expression [255, 256].

Interestingly, the 5-HT_{1A} receptor is not only influenced by BDNF, but is also associated with epigenetic modification of the *BDNF* promoter and mediating *BDNF* expression [257]. Accumulating evidence supports a loop where increased 5-HT signalling through the 5-HT_{1A} receptor leads to increased *BDNF* expression, which in turn leads to BDNF regulation of the serotonergic gene expression and a mature neuronal phenotype [249, 258]. NDD associated genes such as the methylation reader protein MECP2 (associated with Rett Syndrome and ASD primarily) may also contribute to this cycle in a complex fashion, with loss of rodent MECP2 independently associated with both decreased *BDNF* expression and increased 5-HT_{1A} signalling [259, 260]. 5-HT signalling may also influence MECP2 however, and their relationship may be brain region specific, as demonstrated by altered phosphorylation of MECP2 following SERT blockade specific to the nucleus accumbens [261]. MIA-induced disturbances in this complex system have significant consequences on the development of serotonergic neurons. Understanding of how foetal MECP2 specifically is affected by MIA is limited, however postnatal hypomethylation of rodent *Mecp2* has been observed following MIA at E9 [262]. MECP2 has also been implicated in anti-inflammatory responses within the placenta in pre-eclampsia models [263]. BDNF alterations in MIA models are more commonly reported, with reduced *BDNF* expression demonstrated in many tissues following poly I:C administration including the hippocampus, frontal cortex, and placenta [264-267].

1.3.1.3. General Neurodevelopment and Connectivity

During prenatal development, both prior to and during serotonergic innervation of the brain, 5-HT plays a role in regulating broader neurodevelopment [268, 269]. This includes influencing neuronal proliferation, migration, and maturation processes. Rodent models with reduced serotonin production for example display delayed cortical neuron maturation, linked to reduced proliferation in upper cortical layers [270, 271]. In contrast, 5-HT signalling promotes embryonic glutamatergic neuron survival in embryonic rat neuronal cultures [272]. Outside of its previously discussed effects in serotonergic neurons, BDNF is also implicated in the wider context of proliferation and survival of neurons in the embryonic brain [249]. 5-HT may also contribute to

neurogenesis and neuronal differentiation through modulation of transcription factors, such as Neurogenic differentiation factor 1 (NeuroD1) through microRNA miRNA-124 [273, 274]. NeuroD1 activity is implicated in NDDs and related comorbidities such as epilepsy, through altered glutamatergic and GABAergic neuron differentiation and postnatal neuronal plasticity [275]. NeuroD1 has also been implicated in reducing neuroinflammation through reversing astrocyte activation [276]. Limited research has assessed NeuroD1 in MIA models, although stress and environmental enrichment (both mediators of NDD/NPD-like outcomes in MIA) both influence NeuroD1 activity [277-279].

Importantly, immature neurons throughout the brain transiently express SERT, allowing temporary 5-HT uptake and signalling without requiring endogenous 5-HT production [268, 280]. 5-HT receptors may also be transiently expressed in neurons across brain regions, such as the 5-HT_{3A} ligand-gated ion channel; exogenous 5-HT concentration and transient 5-HT signalling in immature neurons are thought to influence growth cone dynamics and direct axonal and dendrite migration [268, 281]. Exogenous 5-HT concentration through 5-HT₆ is also associated with both increased and decreased migratory speed, exemplified in cortical glutamatergic and GABAergic neurons [282, 283]. Riccio et al. demonstrated increased 5-HT at approximately E17 in the mouse is associated with reduced GABAergic interneuron migration and altered innervation of cortical layers at the somatosensory cortex [282]. Interestingly, Vasistha et al. demonstrated that MIA can also induce altered GABAergic innervation of the somatosensory cortex, with reduced innervation at E14 and altered innervation of cortical layers at E17 [284].

Dendritic morphology and synaptic connectivity following neuronal migration are also influenced by 5-HT, contributing to the altered regional hyper- and hypo-connectivity seen in NDD/NPDs [268, 269]. SSRI exposure and altered 5-HT_{1A} expression for example has been shown to lead to dysregulated arborisation of dendritic spines from pyramidal neurons during prenatal development [268, 285]. Typical synaptogenesis in the hippocampus and prefrontal cortex have also been demonstrated to be mediated by serotonergic signalling [286-288]. 5-HT is also known to modulate the activity of microglia, immune cells that may be activated to induce both pro- and anti-inflammatory signalling pathways; 5-HT₂ receptor activity in microglia for example is required for typical neuronal circuit maturation [289-291]. Microglia have a significant

role in synaptic development and pruning throughout life, including the engulfing and remodelling of synapses in the developing brain [232, 292]. As may be expected, inflammatory microglial activity is heavily implicated in the foetal MIA response and associated with MIA-induced changes in synaptic connectivity/function [293, 294]. Interestingly, inflammatory microglial cytokine production and a route of microglia-induced dendrite damage are partially mediated by microglial expression of the 5-HT- and BDNF-associated *MECP2* [295, 296]. In addition to 5-HT modulation of microglia, microglial inflammation may also influence activity of the serotonergic system by rapidly increasing SERT expression in neighbouring neurons and reducing extracellular 5-HT availability; this has even been demonstrated in adult rodents following LPS injection, however has not been assessed in the foetal brain during MIA [297, 298]. Much of the information surrounding the bidirectional relationship between 5-HT and microglial activation comes from postnatal research, however these findings further suggest that the role of serotonergic signalling in gestational MIA models warrants direct investigation [223].

1.3.2. Gestational Changes to Serotonin Production and Availability

Changes to serotonergic neuron development and activity are closely intertwined with the availability of 5-HT in developing tissues. Both increases and decreases in 5-HT production have been demonstrated across the small pool of existing MIA literature, with methodological variation (e.g. MIA and tissue sampling time points) making it difficult to develop a clear picture of the relationship between MIA and 5-HT availability [227, 299-301]. This is exacerbated by an incomplete understanding of gestational 5-HT production and transport mechanisms across tissues during typical development, making it unclear which tissues and time points may be the most impacted by MIA exposure [302].

1.3.2.1. Tryptophan Metabolism following MIA

The general machinery required for 5-HT production is well established. 5-HT is derived from the amino acid tryptophan, which may be metabolised through either the 5-HT or kynurenine (KYN) pathways (Figure 1.6) [303]. Tryptophan hydroxylase (TPH) is considered to be the rate-limiting enzyme in 5-HT production, responsible for the

conversion of tryptophan to 5-hydroxytryptophan (5-HTP). Two major isoforms of TPH exist in humans and rodents; TPH1 contributes to production of peripheral/circulating 5-HT production in locations such as the pineal gland and enterochromaffin gut cells, whereas TPH2 is expressed within the raphe nuclei of the midbrain/medulla and produces 5-HT for the brain [304]. Following 5-HTP production, dopa decarboxylase (DDC) rapidly converts 5-HTP to 5-HT [303]. Finally, excess 5-HT may be metabolised to melatonin, or to 5-hydroxyindole acetic acid (5-HIAA) by monoamine oxidase enzymes. Monoamine oxidase A (MAOA) has a high affinity for 5-HT and is the most common isoform associated with 5-HT metabolism [305]. MAOA is found not only across foetal tissues, but within placental syncytiotrophoblasts and amniocytes within the amniotic fluid [306, 307]. 5-HIAA is a terminal product and is excreted through amniotic fluid/maternal circulation during gestation [302].

The rate of tryptophan metabolism overall, as well as the balance between 5-HT and KYN production, may have direct implications for MIA models. Approximately 90-95% of tryptophan metabolism occurs through the KYN pathway, with only approximately 5% through the 5-HT pathway [303]. Importantly, altered concentration of many KYN pathway metabolites have been implicated in MIA models. Increased KYN has been repeatedly observed in MIA studies, across a broad range of rodent tissues including maternal plasma, foetal plasma, foetal brain, and placenta [227, 308-310]. Increased KYN metabolite kynurenic acid (KA) has also been observed across rodent foetal tissues by Murakami et al and Saidi, with the placental KA increases reported by Saidi driven by female offspring [309, 310]. Down an alternate chain in KYN metabolism, increases in 3-hydroxykynurenine (3-HK) across foetal brain and placental tissue, as well as placental quinolinic acid (QA), have been further demonstrated in MIA models [308, 310]. As KA is considered neuroprotective, while 3-HK and QA are considered neurotoxic, the ratio of metabolites is also of interest. MacDowell et al. for example demonstrates an increased cytotoxicity ratio (QA/KA) in the frontal cortex of juvenile mouse offspring following gestational poly I:C administration, highlighting the potential for long-term alteration of tryptophan metabolism following MIA [299].

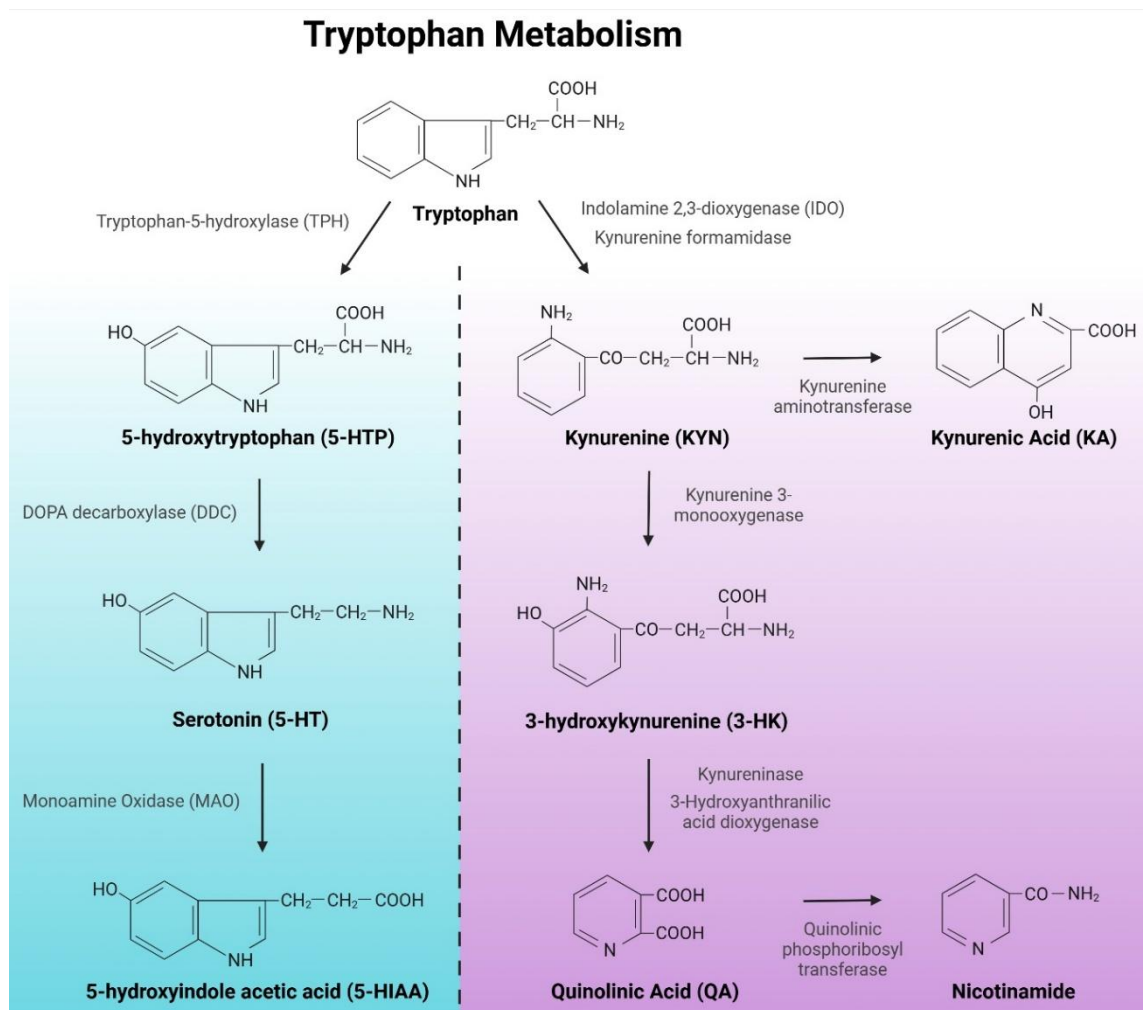


Figure 1.6: The serotonin and kynurenine production pathways. Pathways are simplified to include main metabolites and enzymes of interest in MIA models. Created in <https://BioRender.com>.

The increased KYN metabolic pathway activity typically seen following MIA may also contribute to a decrease in 5-HT production, as inflammatory cytokine-driven increases in indolamine 2,3-dioxygenase (IDO) enzyme activity alters the balance between the 5-HT and KYN metabolic pathways [303, 311]. MacDowell et al. also examined 5-HT metabolism and found decreased 5-HT and altered 5-HT receptor expression in the juvenile frontal cortex, suggesting dysregulation of the serotonergic signalling pathway [299]. Work by Williams et al. using a rabbit MIA model indicated that MIA was associated with both increased metabolism through the KYN pathway and reduced foetal 5-HIAA [312]. It has been suggested that increased IDO activity leads to a decrease in 5-HT production through tryptophan depletion, although conflicting evidence demonstrating increased or unaltered tryptophan availability

alongside KYN/5-HT changes are common [267, 299, 309]. Recent work by Abad et al. has provided additional evidence to support this theory. Human placental explants exposed to poly I:C demonstrated decreases in tryptophan, 5-HT, 5-HIAA and KA demonstrated alongside increased IDO activity and KYN production [300]. Intrauterine administration of LPS to individual foetoplacental units also results in increased KYN pathway metabolism, decreased placental 5-HT, and a cooccurring downregulation of TPH expression in the placenta and foetal brain [301].

Decreased production of 5-HT could have significant effects on the regulatory 5-HT activity within the developing foetal brain. It is also interesting to note that the 5-HT precursor 5-HTP has been demonstrated to reduce proinflammatory responses such as IL-6, IL-17, and TNF- α production in mouse splenocytes [313]. It is unclear whether 5-HTP may have a relevant anti-inflammatory effect within an MIA context, as 5-HTP is typically reported to be rapidly metabolised to 5-HT by DDC and has only rarely been reported in MIA research [227, 303, 314]. Decreased metabolism along the 5-HT pathway driven by a shift towards KYN production however would suggest a reduction in even transient 5-HTP concentrations in foetal tissues, which may contribute to foetal inflammation. Any effect of altered 5-HT metabolism however should also consider the gestational stage of the foetus, as the source of 5-HT available to the foetus changes across gestation [302, 315].

1.3.2.2. Serotonin Availability in Early and Mid-Gestation

The developing foetal brain requires 5-HT throughout gestation to support neurogenesis and brain patterning, however evidence from both rats and humans demonstrate that the foetus only produces endogenous 5-HT in late gestation after the dorsal raphe develops and serotonergic axons extend to innervate the cerebrum [315]. Although serotonergic neurons (characteristically expressing *TPH2* and other serotonergic genes) extend into the developing rat brain between E12-E16, this stage of development is heavily influenced by exogenous 5-HT [210, 227, 315]. Prior to foetal production in both human and rodent gestation, 5-HT is supplied to the developing foetus by either the maternal bloodstream, the placenta, or a combination of the two (Figure 1.7). It is currently debated whether maternally or placentally derived serotonin is the primary exogenous source of 5-HT in early foetal development [227, 302, 315, 316].

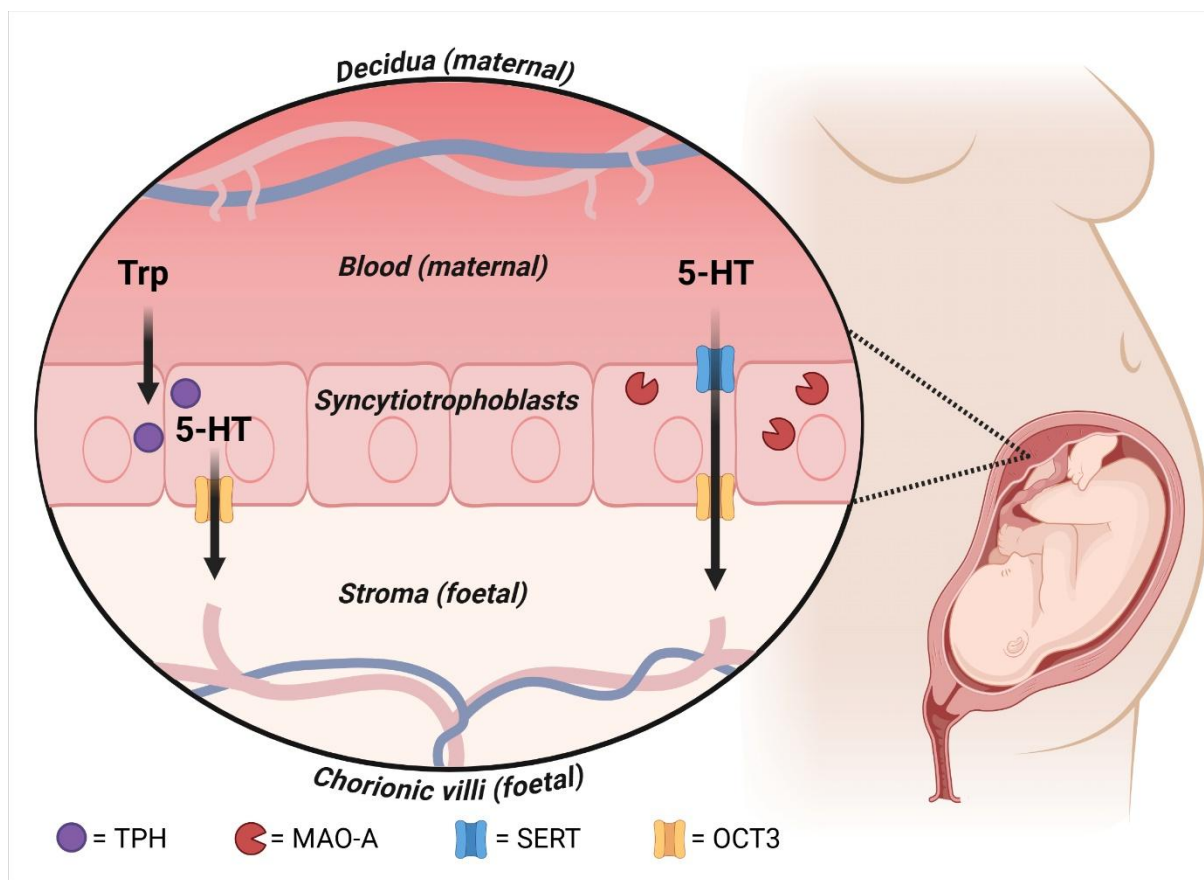


Figure 1.7: Potential serotonin sources during early foetal development. Tryptophan hydroxylase (TPH) expression suggests 5-HT can be produced by placental syncytiotrophoblasts at the foetal-maternal interface, utilising maternal tryptophan (Trp). Serotonin transporter (SERT) and organic cation transporter 3 (OCT3) expression indicates a transport pathway for maternal 5-HT across syncytiotrophoblast layer(s), however monoamine oxidase A (MAOA) expression may prevent significant maternal 5-HT transfer. Created in <https://BioRender.com>.

Studies that support maternal 5-HT as the primary reservoir for early foetal development have focused on the potential for placental uptake of 5-HT and alterations in maternal 5-HT homeostasis. Primary cultured human trophoblast cells exhibit the ability to take up 5-HT, suggesting 5-HT may be absorbed by placental cells during gestation [307]. This uptake was prevented by the SSRI citalopram, indicating SERT activity is required for external 5-HT to cross the placenta and SSRIs may impact 5-HT availability during pregnancy. Human placental explants have also

demonstrated the ability to take up exogenous 5-HT in a SERT-dependent manner, and transport 5-HT across cell types [316]. Interestingly, trophoblast cells are foetal in origin; alterations in foetal SERT expression/activity therefore may influence the availability of maternal 5-HT during early development. Maternal SERT activity has also been implicated in 5-HT availability in rodent models, with pups from SERT knockout dams showing reduced forebrain and placental 5-HT compared to pups from wildtype dams [315]. SERT knockout rats have negligible circulating 5-HT in whole blood, similarly to *Tph1* knockout dams; offspring from *Tph1* knockout dams exhibit altered neurodevelopment and reduced cortical neuronal proliferation, suggesting maternal 5-HT is required for typical brain development [270]. Similarly to SERT knockouts, SERT Ala56 mutant mice also show decreased 5-HT in the forebrain and placenta of developing pups, however interestingly these dams typically show increased 5-HT in peripheral blood [317]. These models together indicate that disrupted maternal 5-HT homeostasis may alter 5-HT availability to the foetus, however the direct mechanism of this is unclear.

Placental synthesis of 5-HT is a relatively recently popularised idea in the literature, with Bonnin and colleagues providing strong evidence for this process in 2011 [315]. *TPH1* was shown to be expressed in mice placental syncytiotrophoblasts between E10.5-E14.5, suggesting a capacity for 5-HT production. This is supported by additional independent work indicating 5-HT production machinery may be upregulated in the mouse placenta as early as E7.5-E9.5, and in the rat from E10-21 [318, 319]. Further ex vivo study using mouse placental perfusions revealed a sharp increase in 5-HT in foetal circulation following maternal tryptophan injection, but far less of an increase following maternal 5-HT injection [315]. Subsequent research laboratories have replicated similar findings, confirming that increased maternal tryptophan results in an increase in expression of placental 5-HT production machinery and an increase in foetal forebrain 5-HT [227, 320].

The existence and high expression of MAOA in syncytiotrophoblasts has been suggested as further evidence against maternal 5-HT being the primary pool of 5-HT, through degrading any transferred maternal 5-HT before it reaches foetal circulation [307, 315]. MAOA at the foetal-maternal interface however may also be involved in maintenance of 5-HT concentration within the foetoplacental unit, rather than solely preventing maternal 5-HT transfer [302, 321]. 5-HT in foetal circulation may be

absorbed by the placenta through organic cation transporter 3 (OCT3) in foetoplacental endothelial cells, and further transported through placental cell layers through a combination of OCT3 and SERT activity [322]. MAOA in syncytiotrophoblasts in this context may then metabolise excess 5-HT from foetal circulation before excretion through maternal circulation [321]. This demonstrates the potential for the placenta to play a bi-directional role in transporting 5-HT and maintaining foetal 5-HT concentration irrespective of production source. It should also be noted that OCT3 5-HT uptake was demonstrated to be far stronger in male fetuses than in females; as sex differences in NDD/NPDs are common, this further highlights gestational serotonin regulation as an interesting avenue for further investigation [322].

It should also be considered that a combination of maternal and placental 5-HT may contribute to early foetal development, for instance as compensatory mechanisms ensuring 5-HT availability. Key research supporting a primary maternal source of 5-HT also reports no detectable *TPH1* expression within placental trophoblasts in their model, suggesting maternal 5-HT transport may occur in the absence of placental production [316]. Interestingly, SERT knockout mice placentas were recently shown to express *TPH1* and at equivalent rates to wild-type controls, with an increased expression of the secondary 5-HT catalytic enzyme, *DDC* [323]. This suggests a potentially faster conversion to 5-HT from available 5-HTP in the absence of significant maternal 5-HT transport. Depletion of 5-HTP may leave SERT knockout placentas at higher risk to the effects of MIA, as 5-HTP may have an anti-inflammatory effect as mentioned previously, although it is unclear whether a significant pool of 5-HTP would be present during gestation to have this effect [314].

1.3.2.3. Serotonin Availability following MIA

Amid current MIA literature, there is limited evidence for both increases and decreases in 5-HT availability and signalling within the foetoplacental unit as a result of maternal infection (Table 1.2). This includes discrepancy on whether MIA may induce increases or decreases in foetoplacental 5-HT, as well as the expression of transporters and metabolism machinery that may contribute to the change in 5-HT availability. Notably, there is significant variability in the studies available; Species, gestational time points assessed, and MIA administration/inflammatory response in particular may all contribute to the varied changes in 5-HT observed. For example, Goeden and

Reference	MIA Details	Gestational Age	5-HT concentration	5-HT metabolism	5-HT transport
Abad, 2025 [301]	LPS; 10ug, intraamniotic	MIA at E18 (rat), sampling +24h	↓ Placental 5-HT	↓ Placental Tph1 ↓ Foetal Tph2 ↑ Foetal MaoA	↓ Placental Sert, Oct3 ↓ Foetal Sert, Oct3
Bondarenko, 2024 [319]	LPS; 25µg/kg, 250µg/kg, route unknown	MIA at E12-20 (rat), sampling +24h	↑ Placental 5-HT at E14 (25ug/kg LPS) ↓ Placental 5-HT at E12, 18, 20	n/a	n/a
Goeden, 2015 [227]	Poly I:C; 2mg/kg, i.p.	MIA at E12 (mouse), sampling +24h, +48h	At 48h: ↑ Placental 5-HT ↑ Foetal forebrain 5-HT	At 24h: ↑ Placental Tph1 At 48h: ↓ Placental Tph1 (mRNA) ↑ Placental Tph (activity)	n/a
Zengeler, 2023 [324]	Poly I:C; 20mg/kg, i.p.	MIA at E11-12 (mouse), sampling +3h	Poly I:C alone: ↑ Placental 5-HT Poly I:C + Fluoxetine: ↓ Placental 5-HT	Poly I:C + Fluoxetine: ↑ Placental Tph1 ↑ Foetal Tph1	Poly I:C alone: ↓ Placental Oct3 Poly I:C + Fluoxetine: ↑ Placental Oct3 ↓ Placental Sert ↑ Foetal Sert, Oct3

Table 1.2: Key studies assessing 5-HT availability in the fetoplacental unit in rodent MIA models. Arrows indicate an increase (↑) or decrease (↓) in protein concentration, mRNA expression, and/or activity. Specific outcome is indicated where conflicting outcomes were identified.

Zengeler both demonstrate increases in placental 5-HT concentration after poly I:C-induced MIA at ~E12 in mice, corresponding to a gestational age of ~E14 in rats [227, 324]. While Goeden demonstrates a transient increase in placental *Tph1* expression and an increase in placental Tph activity, Zengeler does not find a poly I:C-associated increase in placental Tph; instead this study demonstrated increased placental *Tph1* expression only in MIA offspring co-exposed to maternal fluoxetine treatment (which in direct contrast to MIA-only offspring, have a significantly reduced placental 5-HT concentration). Work by Bondarenko and Abad have further demonstrated that MIA exposure through LPS in late gestation likely results in reduced placental 5-HT [301, 319]. It is currently unclear what role placental 5-HT production may play in late foetal neurodevelopment, following the development of serotonergic neurons and beginning of foetal brain 5-HT production. There is notably little evidence exploring the effects of MIA across the developmental period where reliance on placental 5-HT decreases (~E15.5 in mice, predicted to be ~E17.5 in rats) [227, 315].

Based on current understanding of typical gestational 5-HT regulation and MIA models, multiple pathways of serotonergic dysregulation may occur in the foeto-placental unit following mid-gestational MIA (Figure 1.8). Firstly, inflammation at the placenta may influence 5-HT management at the maternal-foetal interface. This may include dysregulation of placental 5-HT production (Figure 1.8B), as is seen in early or late gestation [227, 301]. The presence and regulation of maternal 5-HT should also be considered; while it is still unclear how maternal 5-HT is typically regulated at the maternal-foetal interface, maternal 5-HT has been demonstrated to influence foetal development and be taken up by the placenta [316, 317]. Alterations in placental SERT, OCT3, and MAOA for example may result in impaired placental regulation of maternal 5-HT, allowing an influx of typically barred maternal 5-HT into the foetoplacental unit (Figure 1.8A). As previous work has demonstrated that MIA and altered 5-HT concentration in the developing brain may alter the outgrowth of serotonergic neurons, dysregulation of maternal or placental 5-HT may induce further changes to 5-HT production and regulation in the foetal brain (Figure 1.8B-C) [227, 246].

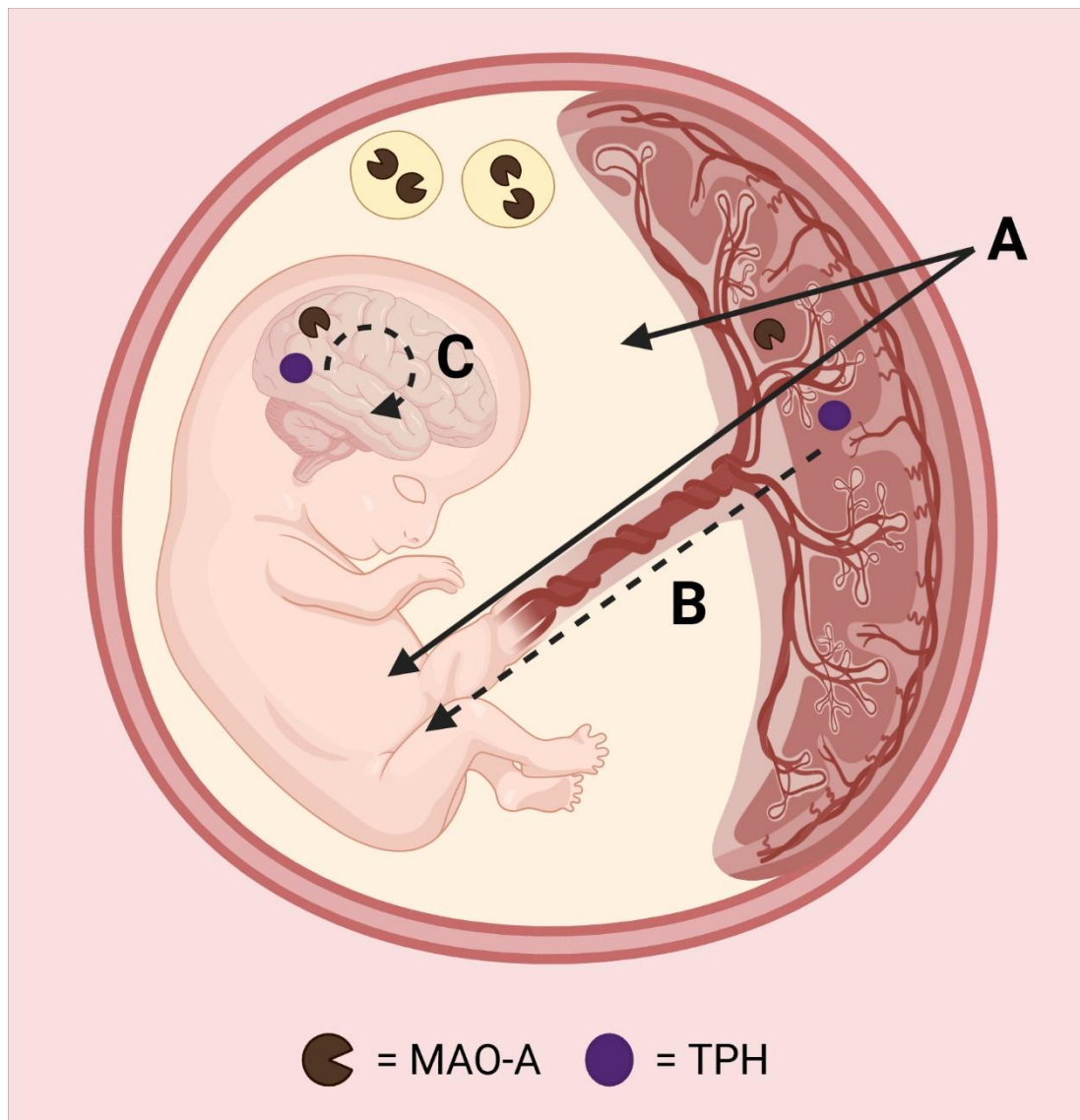


Figure 1.8: Suggested mechanisms by which mid-gestational MIA exposure may disrupt typical 5-HT regulation within the foetoplacental unit. Solid arrows indicate a hypothesised increase 5-HT availability, while dashed arrows indicate potentially increased or decreased 5-HT. Known locations of TPH (indicating capacity for 5-HT production) and MaoA (indicating capacity for 5-HT metabolism) are indicated across tissues. A) Damage to placental tissue or altered expression profiles may allow maternal 5-HT to bypass metabolism by MaoA in syncytiotrophoblasts, flooding foetal circulation and amniotic fluid. B) Placental production and transport of 5-HT may be altered by labyrinth and basal zone damage or signalling changes in response to immune activity. C) Increased foetal neuroimmune activity may delay or alter neuron growth and development, influencing production of 5-HT associated genes and 5-HT regulation. Created in <https://BioRender.com>.

1.4. The Current Thesis

Overall, this thesis aims to further our understanding of the prenatal interactions between MIA and the serotonergic system that may contribute to the incidence and/or symptomology of NDD/NPDs. As described throughout this introduction, there is strong evidence linking maternal infection/inflammation to NDD/NPD incidence and symptomology. Gestational serotonergic activity may contribute to this relationship; not only is 5-HT itself a key neurodevelopmental regulator and independently implicated in NDD/NPDs, but inflammation-associated dysregulation of serotonergic activity, production, and transport has been demonstrated in a growing number of publications. To further understand how MIA may influence the serotonergic system, one area that requires further research is mid-gestational serotonin availability in the foeto-placental unit.

1.4.1. Overview of the MIA in this Thesis

This thesis will explore interactions between MIA and the serotonergic system using a MIA model generated in Sprague Dawley rats. The methodology used in current MIA literature is highly variable, so the establishment of an MIA model requires the careful evaluation of variables including poly I:C administration route, dosage, and the gestational age of both poly I:C administration and tissue sampling [155, 157].

For the model used in this thesis, a dosage of 5 mg/kg via s.c. injection was selected to induce MIA. I.v. administration was unable to be carried out without inducing substantial maternal stress or anaesthesia, which itself may cause confounding pregnancy complications or additional maternal stress [325, 326]. I.p. administration was also selected against, as a lack of technical staff with confidence in training/administering i.p. injections increased the likelihood of poor injection technique, damaging of the uterine horns to reach the peritoneal cavity, and potential confounding dam stress. S.c. administration by contrast to i.v. and i.p. is a simple and rapid procedure, resulting in minimal dam stress and consistent technique.

The 5 mg/kg poly I:C dosage was primarily determined by previous work in our laboratory. While dosage generally ranges between 4-20 mg/kg in published literature, subcutaneous doses tend to fall between 2-10 mg/kg [160]. Factors such as poly I:C

molecular weight and animal caging systems may influence the MIA response and developmental outcomes, so varied doses across laboratory environments should be considered [156, 327, 328]. Rodents in our laboratory previously displayed fever responses 3-6 hours following a 5 mg/kg s.c. dose of poly I:C, therefore 5 mg/kg was chosen as the starting dosage for this study [221].

Poly I:C was administered on E12.5, which was selected based on translatability of tissues and serotonergic system development (see 1.2.2). As mentioned previously, foetal and placental development of rats at E12.5 is approximated human development late in trimester one. This specifically aligns key elements in the serotonergic system, including outgrowth of serotonergic neurons from the raphe nuclei in the developing foetal brain. At this stage of development, the labyrinth zone/chorionic villi are rapidly developing and establishing the foetal-maternal interface, with syncytiotrophoblasts in this region establishing a role in the regulation of 5-HT that includes expression of a combination of 5-HT metabolic and transport machinery.

To investigate changes to the serotonergic system in the foetoplacental unit, sampling was planned for E14.5 and E18.5 (two and six days following poly I:C administration respectively). Two time points were selected to provide multiple windows into development without unreasonably increasing the number of litters required for analysis (as whole litters must be sampled at the same time). As the transition between reliance on exogenous 5-HT to use of foetally produced, endogenous 5-HT occurs ~E15.5 in mice (likely translating to ~E17.5 in rats), these sampling points were designed to enable assessment of foetoplacental tissues before and after this expected transition [315]. Furthermore, sampling relatively late in gestation (E18.5) allowed for comparison with existing research in the rat fetus by Abad et al. (E19) [301].

1.4.2. Thesis Aims

Using the MIA model with the dosage and timing specifications described above, this thesis will focus on potential changes to serotonergic availability in the foetoplacental unit. Based on current literature, MIA may lead to a number of changes in foetoplacental serotonergic availability including an influx of maternal 5-HT, altered 5-HT production and/or transport in the placenta, and altered 5-HT production and/or transport within the foetal brain (Figure 1.8). The aims in this thesis therefore present a broad approach to exploring these possibilities, designed to demonstrate which pathway(s) are most likely to be impacted by mid-gestational MIA. As MIA literature is also highly variable, this thesis will continuously focus on developing transparent methodology that future researchers may reproduce in similar models to expand our understanding of serotonergic dysregulation following MIA. The specific experimental aims of this thesis are as follows:

- Validate the MIA model used in this thesis (*Chapter 3*).
- Assess 5-HT concentration in the amniotic fluid following MIA (*Chapter 4*).
- Measure mRNA expression of 5-HT production, transport, and associated neurodevelopmental genes in the foetoplacental unit following MIA (*Chapter 5*).
- Investigate epigenetic regulation of SERT following MIA (*Chapter 6*).

Chapter 2:

Final Methods

2.1. Animal Use

2.1.1. General Animal Information

Ethics approval for this project was provided by the Victoria University of Wellington Animal Ethics committee under the application number 29971. All animals had access to food and water ad libitum for the duration of the study, as well as cage enrichment in the form of a wooden chew block (with an opaque tunnel and mezzanine platform provided as further enrichment outside of the paired mating stage). Animals were kept on a reverse-light cycle (light from 7pm-7am) with stable room temperature ($21^{\circ}\text{C} \pm 2^{\circ}\text{C}$) and humidity (55-60%).

For the duration of this study, all animals were housed in polysulfone cages under an individually ventilated caging system. Housing consisted of 'small' or 'large' cage systems; small cages were only used for the temporary housing of breeding males, while large cages with or without mezzanine platform were used for the remainder of the study. Small cages were 35.6 cm long x 48.5 cm wide x 21.8 cm high, with a floor area of 870.9 square centimetres. Large cages were 38.9 cm long x 56.9 cm wide (front) x 26.2 cm high, with a floor area of 1181 or 1574 square cm dependent on mezzanine inclusion.

2.1.2.1. Total Animal Use

Overall, 58 breeding animals were utilised in this study (Table 2.1). Sixteen adult males were borrowed from the Sprague Dawley breeding colony for use in the timed mating protocol, and subsequently returned to the facility after use in this study. Forty-two dams were utilised overall and administered an injection of either poly I:C or saline vehicle, with 27 of these produced a litter with pups sampled in this study. Across all dams, a total of 15 were excluded; three dams were not used for breeding and were utilised as pilot animals for injection and blood sampling training, one dam was excluded due to producing a single pup rather than a litter that could be sampled appropriately, and the final 11 failed to produce a litter at all (see Chapter 3, Figure 3.2 for further assessment).

As tissue samples were collected at embryonic time points, all treated dams and generated offspring were sacrificed during this study. Tissue collection protocols

balanced ethical, time, and statistical concerns by sampling between 6-9 offspring from each litter for use in multiple subsequent experiments; this compromises between common approaches that sample 1-2 pups per litter (maximising litter number for statistical power at a high animal cost) or sampling of entire litters (maximising animal use but also maximising litter bias and false positive error within subsequent assays) within a reasonable time frame for this thesis [329]. In cases where litter size fell between 2-6, all available offspring were sampled. This approach created a tissue library for the work presented in this thesis and future studies.

Animal Type	E18.5 PI:C	E18.5 SAL	E14.5 PI:C	E14.5 SAL	Sub- total	Excluded PI:C	Excluded SAL	29971 Total
Adult Male	16				16	0		16
Adult Female	6	9	7	5	27	14	1	42
Offspring (sampled)	46	70	54	45	215	0	0	215
Offspring (unsampled)	37	46	33	22	138	1	0	139
Total	105	125	94	72	396	16		412

Table 2.1: Summary of animals used under ethics approval 29971. Totals include breeding animals and all offspring in the litters generated by breeding pairs discussed in this thesis. PI:C = poly I:C treatment group, SAL = saline treatment group.

2.1.2. Generation of an MIA Model

Naïve Sprague Dawley females of breeding age (between PND60-90) were paired with male Sprague-Dawley breeding stock (PND70+) in large housing (without mezzanine) with a slightly raised, custom-sized metal grate flooring insert in replacement of standard bedding (Figure 2.1). Once per day, cages were changed and inspected for vaginal sperm plugs to indicate successful mating. If a minimum of one plug was identified, the breeding pair would be separated to new home cages (with full bedding and enrichment) and the date would be designated day E0.5 of pregnancy. Lack of sperm plugs resulted in pairs remaining co-housed in fresh

breeding cages, for a maximum of 5 unsuccessful days, until separation and exclusion from the study if no plugs were identified.

Breeding pairs were managed in 4 cohorts across a 1.5 year period. The same researchers and support staff maintained treatment and sampling (Alex Lister with either Joyce Colussi-Mas or Mahima Yapa Nanayakkara), as well as dissection (led by Alex Lister and/or Dr Darren Day), across all animals to minimise variation in technique and animal stress.

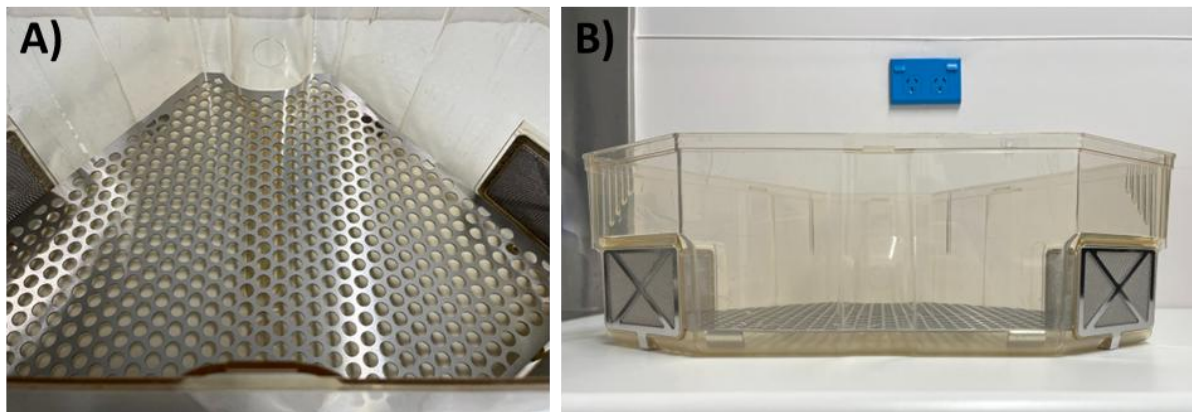


Figure 2.1: Metal insert flooring used for timed breeding protocol. Grate shown in large cages from overhead (A) and front-facing (B) viewpoints.

2.1.2.1. Maternal Treatments

Dams were assigned to a treatment group shortly after successful pairs were separated. Dam allocation to treatment was determined using the following pre-determined guidelines:

- a. Dams from the same litter ('sister' dams) were separated across treatments.
- b. If possible/relevant, dams paired with the same male were separated across both treatment groups.
- c. Remaining dams split across treatments at random. Cohorts one and two were allocated evenly across treatment, however cohorts three and four were predominately assigned to the poly IC treatment (~70%) to compensate for a low success rate in producing poly I:C litters (see Chapter 3, Figure 3.2).

A single treatment was administered as a subcutaneous injection to the flank on E12.5. For the MIA group, this consisted of room temperature polyinosinic:polycytidylic acid (poly I:C) made up to 5mg/mL in sterile 0.9% saline solution. Injection volume was 1mL/kg, resulting in a final dosage of 5 mg/kg. Control group dams received an equivalent 1mL/kg injection of the exact 0.9% saline solution used for poly I:C resuspension.

Poly I:C was purchased from Sigma-Aldrich (P1530-100MG, source #12190304, batch #0000067639), with the same bottle used across all MIA animals. Molecular weight of the poly I:C provided was assessed briefly by agarose gel electrophoresis, using the same method presented fully in section 2.5. The poly I:C used in this thesis has a rough length of 100-200 bp (Figure 2.2), which is considered to be low molecular weight based on the ranges for high (~1.5 – 8 kb) and low (~0.2 – 1 kb) molecular weight poly I:C listed for InvivoGen poly I:C products [330, 331] which are commonly used to compare immunological differences in molecular weight [156].

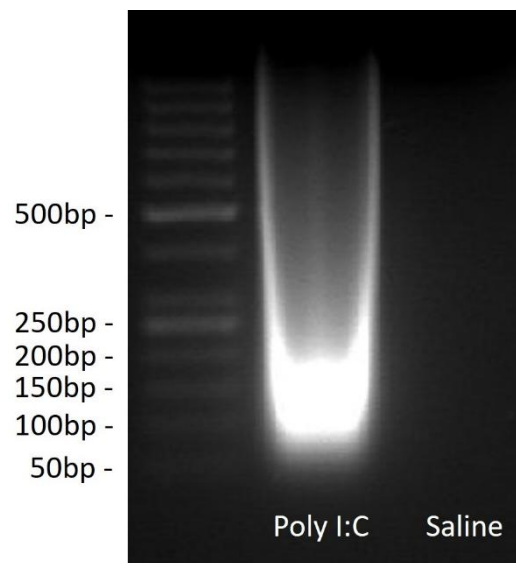


Figure 2.2: The poly I:C used in this thesis is of low molecular weight. Poly I:C is presented next to saline vehicle and GeneRuler 50 bp DNA ladder (#SM0371, ThermoFisher Scientific, USA).

2.1.2.2. Maternal Blood Sampling

A blood sampling protocol was developed for this study in conjunction with Dr Joyce Colussi Mas and Mahima Yapa Nanayakkara, with ethics approval under the first amendment of the ethics application for this project. This protocol was developed based on guidelines for saphenous vein sampling from the NC3Rs website [332].

This protocol requires two researchers: a handler and a sampler. Dams were removed from the home cage and restrained within a towel by the handler, with the base of one hind leg held firmly for stability and to promote bulging of the saphenous vein. The sampler identified the saphenous vein, then lightly swabbed the area with 70% ethanol before applying a thin layer of Vaseline with a cotton bud to improve blood droplet formation. A 23 G needle was then used to pierce the vein at a 90° angle, allowing a 20 – 200 µL volume of droplets to flow readily into a collection tube. Blood flow was halted by use of a gauze compress, and the dam was returned to the home cage and monitored until normal movement on the leg resumed (approximately 0-5 minutes). Blood samples were collected in 0.5mL serum collection tubes (#MIT100, Knight Benedikt, Australia), mixed well, and left at room temperature for approximately 1 hour. Tubes were then centrifuged at 10,000 g for 5 minutes, and the serum supernatant was moved to a new microcentrifuge tube and stored at -80°C.

2.1.3. Tissue Collection and Offspring Weight Assessment

Dams were euthanised by CO₂ asphyxiation and decapitation on either E14.5 or 18.5. The uterus of each dam was removed (Figure 2.3A) following euthanasia and placed in a dish of sterile PBS. The number of intact foetoplacental units (amniotic sacs containing a foetus and placenta) was recorded as litter size, before the uterine lining was removed and units were randomly selected from the PBS dish for sampling.

Whole foetoplacental units within the intact amniotic sac were rinsed in PBS and weighed in a dry weighboat (Figure 2.3B). Amniotic fluid (AF) was then extracted from the amniotic sac using a 28 G needle and syringe and stored in a clean microcentrifuge tube. The whole placenta and foetus were then removed from the amniotic sac, rinsed in PBS, and separately weighed in a dry weighboat. Each placenta was further dissected to isolate the foetally-derived placental tissue (fPLA) from maternally-

derived tissue (Figure 2.3C). Finally, the brain of the foetus was dissected from the body. E14.5 brains were extracted whole due to their small size (Figure 2.3D), while E18.5 brains were further dissected into frontal cortex (FC) and posterior brain (pBR) regions (Figure 2.3F). While frontal cortices were separated for future work, the remaining brain tissue (including the midbrain and surrounding tissues) were the focus of work in this thesis. AF and tissue samples were all snap frozen on dry ice before long-term storage at -80°C.

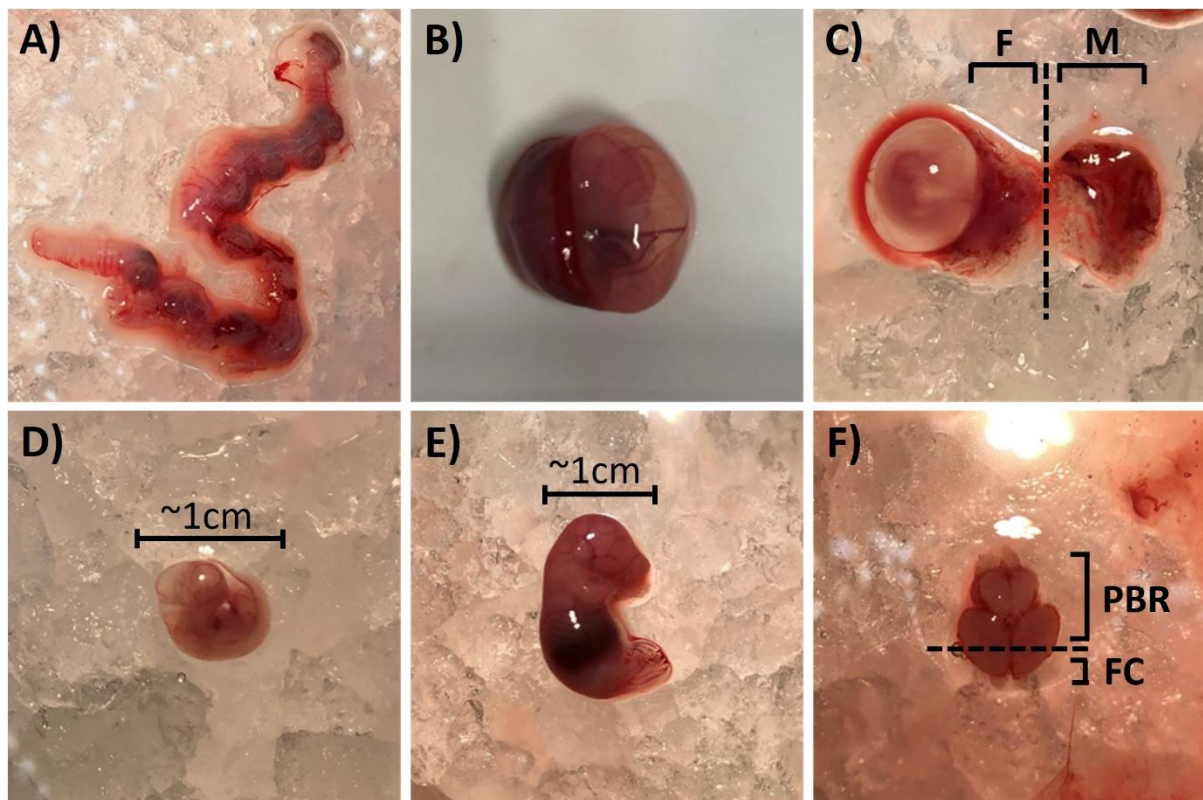


Figure 2.3: Images demonstrating embryonic dissection stages. (A) The removed uterus of a pregnant dam at E14.5. (B) An isolated E18.5 foetoplacental unit. (C) An E14.5 foetoplacental unit with the maternal portion of placenta removed. Placental sections peel apart, demonstrated by curvature of removed maternal portion. Tissue stability at E14.5 is poor, leading to disintegration around tissue edge. Note that the image provides example of tissue structure only and not dissection protocol, where placenta was detached prior to dissection. (D/E) E14.5 and E18.5 fetuses respectively, with 1cm estimated size bar to demonstrate size differences at dissection. (F) E18.5 brain demonstrating regional dissection. F= foetally derived placenta, M = maternally-derived placenta, pBR = posterior brain, FC = frontal cortex.

2.2. Enzyme-linked Immunosorbent Assay (ELISA)

2.2.1. Interleukin-6 (IL-6)

The IL-6 ELISA assay presented in Chapter 3 was performed using the Rat IL-6 Uncoated ELISA Kit (#88-50625, lot 276915-002; Invitrogen, Austria), following kit instructions. Reagents supplied with the kit are listed in Table 2.2 and were used alongside additionally prepared Wash Buffer (1X Phosphate buffered saline (PBS) with 0.05% Tween™-20) and Stop Solution (#SS04, Invitrogen, Austria).

Reagent (kit #88-50625)	Reagent Specifications provided
Assay Buffer A (20X)	PBS with 1% Tween™-20 and 10% BSA
Capture Antibody (250X)	purified anti-rat IL-6 monoclonal antibody
Coating Buffer (10X)	10X Phosphate buffered saline (PBS)
Detection Antibody (250X)	Biotin-conjugated anti-rat IL-6 monoclonal antibody
Detection Enzyme (100X)	100X Streptavidin-HRP concentrate
Standard	Lyophilised recombinant rat IL-6
Substrate Solution	Tetramethylbenzidine (TMB) solution

Table 2.2: Reagents provided in the Rat IL-6 Uncoated ELISA kit #88-50625.

A Corning™ Costar 9018 96-well plate was coated with coating buffer (1X in deionised water) with capture antibody (1X). The plate was sealed and incubated overnight at 4°C, before aspirating and washing wells twice with Wash Buffer. Wells were blocked with Blocking buffer (Assay Buffer A, 2X) before a second overnight incubation at 4°C. Wells were washed twice before loading of 50 µL maternal serum and AF and 50 µL of Assay Buffer A (1X) to a final well volume of 100 µL. IL-6 Standard (reconstituted to 4000 pg/mL in distilled water) was serially diluted in Assay Buffer A (1X) to final concentrations of 2000, 1000, 500, 250, 125, 62.5, 31.75 pg/mL alongside 0 pg/mL blank wells. All wells were run in duplicate aside from maternal serum samples, which were typically unable to be run in duplicate due to limited sample volumes.

After loading samples, 50 μ L Detection Antibody (1X, diluted in 1X Assay Buffer A) was added to each well before the plate was sealed and incubated for two hours at room temperature secured to an IKA Vibrax orbital shaker (VXR-S1-VX7, IKA-Werke, Germany) at ~400 rpm for agitation. Wells were aspirated and washed six times before the addition of 100 μ L Detection Enzyme (1X, diluted in 1X Assay Buffer A) and incubation for one hour again at room temperature with agitation. Another six washes preceded the addition of 100 μ L Substrate Solution, which was left at room temperature until colour was visibly developing in concentrated standard wells (approximately 7 minutes) and 100 μ L of Stop Solution was added. Wells were read at 450 nm and 570 nm using a SPECTROstar Nano microplate reader (BMG LABTECH; Germany), with 570 nm values subtracted from 450 nm values as recommended. Concentrations for each sample were determined based on the accompanying standard curve generated from that plate (Figure 2.4). Conversions for sample dilution (x2) and averages across duplicates were calculated where relevant before final analysis (Figure 3.1).

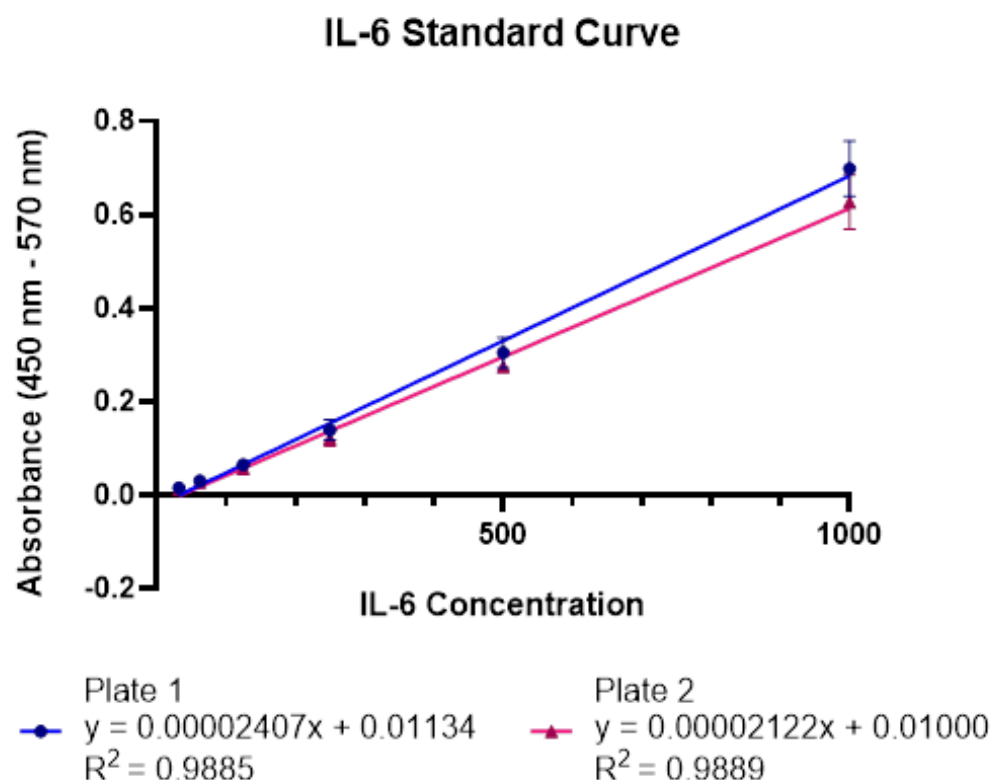


Figure 2.4: Standard curves generated for the two IL-6 ELISA plates used to assess maternal serum and amniotic fluid. Individual formulae and coefficients of determination (R^2) are presented below graph. Error bars indicate standard deviation.

2.2.2. C-Reactive Protein (CRP)

The CRP ELISA presented in Chapter 3 was performed using the Rat CRP ELISA Kit (#ERCRP, lot 739083122, Invitrogen, USA). Minimal reagent information is published alongside kit instructions, and all required reagents (aside from deionised water) were provided. Assay was performed in alignment with kit instructions.

Lyophilised, recombinant Rat CRP standard was reconstituted in Assay Diluent (1X in deionised water) to a concentration of 200 ng/mL, with a standard series of 60, 24, 9.6, 3.84, 1.536, 0.614, and 0.246 ng/mL diluted in Assay Diluent as directed. As only one pre-coated plate was available for this assay, maternal serum samples were diluted 20,000, 40,000, 80,000, and 160,000-fold in Assay Diluent to ensure a working concentration was plated.

Using the provided pre-coated 96-well plate, standards were plated in duplicate with Assay Diluent as 0 ng/mL blank wells, alongside four wells for each maternal serum sample (4x dilutions). Relevant standard/sample (100 μ L) was added to each well before overnight incubation at 4°C secured to an IKA Vibrax orbital shaker (VXR-S1-VX7, IKA-Werke, Germany) at ~400 rpm for agitation. Wells were then aspirated and washed four times with Wash Buffer (1X in deionised water) before addition of 100 μ L of Rat CRP biotin conjugate (1X in Assay Diluent) and incubation for one hour at room temperature with agitation. The plate was again washed four times and 100 μ L of freshly diluted Streptavidin-HRP (1X in Assay Diluent) was added for 45 min incubation at room temperature with agitation. Four additional washes preceded adding 100 μ L of TMB Substrate Solution per well, covering plate in black bag to protect from light, and incubation for 30 min at room temperature with agitation. Finally, 50 μ L of Stop Solution was added to each well.

Wells were read using the SPECTROstar Nano microplate reader (BMG LABTECH; Germany), at 450 nm. A standard curve was generated using a four parameter algorithm as recommended (Figure 2.5). CRP concentrations in wells containing maternal serum were calculated along this curve and multiplied by the relevant dilution factor. 160,000 fold serum dilutions were below the detection threshold of the assay, however the 20,000, 40,000, and 80,000-fold dilutions were averaged as triplicates after accounting for dilution factor to produce the serum CRP concentrations analysed in Chapter 3 (Figure 3.1).

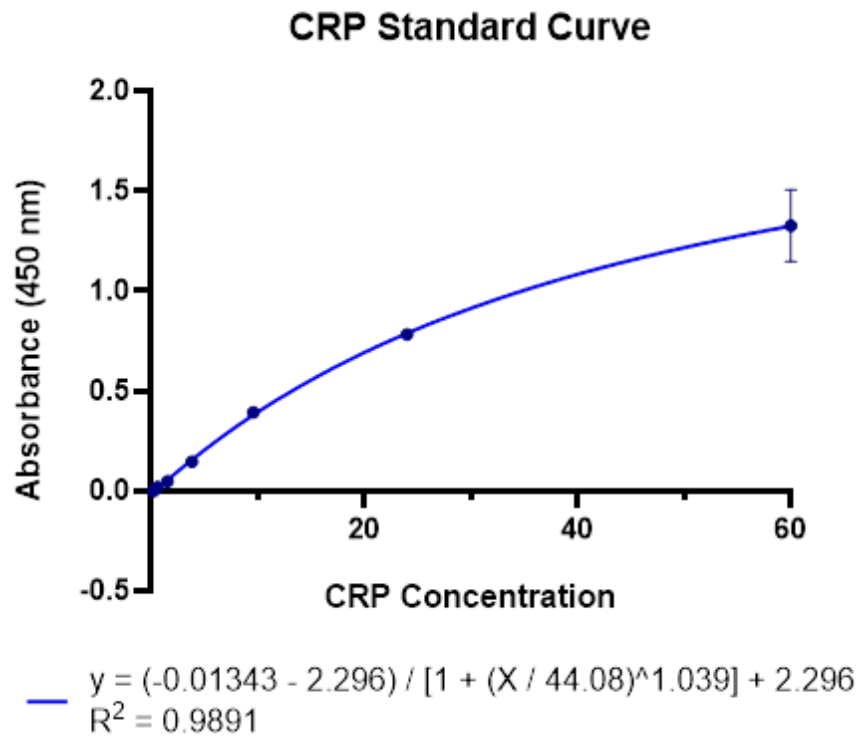


Figure 2.5: The standard curve generated for CRP used to assess maternal serum. Curve formula and coefficient of determination (R^2) are presented below graph. Error bars indicate standard deviation.

2.3. Liquid Chromatography – Mass Spectrometry (LC-MS)

2.3.1. Standard and Sample Preparation

A standard stock of 5-HT (#H9523; Sigma-Aldrich, USA) and 5-HIAA (#H8876; Sigma-Aldrich, USA) was initially produced at 5mM in deionised water. Serially diluted aliquots were stored at -80°C for the duration of methods development, with a fresh aliquot used for each test/batch of samples prepared. Dr van de Wetering provided 2 mL of 1 mM D2-5-HT, which was diluted to 2 µM and stored in 1 mL aliquots for fresh use as above. The D2-5-HT synthesis protocol was carried out in the School of Chemical and Physical Sciences at Victoria University of Wellington, and previously published as a supplementary methods protocol in reference [333].

Ice-cold acetonitrile (ACN) was added to 20 µL of individual AF samples at a 2:1 ratio and briefly vortexed before the addition of high-performance liquid chromatography (HPLC) grade water, D2-5-HT, and 5-HT standards to a final volume of 80 µL and standard concentrations of 300 nM. A further 20 µL from each sample was added to an AF pooled sample to create a consistent background matrix for standard curve vials, with 20 µL of pooled AF being added to ACN and D2-5-HT as above. 5-HT and water ratios varied to create a standard series of 375 nM, 330 nM, 315 nM, 300 nM, 285 nM, 270 nM, and 225 nM 5-HT. Individual samples and pooled standards were kept on ice for approximately 20 minutes during this process and formed a visible white protein precipitate prior to centrifugation at 4°C for 10 min at 12,000 g.

Using 2 mL glass LCMS vials (part #5190-9062; Agilent Technologies, USA) with 6 x 25 mm, 50 µL glass, spring bottom inserts (cat. #4005BS-625; JG Finneran, USA), 20 µL of AF supernatant was added to 10 µL of 80 µM sodium borate buffer (#B3545; Sigma Aldrich, USA). Vials were briefly vortexed before 10 µL of freshly prepared 2% [v/v] BzCl in cold ACN was added to each vial. Vials were then vortexed for 3 minutes. Samples were kept chilled at 4°C within the autosampler chamber until injection into the LC-MS, which occurred a maximum of 16 hours after sample preparation dependent on position within the batch worklist sampling order).

2.3.2. LC-MS Procedure

LC-MS analysis of AF samples was carried out using an Agilent 6530 quadrupole time-of-flight (Q-TOF) LC-MS with a Jet-stream electrospray ionisation (ESI) source (G6530CA; Agilent Technologies, USA). For liquid chromatography, an InfinityLab Poroshell 120 SB-Aq, 2. 1 x 100 mm, 2. 7 μ m reversed-phase chromatography column (685775-914; Agilent Technologies, USA) was installed with a matched InfinityLab Poroshell 120 SB-Aq, 2. 1 mm, 2. 7 μ m, UHPLC guard pre-column (821725-924; Agilent Technologies, USA) and held at 36 °C during analysis. Attached eluent bottles contained 99.9% water/0.1% ammonium formate (Eluent A) and 99.95% acetonitrile/0.05% formic acid (Eluent B). LC-MS analysis was performed on positive ion mode with the following specifications: gas temperature, 275 °C; gas flow, 5 L/min; nebulizer, 30 psi; capillary voltage, 2750 V; nozzle voltage, 0 V; fragmentor voltage, 130 V; acquisition rate, 2 spectra/s; mass range, m/z 100 – 1000. An additional tandem LC-MS (LC-MS/MS) method was run to confirm 5-HT peak identity the same specifications, with additional details as follows: Isolation width, medium (~4 amu); Fixed collision energies, 10, 20, 40; MS/MS preferred ion, 385.15 m/z . Acquisition methods reports are available in Appendix B.

Autosampler injection volume was set to 10 μ L (of a 40 μ L total reaction volume). Blank vials (containing 100% ACN) were injected as a column wash at the beginning and end of batch worklists, and as every fifth injection within the worklist. The liquid chromatography gradient protocol was as previously published [333]: 30% eluent B for 0.5 min, 97.5% eluent B at 9 min, held at 97.5% eluent B until 10 min, 50% eluent B at 10.5 min, 97.5% eluent B at 10.7 min, held at 97.5% eluent B until 11.5 min, 30% eluent B at 11.7 min. Flow rate was 0.45 mL/min until 10.7 min, increased to 0.5 mL/min from 10.7 to 11.7 min, and reduced back to 0.45 mL/min at 11.7 min. The post-run time was set to 3.5 min.

2.3.2.1. Sample Quantification and Standard Curve Calculations

Retention time and relative signal of 5-HT/D2-5-HT peaks were initially visualised in standards using Agilent MassHunter Qualitative Analysis software (version B.06.00, build 6.0.388.0; Agilent Technologies, 2012), then quantified and analysed using Agilent MassHunter Quantitative Analysis software (version B.06.00, build 6.0.388.0;

Agilent Technologies, 2008). In addition to compound m/z and standard concentration specifications, 5-HT isotopic dilution ratio was set to $R_x = 24.0963$ [334]. Standard linear, unweighted calibration curves were then automatically generated where the x axis is standard concentration and the y axis is 5-HT signal relative to D2-5-HT signal. Slope and y-intercept were used to calculate x-intercept where $y=0$, yielding the 5-HT signal contributed by the pooled AF sample. Standard concentrations were subsequently adjusted to include AF 5-HT concentration and samples were quantified; automatic peak integration was primarily used to reduce bias, with only three peaks overall being manually integrated where clear errors were present.

Uncertainty of sample 5-HT concentrations calculated by multiplying the standard deviation in analyte concentration (s_x) by 2.179 (the t value associated with 95% confidence with the 12 degrees of freedom from the standard curve), with s_x calculated by:

$$s_x = \left(\frac{s_y}{m} \right) \sqrt{\frac{1}{k} + \frac{1}{n} + \frac{(y - \bar{y})^2}{m^2 \times \sum_{i=1}^n (x_i - \bar{x})^2}}$$

Where s_y = standard curve y error, m = standard curve slope, k = number of sample measurements, n = number of standard curve levels, y = mean standard relative signal, $\bar{y}$ = mean standard relative signal, x_i = individual concentration of all individual standards, and $\bar{x}$ = mean concentration of standards. Noting that section highlighted in grey reflects the numerator of the standard deviation calculation of standards, value substitutions for the example calculations provided for E14.5 were as follows:

$$s_x = \left(\frac{0.021287}{0.001397} \right) \sqrt{\frac{1}{2} + \frac{1}{7} + \frac{(y - 0.23884)^2}{(0.001397)^2 \times 3375}}$$

2.4. Nucleic Acid Isolation

2.4.1. Combined DNA and RNA Isolation

A rapid lysis protocol using the SingleShot Cell Lysis Kit (#1725080; Bio-Rad Laboratories, USA) was used in this thesis to extract DNA and RNA concurrently from embryonic tissue samples. Small samples of homogenised tissue (approximately 10 μ L) were placed in a microcentrifuge tube containing 44 μ L of lysis buffer and 1 μ L proteinase K. Samples were triturated and incubated at room temperature for 10 min, 37°C for 5 min, and 75°C for 5 min. All lysis samples were kept on wet ice prior to further treatment or polymerase chain reaction (PCR).

2.4.2. RNA Isolation

For RNA isolation, whole tissue samples retrieved from -80°C were first homogenised. E14.5 placenta, E18.5 placenta, and E18.5 brain samples were added to TE buffer at a 1:2 ratio (e.g. 70 mg placental tissue in 140 μ L TE) with two stainless steel, nuclease free beads (NEXSSB32-RNA; Next Advance, USA). Samples were kept on ice and vortexed in 10 second bursts for a total of approximately 1 minute until homogeneous. In contrast, E14.5 brain samples were readily homogenised by trituration.

Homogenised tissue (50 μ L) was added to a new microcentrifuge tube containing 500 μ L TRIzol (#15596018; ThermoFisher Scientific, USA) and further triturated. Samples were incubated at room temperature for 5 min before addition of 100 μ L chloroform and immediate vortexing to produce an opaque pink solution. Tubes were centrifuged at 4°C for 15 min at 12,000g, forming three layers; a colourless aqueous upper layer, a thin white protein layer, and a lower pink phenol layer. Approximately 200 μ L of the aqueous phase was transferred to a new microcentrifuge tube, avoiding any transfer of other layers. Half volume molecular grade ethanol was added to each sample and inverted to mix.

Samples were then added to GeneJet RNA Cleanup and Concentration Micro Kit columns (K0842; ThermoFisher Scientific, USA). Following kit instructions, samples were centrifuged for 1 min at 14,000g. Repeat centrifugation steps were repeated after each of the following washes: 700 μ L wash buffer one, 700 μ L wash buffer two, and

700 µL wash buffer two again. An additional empty spin removed any excess wash buffer before a final addition of 20 µL of nuclease free water for elution of purified RNA. RNA concentration and purity was assessed using a Nanodrop ND-1000 Spectrophotometer (ThermoFisher Scientific, USA). Aside from approximately half of E18.5 brain samples which were measured a single time, samples concentrations were measured in triplicate prior to reverse transcription.

2.5. Sex Determination Assay

Following DNA extraction (see 2.4.1), pup sex was determined using a quantitative real-time PCR assay adapted from Dhakal and Soares [335] and previous work in our laboratory by Dr Thorne [63]. Dhakal and Soares initially published a primer pair targeting a 250 bp amplicon from the lysine demethylase 5d (*Kdm5d*) gene located on the *Rattus norvegicus* Y chromosome (forward: 5'-TTGGTGAGATGGCTGATTCC-3'; reverse: 5'-GGTTTCTTAAACCGTCGCC-3'), however Dr Thorne identified a secondary 276 bp product amplifying the paralogue *Kdm5c* on the X chromosome (Figure 2.6A) [63, 335]. PCR amplification of samples from male rat pups (with both an X and Y chromosome) therefore produced two distinct products, while amplification of DNA from female pups only produced a single 276 bp amplicon.

DNA samples were diluted 100-fold and 1 µL was added to 2 µL SolisFast qPCR Mix (#28-46-00001; Solis Biodyne, Estonia), 1 µL of a 5 µM primer stock (containing both forward and reverse primers), and 6 µL nuclease free water for a final volume of 10 µL. PCR reactions were combined in 0.1 mL tubes for use in a magnetic induction qPCR cycler (mic; Bio-Molecular Systems, Australia). Mic cycling parameters were as follows: 95°C for 2 min; 95°C for 30 s, 55°C for 10 s, 72°C for 42 s repeated for 35 cycles; and high resolution melt curve generation using a 70°C - 95°C gradient at 0.2°C/s (Figure 2.6B). Following amplification, product size and number were verified by gel electrophoresis. Agarose gels (2%) were prepared by microwave heating 75 mL Tris-Acetate-EDTA (TAE) buffer with 1.5 g agarose until fully dissolved, before adding 7.5 µL SYBR safe DNA gel stain (S33102; Invitrogen, USA) and pouring in to gel mould to set at room temperature. Solidified gel was positioned for loading in an Owl EasyCast B2 Mini Gel Electrophoresis tank (ThermoFisher Scientific, USA) which

was subsequently filled with TAE buffer. Xylene-cyanol loading dye (2 μ L of 6X stock) was added to PCR products and triturated prior to loading 10 μ L per well. Gels were run for approximately 50 minutes at 100 V and immediately imaged (Figure 2.6C). All sample sexes were assessed by both melt curve and gel electrophoresis, with no instances of conflicting results.

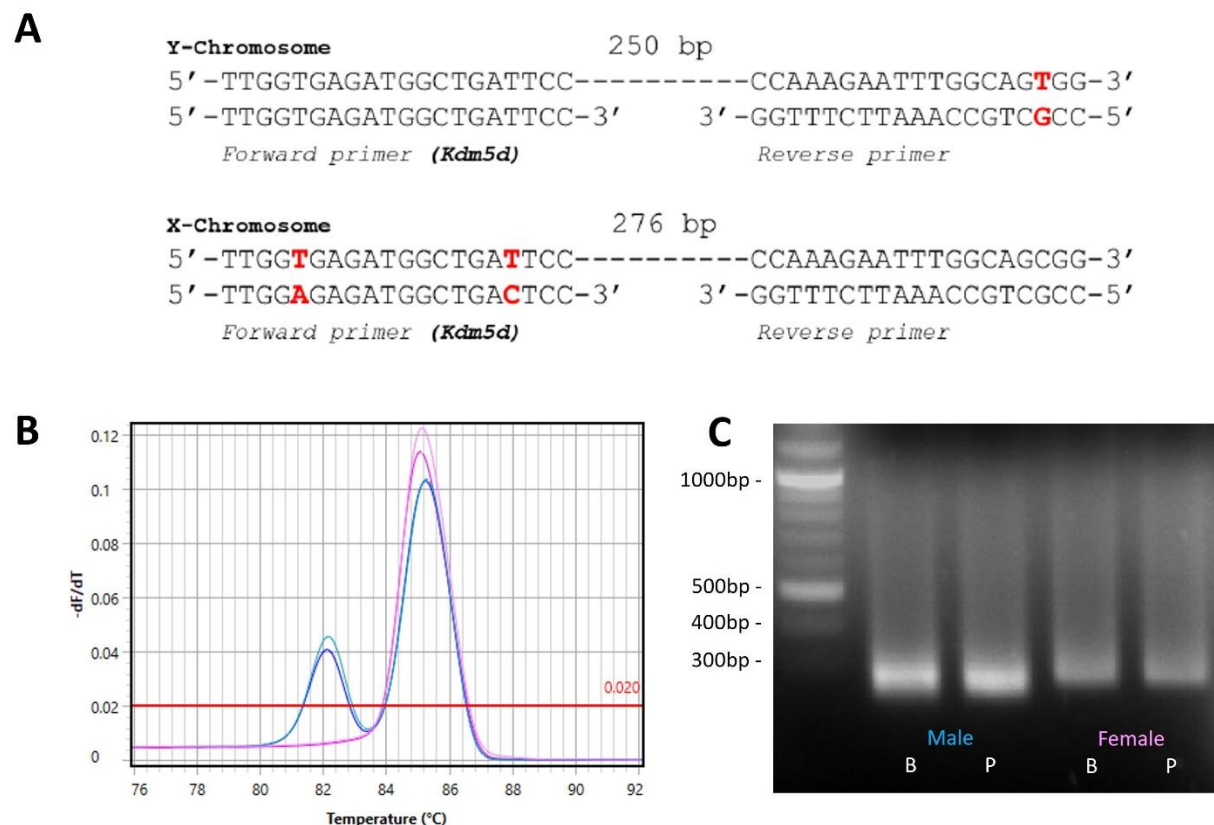


Figure 2.6: Details of the sex determining assay used in this thesis. A) Overview of the amplicons produced by *Kdm5d* primer binding on both Y chromosome *Kdm5d* and X chromosome *Kdm5c*, as originally presented in Figure 4.6 of Dr Thorne's PhD thesis [63]. Melt curves (B) and bands following gel electrophoresis (C) are presented for one male (blue) and one female (pink) pup assessed in this thesis, with the male samples exhibiting a second, slightly shorter product. Paired brain (B) and foetal placenta (P) tissue samples are shown to demonstrate consistent results across tissue types.

2.6. Digital Droplet PCR (ddPCR) Assays

2.6.1. Selfie- Digital Droplet PCR

2.6.1.1. Paired Sample Preparation

Following the isolation of DNA and RNA in 2.4.1, samples underwent primer pre-annealing for gene-specific reverse transcription. A 2.25 μL volume of combined DNA/RNA sample was added to 9 μL of a reverse transcription (RT) primer mastermix, before incubation at 70°C for 1 min and then decreasing temperature to 4°C at a slow ramp rate. This primer mastermix included a 5 μM concentration of every reverse primer (see Table 2.3 for target primer sequences).

Following primer pre-annealing, 5 μL of sample was added to two microcentrifuge tubes along with 10 μL of an RT mix containing (per sample pair): 1.2 μL RNaseOUT Recombinant Ribonuclease Inhibitor (#10777019; Invitrogen, USA), 1.2 μL 100 mM DTT (#18080093; Invitrogen, USA), 2.3 μL 10 μM dNTP mix, 4.7 μL 5X RT buffer (#EP0751; ThermoFisher Scientific, USA), and 13.6 μL nuclease free water. One tube from each pair was designated RT+ (undergoing reverse transcription) and further included 0.5 μL Maxima H Minus Reverse Transcriptase (#EP0751; ThermoFisher Scientific, USA), while the other tube was labelled RT- (receiving no reverse transcriptase, instead an additional 0.5 μL RT buffer). Paired samples were both incubated at 58°C for 30 min, then 90°C for 3 min.

Samples further underwent a restriction digest step to cleave DNA into smaller segments for better inclusion in droplets during ddPCR setup. To every sample tube, 1 μL was added of a restriction digest mix containing a fresh 1/10 dilution of HindIII (#ER0505; ThermoFisher Scientific, USA) in accompanying R buffer. Samples were incubated at 37°C for 1 hour, then 65°C for 20 min prior to 100-fold dilution in nuclease-free water for ddPCR.

2.6.1.2. Primer Design

The primers designed for selfie-ddPCR are presented in Table 2.3. All primers were either originally designed using the PrimerBLAST tool on NCBI, designed by another lab member, or sourced from published literature [336]. Primers were selected to produce a short amplicon as is optimal for ddPCR (targeted between 70-250 bp), with a predicted primer melt temperature of approximately 64°C. Target gene primers were required to bind within exonic sequences to amplify both genomic (g)DNA and complementary (c)DNA. All primers and amplicons were also screened to not include the 5'-A|AGCTT-3' 3'-TTCGA|A-5' restriction site of HindIII, to be compatible with HindIII digestion during sample preparation.

All primers were initially screened by ddPCR using a temperature gradient to determine the optimal annealing temperature for each primer pair (see 2.6.1.4 for cycling protocol). Primer pairs that failed to amplify target DNA/cDNA well at any temperature were redesigned. All primer sets presented in Table 2.3 produced a single clear band of positive droplets with minimal rain when the PCR annealing step was run at either 61.5°C or 64°C, separating target primers into two groups with distinct annealing temperatures but otherwise identical cycling parameters.

Following gradient ddPCR, primers were further validated using qPCR, melt analysis, and gel electrophoresis to confirm the specificity of each primer pair. qPCR cycling and gel electrophoresis methods were as presented in section 2.5, aside from different annealing temperatures as appropriate for each primer pair. Amplification from all primer pairs resulted in a single melt peak and band corresponding with the expected amplicon size (Figure 2.7).

Primer Set (Gene)	Primer Sequence	Amplicon Length (bp)	Annealing Temp. (°C)	Location (NCBI reference)	Template	Source
Bdnf9 (Bdnf)	FW: CTTCCGGTTGCATGAAGGCTG RV: GTCAGACCTCTCGAACCTGC	135	64	96212020 – 96212154 (NC_051338.1)	gDNA + cDNA	Lisa Pugh, 2022*
Ddc v2 (Ddc)	FW: GGCTGCTCGGACTAAAATGATCC RV: ACTTTTCCATGAGAGCAGCTTGTG	77	61.5	398 – 474 (NM_001270853.1)	gDNA + cDNA	Original
MaoA (MaoA)	FW: TTAGTGCCATCCCACCGATT RV: TAACAGCCCCCATTGGAAGA	102	61.5	809 – 910 (NM_033653.1)	gDNA + cDNA	Original
Mecp2 v2 (Mecp2)	FW: GGGGTAGAAAGCCTGGGAGT RV: TCAATGCTGACGGTTTCCCG	142	61.5	1020 – 1161 (NM_022673.2)	gDNA + cDNA	Original
NeuroD1 v3 (NeuroD1)	FW: ACAGCTCCCATGTCTTCCAC RV: AAAGGAAGGGCTGGTGCAAT	104	64	791 – 894 (NM_019218.3)	gDNA + cDNA	Darren Day, 2022*
Oct3 (Slc22a3)	FW: CACAAGTGCCGTGGTGTATC RV: CACGAGCCCCGAGATAAAGA	91	64	1453 – 1543 (NM_019230.1)	gDNA + cDNA	Original
Sert v3 (Slc6a4)	FW: GCACGAACTCCTGGAACA RV: TGGAATGCAGCGTCCAG	75	61.5	62338898 – 62338972 (NC_086028.1)	gDNA + cDNA	[63]
Tph1 v1 (Tph1)	FW: CAAACTCTACCCAACCCACG RV: GGGACGTTGTCTTCCCGATA	90	64	666 – 755 (NM_001100634.3)	gDNA + cDNA	Original
Tph2 v3 (Tph2)	FW: TGCTCCATATTGAATCCAGGAGGT RV: GGTCTGAACTTGAGCAACTGAAT	119	64	374 – 792 (NM_173839.3)	gDNA + cDNA	Original

Table 2.3: Primer pairs designed for selfie-ddPCR. * = designed by lab member, not otherwise published; bp = base pairs.

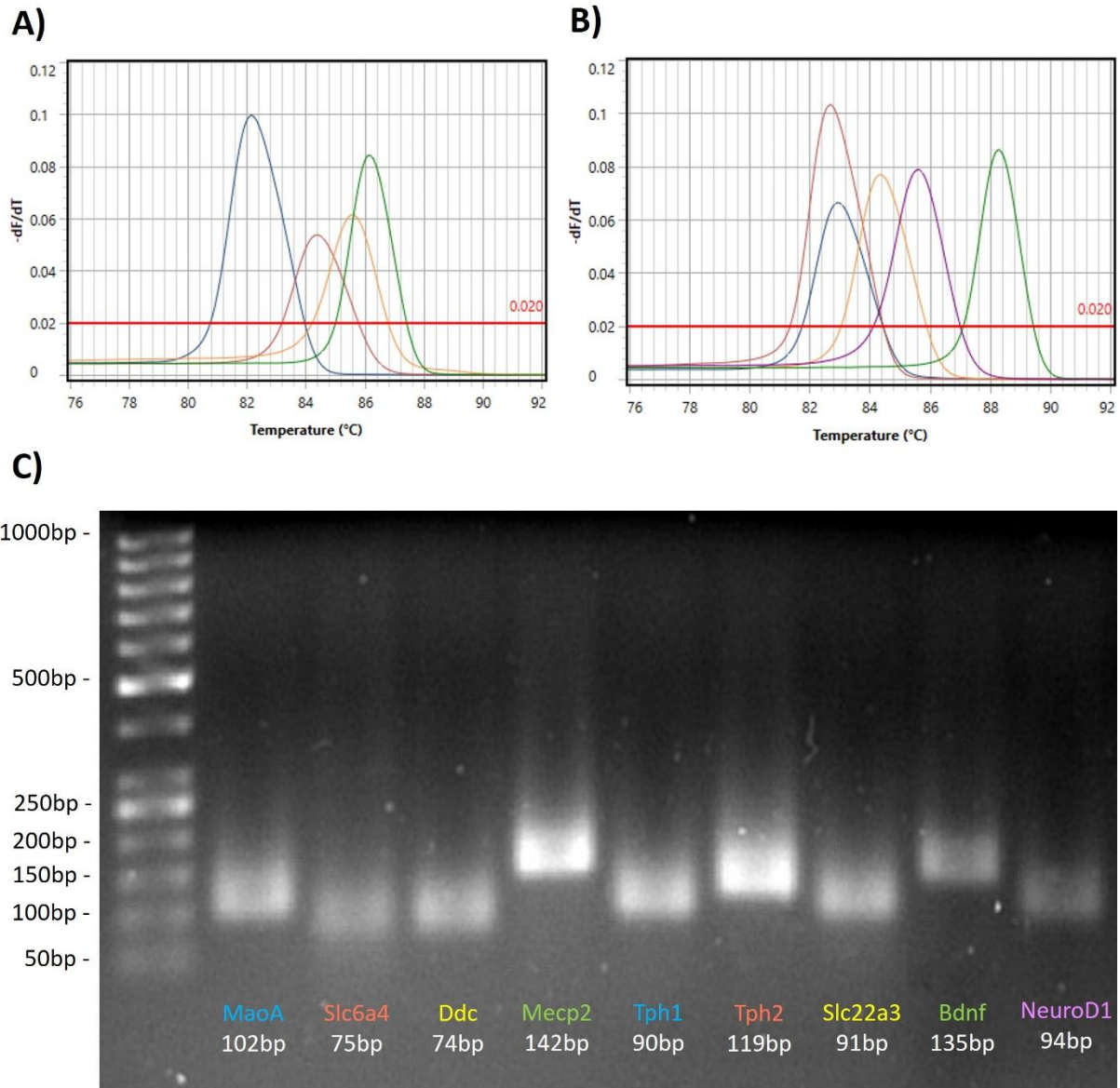


Figure 2.7: Target gene primer sets each amplify a single product that align with expected amplicons. (A) Melt peaks from primer sets used at 61.5°C; MaoA (blue) = 82.16°C, Slc6a4 (red) = 84.38°C, Ddc (yellow) = 85.58°C, Mecp2 (green) = 86.16°C. (B) Melt peaks from primer sets used at 64°C; Tph1 (blue) = 82.94°C, Tph2 (red) = 82.69°C, Slc22a3 (yellow) = 84.36°C, Bdnf (green) = 88.29°C, NeuroD1 (purple) = 85.60°C. (C) PCR products from melt analysis presented in A and B separated by gel electrophoresis, with target gene name and expected size annotated below each band.

2.6.1.3. *ddPCR Reaction Set-up and Droplet Generation*

All ddPCR reactions were first prepared in generic PCR tubes to a final volume of 22 μL , comprised of 1 μL of sample DNA/cDNA, 1.2 μL primer (from the relevant 5 μM working stock containing forward and reverse primers), 11 μL EvaGreen Supermix (#1864035; Bio-Rad Laboratories, USA), and 8.8 μL nuclease free water. Reactions were triturated and added to the sample wells of a DG8 Droplet Generator Cartridge (#1864008; Bio-Rad Laboratories, USA). QX200 Droplet Generation Oil for EvaGreen (#1864006; Bio-Rad Laboratories, USA) was then added to excess ($\sim 70 \mu\text{L}$) to cartridge oil wells, before fitting of a DG8 Gasket (#1863009; Bio-Rad Laboratories, USA) over cartridge wells and insertion of the cartridge into a QX200 Droplet Generator (#1864002; Bio-Rad Laboratories, USA). Following droplet generation, approximately 40 μL from the cartridge droplet wells were slowly extracted at a 45° angle using an Eppendorf multichannel pipette with Dualfilter T.I.P.S. pipette tips (#0030078543; Eppendorf, Germany) and transferred to a ddPCR 96-well plate (#12001925; Bio-Rad Laboratories, USA). Plates were sealed with pierceable foil heat seals (#1814040; Bio-Rad Laboratories, USA).

2.6.1.4. *ddPCR Thermocycling and Droplet Reading*

Sealed ddPCR plates were transferred to a C1000 Touch Thermal Cycler with a 96-well, deep well module (Bio-Rad Laboratories, USA) for amplification. ddPCR temperature gradients using rows or 8 identical reactions were performed to optimise all primer sets. The following cycling parameters were used for gradient PCRs: 95°C for 5 min; 95°C for 30 s, 59°C – 66°C gradient for 1 min 10 s repeated for 40 cycles; 4°C for 5 min; then 90°C for 5 min at a slow ramp rate (2°C/s). For data collection, cycling parameters were: 95°C for 5 min; 95°C for 30 s, (primer-specific annealing temperature) $^\circ\text{C}$ for 1 min repeated for 40 cycles; 4°C for 5 min; then 90°C for 5 min at a slow ramp rate (2°C/s).

Following amplification, ddPCR plates were transferred to a QX200 Droplet Reader (#1864003; Bio-Rad Laboratories, USA) for analysis. The quality of ddPCR data was visually assessed in every well using the following criteria:

- A minimum of 10,000 droplets per well were counted (ensuring the Poisson statistics used to calculate DNA concentration are operating accurately);

- A clear band of positive droplets at a consistent amplitude (suggesting a single product amplifying with reasonable efficiency; amplitude typically between 10,000 – 25,000) was visible, clearly distinguishable from a band of negative droplets at a consistent amplitude (typically between 0 – 10,000, dependent on baseline noise such as primer sequence and concentration);
- Minimal 'rain' (droplets falling between the positive and negative band, which may indicate non-specific amplification, degraded target transcript, or target amplification with poor reaction efficiency) was present, with clear bands still visible;
- A minimum target concentration of 1 copy/μL was required for further analysis, roughly equating to 20/20000 positive droplets that could be identified as forming a band;
- No signs of technical issues that compromised droplet amplitude, such as droplet shearing due to microplastics.

Wells were thresholded slightly above the negative droplet band, including any rain as positive droplets as recommended by Bio-Rad representatives and MIQE guidelines [337]. For selfie-ddPCR, the target concentration produced from amplification of an RT- sample represented copies of genomic (g)DNA. The paired RT+ sample however included both gDNA and reverse transcribed (c)DNA; target concentration from RT+ sample wells therefore included both gDNA signal and original RNA transcript signal. As previously performed within our laboratory and published research, concentration from RT+ and RT- reactions could be used to calculate RNA copies per gene copy, or further calculate copies per cell (assuming single nuclei, diploid cells):

$$\text{copies per gene copy} = \frac{[RT+] - [RT-]}{[RT-]}$$

$$\text{copies per cell} = \frac{[RT+] - [RT-]}{[RT-]} \times 2$$

2.6.2. Reverse Transcription- Digital Droplet PCR (RT-ddPCR)

2.6.2.1. DNase Treatment and Reverse Transcription

Following RNA isolation and concentration measurement (see 2.4.2), 2.5 μ L each of DNase-I (#18068015; Invitrogen, USA) and accompanying 10X buffer were subsequently added to each sample, with nuclease-free water up to 25 μ L total volume. RNA was incubated at room temperature for a maximum of 15 minutes, then at 65°C for 10 minutes with 2.5 μ L 25 mM EDTA. Similarly to 2.6.1.1, gene-specific reverse transcription was then performed. A reverse transcription primer mastermix containing all target and reference gene reverse primers at 5 μ M concentration was used for a primer pre-annealing step, where 4 μ L of primer mastermix, 1 μ L 10 μ M dNTPs, and up to 5 μ g DNase-treated RNA were combined with nuclease-free water to a total volume of 15 μ L. Samples were incubated at 65°C for five minutes then snap chilled on ice for three minutes. To each sample, 0.5 μ L of RiboLock RNase inhibitor (#EO0381; ThermoFisher Scientific, USA), 0.5 μ L of Maxima H Minus Reverse Transcriptase, and 4 μ L 5X RT buffer were added before incubation at 58°C for 30 min. A final incubation of 90°C for 3 min inactivated the reverse transcriptase, and samples were subsequently diluted with nuclease-free water (with a dilution factor between 25-100) before ddPCR.

2.6.2.2. Reference Gene Primer Design

In addition to the target primers previously presented in Table 2.3, a pool of reference gene primers was created and later assessed by Mic qPCR (Table 2.4; Figure 2.8). All primers produced a single product of expected size, aside from Gapdh primers producing a product approximately double the expected length previously reported [338]. Although ddPCR data allows absolute quantification (and can be normalised to input cDNA concentration rather than expression of reference mRNA), reference genes were initially used for normalisation to provide an internal control for each sample (minimising variance due to technical error across multiple sample preparation and ddPCR reaction setup steps). In contrast to target primer design, reference primers targeted sequences including exon boundaries wherever possible to exclude gDNA amplification as is recommended for reverse transcription PCR (RT-PCR). From the pool of available reference primers, three were selected per tissue type to best

normalise mRNA expression (see 5.3.1.1 for results). At least one primer was required at 61°C and 64°C to ensure a reference gene was assessed alongside target reactions prepared simultaneously.

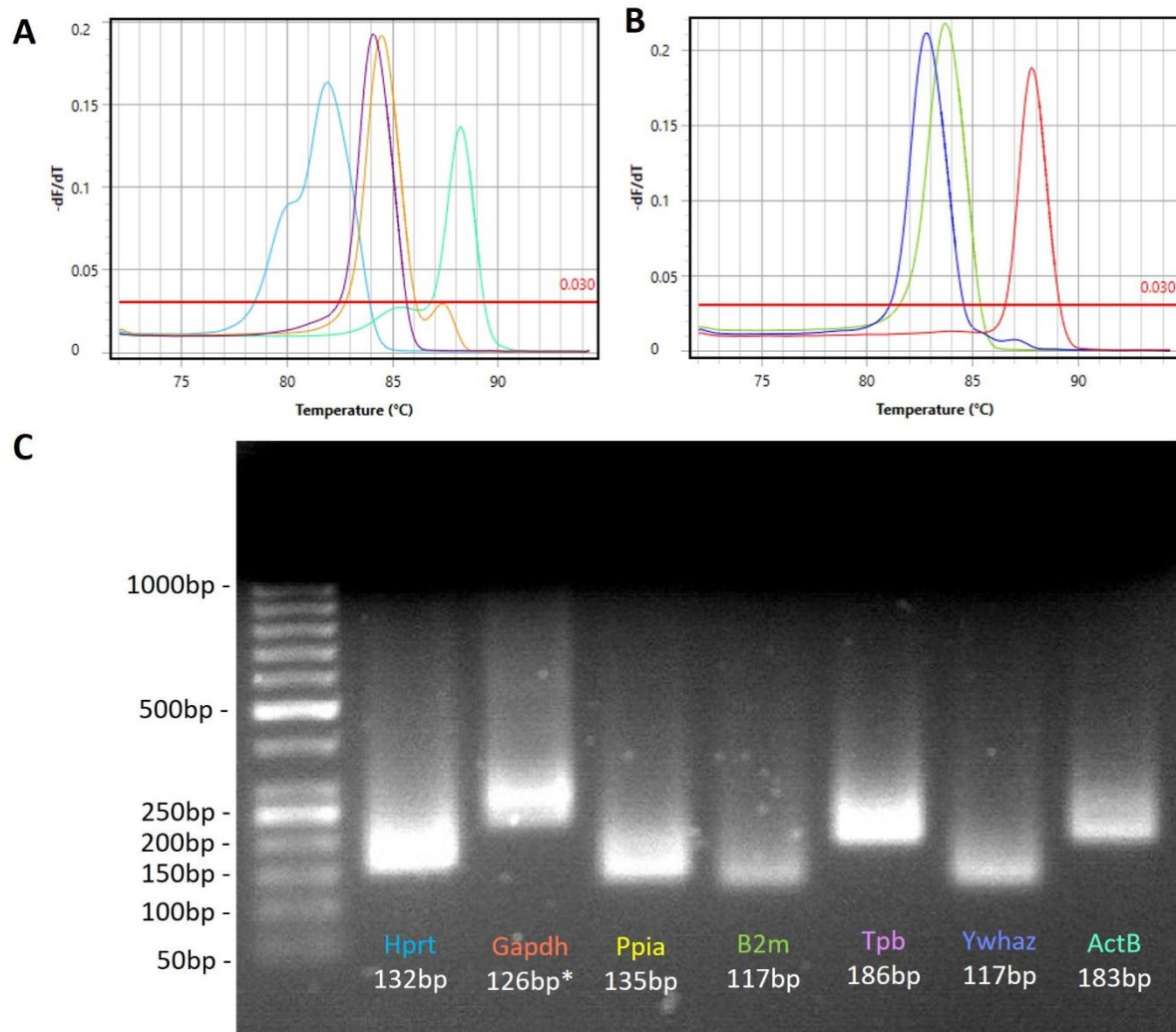


Figure 2.8: Amplification and resulting amplicon size of reference gene primer pairs. (A) Melt peaks from primer sets amplified at a 61°C Mic qPCR annealing temperature; Hprt (blue) = 81.91°C, Tbp (purple) = 84.10°C, Ppia (yellow) = 84.49°C, ActB (teal) = 88.23°C. (B) Melt peaks from primer sets amplified at a 55°C Mic qPCR annealing temp; Ywhaz (dark blue) = 82.82°C, B2m (green) = 83.72°C, Gapdh (red) = 87.81°C. (C) PCR products from melt analysis presented in A and B separated by gel electrophoresis, with target gene name and expected size annotated below each band. * = product that does not match expected size.

Primer Set (Gene)	Primer Sequence	Amplicon Length (bp)	Annealing Temp. (°C)	Location (NCBI reference)	Template	Source
ActB (ActB)	FW: CATCTATGAGGGTTACGCGCTC RV: TGCTCGAAGTCTAGGGCAACAT	183	61.5	570 – 752 (NM_031144.3)	gDNA + cDNA	Original
B2m (B2m)	FW: CCCACCGAGACCGATGTATATG RV: CCAGATGATTCAGAGCTCCATAG	117	64	294 – 410 (NM_012512.2)	cDNA	Original
Gapdh (Gapdh)	FW: AGCTGGTCATCAACGGGAAACC RV: CCTTCTCCATGGTGGTGAAGAC	126	61.5	266 – 391 (NM_017008.4)	cDNA	[338]
Hprt (Hprt1)	FW: TCCTCATGGACTGATTATGGACA RV: TAATCCAGCAGGTCAGCAAAGA	132	61.5	177 – 308 (NM_012583.2)	cDNA	[63]
Ppia (Ppia)	FW: GGCAAATGCTGGACCAAACAC RV: AAACGCTCCATGGCTTCCAC	135	61.5 or 64	342 – 476 (NM_017101.1)	cDNA	[339]
Tbp (Tbp)	FW: GCTGCAGTCATCATGAGAATAAGAG RV: CACCATGTTCTGGATCTTGAAGT	186	64	794 – 979 (NM_001004198.1)	cDNA	[339]
Ywhaz (Ywhaz)	FW: GTCGTACAAAGACAGCACGCTA RV: GCCGGTTAATTTTCCCCTCCTT	117	64	688 – 804 (NM_013011.4)	cDNA	Original

Table 2.4: Reference gene primer pairs developed for RT-ddPCR. bp = base pairs; ActB = Actin, beta; B2m = beta-2 microglobulin; Gapdh = Glyceraldehyde 3-phosphate dehydrogenase; Hprt = hypoxanthine phosphoribosyltransferase 1; Ppia = peptidylprolyl isomerase A; Tbp = TATA box binding protein; Ywhaz = tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein, zeta.

2.6.2.3. ddPCR Analysis

Sample reactions were prepared and amplified as per 2.6.1.3 and 2.6.1.4, with target genes run alongside a total of 3 reference genes for each sample. In contrast to initial selfie-PCR experiments, wells were manually thresholded at a set percentage of the distance between the positive and negative bands (Figure 2.9A). All assessed genes in E18.5 tissues were thresholded at 10% above the negative band, while two genes in E14.5 brain tissue (*Ddc* and *Tbp*, 15%) and eight genes in E14.5 placental tissue (*Hprt* and *Ppia*, 15%; *Oct3*, 20%; *Bdnf* and *MaoA*, 30%; *Sert* and *Mecp2*, 40%; *Ddc*, 60%) were thresholded at higher ratios due to reduced distance between bands and increased thickness of the negative band (Figure 2.9B). This was due to increased input sample concentrations (25-50 -fold dilutions) increasing background noise in negative droplets. Using a set ratio threshold for targets rather than a specific amplitude ensured that slight shifts in negative band amplitude (sometimes noted over the span of data collection) did not lead to increased rain being designated as negative, or noise from the negative band being designated as positive (Figure 2.9C).

Consideration was further given to uncertainty of concentration estimates at extreme concentrations. Samples where less than 5% of droplets were positive (1000/20,000), or less than 1% of droplets were negative (200/20,000), results in increased uncertainty in concentration estimates (>5%) [337]. Uncertainty increases at low concentrations; as a result, low concentration targets that had positive droplet counts consistently lower than 5% across samples were highlighted during analysis. As mentioned previously, samples with a target concentration less than 1 copy/ μ L (approximately 0.01% positive droplets) were considered too low to reliably quantify and excluded from further analysis [337, 340].

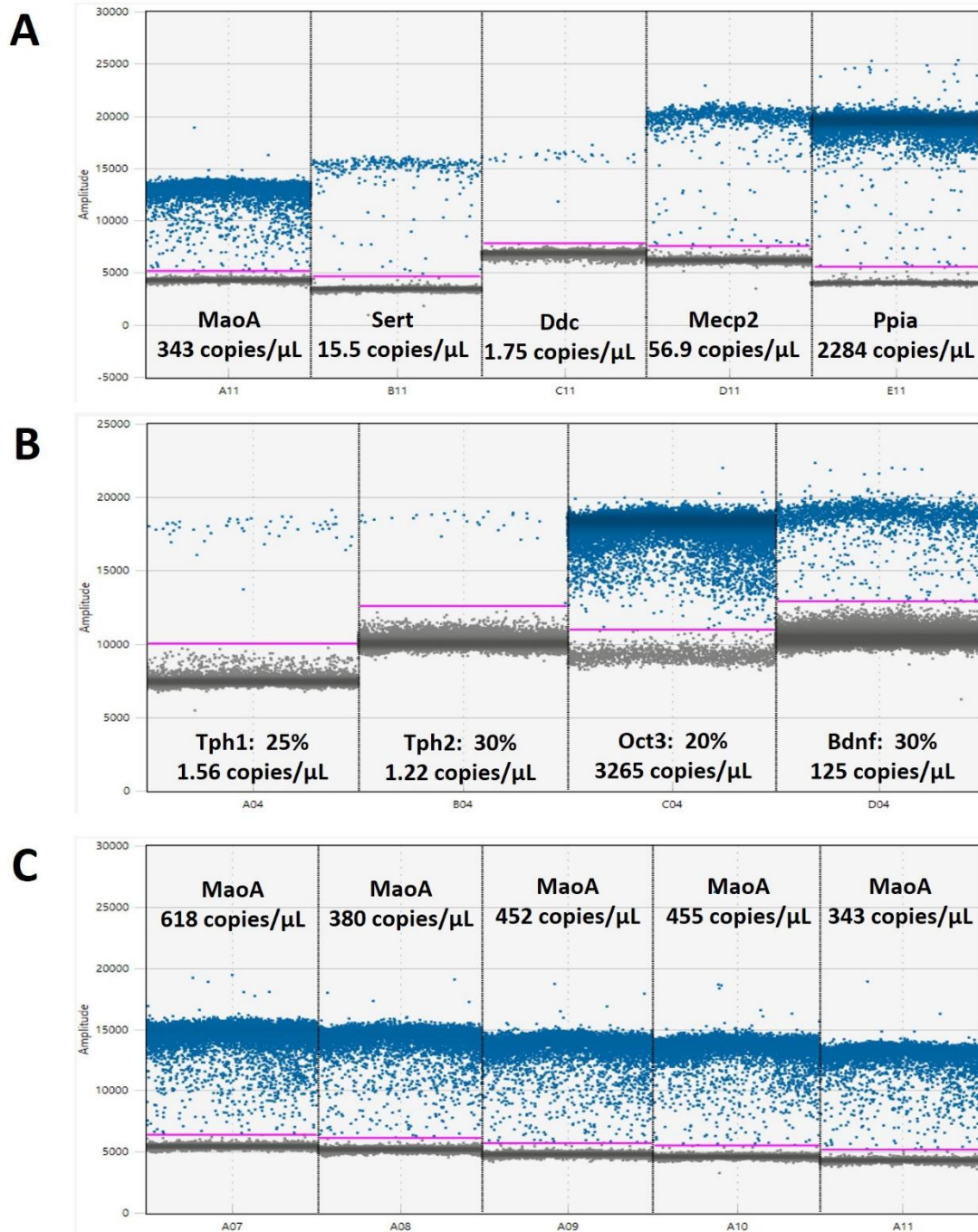


Figure 2.9: Examples of ddPCR thresholding and concentration ranges. A) 10% thresholds across multiple targets in the same E18.5 fPLA sample, labelled with target gene name and concentration. B) Examples of increased target thresholds in E14.5 fPLA samples following increased cDNA input in an attempt to increase low Tph1/Tph2 signal. Increased background noise and near saturation of other targets is observed. C) Example of overall amplitude drift, observed in MaoA amplification across a single ddPCR run of E18.5 fPLA samples.

2.6.3. Methylation Sensitive Restriction Enzyme Assay

The methylation sensitive restriction enzymes (msREs) BstUI (#ER0921; ThermoFisher Scientific, USA) and HpaII (#ER0511; ThermoFisher Scientific, USA) were used to treat DNA extracted from E18.5 brain samples. Reactions containing 2 μ L DNA, 1 μ L msRE (either BstUI or HpaII), 2 μ L accompanying buffer (R buffer for BstUI, Tango buffer for HpaII), and 15 μ L nuclease free water were prepared on ice and briefly vortexed before incubation at 37°C for 5 hours. Samples were finally incubated at 65°C for 20 minutes to inactivate msREs as recommended prior to digital droplet PCR (ddPCR). An additional aliquot of non-digested DNA was equivalently diluted 10-fold to act as a reference sample.

Restriction digested DNA was amplified by ddPCR as described in 2.6.1.3 and 2.6.1.4. Primers shown in Table 2.5 were designed to target three amplicons from the putative *Rattus norvegicus Slc6a4* CpG island; one Full Amplicon (503 bp) and two internal amplicons (Amplicon 1 = 311 bp; Amplicon 2 = 221 bp). Working stocks for each primer pair were created to a concentration of 5 μ M. Temperature gradients were performed on identical reactions of undigested DNA to identify the optimal working temperature for each primer pair. Data was collected using only Amplicon 2, at an annealing temperature of 61°C. Each pup sample was run over 3 ddPCR reactions; undigested DNA, BstUI digested DNA, and HpaII digested DNA. To calculate the percentage of fully methylated (thus indigestible) target sequence, the BstUI-digested and HpaII-digested concentration was divided by the concentration of the undigested control.

Primer	Sequence	Amplicon
Full Forward	AAGGGAAGAAGTGAAGCTCGTTT	Full, Amplicon 1
Internal Reverse	CCCTCACATGGTCTGATCTCTAGATA	Amplicon 1
Internal Forward	GGCTATCTAGAGATCAGACCATGT	Amplicon 2
Full Reverse	AAAGAAGCGTGGGTTAGAGACC	Full, Amplicon 2

Table 2.5: Primer sequences designed for the methylation sensitive restriction enzyme assay used in Chapter 6.

2.7. Statistical Analyses

Similarly to Careaga et al, statistical analyses of offspring data in this thesis were approached with careful consideration of data independence [341]. Pups within a litter were not independently assigned to treatment groups (due to administration of treatment to the mother), however differ in other variables of interest (e.g. sex), preventing simple analysis of pups as either independent data points or biological replicates. As such, data generated from offspring tissues was typically assessed by first structuring the data in a Linear Mixed Model (LMM) with litter as a random variable nested within treatment group. Where possible, risk of litter bias was also minimised experimentally by subsampling a limited and consistent number of pups per litter (e.g. $n=2$ for AF LCMS, $n=2-3$ for brain and placental ddPCR).

LMM and subsequent analysis of variance (ANOVA) analyses were performed in RStudio using the lme4 and lmerTest packages [342-344]. Treatment and offspring sex were included as fixed effects of interest, although sex was often unbalanced within data as offspring sex was unknown for the majority of experimental work. The heteroskedasticity and normality of data in all LMM models were assessed visually by plotting residuals, complimented by Kolmogorov-Smirnov and Levene's tests within the DHARMA package that further assessed the risk of unequal variances between sex/treatment groups [345]. In cases where assumptions were not met, log transformations were tested and/or weighted models were created using the nlme package to improve assumptions [346]. In the case of weighted models, fixed, interaction, and random effects remained the same, with variances allowed to differ by treatment, sex, or treatment and sex. The most appropriate weighted model of all options was subsequently chosen by selecting the model with the lowest Akaike information criterion (AIC). ANOVA modelling was not possible using a weighted model, therefore p values and statistics tables reflect only the weighted LMM statistics. For all analyses, full LMM and ANOVA tables are reported in Appendices relevant to the chapter of interest.

Using the emmeans package, pairwise differences between groups were directly assessed to complement these findings and determine group differences for significance bars presented in graphed data [347]. Only relevant pairwise comparisons were assessed, identifying sex differences within treatment groups, and treatment

differences within sex groups ($n = 4$ comparisons). All four relevant pairwise comparisons were carried out for every data set with pairwise comparisons presented, however only significant comparisons are presented in figures to improve readability. Post hoc p values were adjusted for multiple comparisons using the Bonferroni correction method. Where no significant main sex effect (or treatment*sex interaction) was identified by LMM, sexes were typically grouped in graphs. The majority of graphs presented in this thesis, all correlation analyses, all analysis of maternal cytokine data (including ANOVA analyses and assumption testing), and simple statistical tests (e.g. Fishers exact Chi-squared test, Mann-Whitney U test) were performed in GraphPad Prism [348].

Chapter 3:

Characterisation of the MIA Model

3.1. Introduction

As discussed in Chapter 1, the MIA model has been used extensively to investigate the biological mechanisms associated with NDDs and NPDs. Rodent research using the MIA model can be found across fields such as immunology, developmental biology, neuroscience, and psychology, assessing areas ranging from *in utero* development [349, 350] to postnatal cognition and communication [351, 352].

The complexity of embryonic development and diversity of approaches to MIA modelling in rodents has led to significant variability within published research [353]. Even when focusing on a single immunogenic reagent such as poly I:C, the number of doses [154, 354], dosage [355, 356], gestational day of administration [357, 358], administration method [359-361], and reagent batch variability [327, 341] vary between studies and may contribute to differences in maternal or foetal immune response and developmental outcomes. Further variables associated with changes in immune activity or MIA specifically include animal strain [244, 362], housing conditions [328], handling procedures [363, 364], and dam parity [365]. Even laboratories working in tandem with identical immunogens and protocols have identified differences in offspring outcomes between researchers, suggesting that potentially overlooked laboratory conditions may be interacting with MIA outcomes [156].

Review articles over the last 20 years have attempted to address concerns of reproducibility and comparability of published research by emphasising the need for extensive methods reporting [155, 353, 366]. Incorporating validation measures to characterise the maternal immune response and general development of offspring is also strongly recommended, to improve comparability between studies without relying on dosage or immunogenic agent alone. Validation measures may include indirect techniques such as observation of sickness behaviour, short-term fever, and maternal weight loss, or direct measurement of inflammatory cytokines in maternal circulation. Both indirect and direct approaches have important limitations: indirect measures are only suggestive of immune responses (and may not reflect dose-dependent changes in cytokine response) [367], however direct measures involve more invasive procedures that add to confounding maternal stress [325, 368]. While both approaches are useful indicators that dams successfully respond to poly I:C administration, direct

cytokine analysis provides additional insight into the driving mechanisms underlying MIA outcomes.

3.1.1. Chapter Aims

The overall goal of this chapter is to highlight key aspects of the MIA model used in this thesis and validate the MIA response to poly I:C exposure, focusing on suggestions and guidelines that aim to improve reproducibility in the field. Detailed methods on animal conditions and treatments are presented in the previous chapter, as well as in Appendix C which utilises the detailed MIA-specific 'Animal Research: Reporting of In Vivo Experiments' (ARRIVE) guidelines recommended at the American College of Neuropsychopharmacology conference in 2017 [155]. This chapter builds upon the thorough reporting of experimental conditions, and considers the evolving recommendations for MIA literature through two aims:

1. Measure key cytokine concentrations in maternal blood samples following poly I:C administration
2. Assess gross changes to litter success and growth following poly I:C administration.

Aim one was achieved through the development of a minimally-invasive saphenous vein blood sampling protocol, isolation of serum, and measurement of inflammatory markers IL-6 and CRP from maternal serum by ELISA (see 2.1.2.2 and 2.2 for methods). As discussed in Chapter 1, IL-6 is a common marker of inflammation following poly I:C administration, and both acute and chronic IL-6 exposure leads to MIA-associated outcomes in rodent models [176, 177]. IL-6 is expressed following poly I:C-activated TLR3 signalling [369], however variability has been previously observed in IL-6 response timing and intensity [157, 370]. CRP is less commonly utilised as an indicator of MIA in rodent models, however is well-supported in humans as a marker of MIA with an increased risk of neurodevelopmental disorders in offspring [127, 371]. CRP expression is primarily induced by circulating IL-6, with additional cytokines such as IL-1 enhancing the CRP response, leading to activation of the complement pathway and further cytokine expression [372-374]. As CRP is expressed approximately 100x higher in rodents than humans [375], and expressed later in the inflammatory

response, assessment of IL-6 and CRP together improves the likelihood that inflammation will be observed in poly I:C treated dams during the sampling window in this study. Saline controls however are predicted to show no increase in either IL-6 or CRP, remaining equivalent to baseline serum concentrations.

Aim two was addressed during the tissue collection process, by recording the number of successful litters and investigating the weight of foetal tissues upon sacrifice. As decreased litter survivability and decreased offspring/placental weight have also been associated with the MIA response following poly I:C administration [327, 370, 376], we subsequently assess offspring growth as indicators of MIA in this model.

3.2. Results

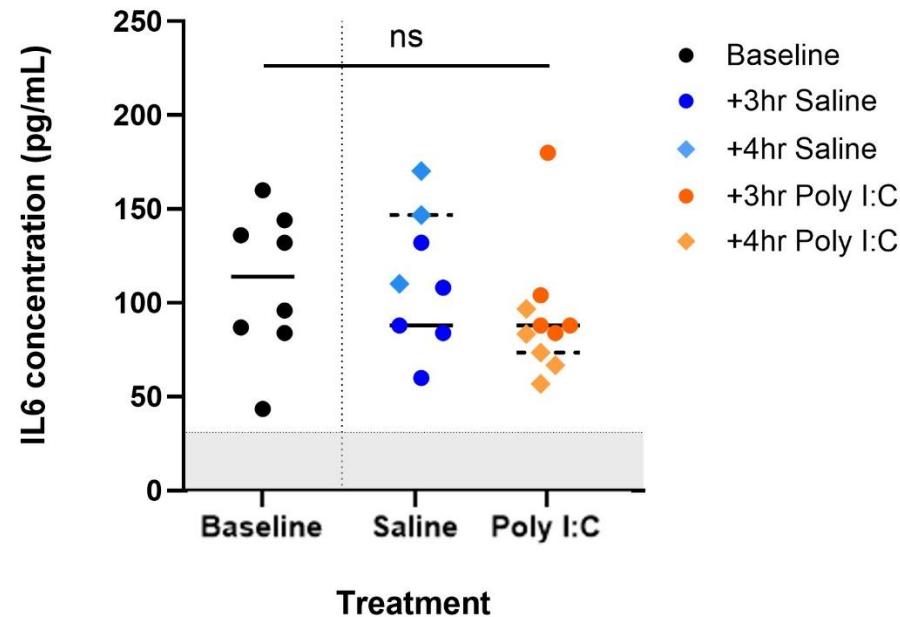
3.2.1. Inflammatory Markers in Maternal Serum

Maternal blood sampling was attempted only twice from every dam: once at approximately E9.5-10.5 as a baseline measurement, then again at E12.5 after injection with either poly I:C or saline vehicle. The second sample was taken either 3 or 4 hours post-injection, resulting in 5 distinct groups of samples: Baseline, Saline +3hrs, Saline +4hrs, Poly I:C +3hrs, and Poly I:C +4hrs. Within-subjects analysis of cytokine concentrations was planned initially, to identify concentration changes post-injection for each individual dam. Sampling was not always successful however, preventing this analysis; some attempts produced a low blood volume (e.g. <40 μ L), or no blood was collected at all. Samples were instead chosen at random for ELISA, with 3-4 paired samples occurring by chance in each ELISA group (see Appendix D for identification of paired samples). Only samples from dams that produced a litter (see Figure 3.2) were selected for analysis.

IL-6 concentration in maternal serum was assessed at both 3 and 4 hours post-injection, with CRP assessed only at 3 hours post-injection (Figure 3.1). No significant changes were observed in IL-6 concentration (Figure 3.1A). This included assessment of treatment group and sampling time as main effects by 2-way ANOVA ($F(2,22) = 0.75$, $p = 0.4829$; $F(1,22) = 0.0037$, $p = 0.9519$ respectively) or any pairwise differences between treatment/time point groups (all $p > 0.2$ prior to corrections for multiple comparisons). Poly I:C samples fell in a lower cluster compared control groups, particularly driven by samples taken at +4hrs, however almost all post-injection values fell within the pre-injection baseline range.

By contrast, 1-way ANOVA revealed a treatment effect on CRP concentration ($F(2, 13) = 9.19$, $p = 0.0033$). Pairwise comparisons with Tukey's post-hoc correction revealed a significant increase in CRP in Poly I:C injected dams compared to baseline measurements (adjusted $p = 0.0024$; Figure 3.1B), however this was not significant in comparison to saline controls (adjusted $p = 0.1816$). CRP in saline dams was not significantly increased compared to baseline (adjusted $p = 0.1652$), however there was a slight trend of increased concentration after saline injection compared to baseline, and poly I:C injection compared to saline injection.

A) IL-6 concentration: 3-4hrs post-injection



B) CRP concentration: 3hrs post-injection

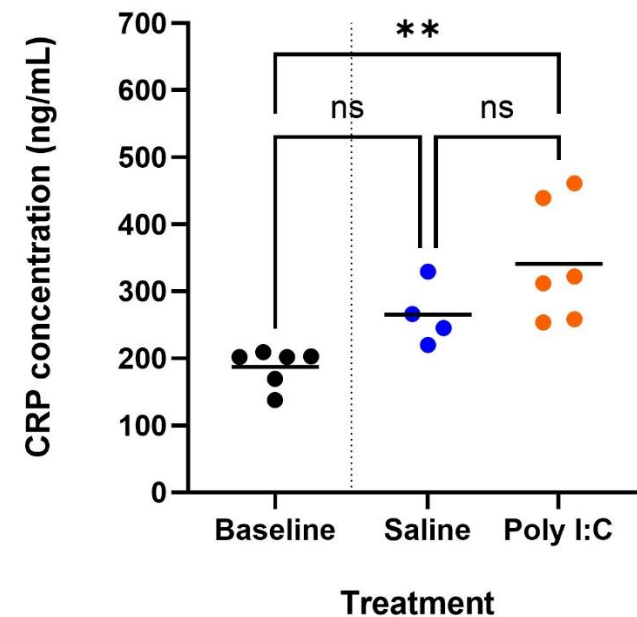


Figure 3.1: Maternal serum shows increased CRP concentration following poly I:C injection compared to baseline, but not IL-6. (A) IL-6 serum concentrations were measured at baseline, then 3 hours and 4 hours post-injection in both poly I:C and saline injected dams. N = 8 baseline, 4 saline+3hr, 3 saline+4hr, 5 poly I:C+3hr, 5 poly I:C+4hr. (B) CRP concentration measured 3 hours after injection. N = 6 baseline, 4 saline, 6 poly I:C. Full or dashed mean line indicates +3hr or +4hr mean respectively. Grey background shading indicates range below the ELISA limit of quantification (IL-6 = 31 pg/mL, CRP = 0.2 ng/mL).

3.2.2. Litter Production

Animal breeding was performed across 4 cohorts, with a maximum of 16 dams able to be paired at once. Cohort 1 included 2 dams as a trial, cohorts 2 and 3 each included 16 dams, and cohort 4 included an additional 8 dams to compensate for a low number of poly I:C litters produced in cohorts 2 + 3.

A total of 39/42 pairings successfully produced plugs and were assigned to a treatment group, while 28 treated dams successfully produced pups on the date of dissection (Figure 3.2A). Of the 11 dams which did not produce a litter, almost all (10/11) had been designated to the poly I:C group. Comparing the total frequency of unsuccessful litters using Fishers exact Chi-squared test, litter success significantly differed between treatments ($p = 0.0277$). With 6.67% and 41.67% of saline and poly I:C dams failing to produce a litter, relative risk between treatments indicated an unsuccessful litter was 6.25 times more likely to occur in the poly I:C group compared to controls. There was no additional effect of cohort on the frequency of unsuccessful litters in the poly I:C treatment group ($p = 0.6882$).

High variability was observed in litter size irrespective of treatment, which ranged between 1 and 20 pups per litter (Figure 3.2B). Litter size distinctly fell in a bimodal distribution; five litters were considered small (< 7 pups) while the rest were large (> 10 pups). There was no significant difference in litter size between poly I:C and saline treatment groups using the non-parametric Mann-Whitney U test, either when assessing total litter production ($U = 88.5$, $p = 0.6728$), or assessing the larger cohorts 2 and 3 individually ($U = 13$, $p = 0.6242$; $U = 12$, $p = 0.7394$ respectively). No significant treatment effect was observed in frequency of producing small litters either ($p = 0.3577$), even though a higher proportion of the poly I:C litters were considered small (10% saline compared to 28.57% poly I:C).

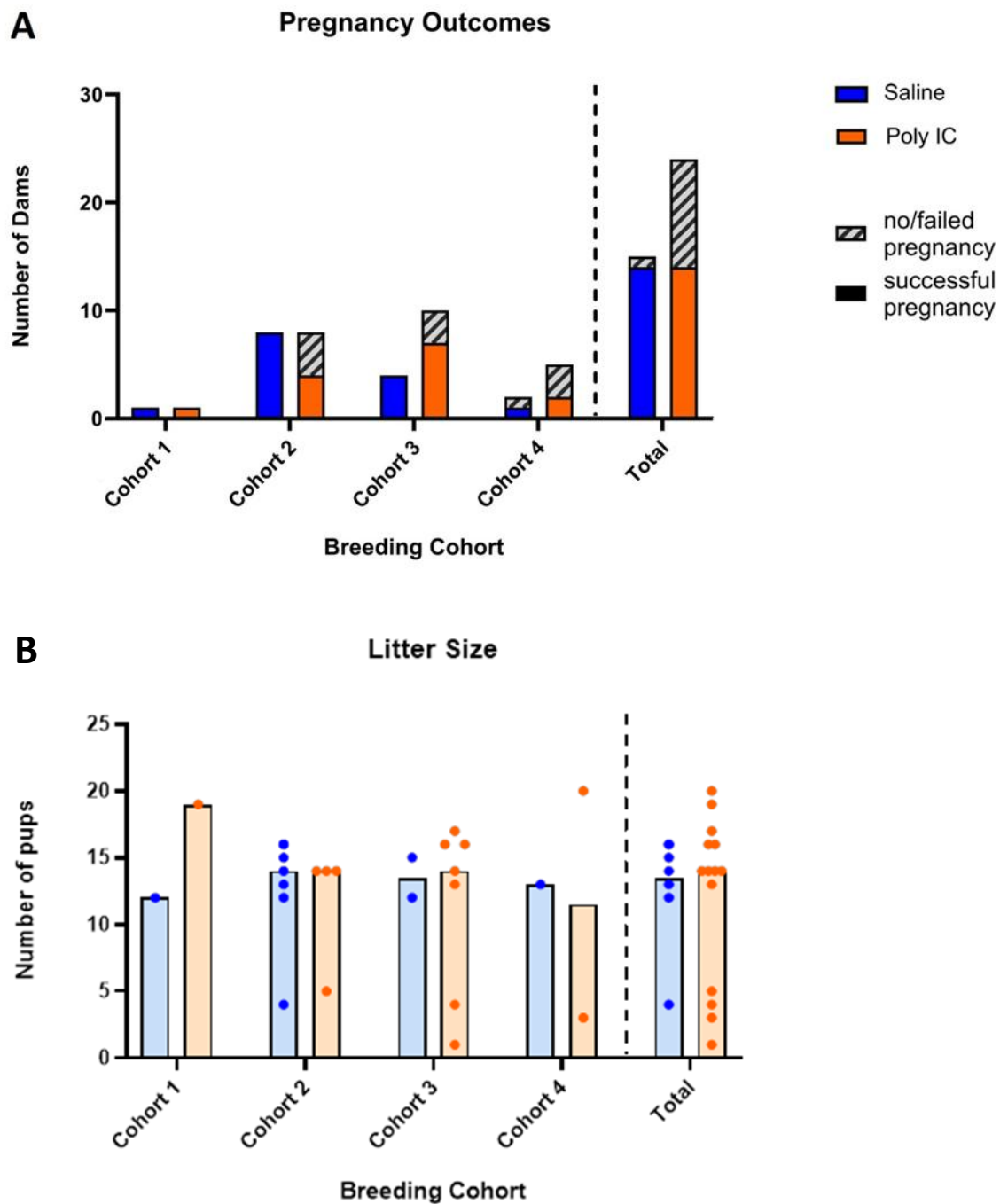


Figure 3.2: Litter production differs between saline and poly I:C treatment groups. (A) Ratio of successful (solid coloured bar) to unsuccessful (grey striped bar) breeding pairs across all 4 cohorts, separated by treatment group, with accumulative total shown to the right of dashed line. (B) Number of pups generated in each successful litter, with bar to indicate median litter size. Accumulative total shown to the right. Blue = saline, orange = poly I:C.

3.2.3. Offspring Development

Whole amniotic sac, whole placenta, and whole foetal body weights were recorded for the majority of pups sampled in this study. Litters S1 and P1 (sampled at E18.5, both litters of cohort 1) were sampled before the decision to record offspring weight and thus not able to be weighed, and S10 (the first litter sampled at E14.5) was not weighed in error. Following weight measurement, the relationship between foetus and placenta was explored by calculating foeto-placental ratio and determining the correlation between foetal and placental weight. In total, 56 pups from 11 litters were assessed at E14.5 (37 pups from 7 poly I:C litters) and 73 pups from 13 litters were assessed at E18.5 (29 pups from 5 poly I:C litters).

The summary data derived from all assessed pups, including overall distribution and the average weight for each litter, is presented in Figure 3.3 (all individual data points are presented in Appendix D). Weight of the total amniotic sac, foetus, and placenta did not show overt treatment differences when graphed, however litter variance was apparent across all measurements. This was most striking at E18, with one saline litter (S4) and two poly I:C litters (P3 and P10) clustered significantly higher in total weight, foetal weight, and foeto-placental ratio (Figure 3.3B). These litters also had the highest average placental weight at E18.5, however did not cluster separately from the other litters examined. The heavier pup weights in these litters were not associated with a small litter size – S4, P3, and P10 were categorised as large litters, with litter sizes of 16, 14, and 17 respectively. Across all tissue type, age, and treatment combinations, intra-litter variability appeared to be the major factor influencing the spread of raw weight data.

Given the significantly increased failure rate in generating poly I:C litters, outlying litters were not removed from analysis. To account for litter variance across weight measurements, linear mixed modelling (LMM) was used to structure the data and include litter as a random effect. Sex, treatment, and sex*treatment interaction effects were assessed, with each time point assessed separately. A summary of effect significance can be seen in Table 3.1.

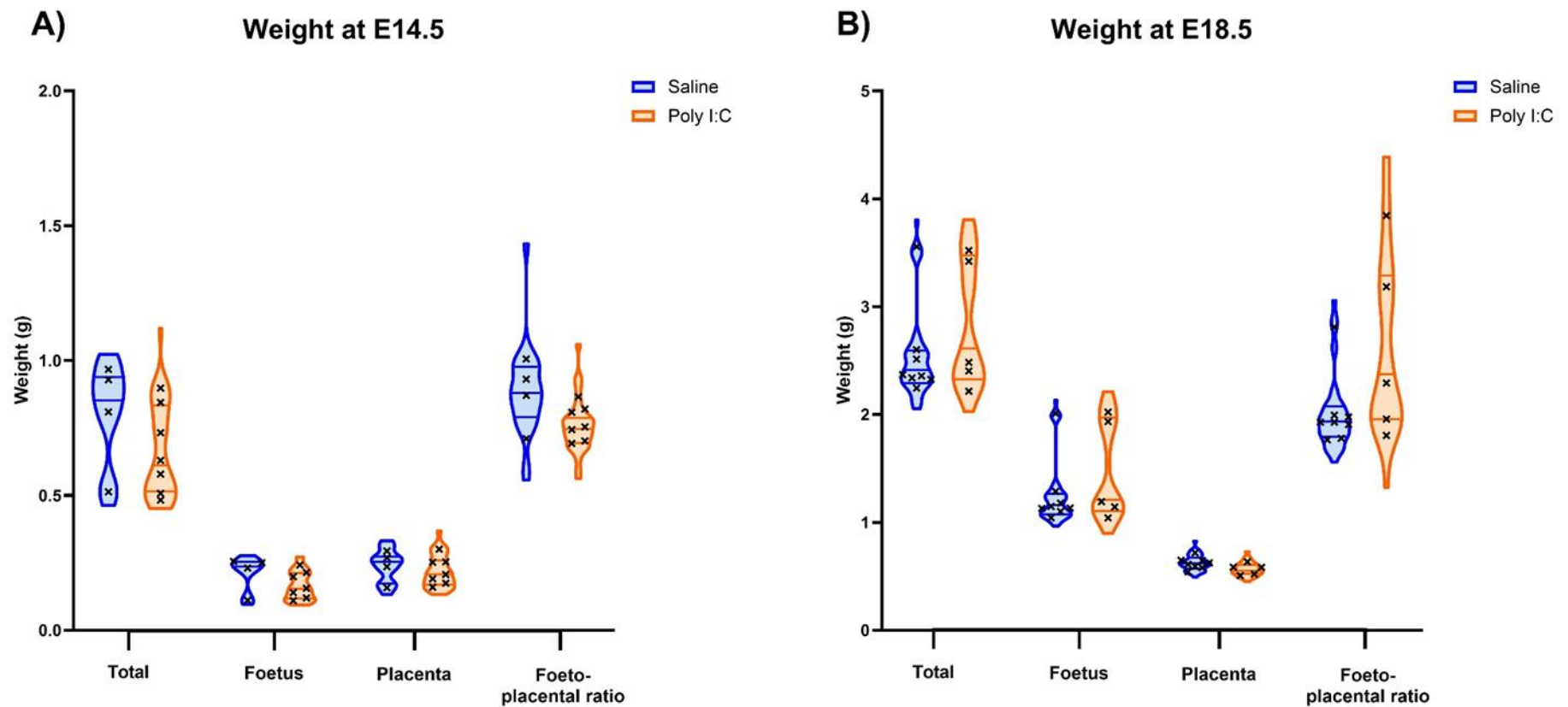


Figure 3.3: High litter variance is present across offspring weight measurements. Weight of the total amniotic sac, foetus, placenta, and calculated foeto-placental ratio is presented from offspring at E14.5 (A) and E18.5 (B). Violin plots are derived from all data available from individual pups, representing data density overall. Scatterplots overlaying each violin show the average weight of each litter contributing to the violin, indicated with x icons.

Age	Tissue	Fixed Effects <i>p</i> -values		
		Sex	Treatment	Sex*Treatment
E14.5	Total	0.7773	0.2395	0.4993
	Foetal	0.4680	0.2590	0.9122
	Placental	0.6931	0.5115	0.5695
	Foeto-placental ratio	0.3111	0.0867	0.4431
E18.5	Total	0.0028	0.3838	0.7581
	Foetal	0.0102	0.3562	0.9706
	Placental	0.0071	0.1230	0.1118
	Foeto-placental ratio	0.7719	0.1007	0.2226

Table 3.1: Significance summary of fixed effects explored in offspring weight data using LMM models. Yellow and green shading highlights near-significant ($p \leq 0.10$) and significant ($p \leq 0.05$) *p*-values respectively. Full accompanying LMM tables can be found in Appendix D .

3.2.3.1. Sex and Sex*Treatment Interaction Effects

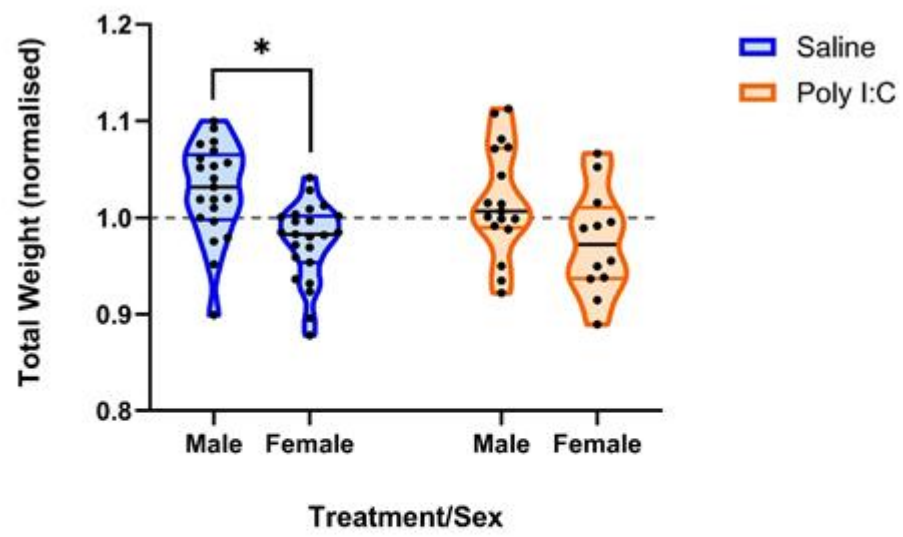
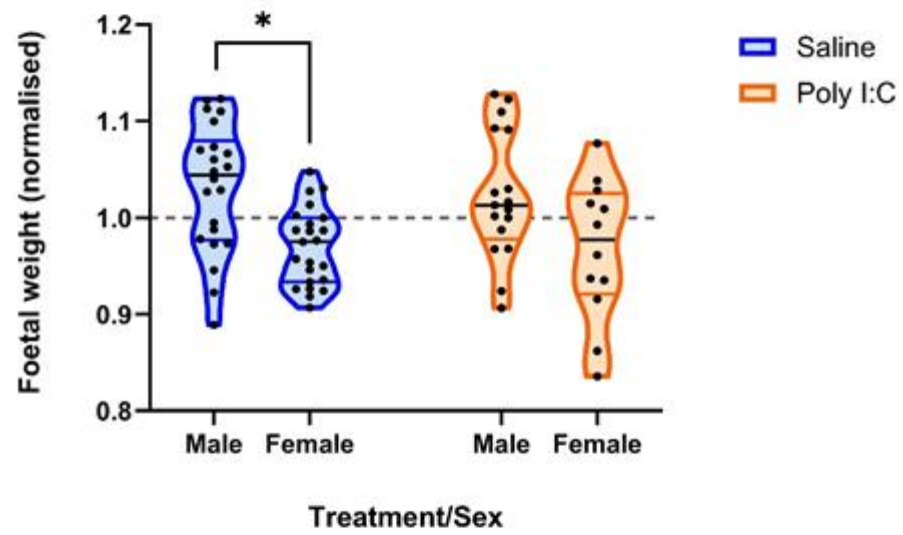
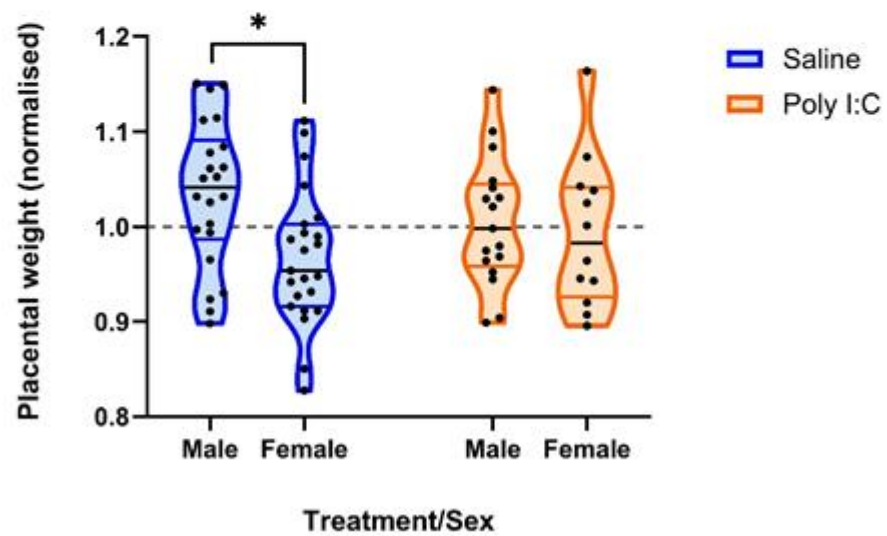
No significant sex effect, or sex*treatment interaction, was observed at E14.5 in any weight assessed (Table 3.1). Significant sex effects were observed however at E18.5 in total, foetal, and placental weight ($t(58) = 3.12$, $p = 0.0028$; $t(58) = 2.65$, $p = 0.0102$; $t(61) = 2.79$, $p = 0.0071$ respectively). To further explore these effects, pairwise comparisons were generated between each relevant sex-treatment combination (Table 3.2). While no significant sex*treatment interactions were observed in total, foetal, and placental weight at E18.5 (all $p > 0.1$, Table 3.1), treatment groups remained separate during sex comparisons due to the role of treatment as an essential variable of interest for this chapter. Adjusting for litter variance was necessary to depict sex differences visually, so weight measurements for each individual pup were subsequently normalised to the average weight for their litter and used to model significant LMM pairwise differences (Figure 3.4).

Pairwise comparisons revealed that sex effects were strongest in saline offspring (Table 3.2, Figure 3.4). In total, foetal, and placental weight, saline male offspring were significantly heavier than saline females after post-hoc corrections ($t(58.45) = 3.12$, adjusted $p = 0.0114$; $t(58.33) = 2.66$, adjusted $p = 0.0409$; $t(61.61) = 2.79$, adjusted $p = 0.0283$ respectively). This effect is observable in litter-normalised data, with the mean weight of saline males always > 1 and the mean weight of saline females always < 1 (Figure 3.4).

Compared to saline, poly I:C offspring followed a very similar visual trend in total and foetal weight (Figure 3.4). Weight measurements from male offspring clustered higher than female littermates, however this was less significant than the sex difference found in saline offspring and narrowly failed to meet corrections for multiple comparisons ($t(58.10) = 2.34$, adjusted $p = 0.0912$; $t(58.07) = 2.39$, adjusted $p = 0.0796$ respectively).

Interestingly, the sex difference observed in the placenta of saline offspring was not mirrored in poly I:C offspring ($t(58.68) = 0.29$, $p = 0.7733$). No sex effect in placental weight was apparent in poly I:C offspring, although initial pairwise comparisons did observe a significant decrease in poly I:C male placentas compared to saline controls. This effect failed to survive corrections for multiple comparisons ($t(13.57) = 2.15$, adjusted $p = 0.1991$).

Figure 3.4: Saline offspring show significant weight differences at E18.5 when compared across sex. Offspring weights normalised to corresponding litter averages illustrate significant pairwise comparisons generated by LMM of total (A), foetal (B), and placental (C) weight. Pairwise comparisons were carried out between sexes within a treatment, as well as across treatment within sex. Only statistically significant comparisons are shown on figure. Mean for each group is represented by black horizontal bars. * = adjusted p value < 0.05 .

A**B****C**

Tissue	Comparison	Estimate	SE	df	T ratio	Initial p-value	Adjusted p-value
Total	Saline Male – Saline Female	0.1524	0.0489	58.45	3.116	0.0028	0.0114
	Saline Male – Poly I:C Male	-0.2493	0.2896	11.32	-0.861	0.4070	1.0000
	Saline Female – Poly I:C Female	-0.2722	0.2902	11.42	-0.938	0.3677	1.0000
	Poly I:C Male – Poly I:C Female	0.1295	0.0554	58.10	2.339	0.0228	0.0912
Foetus	Saline Male – Saline Female	0.0828	0.0312	58.33	2.655	0.0102	0.0409
	Saline Male – Poly I:C Male	-0.2091	0.2174	11.23	-0.962	0.3562	1.0000
	Saline Female – Poly I:C Female	-0.2074	0.2177	11.30	-0.953	0.3607	1.0000
	Poly I:C Male – Poly I:C Female	0.0845	0.0353	58.07	2.394	0.0199	0.0796
Placenta	Saline Male – Saline Female	0.0442	0.0159	61.61	2.786	0.0071	0.0283
	Saline Male – Poly I:C Male	0.0729	0.0338	13.57	2.154	0.0498	0.1991
	Saline Female – Poly I:C Female	0.0339	0.0344	14.55	0.985	0.3405	1.0000
	Poly I:C Male – Poly I:C Female	0.0053	0.0182	58.68	0.289	0.7733	1.0000

Table 3.2: Pairwise sex effects in offspring weight at E18.5 differ between treatments. All pairwise comparisons between relevant sex/treatment groups were generated for LMMs with sex as a significant main effect. Initial p values and adjusted p values (using Bonferroni's correction for multiple comparisons) are presented alongside accompanying statistics. Green and yellow shading emphasises significance ($p \leq 0.05$) or near-significance ($p \leq 0.10$). Values rounded to a maximum 4 decimal places. df = degrees of freedom, SE = standard error.

3.2.3.2. *Treatment Effects*

No significant treatment effects were observed at E14.5 or E18.5 in any weight measurement (all $p > 0.05$, Table 3.1). Interestingly, a near-significant treatment effect was observed in foeto-placental ratio at E14.5 ($t(9) = 1.93$, $p = 0.0867$) and at E18.5 ($t(11) = 1.79$, $p = 0.1007$).

To explore the relationship between foetal and placental weight further, the correlation between these weights was assessed. At E14.5, foetal and placental weight was strongly significantly correlated in both treatment groups (Figure 3.5B). The relationship between foetal and placental weight was modelled through simple linear regression, and no significant difference in slope between treatment groups was identified by ANCOVA ($F(1, 53) = 2.24$, $p = 0.1408$). The elevation of each regression however was found to be significantly different ($F(1, 54) = 6.64$, $p = 0.0127$), suggesting that a poly I:C pup of a given foetal weight would have a significantly heavier placenta than a saline counterpart of equal foetal weight.

Moving from E14.5 to E18.5, saline exposed offspring continued to display a significant correlation between foetal and placental weight (Figure 3.5C) however the slopes at each time point were significantly different ($F(1,66) = 17.59$, $p < 0.0001$; Figure 3.5A). Poly I:C exposed offspring in contrast did not demonstrate a significant correlation at E18.5 (Figure 3.5C). Unsurprisingly, this resulted in a significant difference between the slopes of poly I:C offspring at both time points ($F(1,57) = 37.79$, $p < 0.0001$) as well as between poly I:C and saline offspring at E18.5 ($F(1,70) = 12.62$, $p = 0.0007$). Correlations and linear regressions were also explored where the heavier E18.5 litters were removed from analysis to ensure the bimodal distribution of foetal weights did not overly skew the relationship (Figures 3.3C, 3.5D). While slopes were slightly changed by the exclusion of heavier litters, the significance of saline and poly I:C correlations at E18.5 remained unchanged. This also extended to regression comparisons, which remained significantly different by treatment ($F(1,52) = 9.35$, $p = 0.0035$). Further correlations of E18.5 weights separated by sex were considerably statistically limited by low sample numbers (excluding heavy litters, poly I:C $n = 8$ females and $n = 9$ males), thus not presented as a main figure, but reflected similar outcomes to those presented in Figure 3.5 (Appendix D).

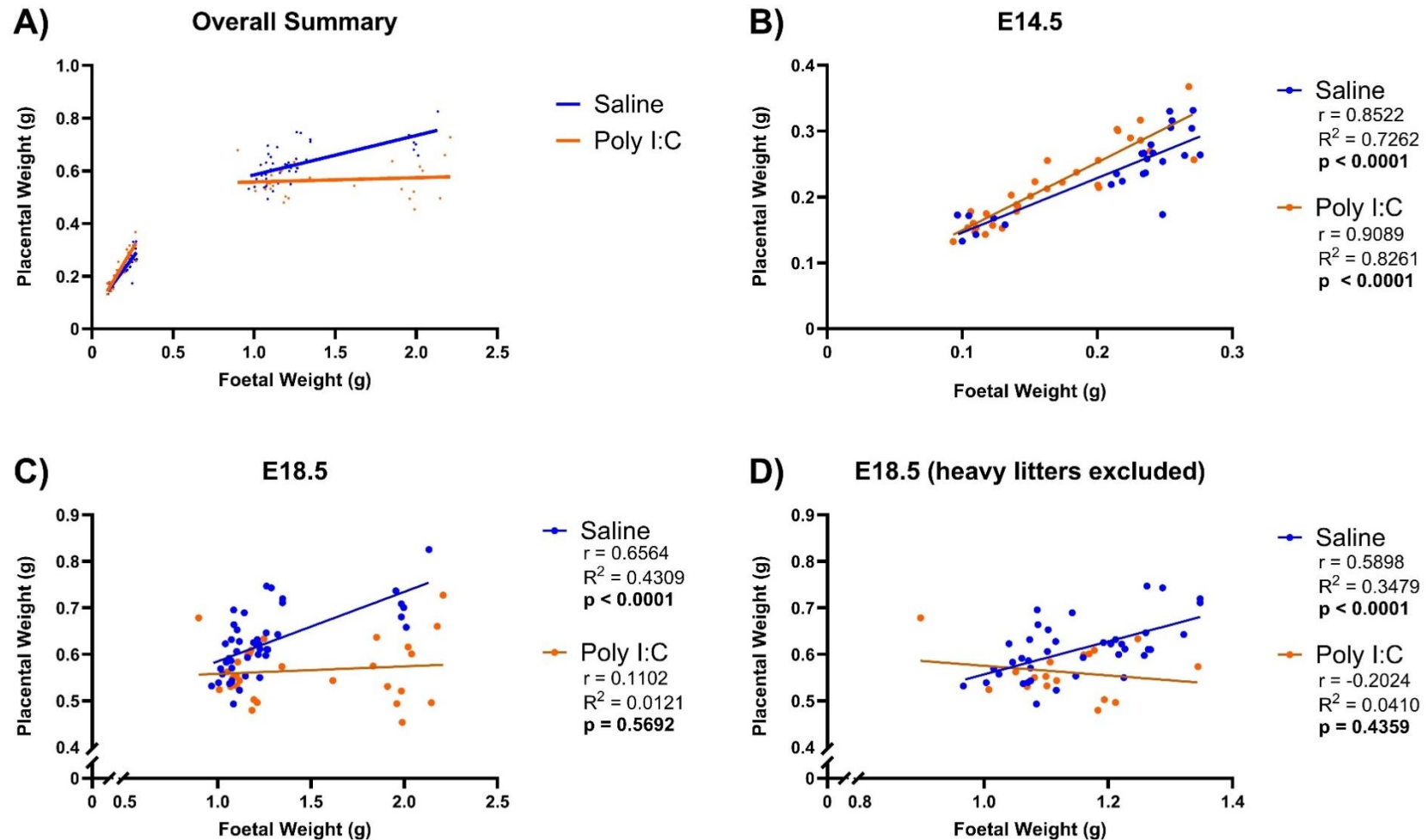


Figure 3.5: Foetal and placental weight correlations differ by treatment and age. Correlations for both E14.5 and E18.5 on the same axis (A), E14.5 (B), E18.5 (C), and E18.5 without the three outlying litters S4, P3, and P10 (D) are presented with accompanying correlation statistics and p values in bold. r = Pearsons correlation coefficient, R^2 = coefficient of determination.

3.3. Discussion

Both of the major aims of this chapter focused on evaluating the general response to poly I:C treatment in this model, using approaches commonly observed and recommended in MIA literature. The maternal response was primarily assessed in this chapter through serum concentrations of IL-6 and CRP, with additional consideration given to the production, size, and overall growth of successful litters. This was not only to validate the MIA model and show that poly I:C successfully generated an inflammatory response, but also to provide a number of measures for comparing this model to other MIA studies with variable methodology.

3.3.1. Poly I:C Increases Markers of Maternal Stress

3.3.1.1. *IL-6 and CRP in Maternal Serum*

Two inflammatory markers were selected for analysis in maternal serum, to provide insight directly into the maternal immune response within the constraints of cost and serum availability. The first was IL-6, which is arguably the most commonly assessed cytokine in maternal serum following poly I:C administration [157, 377]. In our model, serum concentrations of IL-6 did not differ between treatment group, either overall or when isolated into 3 or 4 hours post-injection (Figure 3.1A). This was not in line with our expectation (observing an increase in IL-6 following poly I:C administration), however the secondary target CRP did demonstrate a significant increase 3 hours after poly I:C treatment compared to baseline (Figure 3.1B). This increase was not significantly higher than saline controls, which may indicate the presence of a procedure-related response; low-level inflammation at the injection or blood sampling site may have occurred in both saline and poly I:C treated dams, however the neutral saline vehicle caused no additional response while poly I:C activity as a TLR3 agonist led to elevated inflammation and a further shift in CRP that met significance compared to baseline only.

Importantly, CRP is stimulated downstream of IL-6 production, which suggests serum IL-6 after poly I:C administration may have peaked prior to the sampling at 3 or 4 hours. Timing of the IL-6 response is commonly reported at 3-4 hours after

administration in Sprague-Dawley rats, however IL-6 changes have been observed as early as 2 hours and as late as 6 hours across other rodent species and strains [157].

Outside of species/strain effects, a particularly relevant consideration for timing and intensity of the IL-6 response in this model is poly I:C administration route. While the subcutaneous (s.c.) injections utilised in this study are common when administering lipopolysaccharide (LPS) as an immunogen, the majority of literature utilising poly I:C opts for intraperitoneal (i.p.) or intravenous (i.v.) administration [157]. I.p. or i.v. injections of small molecules are typically thought to enter circulation faster than s.c. injections, however limited information is available investigating poly I:C responses specifically. Early studies comparing administration routes of poly I:C in rats used significantly lower doses than used in current MIA models (typically ≤ 1 mg/kg) and did not highlight a difference in the timing of the immune response, instead demonstrating a significantly reduced/undetectable overall response to poly I:C by s.c. injection. No publications assessing maternal IL-6 after s.c. injection of higher poly I:C doses were available until recently, with two studies by Taskiran and colleagues administering 10 mg/kg poly I:C to Wistar rats by s.c. injection (as confirmed by personal email correspondence, due to inconsistencies in published methods) [146, 147]. These studies both found that maternal IL-6 was significantly increased only 2 hours following poly I:C administration, aligning with the suggestion that maternal IL-6 concentration may have peaked and abated prior to the sampling time point used in this thesis.

Molecular weight of the poly I:C used in this thesis may have also contributed to the strength and duration of the IL-6 response observed. For example, Lan et al. used a 4 mg/kg dose of poly I:C in pregnant Sprague Dawley rats, and found a significant increase in IL-6 at 3 hours after poly I:C administration that reached 300 pg/mL and abated by 6 hours [378]. In comparison, Careaga et al. (under the same strain/dosage conditions) found no IL-6 increase after low molecular weight poly I:C at either time point, and dramatically higher increases in IL-6 at both 3 and 6 hours following high molecular weight poly I:C (~1000pg/mL and 6000pg/mL respectively) [341]. While Lan et al. does not report molecular weight in their study, the poly I:C utilised in this study is of low molecular weight (see 2.1.2.1). Low molecular weight poly I:C is often associated with a reduced IL-6 response compared to high molecular weight, although both high and low molecular weight poly I:C have been demonstrated to induce

behavioural and molecular changes in offspring [156]. This may in part be driven by additional cytokines not examined here; Careaga et al. also demonstrated that both high and low molecular weight poly I:C were associated with an increase other cytokines 3 hours after injection such as IL-1 β , which is known to enhance the CRP response downstream of IL-6.

Taken altogether, the findings from maternal serum and outside research support the induction of an immune response generated by poly I:C administration, although a fuller characterisation of the immune response specific to these animals was not possible due to the limited amounts of serum. Samples were unable to be compared within pairs, and often of limited volume, due to the ongoing improvement of the saphenous vein blood sampling technique developed for this chapter. No method of blood sampling pregnant dams was established in our facility (short of cannulation, which brings additional health and maternal stress risks) prior to this technique. Future in-house work will be able to build upon these findings and techniques, leading to a more comprehensive cytokine analysis in subsequent studies that should include a panel of additional cytokines such as TNF α and IL-1 β and an earlier sampling of maternal serum at 2 hours post-injection [146, 341] .

3.3.1.2. *Successful Birth Rate and Spontaneous Abortion Risk*

Litter loss is indicative of significant gestational stress in small rodents. Reabsorption of a partial or whole litter (leading to a small litter at birth, or no litter entirely) is a survival mechanism preventing reproduction when the dam is experiencing dangers including physical injury, illness, or poor nutrition. Poly I:C treatment has the potential to induce an immune response that activates this mechanism, with previous evidence from MIA models suggesting a reduced litter size and increased rate of spontaneous abortion may be observed in poly I:C treated dams [139, 327, 379].

While litter size here was not significantly affected by poly I:C, a stark decrease in the number of successfully produced litters was clear in the poly I:C treatment group. This aligns with previous research demonstrating increased litter loss after poly I:C treatment, particularly in batches of poly I:C that induced a stronger cytokine response [327]. It is important to note that the absence of a litter at the time of dissection is not necessarily the result of spontaneous abortion – it is possible that in some cases, vaginal plugs were identified and mating pairs were separated before the dam was

successfully pregnant. No obvious differences were noted upon dissection of dams that would indicate the presence and subsequent resorption of pups rather than a lack of pregnancy entirely, however reabsorption is not always visually apparent and time constraints during dissection sessions prevented thorough inspection for resorption sites such as ammonium sulphide staining [380]. Additionally, factors such as physical stress (from handling during blood sampling and injection procedures) may contribute to litter loss in dams for reasons other than poly I:C treatment [381].

The above considerations are likely to contribute to the number of dams that fail to produce a litter, however it is critical to note that these would also affect both saline and poly I:C treated dams equally. Failed mating and handling-associated litter loss may explain the one saline dam that failed to produce a litter, and by extension a small number of the poly I:C treated dams; however, they would not explain the significant extent to which litters failed to be produced in the poly I:C group specifically. Rather, the significant treatment effect observed here suggests poly I:C administration in this study resulted in an increased risk of foetal reabsorption commonly associated with maternal inflammation [139, 327, 379].

3.3.2. Sex and Treatment Influence Growth of the Foetoplacental Unit

3.3.2.1. Initial Sex Effects

Weight of the total foetoplacental unit, foetus, and placenta were investigated to assess general growth of foetal tissues after poly I:C administration. Curiously, sex was found to have a significant effect on all raw weight measurements at E18.5 (Table 3.1), with smaller weights observed in females. While it is common for adult Sprague Dawley females to be smaller than male littermates, pup weight is very similar between sexes prior between PND1 - PND30 [382]. The significantly reduced female tissue weights at E18.5 were not previously observed at E14.5, indicating overall female development diverges from males in a short window after E15 but does not necessarily persist at birth around E21. While no pups were sampled and/or weighed postnatally in this study, previous work with Sprague Dawley rats in our laboratory have not identified significant weight differences between sexes after birth, even by weaning

age (~PND21) [383]. Looking into this period further, research from Kalisch et al. highlighted a similar effect in placental development of Sprague Dawley rodents at E15 and E20 [188]. Placental weight was reduced in females at E15 and E20, however foetal weight was only reduced in females at E20. Female placentas exhibited reduced vascularisation in the foetally-derived labyrinth zone, hypothesised to be the result of increased metabolic demand and thus nutrient requirements in males during the development of testes and increased androgen production.

Similarly to saline controls, poly I:C offspring exhibited a trend of reduced overall and foetal weight in E18.5 females (Figure 3.4). Interestingly, the trend was not present in the placenta – while saline females exhibited a significantly smaller placenta than saline males, placentas from poly I:C male and female offspring were indistinguishable. This is likely due to the trend of male poly I:C placentas being smaller than male saline placentas, which was found to be a significant reduction prior to (but not following) post-hoc corrections (Table 3.2). A downwards shift in placental weight in male offspring following poly I:C would diminish the underlying sex difference between male and female placental weights seen in control animals, leading to a loss of significance. This would also align with the hypothesis presented in Kalisch et al that male fetuses rely on increased placental efficiency and are likely the offspring that are first affected by adverse maternal conditions during pregnancy [188].

3.3.2.2. Placental Changes Following Poly I:C Exposure

Placental efficiency has previously been explored in MIA research and continues to be assessed in this study, through calculations including foeto-placental ratio and the correlation between foetal and placental weight [195]. The near-significant treatment effect observed in foeto-placental ratio (Table 3.1) led to identifying significant differences in the correlations between foetal and placental weight at both time points (Figure 3.5).

At E14.5, placental weight was highly correlated to foetal weight in both poly I:C and saline offspring, although a difference in the regression intercept suggested that poly I:C offspring at E14.5 tend to have heavier placental weight than saline counterparts (Figure 3.5). At E18.5 however, only saline offspring show a correlation between foetal and placental weight. Poly I:C placental weight also visually clusters lower than in saline offspring, especially at higher foetal weights, suggesting that placental weight

at E18.5 may have plateaued relative to foetal growth or be otherwise delayed at this developmental stage following poly I:C. While overall treatment effects on foetal and placental weights were not significant by LMM, this suggests that more subtle changes in the developmental trajectory of offspring are present.

Taken altogether, these findings would suggest that poly I:C exposure at E12.5 is associated with a subtle shift towards increased placental weight at E14.5, yet a tendency for lower placental weight at E18.5 that is no longer coupled with foetal growth as observed in saline controls. Overall foetal weight appears to be only minimally affected by poly I:C exposure, which does not significantly differ from saline offspring by main effects analysis or in pairwise comparisons - the significant sex disparity seen in the foetal weight of saline offspring is however weaker in poly I:C offspring. This may be a result of reduced placental efficiency, with a smaller placental mass in particularly poly I:C males resulting in a less stark contrast between male and female offspring [188]. Previous MIA research has reflected these outcomes; Kowash et al for example found significant reductions in placental weight following poly I:C during late gestation, as well as a reduced brain weight in males, however overall foetal weight was significantly affected sex and not by treatment group [370]. While poly I:C was administered at GD15 and these weights were obtained at GD21, the Wistar rats in the Kowash study have a 2- day longer gestational period than the Sprague-Dawley strain in this study (23 vs 21 days) and are likely at a very similar developmental stage compared to the E18 animals in this study.

3.3.3. Considerations for the use of the MIA model in this thesis

3.3.3.1. Statistical Approach

As seen in the offspring weight data presented, litter variance has the potential to lead to high variance or skewed data that risks confounding the effect of treatment (Figure 3.3). The underlying association between offspring in a litter threatens the independence assumption between offspring and should be examined in statistical analysis of MIA models. Some suggested approaches utilise litter as the sample number rather than offspring, by either sampling one pup per litter or averaging all

littermates for one final value per litter [155]. Aside from ethical questions in generating litters for the use of a single pup, factors such as sex demonstrate that important differences between littermates exist; in this model, sex was a highly significant influence on offspring weight that would not have been identified if littermates were simplified to analysis as technical replicates to be averaged (Figure 3.4). The weight data analysed in this chapter highlights the importance of between- and within- litter offspring differences, supporting the use of LMM analysis in this and future chapters to balance these factors during data analysis.

3.3.3.2. *Implications for Later Chapters*

Across the measures examined in this chapter, poly I:C administration has been shown to influence gestational development in ways comparable to published MIA literature. Increased maternal CRP (compared to baseline only and not saline controls), increased litter loss, and altered placental growth at E18.5 support the underlying assumption of this thesis that the model presented did successfully generate an MIA response. While future work should build upon these techniques to further characterise immune activity, the information presented does provide useful context for understanding the general poly I:C response in these animals.

Chapter 4:

Quantification of 5-HT in Amniotic Fluid Following MIA

4.1 Introduction

Amniotic fluid (AF) acts as a dynamic, buffering fluid that surrounds the developing foetus and placenta within the amniotic sac [384]. Physically, AF offers a protective cushion for the developing foetus and pressure to support lung and muscle development. Chemically, AF acts as a transfer medium for waste products and a pool of growth factors, hormones, electrolytes, vitamins, and trace elements that support the development of increasingly complex tissues [385]. Transfer of solutes from foetus to AF is bi-directional, either through diffusion of solutes directly across cell membranes (in early gestation) or through swallowing and excretion of AF through the developing foetal digestive system (in late gestation) [384]. As a result, AF is a complex yet useful matrix for assessing foetal nutrition and health.

AF is hypothesised to act as a buffering site for excess 5-HT, although only a small number of studies have reported investigating 5-HT concentration within human or rodent AF [306, 386]. Previous literature has shown that human amniocytes have the capacity to take up and metabolise 5-HT through SERT and MAO-A activity, producing 5-HIAA which may be excreted through maternal circulation [306]. Numerous studies have also demonstrated SSRI transfer from maternal circulation to the AF, although the effects of this on AF 5-HT concentrations have not been addressed [387-389]. While previous MIA research has assessed 5-HT concentration within the foetal brain and placenta (tissues that are known to produce 5-HT), no known MIA studies have investigated AF 5-HT [227, 301]. In line with the role of AF as a buffer system, MIA-associated increases in 5-HT within the foetoplacental unit (such as an influx of maternal 5-HT, or the increased placental 5-HT production seen by Goeden and colleagues) may also be evident within AF [227].

In the foetoplacental unit, reliable quantification of 5-HT remains challenging. Some studies have previously used antibody-based methodology such as ELISA to detect 5-HT in brain and placental tissue [324]. However, commercially available serotonin antibodies have repeatedly demonstrated poor specificity and/or are highly sensitive to small variations in tissue preparation [390, 391]. A more commonly used technique to quantify 5-HT in biological matrices is high-performance liquid chromatography (HPLC), where analytes in solution (injected in the liquid, mobile phase) are separated through unique size, charge, and/or affinity interactions within the chromatography

column (containing a solid stationary phase) [227, 315]. This approach however often shows limited sensitivity and specificity for small, reactive metabolites such as 5-HT, particularly in complex matrices where multiple metabolites may share the same retention time or may interact with each other to produce inconsistent retention times [392]. This may be improved by liquid chromatography mass spectrometry (LC-MS), which utilises a liquid chromatography column (as seen in HPLC) paired with a mass spectrometer for detection. Target analytes are able to be extracted from the total ion chromatogram by their mass to charge ratio (m/z), and/or fragmented prior to detection using tandem mass spectrometry (LC-MS/MS), allowing improved target identification and specificity [393]. While there is currently a lack of research specifically assessing 5-HT in AF by LC-MS, LC-MS has been used to quantify 5-HT in biological matrices, including serum, placental homogenate, and rat brain microdialysate [301, 333, 393]. This includes research performed at Victoria University of Wellington using equipment and reagents readily available, which provides a promising starting point for exploring 5-HT and its metabolites in AF in this thesis [333, 334].

4.1.1 Chapter Aims

MIA may lead to altered foetal 5-HT exposure, potentially through influx of maternal 5-HT or altered 5-HT synthesis within the placenta or foetal brain. The overall goal of this chapter was to investigate whether changes in the concentration of 5-HT and the metabolite 5-HIAA could be detected in AF following MIA, indicating changes in serotonin availability and metabolism within the foetoplacental unit. To achieve this, the two main aims of this chapter were to:

1. Adapt a protocol for the detection of 5-HT and 5-HIAA in AF by LC-MS
2. Quantify 5-HT and 5-HIAA in AF from MIA-exposed offspring and saline controls at both E14.5 and E18.5.

Aim one is described in section 4.2, carried out through modification of the method for 5-HT and 5-HIAA detection in mouse microdialysate samples by LC-MS developed by Dr Ross van de Wetering [333, 334]. Following the establishment of a reliable method of quantification for 5-HT, data collection in MIA-exposed and control offspring was carried out to address aim 2 and is presented in section 4.3.

4.2 Methods Optimisation

4.2.1 Sample Preparation and Derivatisation

Methods were primarily adapted from Chapter 4 of Dr van de Wetering's doctoral thesis, which outlined an initial benzoyl chloride derivatisation process optimised for rat brain microdialysate samples [334]. Briefly described, 15 μL dialysate samples were isolated and spiked with 2.5 μL of an isotopically labelled internal standard, D2 5-HT (100 nM). Samples were derivatised by the addition of 10 μL 2% [v/v] benzoyl chloride (BzCl) in acetonitrile (ACN), with 7.5 μL sodium tetraborate buffer (BB; 100 nM) to ensure the required basic pH, and vortexed for 3 minutes. Derivatisation of 5-HT, 5-HIAA, and D2-5-HT results in the addition of benzoyl groups to reactive amine and hydroxyl groups, improving the consistency of ionisation and chromatography column retention (Figure 4.1) [393, 394].

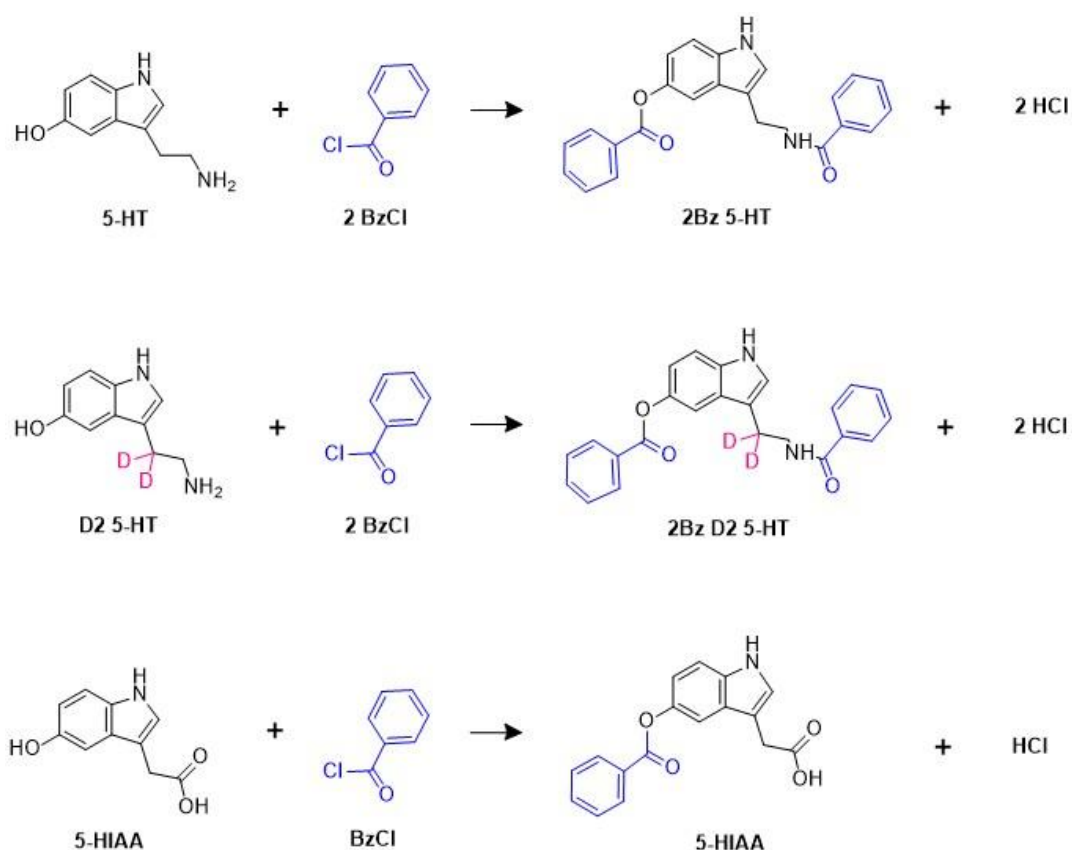


Figure 4.1: Benzoyl chloride derivatisation of 5-HT, D2-5-HT, and 5-HIAA. Colour denotes key differences across molecules: blue highlights benzoyl chloride/new benzoyl groups following derivatisation, while pink highlights the isotopic labelling of D2-5-HT. All reactions produce hydrochloric acid (HCl) as a byproduct.

Initial tests were carried out using high-concentration standards (between 1-100 μM) prepared in water and thawed AF previously stored at -80°C (centrifuged at 4°C , 15 min, 3500g prior to use [395]), according to the general derivatisation protocol previously described. After derivatisation, samples were assessed by LC-MS as previously described through a reverse phase HPLC column, Jet-Stream electrospray ionisation (ESI), and a quadrupole time-of flight (Q-TOF) mass spectrometer (column details and LC-MS run parameters can be found in section 2.3.2) [333, 334]. Following derivatisation and ionisation in positive ion mode, 5-HT, D2-5-HT, and 5-HIAA had m/z ratios of 385.15, 387.15, and 313.12 respectively [334, 393].

The results of early testing were highly inconsistent. Within the 11 minute detection window, extracted chromatograms at the m/z ratios for 5-HT, D2-5-HT, and 5-HIAA all exhibited an inconsistent number of peaks, with peaks across standards failing to appear at consistent retention times. Comparison between standards prepared in water and standards prepared with AF were particularly poor, suggesting that standards prepared in water alone may not be accurate for later quantification of 5-HT/5-HIAA in AF samples. Variability continued to be an issue when focusing on standards prepared in AF however, even in serially diluted 5-HT/5-HIAA standards with the same AF background sample (Figure 4.2). A standard at stably decreasing concentrations should be identifiable through a peak at a stable retention time across all standards, with peak height/area decreasing relative to concentration. This pattern was not apparent at this stage, with peaks either shifting or disappearing and reappearing across the dilution series.

At this stage it was difficult to determine which peak(s) in their respective chromatograms may reflect 5-HT, D2-5-HT, and 5-HIAA. It is possible that multiple peaks represented elution of other ions at the same m/z, or repeated elution of 5-HT/5-HIAA at multiple retention times. Regardless of true peak identity, the lack of consistency seen in standards prepared from the same AF aliquot suggested that poor derivatisation, degradation, and/or transient interactions with other molecules in the AF matrix were driving inconsistency in the elution and detection of metabolites. It was also likely that these factors were related; incomplete or delayed derivatisation for example may have driven changes in pH within the sample (through HCl production), contributing to differences in protein/column interactions and variable retention and/or ionisation.

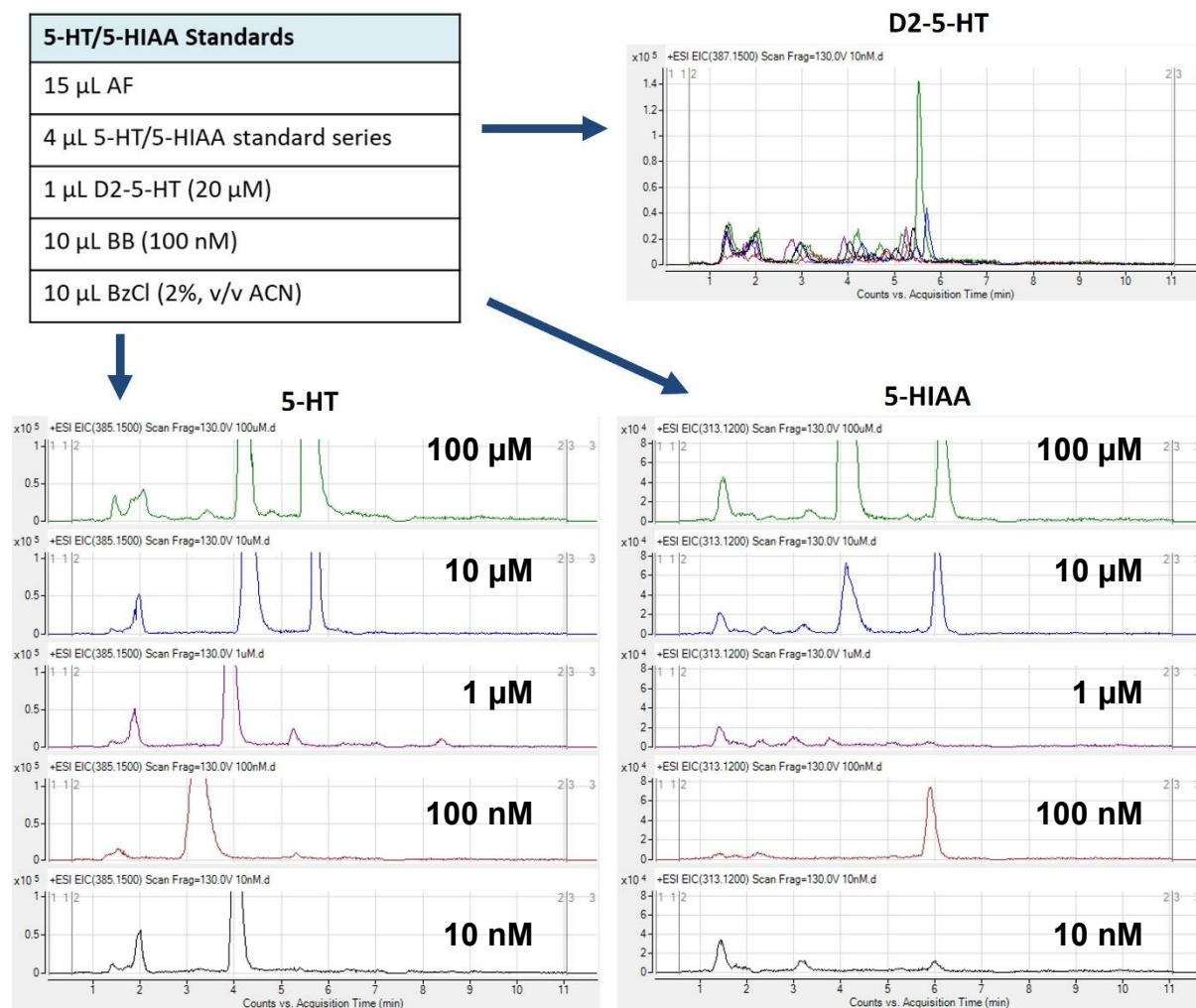


Figure 4.2: Example of initial testing to detect 5-HT and 5-HIAA in amniotic fluid. Extracted Ion Chromatograms (EIC) derived from serially diluted 5-HT/5-HIAA standards after derivatisation: presented top to bottom, 100 μ M (green), 10 μ M (blue), 1 μ M (purple), 100 nM (red), 10 nM (black). All vials contained a fixed concentration of internal standard D2-5-HT (1 μ M), with 387.15 m/z (D2-5-HT) chromatograms for each vial overlaid rather than listed. Only the 100 μ M standard shows a clear 387.15 m/z peak at approx. 5.5 min, with considerable noise present across standards. Both 385.15 (5-HT; y axis scale fixed at 1×10^5) and 313.12 (5-HIAA; y axis scale fixed to 8×10^4) m/z chromatograms exhibit relatively consistent, small peaks prior to 4 minutes, with significantly stronger and more variable peaks typically observed between 4-7 minutes. Chromatograms extracted using Agilent MassHunter Qualitative Software B.06.00.

Reliable quantification of target metabolites requires both a stable retention time and a clear, single target peak differentiated from any other background ions. To improve retention time and minimise noise, multiple modifications were tested. The most effective of these approaches was an approximate 100-fold increase in BB concentration, from 25 nM final concentration (10 μ L of an 100 nM stock in 40 μ L reaction volume) to 20 mM (10 μ L of an 80 mM stock) as used in other published BzCl derivatisation protocols [393, 394]. This resulted in clear peaks with relatively stabilised (and nearly identical) retention time for 5-HT and D2-5-HT within a batch of samples injected on the same day (Figure 4.3). Overall signal magnitude and retention time did shift between batches/days, however. Importantly, overlapping peaks of decreasing magnitude could now be identified in batches with serially-diluted 5-HT standard.

Of note, some additional, smaller peaks at 385.15 and 387.15 m/z persisted and may have reflected early elution of 5-HT and D2-5-HT. To explore this, an LC-MS/MS protocol was used to periodically fracture ionised analytes prior to detection and aid in peak identification (Figure 4.4; see 2.3.2 for method details). Previous work by Wong et al. identified derivatised 5-HT to fracture into daughter ions with 105 and 264 m/z, with the 105 m/z daughter ion being the benzoyl group added during derivatisation [393]. As expected, these daughter ions were detected in the large peak, including both 264 and 266 m/z fragments reflecting the daughter ion of 5-HT and D2-5-HT, respectively. Smaller, earlier peaks, however, only presented the 105 m/z fragment with no daughter ions detectable at 264 or 266, suggesting additional molecules within the AF (potentially benzoylated, or eluting simultaneously with excess BzCl) were contributing to these additional small peaks and not inconsistent 5-HT elution.

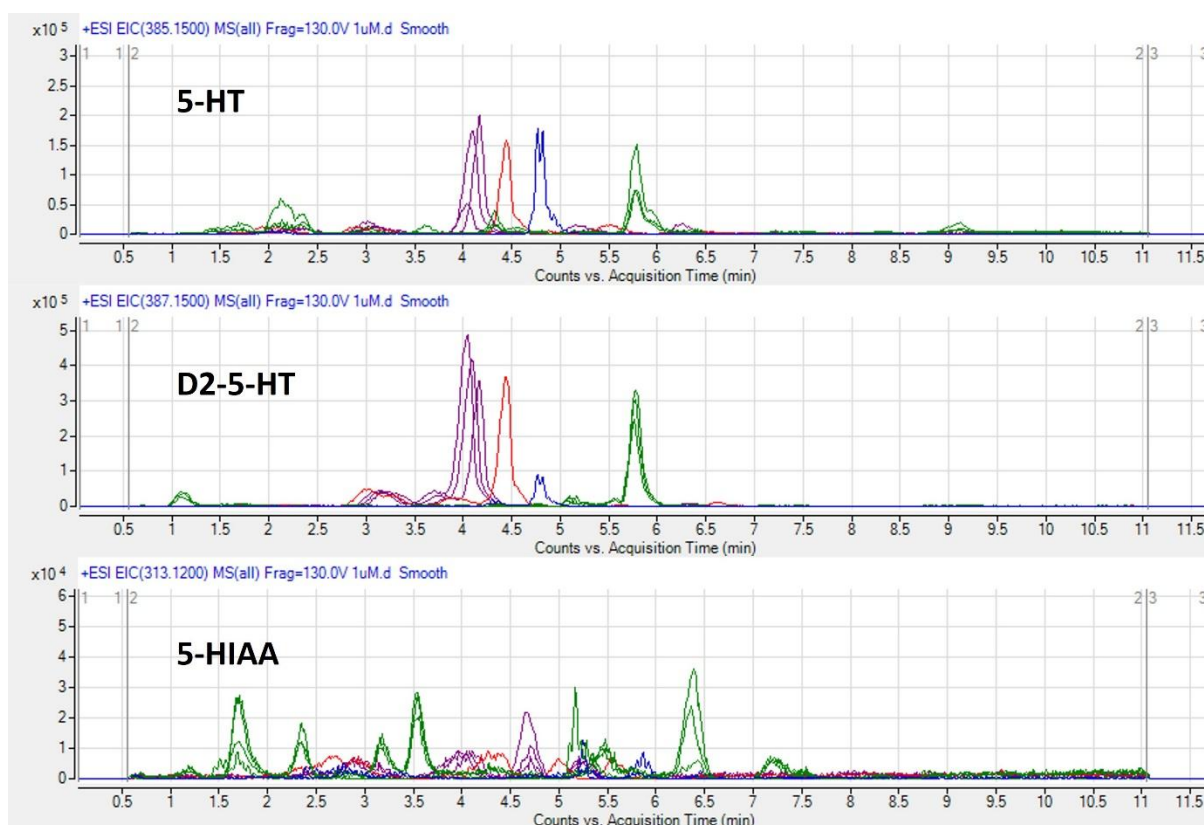


Figure 4.3: Overlaid chromatograms showing the retention time of targets following an increase in buffer concentration. Colour of peaks indicates the day/batch samples were tested on, with the same samples shown across each target. Retention time is demonstrated across batches ($n=4$) and within batches ($n=2$, 3x replicates). Purple: three standards tested on 15/06/2024; concentrations 20, 75, and 100 nM 5-HT/5-HIAA, 300 nM D2-5-HT. Red: one standard tested on 16/06/2024; concentration 75 nM 5-HT/5-HIAA, 300 nM D2-5-HT. Blue: one standard tested on 23/05/2024; concentration 1 μ M 5-HT/5-HIAA, 500 nM D2-5-HT. Green: three standards tested on 17/10/24; concentrations 157.5, 165, 187.5 nM 5-HT/5-HIAA, 150 nM D2-5-HT. Purple and red standards contain aliquots of the same pooled AF sample, blue contains an individual AF sample (poly I:C, male, E18.5 sample), and green standards contain a separate pooled AF sample.

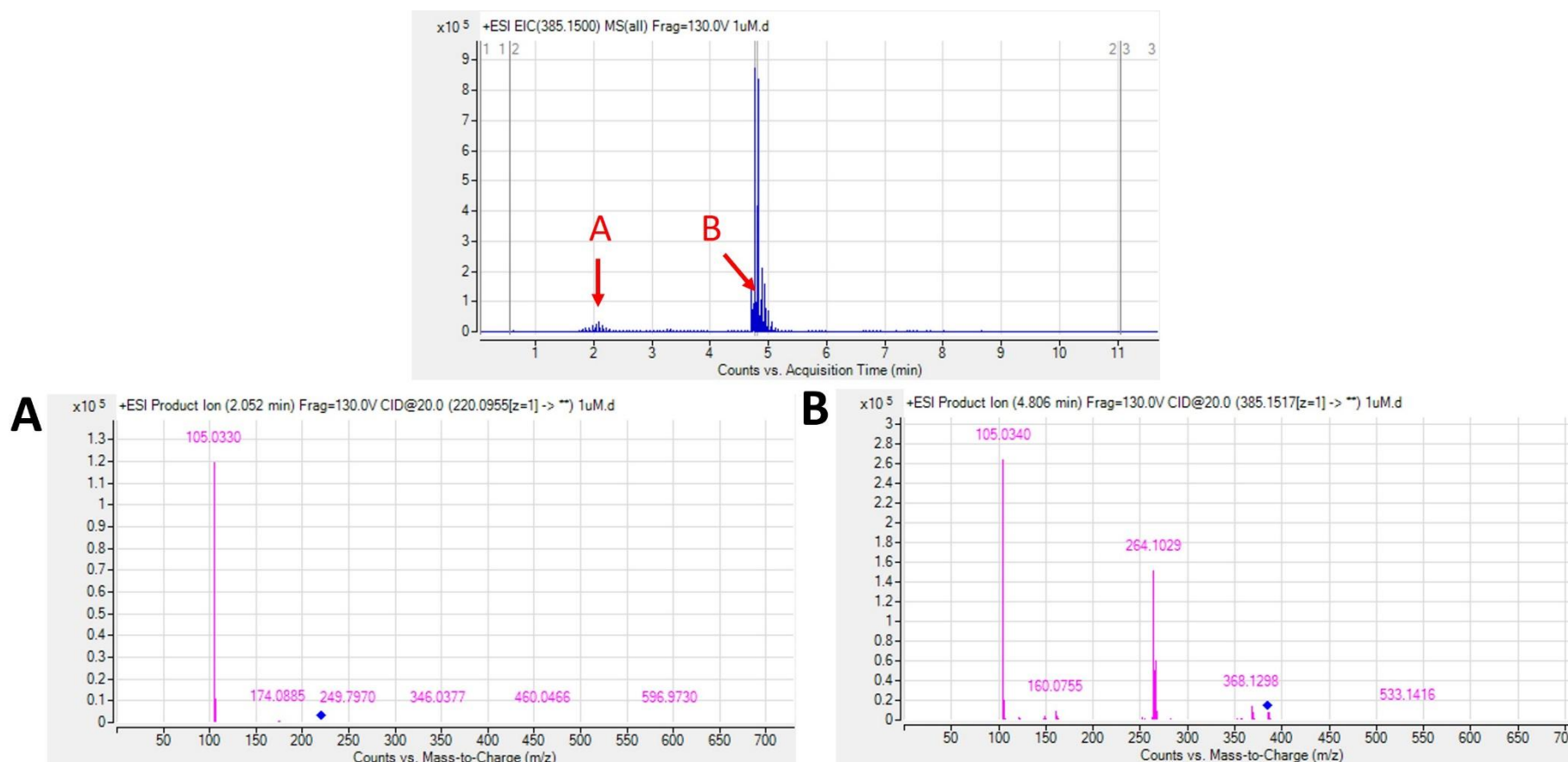


Figure 4.4: Example of LC-MS/MS use to support 5-HT/ D2-5-HT peak identification. Sample presented is identical to blue sample in Figure 4.3, with a concentration of 1 μ M 5-HT and 500 nM D2-5-HT. Fragmentation of all ions present at the time equivalent to peaks A and B show benzoyl daughter ions (105 m/z), however only peak B displays the expected 5-HT and D2-5-HT daughter fragments (264 and 266 m/z, respectively) that indicate 5-HT/D2-5-HT elution at this time point. Blue diamond in A/B plots indicate the m/z ratio of the highest abundance ion detected in the sample at the specified time: A = 220.10 m/z, B = 385.15 m/z.

As seen in Figure 4.3, 5-HT/D2-5-HT overall signal and retention were still variable across batches. Additionally, 5-HIAA detection was not noticeably improved by increased buffer concentration and was highly inconsistent and/or unidentifiable in standards $\leq 1\mu\text{M}$. Further consideration was given to reducing the complexity of the AF matrix across samples, aiming to minimise potential matrix effects such as matrix suppression interfering with 5-HIAA ionisation. Hence, a protein precipitation method was tested from Wong et al; ice-cold ACN was added to a single AF sample at a 4:1 or 2:1 ratio, briefly vortexed, then centrifuged at 4°C for 10 minutes at 12,000g to produce a white protein pellet and clear supernatant [393]. ACN protein precipitation did produce a larger pellet than the original technique and visually improved the clarity of cloudy AF samples. However, no differences in 5-HT, D2-5-HT, or 5-HIAA signal stability or retention were observed across any AF preparation method tested. ACN precipitation at a 2:1 ratio was used for later data collection, to minimise potential accumulation of large proteins within the HPLC column over a long run time while maximising the AF concentration (and therefore underlying 5-HT concentration) within the injected sample.

Previous testing by Dr van de Wetering demonstrated that optimal BzCl concentration may differ between 5-HT and 5-HIAA, with concentrations less than 2% found to improve 5-HT signal but impair 5-HIAA detection and lead to elution at multiple time points [334]. BzCl concentration was also tested here to ensure that suboptimal BzCl availability was not contributing to poor 5-HIAA detection. 1% and 4% BzCl concentrations (v/v ACN) were tested alongside the originally used 2%. Unfortunately, 5-HIAA detectability did not appear to improve with altered BzCl concentrations (Figure 4.5). A large peak was present in 10 μM 5-HIAA standards across BzCl concentrations, but this was not detected in 100 nM and 1nM standards. A secondary peak was also present across standards, which may either represent split elution of 5-HIAA, elution of another ion at 313.12 m/z, or both. No notable changes were observed in 5-HT signal across the tested BzCl concentrations (Figure 4.5).

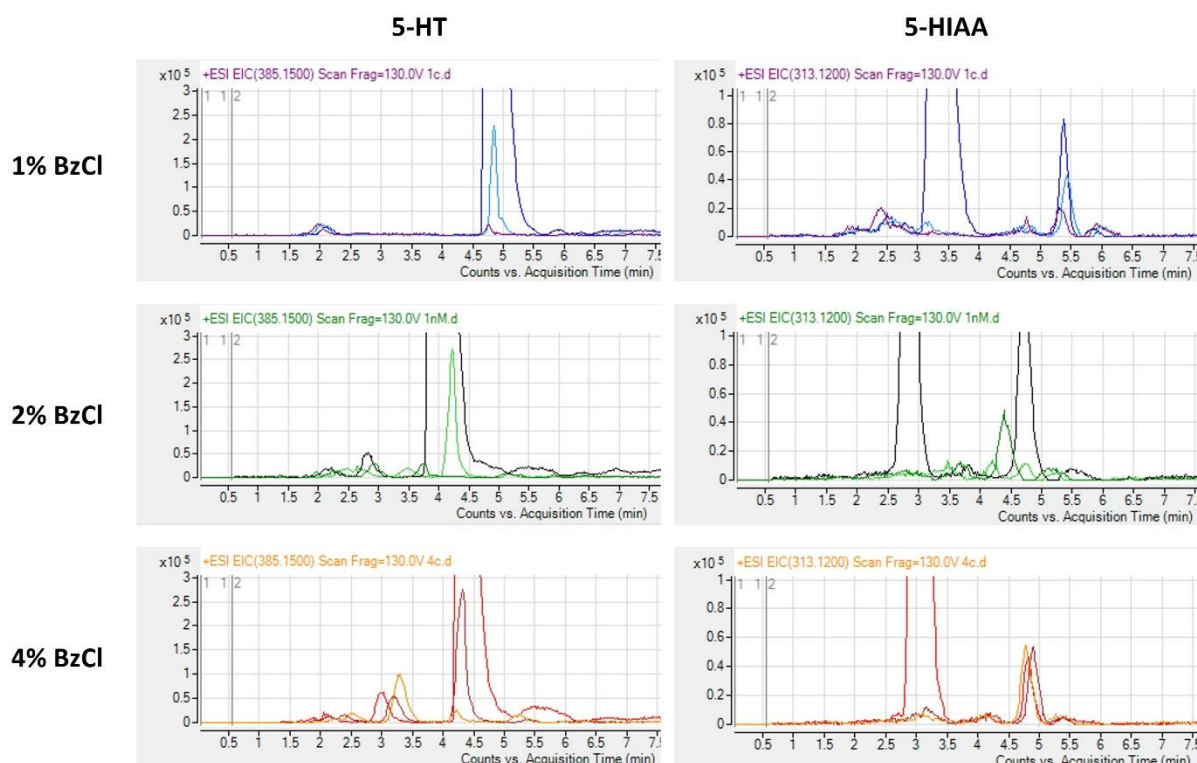


Figure 4.5: BzCl concentration did not improve 5-HIAA detection. Three standards were derivatised per BzCl concentration: 10 μ M (dark blue, black, and red for 1%, 2%, and 4% BzCl concentrations, respectively), 100 nM (light blue, light green, and dark red, respectively), and 1 nM (purple, dark green, and orange, respectively). Extracted ion chromatograms are fixed at 3×10^5 (385.15 m/z, 5-HT) and 1×10^5 (313.12 m/z, 5-HIAA) to present scale at which lower concentration peaks are visible alongside non-target peaks/background noise.

At this stage, it was likely that the lack of reliable 5-HIAA signal observed was due to poor ionisation of 5-HIAA and/or matrix suppression. While 5-HIAA may be detected in positive ion mode through derivatisation and the formation of a positively charged ammonia adduct ion, 5-HIAA may also be assessed in negative ion mode with the deprotonation of the carboxylic acid and negatively charged carboxylate ion formation [393, 396]. As benzoyl chloride derivatisation requires a basic pH (and high buffer concentrations were necessary to stabilise retention of 5-HT/D2-5-HT), basic conditions would promote carboxylate ion formation in solution and may have contributed to poor positive ionisation. Although Wong et al. were able to detect

derivatised 5-HIAA in positive ion mode as low as 0.7 nM in pure derivatised standards, interactions within the AF matrix and technical differences between mass spectrometers likely prevented detection of 5-HIAA at nanomolar concentrations in this study [393]. Significant methodological changes, such as complete removal of the AF matrix (e.g. evaporation and resuspension in ACN, water, or methanol) or additional filtering to further purify samples, may have improved 5-HIAA ionisation and signal. However, minimising the risk of target degradation/loss would have required extensive time and sample investment to explore. Altering the AF matrix would also have required further optimisation of 5-HT and D2-5-HT, which now presented a relatively stable retention time within batches. As a result, 5-HT detection was prioritised as the main metabolite of interest moving forward into bulk data collection. 5-HIAA was still included in standard stocks, however failed to be consistently detectable during the remainder of work presented in this chapter.

4.2.1 Initial Quantification of 5-HT

Moving forward with 5-HT and D2-5-HT, initial data were collected across a range of samples across all variables of interest (treatment, age, sex). Batches comprised of approximately 20 samples were prepared along with a standard curve at final concentrations of 100, 75, 50, 40, 30, 20, 10, 5, 1, and 0 nM. Standards were prepared with aliquots of a large, pooled AF sample derived from all samples within the batch, ensuring a stable concentration of endogenous 5-HT across all standards. Using a standard addition approach, the endogenous 5-HT concentration within standards could be quantified from the y-intercept and slope of the standard curve and added to standard concentrations for accurate quantification of experimental samples [397, 398]. 200 nM D2-5-HT concentration was used as a consistent internal standard across all samples for normalisation.

The cause of previously observed batch differences in retention time/signal was unclear, although suspected to be due to a combination of factors, including small changes in solvent concentration, ambient temperature fluctuations, or pressure changes from replaced HPLC tubing [399]. These factors were difficult to control for, largely driven by the age of the LC-MS equipment and high usage. To minimise technical variation from inter-day effects, samples were stratified to be run in a single,

76hr protocol comprising 3 prepared sample batches run consecutively. Samples were blinded during assessment to prevent treatment/age bias (sex of samples had not yet been determined). However, each set of 4 samples run between blank washes included one saline and poly I:C exposed pup at each age. Replicate standard injections were also taken every 12-15 hours. This structure ensured that any periodic changes in the LCMS equipment during the long protocol, batch effects, or tubing failure and early termination of the run, did not result in biased measurement of any one experimental group. Samples and replicate standards were freshly prepared every 24 hours, with all reactions kept at 4°C in a temperature-controlled compartment during autosampling.

Initial analysis identified significant inconsistency again in 5-HT/D2-5-HT retention time, particularly in the first batch (ranging over the 11-minute detection window), with the second batch being the most consistent (over a ~30-second range). This was later attributed to a faulty water pump, resulting in fluctuations in pressure over time and an inaccurate water flow rate that produced incorrect eluent ratios in the mobile phase throughout data collection. Focusing on the most stable period of data collection (batch 2), there was no clear difference in 5-HT concentration following MIA (Figure 4.6A). There was an apparent shift towards decreased 5-HT concentration at E18.5 compared to E14.5, however, possibly aligning with a shift in exogenous 5-HT supply to endogenous foetal 5-HT production. As previously mentioned, assessment of 5-HT in AF is rarely reported. Salas et al. did report a rat AF 5-HT concentration of 28.4 nM (5.0 ± 3.0 ng/mL), although the embryonic age and specific methodology associated with this concentration was not reported [386]. This concentration is similar to the average observed in E14.5 AF within this data set (24.3 nM), as well as the calculated concentration of the pooled AF sample used to prepare standards (24.7 nM).

Statistical analyses were not performed on this data due to concerns with data reliability. In addition to inconsistent retention time, notable variability was observed in target signal and signal:noise (s:n) ratios between duplicate injections from the same vial (Figure 4.6B-D). Second injections were not directly associated with an increase/decrease in D2-5-HT and/or 5-HT signal but were commonly found to have a decreased s:n ratio. As shifts in D2-5-HT signal mirrored shifts in 5-HT signal relatively consistently, this effect did not cause large variability in the standard curve (e.g. vertical shifting of the standard curve as relative 5-HT signal increased or

decreased). Duplicate injection variance likely resulted from technical issues with the faulty water pump, where inconsistent pressure and solvent ratios resulted in changes to elution of numerous different targets within the complex AF matrix. Subsequent buildup of specific compounds from initial injections not efficiently cleared from the column may have led to increased background noise in second injections prior to routine blank washes, as well as inconsistent matrix suppression/enhancement of signal.

Regardless of variability, samples and standards also exhibited a very low s:n for 5-HT. Typically, a s:n ratio of 3 is required for reliable detection, while a ratio of 10 is required for reliable quantification [397]. While a peak at 385.15 m/z was consistently visible across samples eluting at the same retention time as D2-5-HT (providing reasonable confidence in peak identity being 5-HT), the peak signal compared to noise within AF was poor. Standards presented a s:n ratio of between 0.4 and 9.4, with 40% of standards and 100% of samples < 3, which significantly limited the reliability of these data. Improving the s:n ratio to have samples meet the limit of quantification was therefore a focus for subsequent work. To improve detection of 5-HT within AF, experimental samples were spiked with standard 5-HT and D2-5-HT to increase concentrations of each vial by a known 150 nM (Figure 4.7). Following adjustment of the standard curve and quantification of sample 5-HT, the 150 nM 5-HT spike was subtracted from sample concentrations to find the initial 5-HT concentration within individual AF samples. Detailed final methods used to prepare AF and quantify 5-HT by LC-MS can be found in Chapter 2.3.

This spiking method lifted the overall concentration of each vial to a more optimal range for quantification and was adapted from the same standard addition principle underlying the standard curves used in this chapter. Standard addition involves adding known concentrations of standard to an experimental sample of unknown concentration, and using the known, spiked concentrations to determine the original target concentration [400, 401]. While standard addition protocols typically involve generating an individual standard curve within each sample, this was not possible here without significantly limiting the number of samples quantified. The approach was instead modified, with the pooled AF sample provided a common matrix for multiple standard addition (the shared standard curve) and a single addition to 150nM was used in individual samples to improve signal.

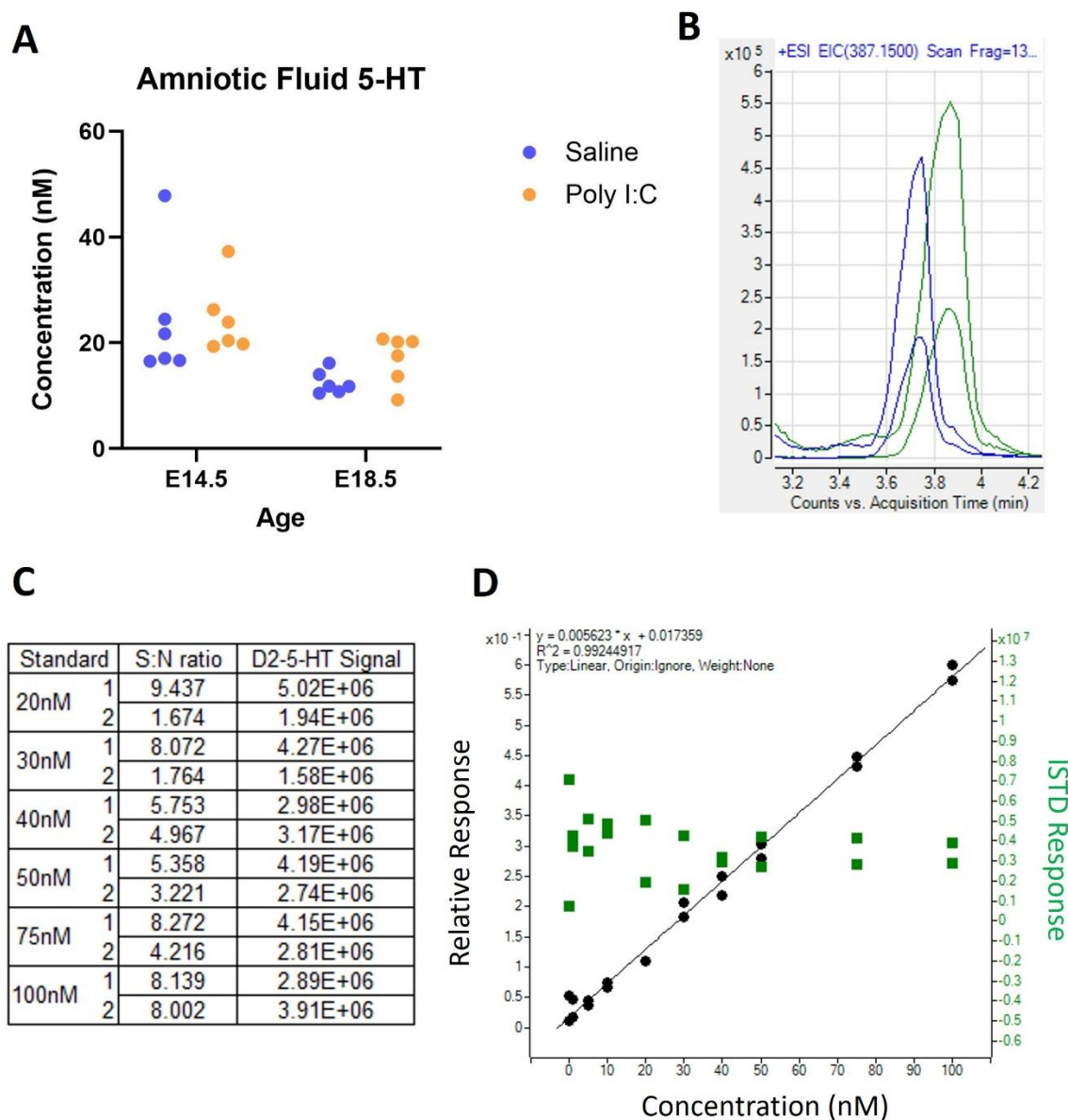


Figure 4.6: Initial assessment of 5-HT within amniotic fluid. A) Calculated concentration of 5-HT across age and treatment groups. B) Example extracted ion chromatograms showing loss of signal in replicate injections (green and blue, 15 hours apart) of 100 nM 5-HT standard in batch 2. Smaller 385.15 m/z (5-HT) and larger 387.15 m/z (D2-5-HT) peaks are overlaid. C) Signal:noise ratios and accompanying D2-5-HT response for an example subset of standards (20-100 nM), exhibiting variance across duplicate injections. D) Unadjusted standard curve produced in batch 2. Green squares indicate D2-5-HT response variance across replicates, Black points and line reflect linear standard curve. Solving for x when $y=0$, $-(y \text{ intercept})/(\text{slope}) = -3.0871$, identifying background 5-HT concentration within prepared standards to be 3.0871 nM (24.7 nM accounting for AF dilution of $1/8^{\text{th}}$ final reaction volume).

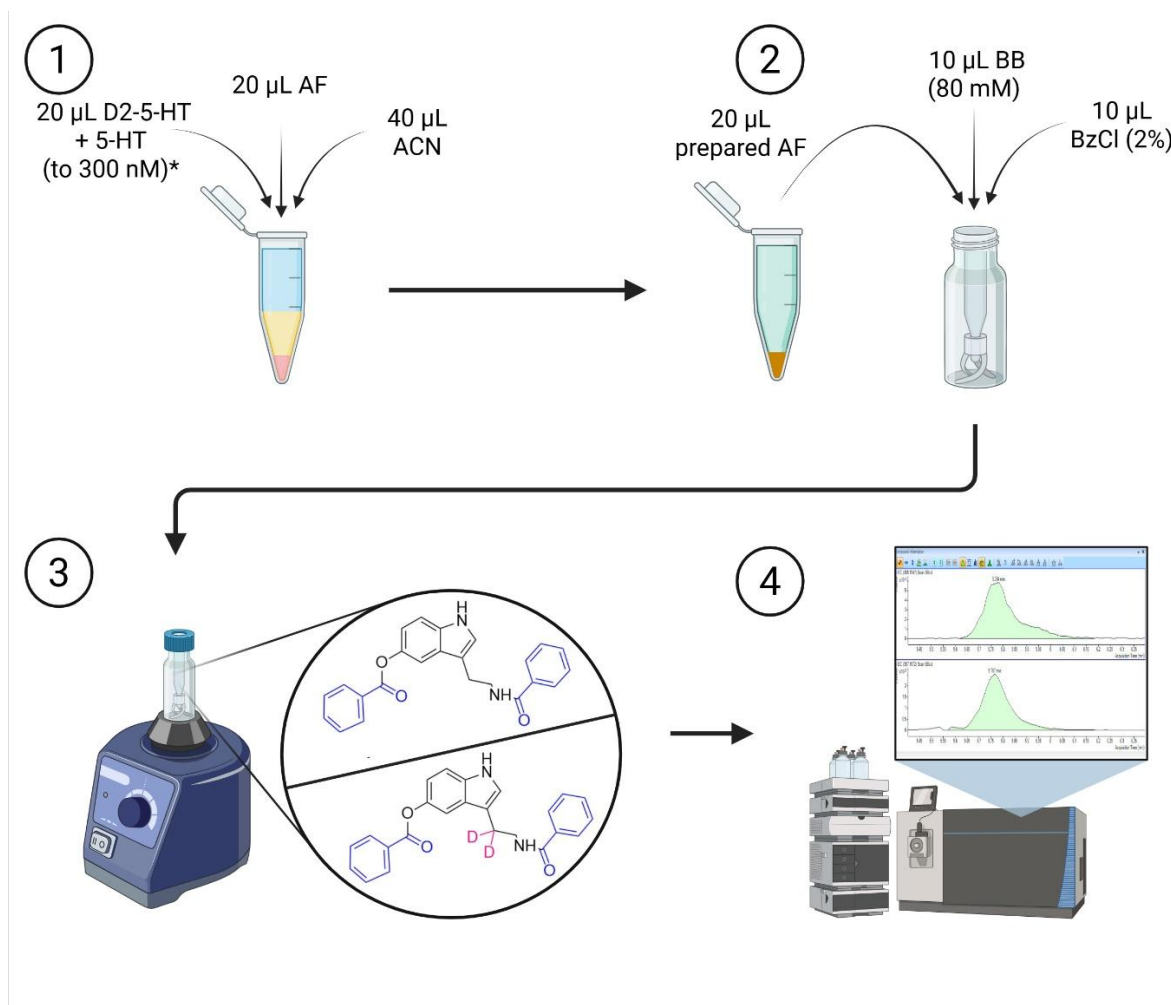


Figure 4.7: Overview of sample preparation method used for final quantification of 5-HT. Ice-cold acetonitrile (ACN) was added to either individual or pooled AF aliquots, and vortexed prior to addition of internal and 5-HT standard (suspended in water) and protein precipitation. Following centrifugation, supernatant was added to borate buffer (BB) and benzoyl chloride (BzCl) for derivatisation. Vials were kept at 4 °C until 10 µL injection into a quadrupole time-of flight (Q-TOF) mass spectrometer equipped with a reverse phase HPLC column and Jet-Stream electrospray ionisation source. Created in <https://BioRender.com>

4.3 Final Results and Discussion

4.3.1 Quantification of 5-HT following Sample Spiking

At both E14.5 and E18.5, no significant differences in AF 5-HT concentration were observed (Figure 4.8). This includes no significant effect of treatment (E14.5: $F(1, 5.23) = 0.12$, $p = 0.7453$; E18.5: $F(1, 1.60) = 0.12$, $p = 0.7677$), sex (E14.5: $F(1, 11) = 1.31$, $p = 0.2760$; E18.5: $F(1, 11.50) = 0.72$, $p = 0.412$), or treatment*sex interaction (E14.5: $F(1, 11) = 0.55$, $p = 0.4731$; E18.5: $F(1, 11.50) = 0.09$, $p = 0.7635$). Direct age comparisons were not performed as E14.5 and E18.5 samples were run in separate batches, which had been a source of continuous variability during the methods optimisation process. In alignment with previous observations, retention time of 5-HT (and co-eluting D2-5-HT) did vary slightly between batches, but remained remarkably stable within batches (E14.5 = 6.043-6.135 minutes; E18.5 = 5.749-5.803 minutes). S:n ratio was also notably improved (see Figure 4.9).

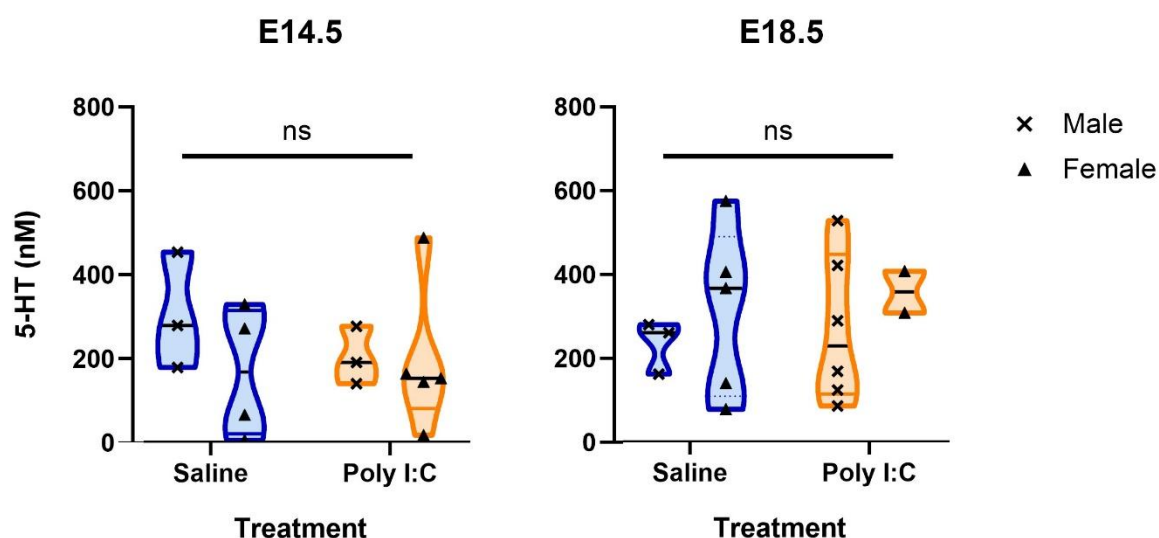


Figure 4.8: MIA did not significantly alter amniotic fluid 5-HT concentration at E14.5 or E18.5. 5-HT concentration presented after correcting standard curve 5-HT concentrations, subtracting 150 nM spike, and accounting for AF dilution in derivatised sample.

LMM was used to structure the data to account for litter bias and limited independence prior to assessment by two-way ANOVA, as carried out in Chapter 3 (Full statistics tables in Appendix E). Normality and equal variances assumptions were assessed: data roughly met these assumptions at E14.5, although small sample sizes for each sex within treatment groups at E18.5 may have limited the power of the LMM ANOVA to differentiate potential sex/treatment effects. While both log transformation and weighted model (to account for unequal variance in sex/treatment groups) were tested, neither method resulted in an improved model fit compared to the standard LMM approach (see 2.7 for further details).

This analysis presented significantly higher estimates of AF 5-HT concentration than expected, based on both initial data collection and the baseline 5-HT concentration of 28.4 nM presented in Salas and colleagues (Figure 4.6) [386]. Baseline 5-HT concentration in this study was determined and adjusted for as previously described from the slope of the standard curve, specifically the x intercept when $y=0$ (Figure 4.9A-B). In E14.5 and E18.5 standard vials, the underlying 5-HT concentration was estimated to be 21.0 nM and 23.8 nM, respectively; accounting for AF dilution, this translated to 167.7 nM and 190.7 nM in the original pooled sample. This concentration was directly reflected in experimental samples, which spanned a relative response range corresponding to the entire length of the upper half of (but not exceeding) the standard curve (150 nM to 187.5 nM prior to adjustment). Samples at the extremes of this original 37.5 nM range, after accounting for AF dilution, represented a 5-HT concentration difference of ~300 nM. A difference persists even when considering uncertainty in sample concentration estimates propagated from error within the standard curve (see section 2.3.2 for formula). Using the highest and lowest samples measured at E14.5 as examples, the final concentration range at E14.5 falls between 6.5 ± 106.5 and 488.6 ± 129.0 nM. Overall, this suggests that the relatively high AF 5-HT concentrations calculated in this study are not likely to be caused by poor accuracy of the standard curve, such as an underlying non-linear signal:concentration relationship preventing accurate extrapolation of baseline concentration. As Salas et al. utilised an HPLC technique with limited sample preparation and/or HPLC method detail reported, it is difficult to gauge how methodological differences may have contributed to the disparity between control concentration.

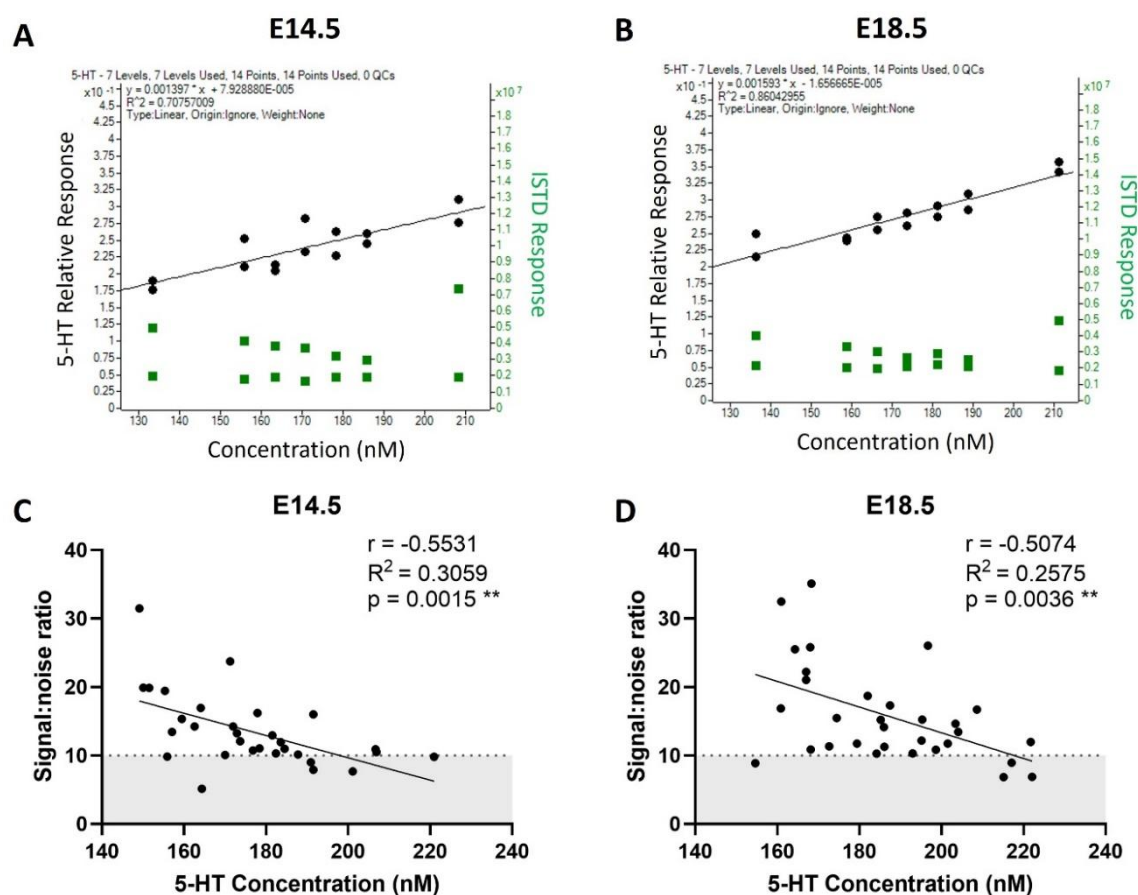


Figure 4.9: Adjusted standard curve uncertainty and signal:noise correlations relevant to 5-HT quantification in experimental samples. Adjusted standard curves for E14.5 (A) and E18.5 (B) used for quantification of experimental samples, after addition of 21.0 nM and 23.8 nM to standard concentrations to account for baseline AF signal. Samples with a higher signal:noise ratio at both E14.5 (C) and E18.5 (D) were more likely to be near or below the quantification threshold, suggesting a higher risk for quantification error. r = Pearson's correlation coefficient, R^2 = coefficient of determination.

As anticipated, spiking experimental samples with 150 nM 5-HT resulted in large improvement in the s:n ratios observed across samples and standards (Figure 4.9C-D). Although s:n ratio within a duplicate pair was still decreased in the second injection, 84% of sample injections (52/62) and 79% of standard injections (22/28) over both batches had a s:n ratio ≥ 10 , with all remaining injections being above 4.6 (surpassing the s:n ratio of 3 that indicates reliable 5-HT detection). The small proportion of injections with a 5-HT s:n < 10 were not excluded from analysis, as this would have resulted in 6 samples being quantified from a single injection, and two samples removed entirely: P9.2 (E14.5, poly I:C, female: 489 nM) and P2.5 (E18.5, poly I:C, male: 529 nM). It is important to note however that lower signal:noise ratios were significantly correlated with increased concentration estimates across E14.5 and E18.5, indicating that the injections with the highest 5-HT concentration estimates may be more susceptible to quantification error (Figure 4.9C-D). S:n ratio was not further associated with other investigated potential sources of error such as sample position in the batch worklist (E14.5 $r = -0.6344$, $R^2 = 0.4025$, $p = 0.0911$; E18.5 $r = 0.4478$, $R^2 = 0.2005$, $p = 0.2658$), suggesting the variation in s:n over the 24 hour batch was not cumulative and gradually influencing 5-HT concentration estimates over time.

4.3.1 Conclusion and Future Directions

Overall, this chapter demonstrates the development of an LC-MS protocol with the capability to quantify 5-HT from rodent AF. Initial technical and matrix challenges contributed to batch inconsistency and poor s:n resolution. However, the optimisation process resulted in a relatively stable retention time, improved s:n ratio for reasonable quantification confidence, and demonstrated specificity through LC-MS/MS analysis with consistent co-elution with a clear D2-5-HT peak as an internal standard. Interestingly, this method reported higher and more variable concentrations than previously described; although Salas et al. found that maternal 5-HTP treatments increased AF 5-HT to an average concentration of 147.5 ± 17 nM, control offspring were reported to have far lower AF 5-HT concentrations at 28.4 ± 17 nM [386]. Unfortunately it is unclear what methodological or sample differences may contribute to the discrepancy in concentrations, as limited detail was reported; the development of this protocol will hopefully encourage future work investigating AF 5-HT, providing

further comparisons to improve understanding of 5-HT concentration and metabolism during gestational development. This may be informative for not only MIA research, but for understanding a range of prenatal conditions/outcomes influencing foetoplacental serotonin and neurodevelopment such as GDM and foetal growth restriction [198, 237, 238].

Previous MIA research has shown an increase in 5-HT within the foetoplacental unit during placental production/regulation at E14.5 [227]. Here, no change in AF 5-HT was observed at E14.5 or E18.5 following MIA, suggesting that long-lasting changes to 5-HT regulation following MIA are not observed within the AF, and/or may be specific to foetal and placental tissues. It is possible that changes to AF 5-HT did occur and were not captured by the research in this chapter; for example, 5-HT concentrations may have increased acutely following MIA at E12.5 but were metabolised by amniocytes and/or absorbed by the foetus or placenta prior to sampling at E14.5 [306, 322]. As rapid metabolism of 5-HT by amniocytes has previously been indicated, inclusion of 5-HIAA in this assay would have shed further light on this possibility [306]. Unfortunately, 5-HIAA was unable to be quantified using this technique with the methods optimised for 5-HT quantification. Although outside the time and sample limitations of this thesis, exploration of further sample purification techniques to reduce noise and potential AF matrix effects would be a valuable path for future research.

Chapter 5:

Foetoplacental mRNA Expression of 5-HT -Related Targets Following MIA

5.1. Introduction

The transcriptome in foetal development is constantly evolving, with cells rapidly dividing and differentiating to form new tissues and structures. Delay or dysregulation of mRNA transcription, induced by changes in the foetal environment, may cause significant changes to cell function and disrupt typical development.

As discussed in the introduction to this thesis, 5-HT production/availability in the rat foetoplacental unit undergoes a regulatory shift estimated to occur at approximately E17.5 (see 1.3.2.2). For the first ~17 days of development, foetally-available 5-HT originates from an exogenous pool regulated by the placenta [227, 315, 322]. While most current research has suggested this 5-HT is produced by the placenta, some studies have failed to observe expression of serotonin production machinery in the placenta and instead support an alternative maternal origin for early 5-HT [316]. Between E12 and E16 serotonergic neurons are detectable within the foetal brain, extending from the dorsal raphe; the establishment of these neurons marks the beginning of 5-HT production capacity within the foetal brain [246, 315]. Notably, research outlining this shift in foetoplacental 5-HT regulation has primarily been carried out in mouse models and not directly modelled in rats [315].

5.1.1. Chapter Aims

The primary aim of this chapter was to explore potential changes in mRNA expression of 5-HT-related target genes in the foetal brain and placenta following MIA on E12.5. As outlined in section 2.1.3, this included four sampled tissues: the E14.5 whole brain (E14.5BR), E14.5 foetally-derived placenta (E14.5 fPLA), the E18.5 posterior brain (whole brain excluding the frontal cortex; E18.5 pBR), and the E18.5 foetally-derived placenta (E18.5 fPLA). Key genes involved in 5-HT transport or the 5-HT metabolic pathway were investigated, alongside secondary targets associated with serotonergic signalling and altered neurodevelopmental outcomes (see 1.3.1 for further discussion). The function of each target and any previous findings observed in rodent MIA models are summarised in Table 5.1.

Expression of these targets was explored using the digital droplet polymerase chain reaction (ddPCR) techniques described in Chapter 2, rather than the more commonly seen quantitative polymerase chain reaction (qPCR) approach used in previous MIA studies. Following reverse transcription of mRNA to complementary (c)DNA, qPCR detects fluorescence within each well at each qPCR cycle, relying upon strong primer efficiency and stably expressed, equally efficient reference gene amplification for relative quantification of target cDNA [402]. In contrast, ddPCR is less reliant on primer efficiency; fluorescence of ~20,000 droplets within each well (labelled as positive or negative during analysis) provides absolute quantification of target cDNA concentration, which may be normalised to a wider range of factors, including input cDNA concentration, genomic (g)DNA concentration, or the amplification of reference genes [337, 340]. Multiple approaches to sample preparation and normalisation were explored and detailed in the following results sections of this chapter, followed by final presentation of target mRNA expression data at E14.5 and E18.5 assessed by LMM.

Target	Selection Rationale		MIA-associated changes in rodent foetoplacental mRNA expression
<i>Tph1</i>	5-HT Synthesis and Metabolism	Rate-limiting enzyme in 5-HT synthesis (peripheral serotonin)	~E12* MIA: increased mRNA (~E13* placenta), decreased mRNA (~E14* placenta) [227] E18 MIA: decreased mRNA (E19 placenta) [301]
<i>Tph2</i>		Rate-limiting enzyme in 5-HT synthesis (brain serotonin)	E18 MIA: decreased mRNA (E19 brain) [301]
<i>Ddc</i>		Enzyme following Tph in 5-HT synthesis	n/a
<i>MaoA</i>		Metabolises 5-HT to 5-HIAA	E18 MIA: increased mRNA (E19 brain) [301]
<i>Sert</i> (<i>Slc6a4</i>)	5-HT Transport	Serotonin transporter	E18 MIA: decreased mRNA (E19 brain, placenta) [301]
<i>Oct3</i> (<i>Slc22a3</i>)		Cation transporter, shown to transport placental serotonin	~E12* MIA: decreased mRNA (~E12* placenta) [324] E18 MIA: decreased mRNA (E19 brain, placenta) [301]
<i>Bdnf</i>	Secondary targets	Neurotrophin strongly associated with serotonergic neuron survival and maturation (see 1.3.1.2)	~E12* MIA: decreased mRNA (~E12* brain) [403]
<i>NeuroD1</i>		Transcription factor involved in neuronal differentiation/plasticity and NDDs, briefly linked to 5-HT and inflammation (see 1.3.1.3)	n/a
<i>Mecp2</i>		Methylation reader protein linked to NDDs, Bdnf, and 5-HT signalling/abundance (see 1.3.1.2).	n/a

Table 5.1: Target genes selected for RT-ddPCR analysis. All primer sequences and associated amplicon validations can be found in section 2.6.1. * = findings were reported in mice: relative embryonic age is -2 compared to rats.

5.2. Results 1: Selfie-ddPCR

Selfie-digital droplet PCR (selfie-ddPCR) allows absolute quantification of cDNA within a sample, internally normalised to DNA copies [63, 404]. This is achieved by concurrent isolation of RNA and DNA, then division of the sample into two; one half undergoing reverse transcription (RT+), the other receiving the same treatments without inclusion of reverse transcriptase (RT– ; Figure 5.1). Target concentration of the RT– sample may then be subtracted from the concentration of the RT+ sample to determine cDNA concentration, further divided to provide mRNA copies/gene copy, and multiplied by 2 to provide copies/cell (excluding in male offspring for targets found on the X chromosome). This technique was successfully used at the beginning of this research to expand previous Masters thesis research, exploring mRNA expression of the Rac1/Kal7/Disc1 signalosome in PND21 hippocampal tissue of Wistar Sert knockout rats following MIA (see Appendix A for results).

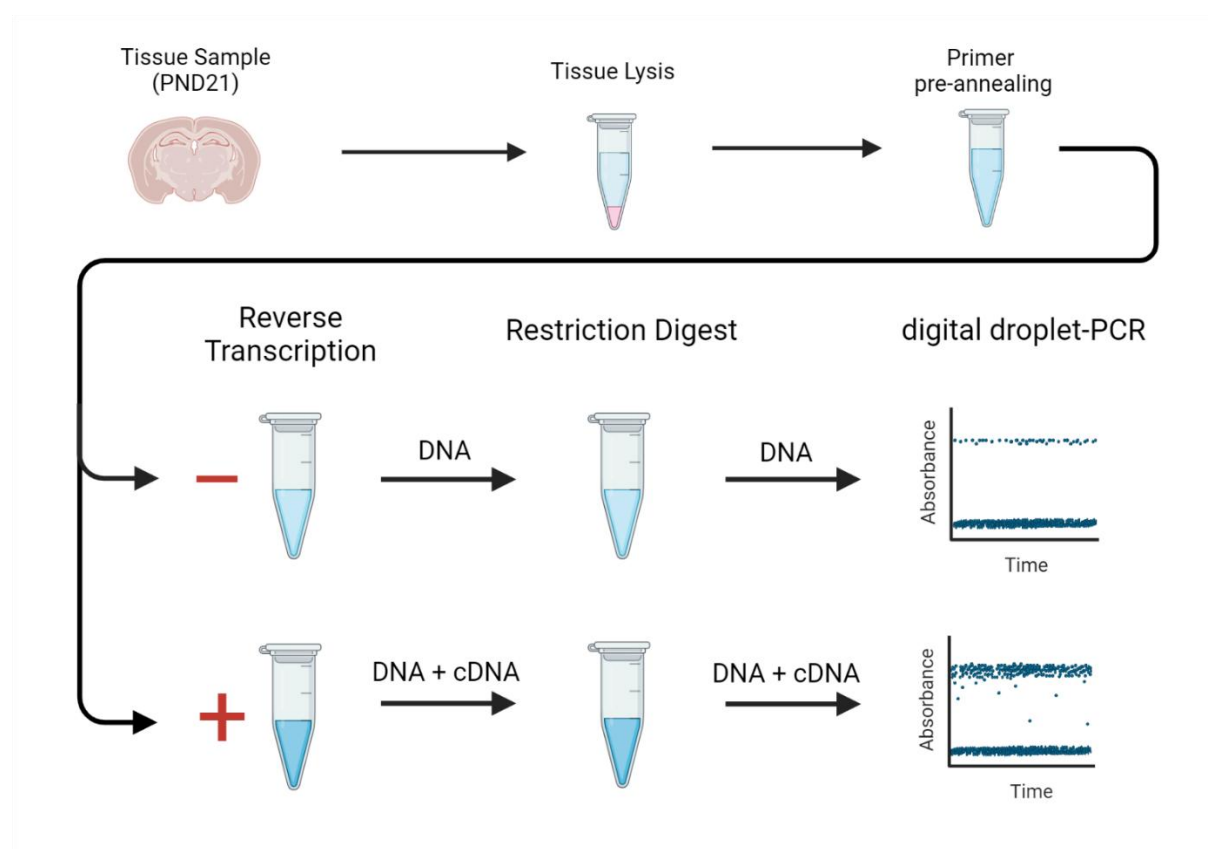


Figure 5.1: Overview of selfie-ddPCR workflow. Created in <https://BioRender.com>.

Unfortunately, translating this technique to embryonic tissue proved to be unsuccessful. Although ddPCR amplification appeared to be working well (i.e. clear and consistent separation of positive and negative droplet bands, limited rain), expression across target genes in E18.5 brain tissue was found to be highly variable (Figure 5.2A). Samples would often cluster by batch of sample preparation, although this wasn't reflected by all targets. Attempts to keep sample preparation conditions as identical as possible (i.e. identical vials of reagents, temperatures of samples and reagents, time taken at each step of method) failed to improve consistency across batches. Replacement of stock reagents with fresh vials did not have any effect, nor did attempts to optimise reverse transcription (e.g. increased reaction time, decreased temperature). Variability did not appear to stem from time spent in -80°C storage or simply inherent sample differences (as tested by aliquots of pre-homogenised sample prepared across multiple batches, which produced inconsistent concentrations).

Changes to the initial homogenisation and lysis of brain tissue had the largest effects on target cDNA concentrations: for example, an increase in the dilution ratio of brain tissue in TE buffer led to increased target cDNA (Figure 5.2B). Unfortunately, increased dilution ratios did not consistently improve the variability observed. In addition to variability, targets that appeared to have a low cDNA concentration (e.g. *Sert*, *Tph1*, and *Tph2*) often demonstrated a lower RT+ concentration compared to the paired RT- concentrations. This resulted in a 'negative' cDNA concentration within the sample. Small negative values in a sample with negligible mRNA expression may have been driven by simple technical variation between aliquots of the split sample, then leading to slightly different gDNA concentrations (e.g. 104 and 105 copies/μL; Figure 5.3C), although this in and of itself raised concerns of the sensitivity of this method for investigating low-mRNA concentration targets. Of additional concern was that a low RT+ concentration may have implied impaired reverse transcription, or sample degradation that influenced both RT+ and RT- aliquots. As a positive control for the gene-specific reverse transcription used in this method, additional tissue types known to express *Tph1* (adult rat pineal gland) and *Tph2* (adult rat dorsal raphe) were tested alongside embryonic brain tissue. Although amplification in embryonic brain tissue remained poor, the existing technique resulted in successful reverse transcription of *Tph1/Tph2* mRNA and amplification of *Tph1/Tph2* cDNA in the expected adult tissues (Figure 5.3).

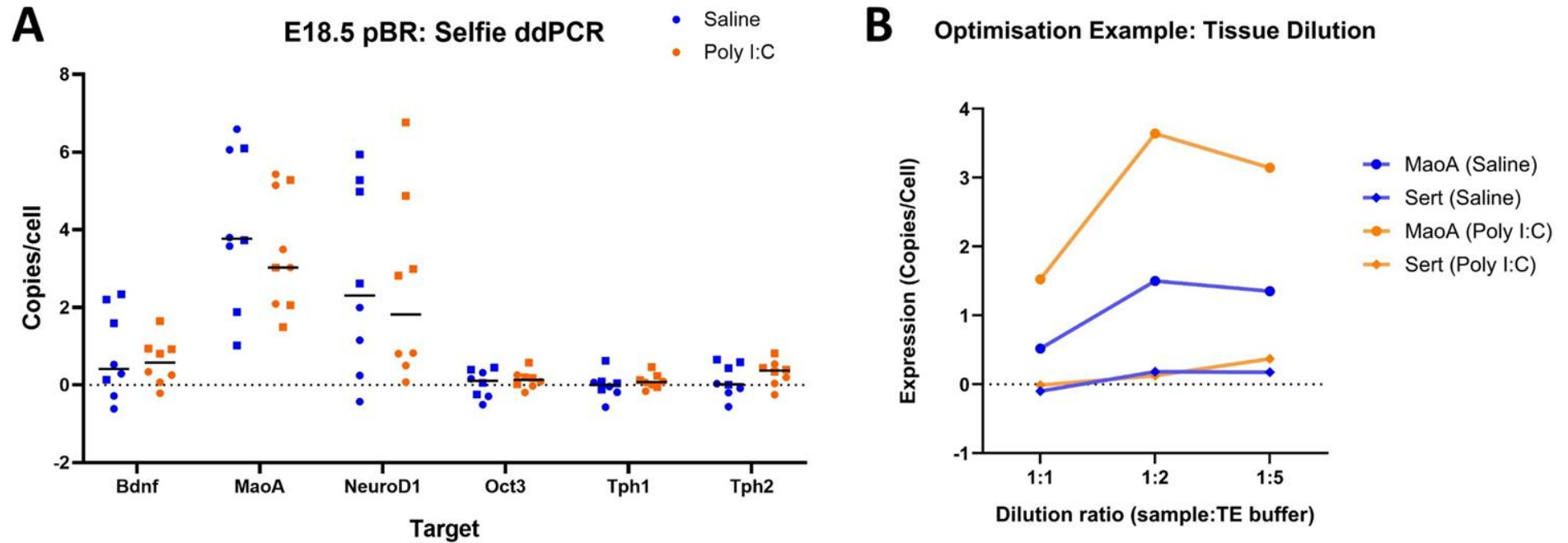


Figure 5.2: Example of selfie-ddPCR data and optimisation within E18.5 pBR tissue. A) Example of initial sample variability across two batches of prepared samples (denoted with square or circle icons) in a range of targets. Multiple samples resulted in negative cDNA concentrations (translating into negative copies/cell) across targets. B) Increasing TE buffer volume to dilute homogenised sample tissue increased cDNA concentration of a high and low concentration target in two example samples (one from each treatment group, repeated at 3 dilution ratios).

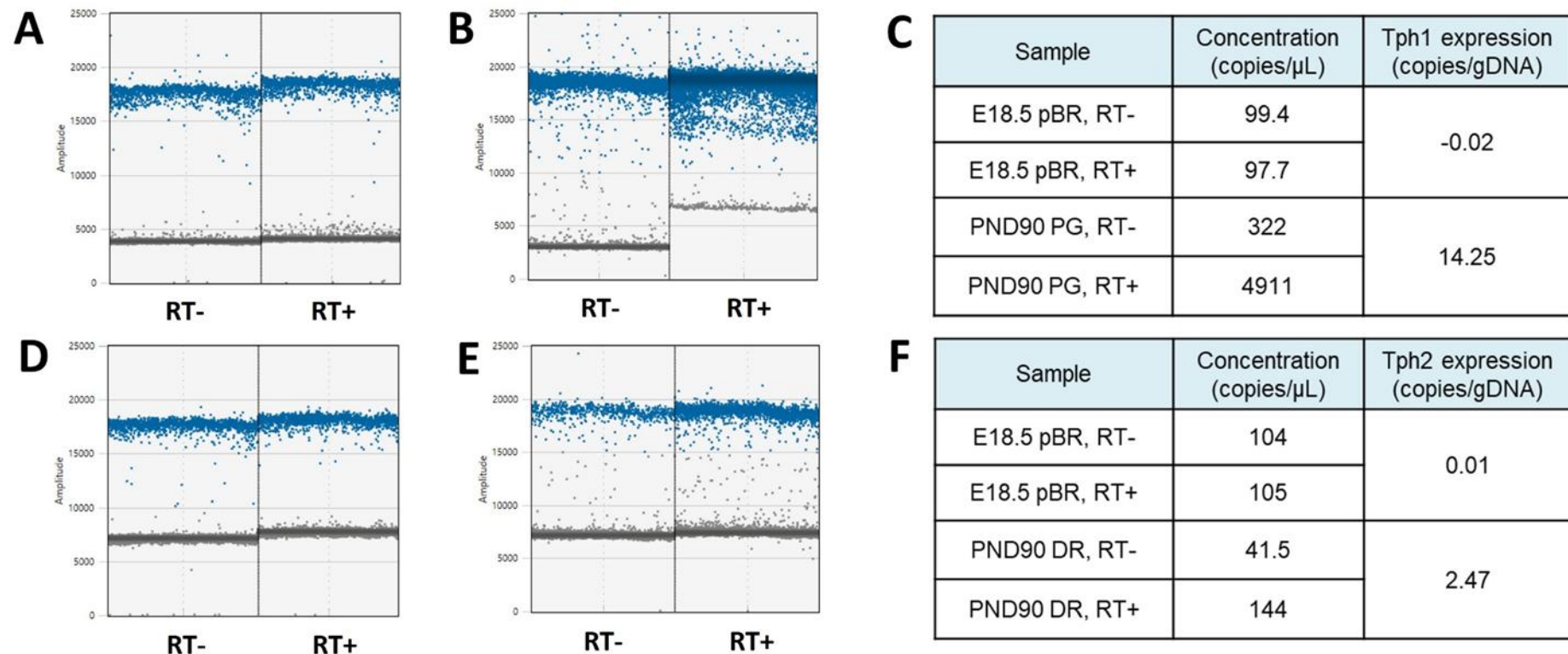


Figure 5.3: Comparisons of *Tph1* and *Tph2* amplification in embryonic and adult (PND90) brain tissue. *Tph1* expression in E18.5 pBR embryonic (A) brain tissue was negligible (C), while *Tph1* expression in the adult pineal gland (B) was considerably higher (C). *Tph2* expression in embryonic brain tissue (D) was similarly low (F), while adult dorsal raphe tissue (E) again showed notably higher expression (F). pBR = posterior brain; PG = pineal gland, DR = dorsal raphe nucleus.

Overall, these findings suggested that use of selfie-ddPCR in embryonic brain tissue faced unique challenges compared to the adult brain tissue previously analysed in this laboratory [63, 405]. Final target cDNA concentrations were heavily influenced by sample dilution prior to cell lysis, which suggested that the initial stages of sample preparation were a major source of the variability in this technique. A key difference between selfie-ddPCR and other RT-PCR techniques is the need for joint isolation of both DNA and RNA from sample tissue [404]. As a result, the brief lysis prior to reverse transcription (rather than phenol and/or column-based methods commonly used to isolate and purify RNA from tissue samples) likely resulted in the persistence of numerous RNases and inhibitors present in higher concentrations in embryonic brain compared to adult tissue [406-408]. While any contaminants are heavily diluted prior to ddPCR amplification, and ddPCR overall is less sensitive to the presence of inhibitors as qPCR, an abundance of contaminants early in sample preparation would increase risk of mRNA/DNA degradation and/or poor reverse transcription efficiency.

While further optimisation for embryonic brain tissue may have improved the viability of this technique (e.g. testing combinations of nuclease inhibitors and tissue dilutions using tissue spiked with known concentrations of purified PCR products), the time and reagent cost required to explore this further was outside the scope of this thesis. Instead, previously established methods of RNA isolation were explored for RT-ddPCR (see 2.4.2 and 2.6 for methods) [221]. As gDNA was no longer present for normalisation of PCR amplification, two new possibilities were considered: normalisation to input cDNA concentration (determined from concentration of isolated RNA), and normalisation to the expression of reference genes [337, 340]. While cDNA concentration was less likely to introduce bias across treatment and sex groups of interest, the risk of technical error from repeated pipetting steps / poor RT efficiency (increasing variability) was of concern, particularly based on the low target cDNA concentrations demonstrated previously [409]. Stable reference gene expression may act as an internal control for each sample and reduce variability from technical error; however, the difficulty in selecting and validating truly stable reference genes (not varying in expression across variables of interest) is a known limitation of RT-qPCR. In an attempt to balance the benefits of both techniques, a cDNA-normalised approach to reference gene selection/validation was chosen, followed by reference gene normalisation of target cDNA concentrations.

5.3. Results 2: RT-ddPCR

5.3.1. Reference Gene Normalisation

5.3.1.1. Initial Screening and Reference Gene Selection

In an effort to control for potential challenges with RNA stability and/or reverse transcription efficiency in selfie-ddPCR, a pool of seven reference gene primers was selected from existing literature in MIA models and studies of embryonic tissue (see section 2.6.2 for primer sequences and validations) for use in RT-ddPCR. To ensure selection of the most stable reference genes in each tissue type, mRNA expression normalised to input cDNA (copies per ng; on average, 2.8 ng per reaction) was quantified in a subset of 6-8 initial samples. Any reference genes that were expressed at too high or low a concentration to be quantified alongside target genes, were highly variable across samples, or appeared to be differentially expressed across treatment groups, were excluded (Figure 5.4). Sex was unknown at this stage and was not included in the assessment. Following any necessary exclusions, combinations of at least three reference genes were tested to find the combination with the lowest coefficient of variance (CV). As target and reference primers were split into two groups for ddPCR amplification (annealing temperatures of 61.5°C and 64°C; see Tables 2.3 and 2.4), the combination of reference primers must also have included one reference in each group. In all tissue types, this requirement was met in the combination with the lowest CV and did not require the use of a more variable reference combination.

Primer pairs selected for each tissue type, alongside the cDNA-normalised expression data used to select the reference primer set, are presented in Figure 5.4. The primer combinations selected for each tissue were as follows: *Hprt*, *Ppia*, and *B2m* in the E14.5 brain (Figure 5.4A; CV = 8.8%); *Gapdh*, *Hprt*, and *Ppia* in the E14.5 placenta (Figure 5.4B; CV = 5.3%); *Gapdh*, *Tbp*, and *Ywhaz* in the E18.5 brain (Figure 5.4C; CV = 8.7%); and *Ppia*, *Tbp*, and *Ywhaz* in the E18.5 placenta (Figure 5.4D; CV = 16%).

Across all tissue types, only one potential reference was excluded due to high variability: *B2m* in the E14.5 placenta, with a CV of 58%. Concentration was a more prominent reason for exclusion. Notably, *ActB* was far higher than targets and fully saturated ddPCR reactions (all droplets being positive). Although samples could be diluted to quantify *ActB* and multiplied for normalisation of targets, the reference primer pool was large enough to provide reasonably stable reference options without the addition of further pipetting steps that may have contributed to inaccuracy. *Hprt* was further excluded in E18.5 brain tissue with the opposite concern, as low concentrations were detected particularly in saline samples. Interestingly, both *Tbp* and *Ywhaz* in the E14.5 brain also showed a treatment bias, with all saline concentrations higher than poly I:C samples. While these were excluded from reference gene selection, they were instead assessed as target genes out of interest to confirm whether a robust treatment effect existed.

Figure 5.4: Selection of reference genes based on 6-8 pilot samples per tissue type. Concentrations are highlighted as a heat map within each column to readily identify any treatment bias (dark red = highest concentration, dark blue = lowest concentration). The geometric mean, standard deviation (std), and coefficient of variance (CV) are presented for each individual primer, as well as the final primer combination selected for each tissue (displayed to the right). Greyed cells indicate exclusions based on concentration, variability, or treatment bias. *Gapdh* = Glyceraldehyde 3-phosphate dehydrogenase; *ActB* = Actin, beta; *Hprt* = hypoxanthine phosphoribosyltransferase 1; *Ppia* = peptidylprolyl isomerase A; *B2m* = beta-2 microglobulin; *Tbp* = TATA box binding protein; *Ywhaz* = tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein, zeta.

A	E14.5 BR: Normalised Expression (copies per ng cDNA)							Hprt + Ppia +B2m
	Gapdh	ActB	Hprt	Ppia	B2m	Tpb	Ywhaz	
Saline	35847.0	saturated	351.2	41656.9	258.5	269.8	17565.5	1557.9
	36324.0		331.7	40245.0	213.8	246.4	13124.2	1418.4
	41440.1		403.3	50613.5	310.7	264.0	13911.8	1851.1
Poly I:C	34556.7		296.7	34466.8	314.7	198.8	12278.2	1476.4
	45207.7		341.9	43683.8	233.6	239.6	12642.9	1516.7
	saturated		336.9	41662.2	286.9	232.5	12140.9	1591.0
mean	38675.1		343.6	42054.7	269.7	241.8	13610.6	1568.6
std	4020.1		31.7	4779.7	37.8	23.2	1863.7	137.9
CV (%)	10.4		9.2	11.4	14.0	9.6	13.7	8.8

B	E14.5 fPLA: Normalised Expression (copies per ng cDNA)							Gapdh + Hprt +Ppia
	Gapdh	ActB	Hprt	Ppia	B2m	Tpb	Ywhaz	
Saline	27532.3	saturated	5897.7	35587.8	3692.1	558.1	15650.6	17945.0
	27492.0		5193.2	43767.7	3459.9	640.5	15499.4	18419.0
	32229.2		4865.5	48674.2	15343.6	668.5	18925.6	19689.1
	23636.3		6832.0	35550.6	3441.6	608.2	16112.2	17905.7
Poly I:C	23519.4		5716.9	34461.7	5588.7	490.5	14019.8	16671.5
	26589.1		4269.6	39700.5	13573.4	516.2	14561.1	16518.1
	25888.2		5705.0	42166.6	5571.9	568.6	15276.8	18398.2
	31537.4		4182.8	45443.3	10069.3	681.4	18254.9	18165.8
mean	27303.0		5332.8	40669.0	7592.6	591.5	16037.5	17964.1
std	3011.6		832.4	4876.0	4466.6	65.4	1602.1	947.2
CV (%)	11.0		15.6	12.0	58.8	11.1	10.0	5.3

C	E18.5 PB: Normalised Expression (copies per ng cDNA)							Gapdh + Tpb + Ywhaz
	Gapdh	ActB	Hprt	Ppia	B2m	Tpb	Ywhaz	
Saline	17478.4	saturated	below quant.	22381.3	921.5	322.5	16737.2	4552.5
	23153.4			30609.0	1390.5	432.7	21945.9	6035.6
	18772.4		7.3	22743.2	971.2	339.3	20057.1	5036.5
Poly I:C	19556.3		14.6	24140.1	1147.5	378.4	17920.5	5446.5
	22101.7		26.6	28056.5	1278.8	365.5	19999.6	5099.6
	19266.6		20.5	27317.9	1440.9	380.7	22537.8	5488.0
mean	20054.8		17.3	25874.7	1191.7	369.9	19866.3	5276.5
std	1955.5		7.2	3007.6	197.0	35.0	2045.1	458.8
CV (%)	9.8		41.6	11.6	16.5	9.5	10.3	8.7

D	E18.5 fPLA: Normalised Expression (copies per ng cDNA)							Ppia + Tpb + Ywhaz
	Gapdh	ActB	Hprt	Ppia	B2m	Tpb	Ywhaz	
Saline	23202.6	saturated	8678.3	16883.7	3945.8	300.0	11794.5	3909.3
	26114.9		8494.4	16496.9	3434.3	339.3	12990.0	4173.7
	28181.8		8826.8	21852.3	5977.6	414.6	14126.8	5039.6
Poly I:C	41144.5		15283.4	28587.8	5896.6	451.7	16657.8	5991.8
	20850.6		7211.9	16604.1	4494.0	338.6	11107.5	3967.4
	29739.1		11912.8	17841.3	4217.3	394.4	12346.7	4429.1
mean	28205.6		10067.9	19711.0	4660.9	373.1	13170.6	4585.1
std	6497.0		2729.7	4375.2	957.5	51.7	1822.8	732.3
CV (%)	23.0		27.1	22.2	20.5	13.9	13.8	16.0

5.3.1.2. Final Reference Gene Analysis

The reference genes selected above were subsequently quantified alongside target genes. As selection of reference genes was based on a small number of samples per tissue type, and was stratified across treatment but not sex, expression of reference genes was re-assessed after data collection and sex determination (again normalised to cDNA concentration). This allowed a more robust assessment of the stability of the reference genes selected for each tissue type. While the reference genes selected for the E14.5 and E18.5 placenta appeared to be stably expressed across fixed effects and did not display any unequal variances between groups (requiring weighted modelling), significant effects were observed in the expression of reference genes within embryonic brain tissue at both time points (Table 5.2, Figure 5.5).

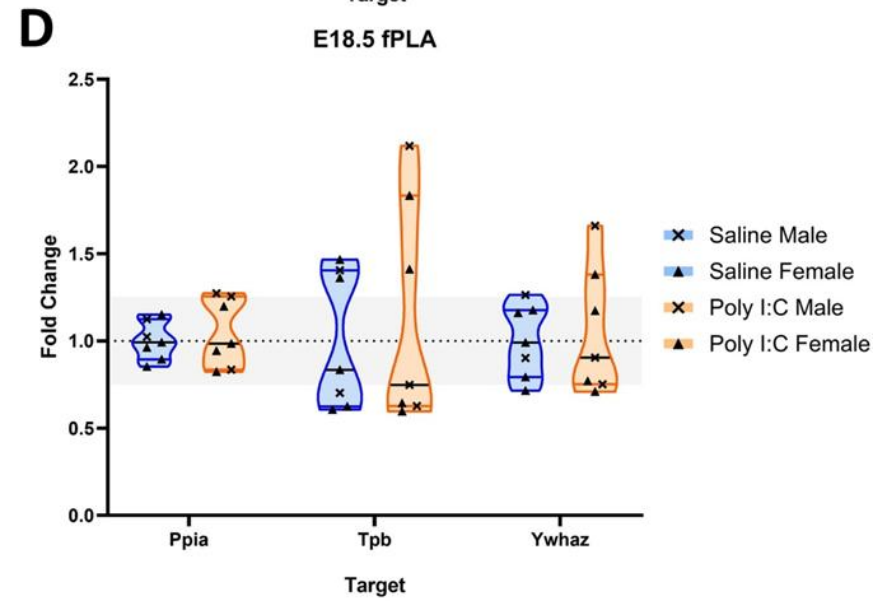
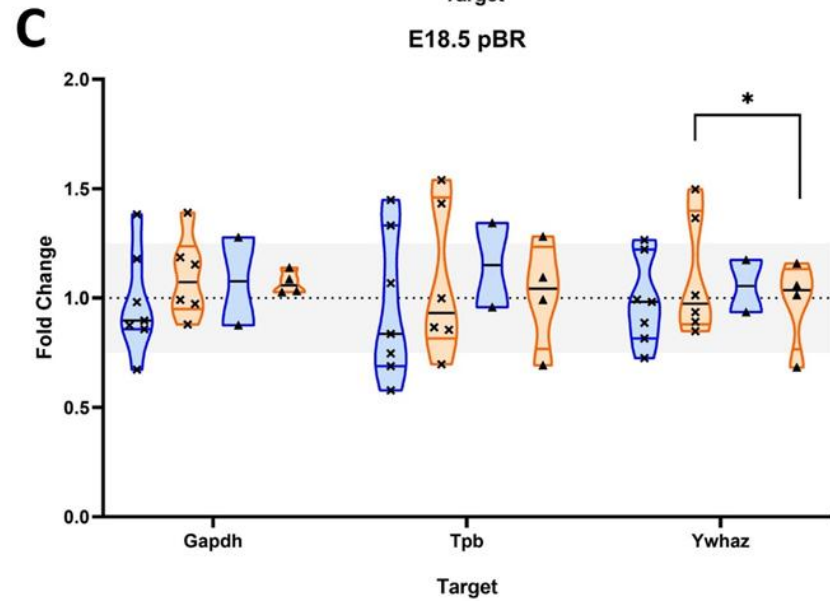
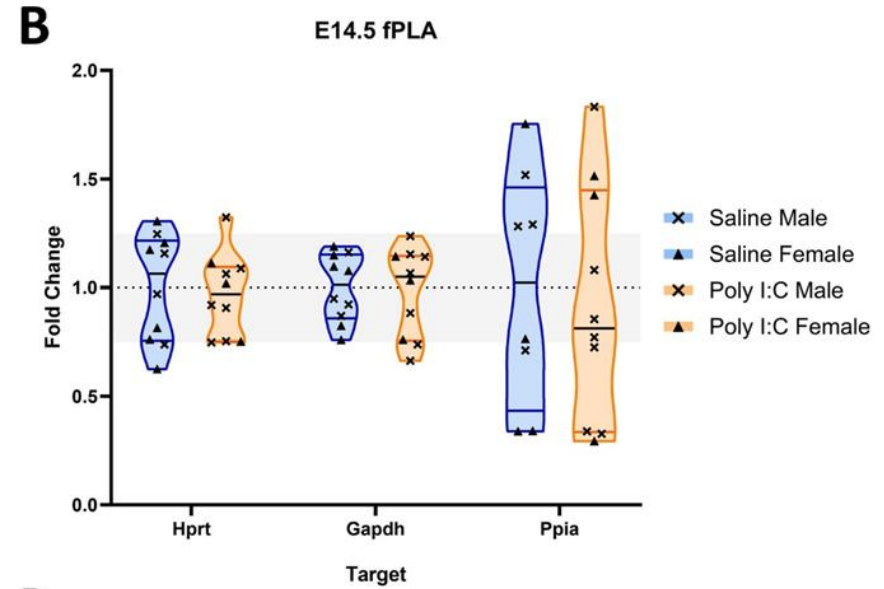
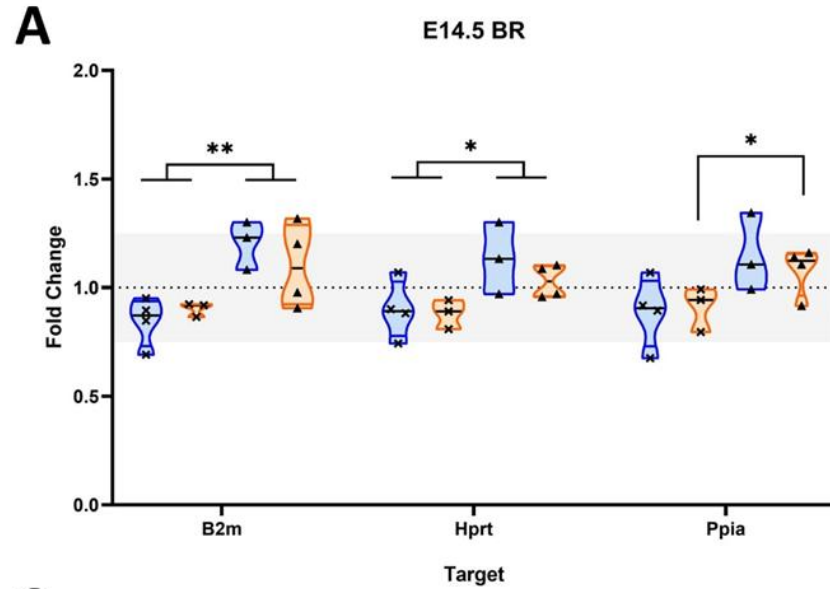
Tissue	Gene	Mean copies per ng cDNA ($\pm$ SEM)	Fold Change from Saline Mean: Fixed Effects p-values		
			Sex	Treatment	Sex*Treatment
E14.5 BR	<i>B2m</i>	240.66 ($\pm$ 12.04)	0.0094	0.7357	0.3566
	<i>Hprt</i>	304.70 ($\pm$ 11.99)	0.0460	0.6622	0.8411
	<i>Ppia</i>	37792.28 ($\pm$ 1689.08)	0.0847	0.8092	0.6031
E14.5 fPLA	<i>Gapdh</i>	27368.23 ($\pm$ 1089.48)	0.8473	0.8736	0.8090
	<i>Hprt</i>	5487.21 ($\pm$ 268.20)	0.7974	0.7335	0.8703
	<i>Ppia</i>	25804.80 ($\pm$ 3338.81)	0.7562	0.8918	0.2618
E18.5 pBR	<i>Gapdh</i>	19073.90 ($\pm$ 789.44)	0.7438	0.6236	0.5004
	<i>Tbp</i>	309.47 ($\pm$ 20.56)	0.4190	0.6324	0.0386
	<i>Ywhaz</i>	17860.91 ($\pm$ 850.86)	0.7303	0.7387	0.0030
E18.5 fPLA	<i>Ppia</i>	18152.39 ($\pm$ 729.57)	0.2010	0.9007	0.9873
	<i>Tbp</i>	237.22 ($\pm$ 30.57)	0.8608	0.7092	0.9622
	<i>Ywhaz</i>	8845.27 ($\pm$ 661.81)	0.5702	0.8640	0.9603

Table 5.2: Concentration and significant LMM effects identified in foetoplacental reference gene expression. Yellow and green shading highlights near-significant ($p \leq 0.10$) and significant ($p < 0.05$) p-values respectively. Full accompanying LMM tables can be found in Appendix G. SEM = standard error of the mean.

Sex differences were apparent in the E14.5 brain; LMM analysis revealed a significant overall sex effect in *B2m* ($t(4) = -4.68$, $p = .0094$) and *Hprt* ($t(10) = -2.28$, $p = 0.0460$), with the sex effect in *Ppia* expression also nearing significance ($t(4) = -2.28$, $p = 0.0847$). For all three references, female offspring had higher expression than male counterparts; when further separating groups based on both treatment and sex, this trend was observed across sexes in both treatment groups (Figure 5.5A). Only the increase in *Ppia* expression in MIA-exposed females compared to male littermates proved significant ($t(4) = -7.74$, adjusted $p = 0.0060$), although the difference between sexes in saline offspring *B2m* expression only narrowly lost significance following adjustment for multiple comparisons ($t(4) = -3.88$, $p = 0.0179$, adjusted $p = 0.0714$).

In contrast to the E14.5 brain, no main sex effects were observed in the E18.5 posterior brain tissue examined. In both *Tbp* and *Ywhaz* expression however, significant sex*treatment interactions were identified (*Tbp*: $t(9.48) = -2.40$, $p = 0.0386$; *Ywhaz*: $t(8.69) = -4.08$, $p = 0.0030$). In *Ywhaz* expression, this extended to a significant pairwise comparison between sexes in poly I:C littermates ($t(8.64) = 3.62$, adjusted $p = 0.0238$; Figure 5.5C). While these groups visually overlap, the estimated marginal means used for statistical comparison are derived from LMMs, which also take litter effects into account. It should be noted that when making pairwise comparisons between sex/treatment groups, the number of pups in each group was limited ($n = 2-7$ per group). The sample size and unbalanced nature of some pairwise comparisons (e.g. saline male and female comparisons within the E18.5 brain) somewhat limited model strength in assessing interactions or small pairwise comparisons, although assessing main effects alone was stronger. The limitations of this are further considered in section 5.4.

Figure 5.5: Expression of reference genes normalised to cDNA concentration. Expression differs across groups of interest in the E14.5 brain (A) and the E18.5 brain (C), while no effects were observed in the E14.5 (B) or E18.5 (D) placenta. Pairwise comparisons were carried out between sexes within a treatment, as well as across treatment within sex. Only statistically significant comparisons are shown on figure: where sex was not significant, sex is combined and denoted by symbol (male = cross, female = triangle). Grey background shading indicates 0.75 – 1.25 fold change range for ready comparison between graph scales. * = $p < 0.05$; ** = $p \leq 0.01$.



The significant sex and sex*treatment interaction effects identified in reference genes used in foetal brain tissue raised concerns over their use in target gene normalisation. For example, in expression of a target gene in the E14.5 brain with no true sex difference, biased normalisation to *B2m*, *Hprt*, and *Ppia* may have lead to relatively lower expression calculated in female offspring and observation of a false sex or sex*treatment interaction effect. As a result, it was decided that normalisation of target mRNA concentrations to these reference genes may be misleading and should not be presented in this chapter (although is available in Appendix F for comparison). Rather, data was normalised to input cDNA concentration for final analysis. Although reference gene bias was not found in the placenta, all tissue types were normalised to input cDNA concentration for consistency. As previously discussed, normalisation to cDNA may be sensitive to technical error and increase data variability, but is less likely to be biased across sex or treatment groups. In addition, wide variability was not consistently seen in reference gene normalisation, suggesting cDNA normalisation was the most appropriate method of data normalisation in this instance.

5.3.2. Target mRNA Expression at E14.5

5.3.2.1. Whole Brain Tissue

Assessment of E14.5 brain tissue found quantifiable expression of 6/9 target genes, including *Tpb* and *Ywhaz* which were included in E14.5 brain analysis due to potential treatment effects (Table 5.3). Potential expression of *Tph1*, *Tph2*, and *Oct3* was detected, but below the cut-off for quantification (less than 1 copy/ μ L within the reaction, approximately $\leq 0.1\%$ positive droplets). *Bdnf*, *Ddc*, *MaoA*, *Sert*, and *Tbp* expression was considered relatively low (less than 5% positive droplets), while *NeuroD1*, *Mecp2*, and *Ywhaz* concentrations were well above working limits. As discussed in section 2.6.1.2, targets with a low proportion of positive droplets are subject to increased Poisson error in concentration estimates, although still typically considered quantifiable until reaching concentrations below 1 copy/ μ L [337, 340, 410].

Following normalisation to cDNA concentration, significant treatment and sex*treatment interaction effects were observed in the E14.5 brain (Table 5.3; full statistical tables in Appendix G). A significant treatment effect was observed in *NeuroD1* expression ($t(6) = 2.64$, $p = 0.0383$), with decreased expression in poly I:C-exposed offspring (Figure 5.6A). This effect appeared to be strongest in females, although when separated into sex/treatment groups for post-hoc comparison, neither males nor females significantly differed by treatment following Bonferroni corrections for multiple comparisons (males: $t(4) = 0.49$, $p = 0.6505$; females: $t(4) = 3.48$, $p = 0.0255$, adjusted $p = 0.1018$). No other target genes demonstrated a significant main treatment effect. No targets demonstrated a main sex effect either, although *Bdnf*, *MaoA*, and *Mecp2* expression neared significance. Both *Bdnf* ($t(5.30) = -2.32$, $p = 0.0652$) and *MaoA* ($t(4) = -2.11$, $p = 0.1022$) showed a non-significant trend of overall expression being increased in females compared to males (Figure 5.6B); when looking at poly I:C-exposed offspring alone, this effect did not reach significance for *Bdnf* expression ($t(4.79) = -2.24$, $p = 0.0779$, adjusted $p = 0.3117$) but did for *MaoA* expression ($t(4) = -8.99$, adjusted $p = 0.0034$). Although *Ywhaz* did not show a significant sex effect either, expression trends mirror *MaoA* expression, which resulted in another significant increase in female poly I:C offspring compared to male littermates ($t(4) = -4.81$, adjusted $p = 0.0344$).

In contrast to *Bdnf* and *MaoA* expression, *Mecp2* demonstrated a significant sex*treatment interaction. Two significant pairwise comparisons were found and survived Bonferroni corrections: expression of *Mecp2* was elevated in the E14.5 brain of saline female offspring in comparison to both male littermates ($t(5.54) = -4.35$, adjusted $p = 0.0232$) and poly I:C-exposed females ($t(6.7) = 3.77$, adjusted $p = 0.0302$). Sert expression also demonstrated a significant sex*treatment interaction ($t(4.31) = 3.24$, $p = 0.0283$), although no significant pairwise differences were identified (all adjusted $p > 0.11$).

Interestingly, the distribution of target expression across saline samples was often skewed by a single male and/or female with high expression (e.g. Figure 5.6A, *Mecp2*; Figure 5.6B, *Ddc*). In all cases where a saline data point is visually distinct from the main body, the sample originated from the same litter (litter S10; female pup S10.4 or male pup S10.5). Importantly, these two samples were not outliers/the samples with highest expression in all targets; this suggested the effects of litter bias are not consistent across targets, highlighting the importance of considering potential litter bias when sampling and analysing data in MIA models.

Target	Mean copies per ng cDNA ($\pm$ SEM)		Fold Change from Saline Mean: Fixed Effects p-values		
	Saline	Poly I:C	S	T	S*T
<i>Bdnf</i>	28.15 ($\pm$ 4.10)	27.52 ($\pm$ 2.07)	0.0652	0.9675	0.7485
<i>Ddc</i>	67.62 ($\pm$ 27.40)	41.11 ($\pm$ 5.18)	0.3881	0.5829	0.7169
<i>MaoA</i>	458.70 ($\pm$ 47.08)	453.68 ($\pm$ 18.57)	0.1022	0.9839	0.7754
<i>Mecp2</i>	1096.05 ($\pm$ 49.02)	912.07 ($\pm$ 34.40)	0.0954	0.1171	0.0041
<i>NeuroD1</i>	1066.89 ($\pm$ 99.69)	813.66 ($\pm$ 28.76)	0.1206	0.0383	0.1294
<i>Sert</i>	24.84 ($\pm$ 9.19)	42.02 ($\pm$ 9.87)	0.4708	0.2931	0.0283
<i>Tbp</i>	233.19 ($\pm$ 12.25)	232.88 ($\pm$ 6.72)	0.1896	0.7877	0.0664
<i>Ywhaz</i>	13249.07 ($\pm$ 818.00)	11772.75 ($\pm$ 282.48)	0.1532	0.1337	0.8207

Table 5.3: Target cDNA concentration and associated LMM effect p-values in the E14.5 brain. Yellow and green shading highlights near-significant ($p \leq 0.10$) and significant ($p < 0.05$) p-values, respectively. Full accompanying LMM tables can be found in Appendix G. SEM = standard error of the mean; S= sex; T = treatment; S*T = sex-treatment interaction.

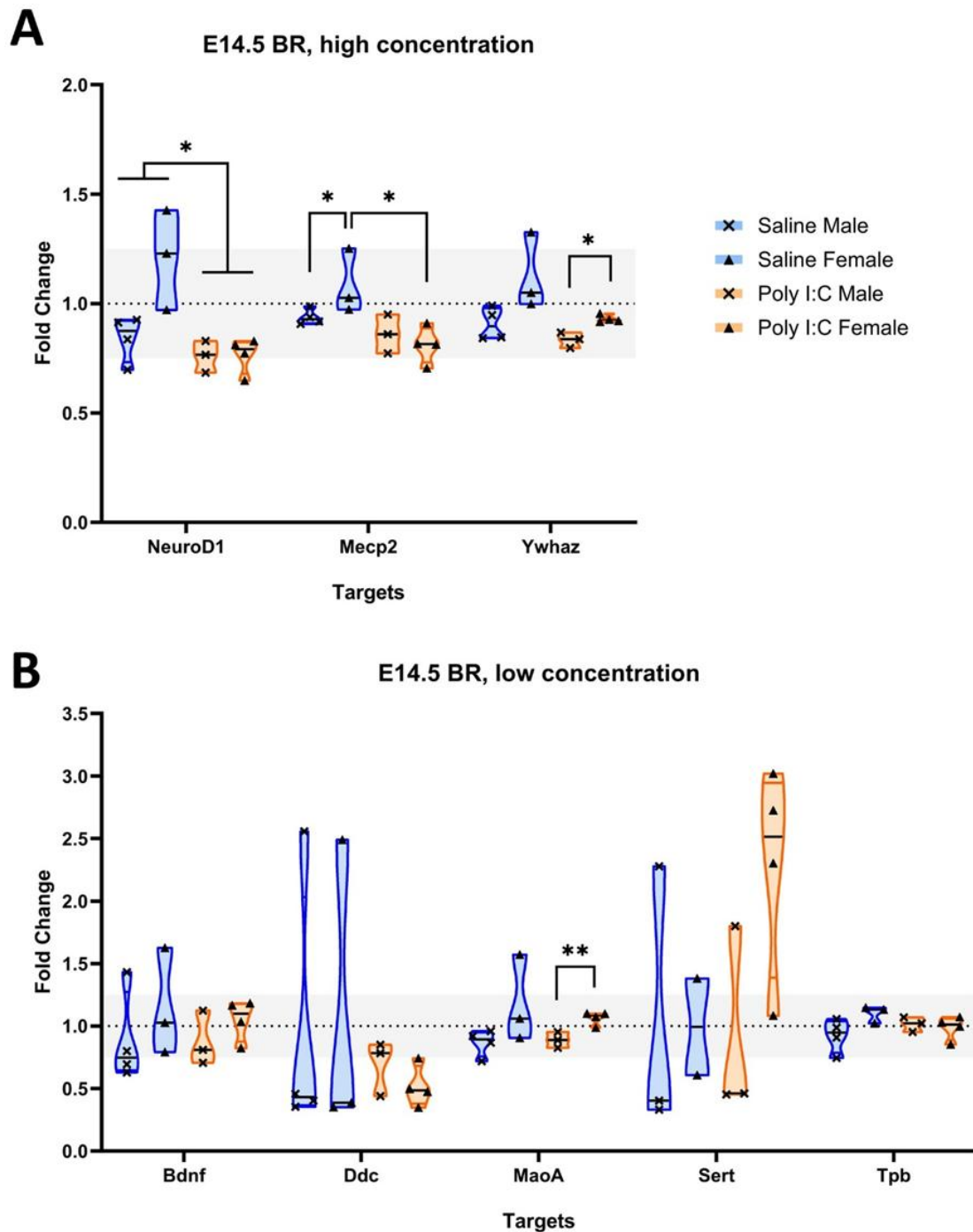


Figure 5.6: Fold change (compared to saline mean) in target expression in the E14.5 brain. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. Only significant main effects and pairwise comparisons are presented, however all relevant sex/treatment pairwise comparisons were made (see Appendix G, Figure G2). * = $p < 0.05$; ** = $p \leq 0.01$.

5.3.2.2. *Foetally-derived Placental Tissue*

All except one target gene (*NeuroD1*) was found to be expressed in the E14.5 placenta, with high concentrations of *MaoA*, *Mecp2*, and *Oct3* present in cDNA samples (Table 5.4). *Tph1* and *Tph2* expression was at the lower edge of quantification, with 7 samples in *Tph1* expression (4 saline, 3 poly I:C) and 8 samples in *Tph2* expression (the same samples as *Tph1*, with an additional poly I:C sample) excluded from analysis as a result of amplification below the required threshold. These samples did not have a lower overall input cDNA concentration relative to samples with quantifiable *Tph1* and *Tph2* expression, nor did any two excluded samples originate from the same litter; as 2-3 samples were tested per litter (overall litter $n = 8$), this did not lead to the exclusion of any litter from analysis. Unfortunately, further increasing input cDNA concentrations also increased background noise and prevented improved quantification (see 722.6.2.3 for example).

One target demonstrated a significant main treatment effect; *Tph1* expression was significantly increased in poly I:C offspring ($t(4) = -2.78$, $p = 0.0496$), with poly I:C samples also showing high variability compared to saline (Figure 5.7B). As a small number of samples across litters were excluded for low *Tph1* expression, this suggested a shift towards increased expression that is variable across pups within a litter. A similar increase in variability was seen in *Ddc* expression in poly I:C pups, although this was not significant as either an overall main treatment effect ($t(5) = -1.68$, $p = 0.1530$) or in pairwise comparisons between sex/treatment groups (all adjusted $p > 0.3$). Expression of *Bdnf* ($t(10) = 2.34$, $p = 0.0412$) and *Oct3* ($t(10) = 2.63$, $p = 0.0251$) exhibited significant sex effects, likely weighted by the non-significant visual difference between saline male and poly I:C female pups (Figure 5.7A,B). No pairwise comparisons in *Tph1*, *Bdnf*, or *Oct3* expression survived corrections for multiple comparisons in offspring across relevant treatment or sex pairings.

Target	Mean copies per ng cDNA ($\pm$ SEM)		Fold Change from Saline Mean: Fixed Effects p-values		
	Saline	Poly I:C	S	T	S*T
<i>Bdnf</i>	78.64 ($\pm$ 10.32)	64.76 ($\pm$ 9.36)	0.0412	0.1030	0.6935
<i>Ddc</i>	12.51 ($\pm$ 2.16)	42.87 ($\pm$ 15.40)	0.3212	0.1530	0.2423
<i>MaoA</i>	2279.80 ($\pm$ 266.65)	2012.43 ($\pm$ 212.47)	0.1394	0.2360	0.8916
<i>Mecp2</i>	385.59 ($\pm$ 22.45)	390.93 ($\pm$ 18.79)	0.3176	0.8916	0.9776
<i>Oct3</i>	1433.05 ($\pm$ 182.99)	1470.72 ($\pm$ 181.59)	0.0251	0.5658	0.9310
<i>Sert</i>	34.89 ($\pm$ 3.19)	46.29 ($\pm$ 4.98)	0.1055	0.7227	0.3745
<i>Tph1</i>	1.48 ($\pm$ 0.23)	3.33 ($\pm$ 0.81)	0.2697	0.0496	0.0526
<i>Tph2</i>	1.39 ($\pm$ 0.58)	1.53 ($\pm$ 0.69)	0.8663	0.9944	0.4547

Table 5.4: Target cDNA concentration and associated LMM effect p-values in the E14.5 foetally-derived placenta. Yellow and green shading highlights near-significant ($p \leq 0.10$) and significant ($p < 0.05$) p-values respectively. Full accompanying LMM tables can be found in Appendix G . SEM = standard error of the mean; S= sex; T = treatment; S*T = sex-treatment interaction.

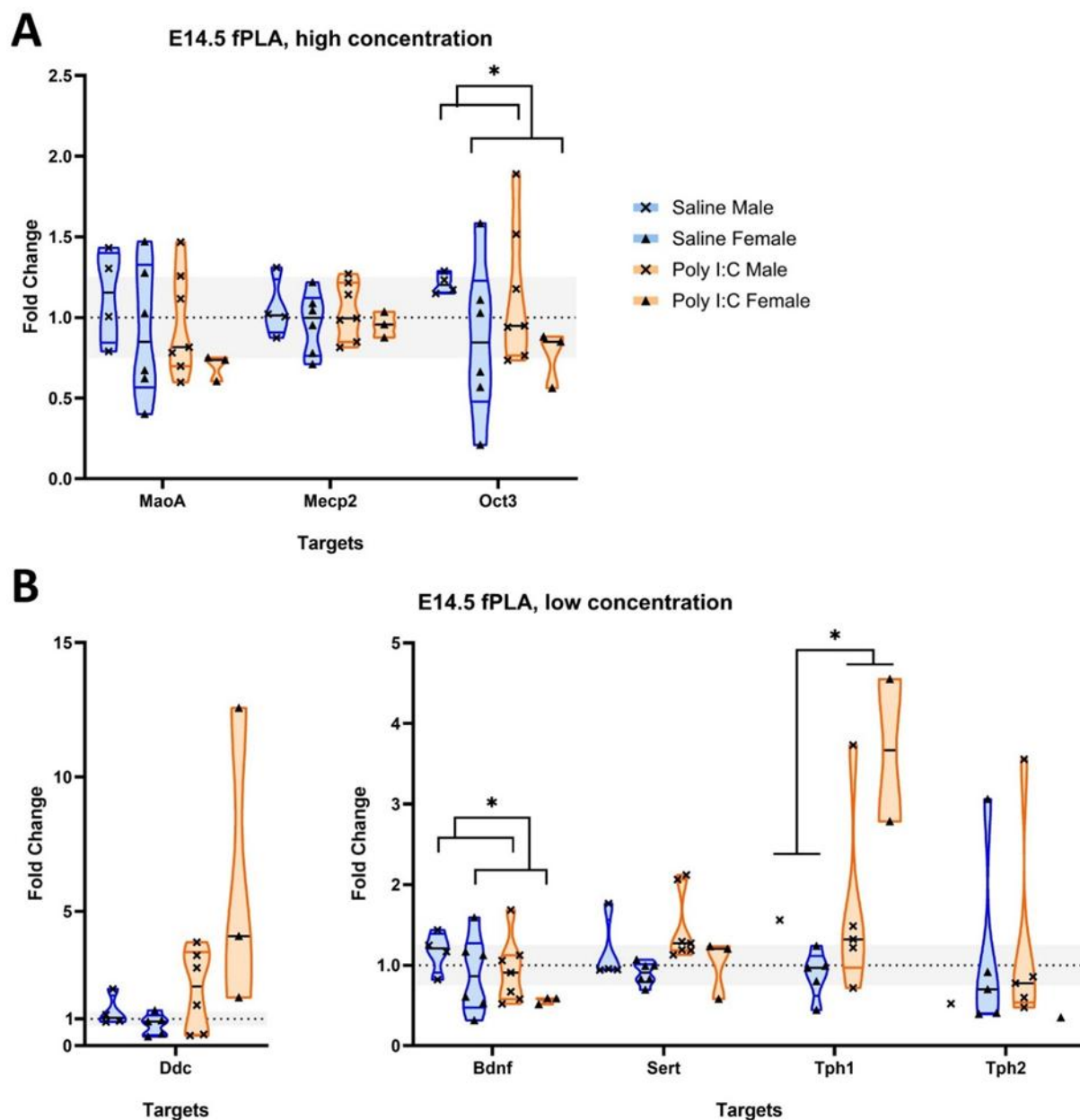


Figure 5.7: Fold change (compared to saline mean) in target expression in the E14.5 foetally-derived placenta. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. Only significant effects and pairwise comparisons are presented on figure, however all relevant sex/treatment pairwise comparisons were made (found in Appendix G, Figure G4). * = $p < 0.05$.

5.3.3. Target mRNA Expression at E18.5

5.3.3.1. Posterior Brain Tissue

Within the E18.5 foetal brain, all targets showed some amplification, but only 7/9 targets were quantifiable: *MaoA*, *Mecp2*, and *NeuroD1* at a relatively high concentration, and *Bdnf*, *Ddc*, *Tph2*, and *Sert* at a lower concentration. As expected, mean *Tph2* expression was increased in the brain at E18.5 compared to E14.5, which was previously expressed at concentrations too low to reasonably quantify. No samples were excluded from these targets for being below the quantification limit.

No significant main effects were observed in this tissue, with all sex/treatment groups showing considerable overlap in target expression (Table 5.5; Figure 5.8). Two significant interaction effects were reported in *MaoA* ($t(8.89) = -2.69$, $p = 0.0252$) and *Mecp2* ($t(9.05) = -3.04$, $p = 0.0140$) expression. While no significant pairwise comparisons were found, these interactions may have been driven by the near-significant trends of increased *MaoA/Mecp2* expression in male poly I:C offspring compared to female poly I:C offspring (*MaoA*: $t(8.77) = 1.93$, $p = 0.0859$, adjusted $p = 0.3438$; *Mecp2*: $t(8.94) = 2.58$, $p = 0.0299$, adjusted $p = 0.1196$) which was not reflected in saline counterparts. Since female samples were relatively few, however, these interactions should be considered with caution.

Target	Mean copies per ng cDNA (± SEM)		Fold Change from Saline Mean: Fixed Effects p-values		
	Saline	Poly I:C	S	T	S*T
Bdnf	68.39 (± 6.26)	79.23 (± 6.77)	0.9016	0.5915	0.4357
Ddc	66.56 (± 8.83)	65.74 (± 5.60)	0.8292	0.9736	0.8820
MaoA	430.55 (± 33.07)	538.73 (± 41.79)	0.7619	0.6049	0.0252
Mecp2	1079.10 (± 48.64)	1177.80 (± 71.43)	0.9017	0.9772	0.0140
NeuroD1	1125.93 (± 60.89)	1180.04 (± 72.55)	0.8194	0.7646	0.4149
Sert	31.96 (± 7.21)	32.74 (± 5.58)	0.9051	0.8287	0.3372
Tph2	85.08 (± 15.62)	70.95 (± 13.68)	0.5055	0.6102	0.9298

Table 5.5: Target cDNA concentration and associated LMM effect p-values in the E18.5 posterior brain. Green shading highlights significant ($p < 0.05$) p-values. Full accompanying LMM tables can be found in Appendix G. SEM = standard error of the mean; S= sex; T = treatment; S*T = sex-treatment interaction.

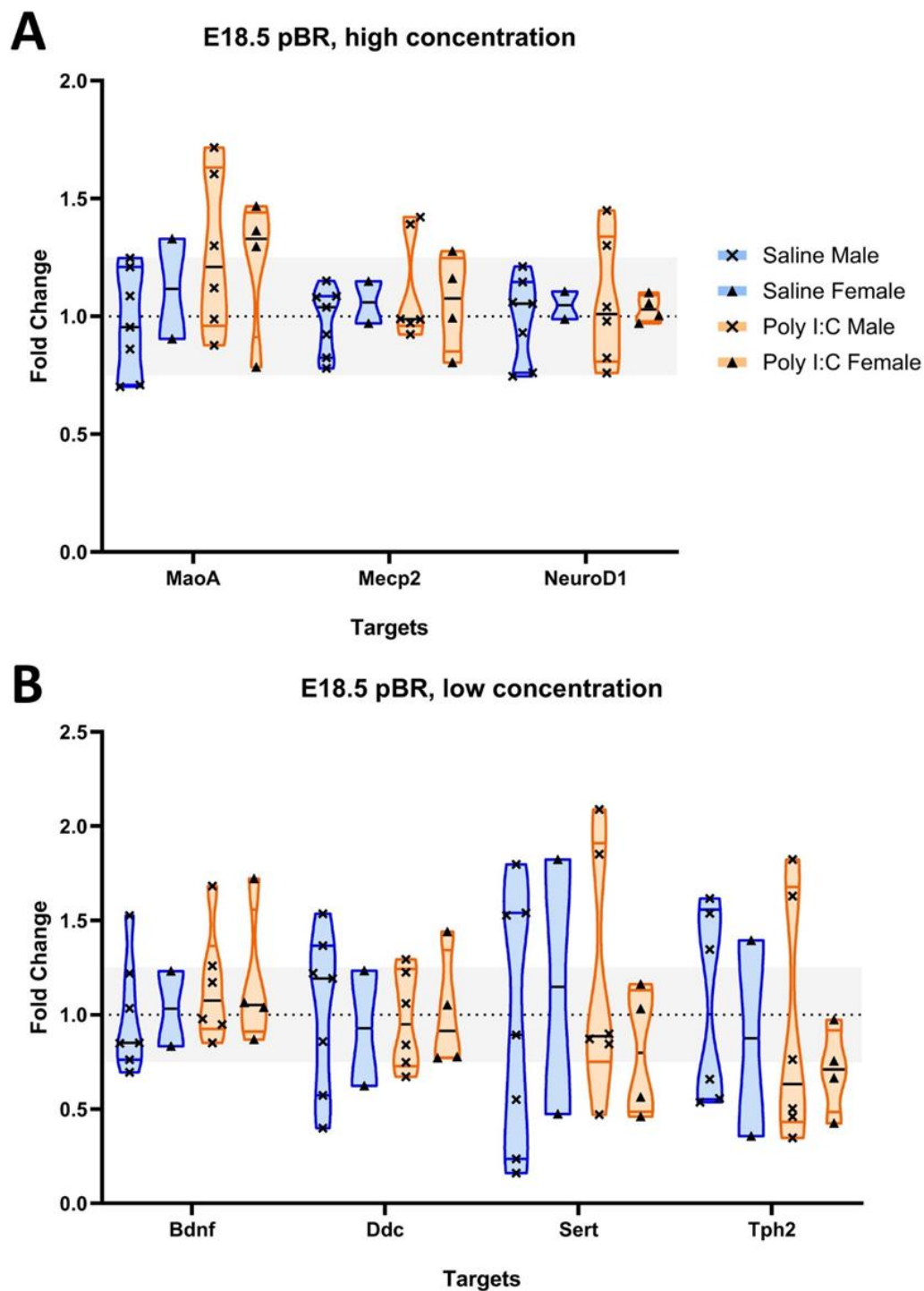


Figure 5.8: Fold change (compared to saline mean) in target expression in the E18.5 posterior brain. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. No significant effects were observed in main effects or pairwise comparisons (see Appendix G, Figure G6).

5.3.3.2. Foetally-derived Placental Tissue

In the E18.5 placenta, 6/9 were quantified; *MaoA* and *Oct3* at high concentrations, and *Bdnf*, *Ddc*, *Mecp2*, and *Sert* at relatively low concentrations (Table 5.6). In contrast to E14.5, *Tph1* and *Tph2* were no longer at quantifiable concentrations – nor close to this threshold, with positive bands unable to be determined or quantified due to less than 10/20,000 positive droplets per well at working concentrations.

No significant sex, treatment, or sex*treatment effects were found in this tissue, although *Bdnf* expression neared a significant sex effect ($t(4) = 2.10$, $p = 0.1042$). This appeared to be driven by a significant decrease in *Bdnf* expression in saline females compared to males ($t(4) = 5.00$; adjusted $p = 0.0301$), which was not reflected in poly I:C offspring (Figure 5.9). Although poly I:C males presented a decreased trend in *Bdnf* expression compared to saline males, this did not reach significance before or after correction for multiple comparisons ($t(4) = 1.98$, $p = 0.1190$, adjusted $p = 0.4759$).

Target	Mean copies per ng cDNA ($\pm$ SEM)		Fold Change from Saline Mean: Fixed Effects p-values		
	Saline	Poly I:C	S	T	S*T
<i>Bdnf</i>	137.72 ($\pm$ 13.08)	124.22 ($\pm$ 15.10)	0.1042	0.1846	0.1980
<i>Ddc</i>	21.02 ($\pm$ 4.13)	15.02 ($\pm$ 2.84)	0.8149	0.2846	0.6601
<i>MaoA</i>	2771.38 ($\pm$ 161.88)	2948.15 ($\pm$ 234.70)	0.2551	0.8040	0.6143
<i>Mecp2</i>	347.39 ($\pm$ 18.48)	384.09 ($\pm$ 26.28)	0.4180	0.3286	0.6888
<i>Oct3</i>	805.41 ($\pm$ 35.32)	902.29 ($\pm$ 106.92)	0.6259	0.5341	0.4875
<i>Sert</i>	89.48 ($\pm$ 6.06)	111.91 ($\pm$ 15.20)	0.6720	0.2178	0.4654

Table 5.6: Target cDNA concentration and associated LMM effect p-values in the E18.5 foetally-derived placenta. Yellow shading highlights near-significant ($p \leq 0.10$) p-values. Full accompanying LMM tables can be found in Appendix G. SEM = standard error of the mean; S= sex; T = treatment; S*T = sex-treatment interaction.

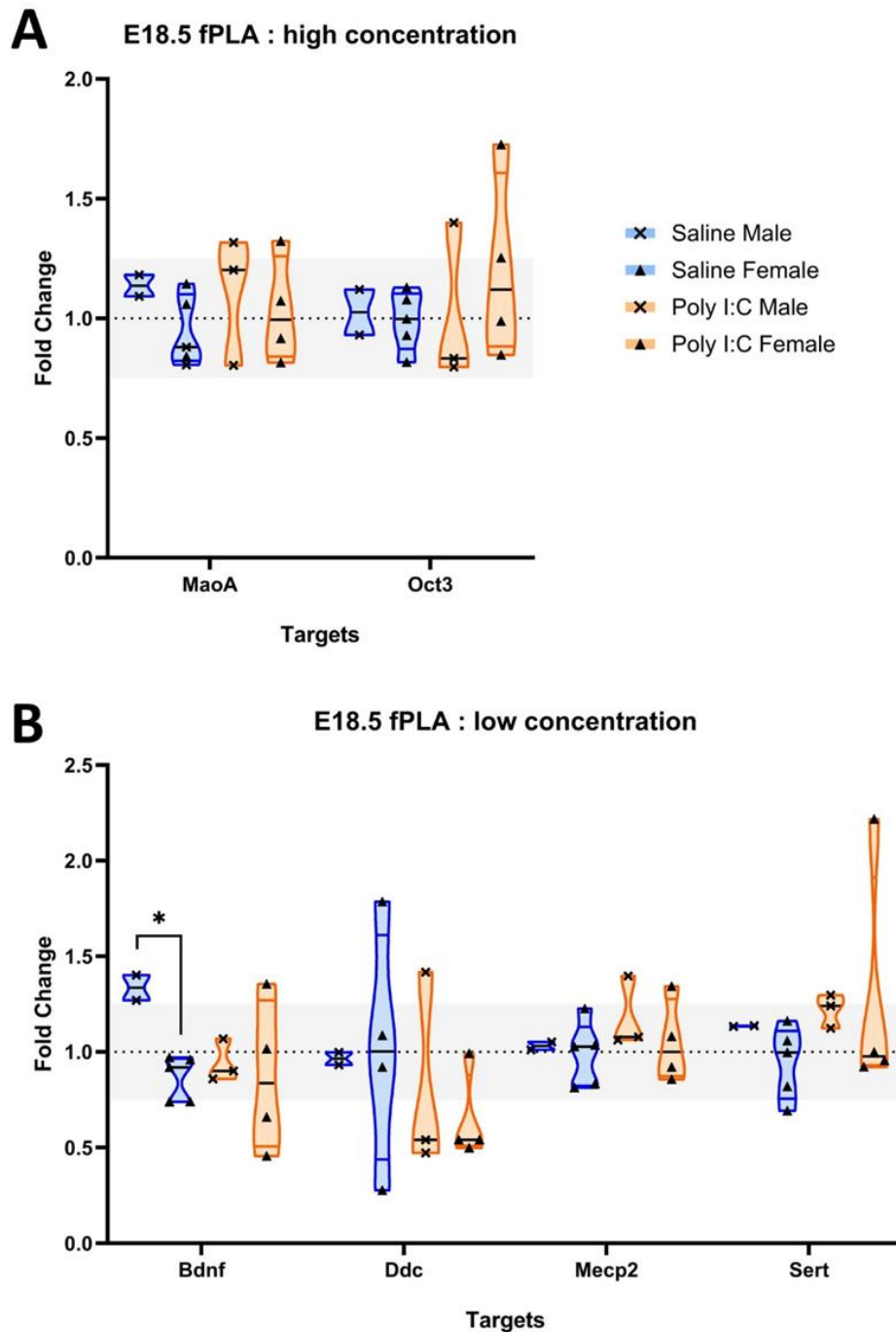


Figure 5.9: Fold change (compared to saline mean) in target expression in the E18.5 foetally-derived placenta. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for comparison between graph scales. Only significant main effects and pairwise comparisons are presented, however all relevant sex/treatment pairwise comparisons were made (Appendix G, Figure G8) * = $p < 0.05$.

5.4. Discussion

5.4.1. Technical Considerations

5.4.1.1. *ddPCR Normalisation in Foetal Tissues*

Multiple approaches to ddPCR quantification of target mRNA expression were explored in this chapter. Firstly, a method for selfie-ddPCR established in juvenile and adult brain tissue was applied to embryonic tissue [63, 405] (Appendix A). This method was found to produce highly variable target cDNA concentrations, potentially through mRNA degradation and/or reverse transcription inhibition (Figures 5.2, 5.3). Issues with translating this technique were unanticipated, although no examples of selfie-ddPCR in foetal brain tissue (successful or otherwise) were able to be identified for method comparison or optimisation. Published ddPCR assays in other tissues with high RNase content, such as the human or rodent pancreas, have not used selfie-ddPCR either and instead used techniques that isolate and purify DNA/RNA specifically [411-413]. Further optimisation of the combined DNA/RNA isolation in RNase-high tissues for selfie-ddPCR would therefore be a novel avenue of exploration, although this was not able to be further explored within this thesis research. As normalisation of target cDNA was not able to be normalised to reference gene expression either, a reliable method of foetal sample preparation for selfie-ddPCR may provide a more stable endogenous reference for future studies.

The risk of unstable reference gene expression is a known limitation in RT-qPCR experiments, and was identified in normalisation of RT-ddPCR data in this chapter [402]. Numerous studies have published tissue-specific and/or model-specific validations of common reference genes, although none have combined both embryonic age, tissue types, and MIA exposure as factors for expression analysis [339, 408, 414, 415]. This emphasised the need for in-house validation of a pool of common reference primers. While initial primer selection was based on a small subset of samples across treatments to avoid clear treatment bias, a large limitation of this early approach was the lack of sex-specific sampling; as a result, the reference genes chosen for use in the foetal brain (particularly *B2m*, *Hprt*, and *Ppia* at E14.5) were not stably expressed across groups. As sex effects were present in control animals, this is important not only for normalisation of this data, but potential bias in other studies

using foetal brain tissue if this effect is replicated across other Sprague-Dawley models. As MIA models often use experimental designs where individual foetal samples are pooled to create one sample per litter, sex differences in reference gene expression may lead to biased treatment effects if sexes are not included equally across litters of different treatments [227]. Careful validation of reference genes across all experimental samples may ameliorate this effect, but may lead to the loss of biologically-relevant sex differences in MIA offspring. Both offspring sex analysis and reference gene validation is specifically stated in the methodology of Goeden et al (with no significant sex differences observed), although the other two studies presented in Table 5.1 did not assess and/or report any foetal sex effects, although do imply some form of in-house reference gene validation was carried out for the reference gene combinations selected [227, 301, 403].

The increased expression of *B2m*, *Hprt*, and *Ppia* in E14.5 female offspring (*Ppia* only reaching significance in poly I:C-exposed pups, with saline offspring following the same trend) is an interesting finding in and of itself (Figures 5.4, 5.5). *B2m* is a crucial component in cell-surface expression of major histocompatibility complex I (MHC I) proteins, postnatally enabling antigen presentation to adaptive immune cells. Prenatally however, *B2m*/MHC I is suspected to influence neurodevelopment, including inhibiting neurite outgrowth in a range of neuronal subtypes [416]. *Hprt* is a catalyst in purine recycling, with disruptions in *Hprt* expression linked to impaired dopaminergic neuron development and reduced postnatal serotonin and dopamine production [417-419]. *Ppia* (also known as cyclophilin A/CypA) has a versatile range of functions within the cell, including mediating protein folding and trafficking, apoptosis, and inflammatory responses [420, 421]. It is possible that sex-specific expression of these factors at E14.5 reflects a general shift in brain development between male and female offspring. As previously discussed in Chapter 3, control group females exhibited reduced foetal and placental weight compared to male littermates at E18.5; while this growth difference was not observed at E14.5 and foetal brain weight was not directly assessed, changes in expression of these genes may reflect a divergence in the developmental trajectory of male and female offspring that leads to more pronounced effects in overall tissue mass at later developmental stages.

5.4.1.2. Sample Size Limitations

Although the technical ddPCR approach used to assess target mRNA expression was significantly improved during this chapter, the largest limitation of the data presented in section 5.3 remains the sample size of separated sex/treatment groups. This primarily stems from resource limitations; the consumables exhausted in attempts to optimise the ddPCR techniques discussed, as well as mechanical issues in droplet generation and droplet reading that required maintenance and testing during the data collection period, resulted in limited consumable availability towards the end of this study. As sample sex was not known during this phase of data collection, sample selection was instead stratified to ensure equal sampling across treatment groups at each time point/tissue of interest. Sex was determined for each brain/placental sample shortly after ddPCR target gene analysis, to prevent any effect of subsampling or the potential loss/degradation of RNA due to prior sample thawing. Collection of some excess body tissue during the original foetal tissue collection process would have allowed sex determining to occur prior to experimental work without risking nucleic acid degradation, and is strongly encouraged for future work.

As a result of unbalanced sex sampling within treatment groups, some comparisons should be considered with caution. Overall treatment effects in every tissue are assessed over 14-20 offspring from at least eight litters ($n = 4-5$ litters per treatment group), which is comparable to the sample sizes presented in published literature [227, 301]. Further dividing treatment groups by sex, however, reduces sample size considerably, particularly where unbalanced sex sampling in E18.5 tissues resulted in only a pair of saline females or saline males in the E18.5 brain or placenta, respectively (Figures 5.8, 5.9). This limits the ability of LMMs to accurately judge variance across groups, although LMMs do perform better than typical ANOVAs in this situation [422]. The significant sex*treatment interactions reported in this chapter should be considered as promising preliminary findings, and would strongly benefit from future work replicating these findings, ensuring that these effects are robust in larger sample sizes.

5.4.2. 5-HT-related Target Expression at E14.5

5.4.2.1. 5-HT Production and Transport

In addition to the sex effects observed in reference mRNA expression in the E14.5 brain, expression of multiple target genes was also significantly different across pups in the E14.5 brain and placenta (Figures 5.6, 5.7).

Most notably in the context of 5-HT production at E14.5, *Tph1* expression was increased in the placenta following MIA. This directly complements previous work by Goeden et al, who found an MIA-associated increase in *Tph1* expression after 24 hours (and increased Tph activity after 48 hours) in the mouse placenta at a similar point in gestational development [227]. Tph isoforms are the rate-limiting enzymes in 5-HT synthesis; while the trend of increased *Ddc* expression did not meet significance, the significant increase in *Tph1* may increase placental capacity for 5-HT synthesis following MIA. Expression of both *Tph1* and *Tph2* was too low to be quantified in the E14.5 foetal brain, which aligns with previous estimates based in mouse models that suggest 5-HT production in the foetal brain is not established until approximately E17.5. *Ddc* expression in the brain was low but quantifiable, which likely reflects the role of Ddc in the synthesis of other neurotransmitters such as dopamine (production of which has been detected in Sprague-Dawley rats at E14) [423].

Transport of 5-HT at E14.5 may also be affected by MIA, likely in a sex-specific manner. Although pairwise comparisons of sex/treatment groups were non-significant, an overall significant sex*treatment effect in *Sert* in the E14.5 brain was identified. This appeared to be driven by a trend of increased expression in poly I:C-exposed females. Generally speaking, a sex-dependent increase in *Sert* expression would suggest an increased capacity for 5-HT uptake and potentially increased 5-HT signalling in the E14.5 foetal brain. Interestingly, *Sert* is expressed by and/or trafficked to a number of cell types within the E14.5 rat brain, including serotonergic neurons, transient expression by non-serotonergic neurons, microglia, and even mitochondria [63, 223, 280]. An MIA-associated change in *Sert* expression may have different implications depending on its localisation; increased *Sert* expression, for example, may be associated with the number and/or growth of extending serotonergic neurons, transcriptional changes based on microglial activity, or increases in mitochondrial copy number requiring increased cellular *Sert* production. In addition to future studies

replicating this finding in larger sample sizes, exploring the localisation of *Sert* mRNA (for example using RNAscope in fixed brain tissue alongside immunohistochemical cell markers such as IBA1 and NeuN) would provide further context for the functional relevance of this change [424].

Previous MIA models have suggested MIA leads to a short-term increase in placental 5-HT synthesis, leading to increased 5-HT accumulating in the foetal forebrain [227, 319]. If the same 5-HT increase is reflected in this model, the sex*treatment interaction in *Sert* expression may indicate a sex-dependent response to increased 5-HT availability. *Oct3* expression may be an interesting factor in this sex-dependent response; although previous studies have identified a significant MIA-associated decrease in placental *Oct3* expression at similar developmental stages, only a significant sex effect was observed in this model (decreased in females; Figure 5.7). Interactions between placental 5-HT synthesis capacity (increased *Tph1* expression following MIA) and placental 5-HT transport (decreased *Oct3* expression in females) may contribute to sex-specific MIA responses in the foetal brain. This is further supported by *MaoA* expression in the E14.5 brain, which was significantly increased in poly I:C-exposed females compared to male littermates (Figure 5.6B). While sex-dependent MIA responses in E14.5 5-HT regulation is an intriguing theory, confirming direct 5-HT concentration changes in placental and brain tissue across sexes would be essential in strengthening this supposition.

5.4.2.2. Secondary Targets

At E14.5, a number of the secondary targets investigated in this model also showed interesting outcomes. For example in the E14.5 placenta, *Bdnf* expression was found to be increased in males compared to females (Figure 5.7B). This is more apparent in poly I:C females, although not significant at current sample sizes. While much of the work on *Bdnf* expression in the rat placenta lacks investigation of sex-specific effects, previous studies in the human placenta have identified sex-specific differences in *BDNF* expression, including decreased placental *BDNF* expression in term female placentas in high-BMI pregnancies [425, 426]. While the overall shift between sexes in this study is subtle, it does reflect previous discussion of a difference in the developmental trajectory between males and females. Placental *Bdnf* signalling promotes growth, particularly of the foetally-derived labyrinth zone, with decreases in

placental *Bdnf* expression associated with IUGR models and human pre-eclampsia [427-429]. As placental weight in these animals was significantly decreased in females at E18.5 compared to male littermates, a shift towards decreased expression of placental growth factors such as *Bdnf* likely contribute to this discrepancy.

In the E14.5 brain, a significant treatment effect in *NeuroD1* and sex*treatment interaction in *Mecp2* was observed (Figure 5.6A). In both cases, saline female offspring appear to have the highest expression; this is most evident in *Mecp2*, where saline female expression is significantly higher than both male littermates and poly I:C females. Although MIA-associated decreases in foetal *NeuroD1* and *Mecp2* expression have not yet been investigated in published literature for comparison, both would suggest altered downstream regulation of neurodevelopment. *NeuroD1* has broad roles in regulating neuronal survival and dendritic spine growth, while *Mecp2* influences the expression or silencing of genes in developing neurons as a methylation reader [275, 430]. Both are therefore promising candidates for future research in MIA models. If decreased mRNA expression of *NeuroD1* in MIA-exposed offspring for example is mirrored by an accompanying decrease in protein, investigating whether impaired neuronal survival and/or connectivity in specific brain regions/neuronal subtypes may identify important mechanisms underlying the MIA response.

The sex*treatment interaction in *Mecp2* may also reflect changes in the serotonergic system, given the relationship previously outlined between *Mecp2*, *Bdnf*, and 5-HT (see 1.3.1.2). In saline controls, increased *Mecp2* expression in females compared to males is not unexpected; *Mecp2* is found on the X chromosome, and previous work has identified increased expression of *Mecp2* in female rodents on PND1 [431]. The comparative decrease in poly I:C females compared to controls is an intriguing preliminary finding, however. *Mecp2* hypomethylation in adolescents has been previously observed following MIA, but no exploration of MIA-associated *Mecp2* methylation or mRNA expression during prenatal development has been published [262]. Decreased *Mecp2* expression in females following MIA would reflect the reduced MECP2 observed in human Rett Syndrome, which is closely associated with developmental delays and ASD [296, 431]. Postnatal *Mecp2* expression is also associated with 5-HT signalling and Sert activity, with fluoxetine administration (preventing Sert uptake of 5-HT) leading to increased *Mecp2* across the brain [432]. As poly I:C-exposed females show a relative trend of increased *Sert* expression and

decreased *Mecp2* expression relative to the other groups assessed, *Mecp2* expression may be an additional indicator of a sex-mediated MIA response in the serotonergic system.

5.4.3. 5-HT-related Target Expression at E14.5

Fewer significant effects in target expression were observed at E18.5, although the majority of these interestingly reflect changes also observed at E14.5. In the E18.5 placenta, the only significant effect of note was a decrease in saline female *Bdnf* expression compared to male littermates (Figure 5.9B). Although sample sizes were small, this aligns with the overall sex effect observed in the placenta at E14.5 (Table 5. 4). As previously discussed, saline females demonstrated a significant decrease in placental weight compared to males; this distinct difference in *Bdnf* mRNA expression at E18.5 mirrors the disparity in placental growth across sexes, and aligns with previous research suggesting *Bdnf* promotes placental growth [425, 429]. Interestingly, the same trends across sexes in *Bdnf* expression and placental weight were not observed in poly I:C offspring (see 3.2.3.1). Further exploration of placental *Bdnf* in MIA models is warranted to identify whether these changes are robust and hold functional significance; this may include investigating whether mRNA expression profiles are replicable in larger samples, whether mRNA expression translates to sex differences in *Bdnf* protein, and accompanying changes in components of *Bdnf* signalling (e.g. the tyrosine kinase B/TrkB and p75 neurotrophin receptors) [433].

Two further sex*treatment interactions were identified in the E18.5 foetal brain; in both *MaoA* and *Mecp2* expression, this appears to be driven by a subtle increase in poly I:C-exposed males compared to females (Table 5.5, Figure 5.8A). These preliminary findings complement potential sex differences in the MIA response at E14.5, further highlighting the importance of including sex as a variable in future developmental MIA studies. If replicable, sex-specific changes in *MaoA* and *Mecp2* may be indicative of persistent changes in serotonergic activity at E18.5, with *Mecp2* potentially highlighting differences in serotonin abundance and *MaoA* suggesting an altered capacity for 5-HT metabolism [259, 432]. Similarly to *Sert*, *MaoA* is expressed and/or localised to microglia, mitochondria, and a range of neuronal subtypes; further exploration of any robust *MaoA* expression changes should also take localisation into account as an

indicator of functional relevance [223, 434, 435]. As *MaoA* catalyses the metabolism of multiple other neurotransmitters such as dopamine and norepinephrine, assessing concentrations of 5-HT and 5-HIAA in foetal brain tissue would further support the functional relevance of changes in *MaoA* expression [435].

5.4.4. Overall Conclusion and Future Directions

Overall, this chapter provides evidence of MIA-associated, mRNA expression changes in genes involved in both broader neurodevelopment and in the regulation of foeto-placental 5-HT availability. Consideration of the combined sex, treatment, and sex*treatment interactions demonstrated in this chapter supports the possibility that an MIA-associated increase in 5-HT availability occurs in rats at E14.5. Furthermore, the preliminary sex*treatment interactions uncovered in the foetal brain at E14.5 (*Sert*, *Mecp2*) and E18.5 (*MaoA*, *Mecp2*) suggest that sex may mediate the foetal response to increased 5-HT. These findings overall align with previous MIA models that identified an increase in placental 5-HT production at ~E14, however newly suggest potential sex effects in the response to increased 5-HT [227, 319]. Future research replicating the sex*treatment interactions presented in this chapter is encouraged. Since sex differences in the diagnosis and presentation of NDD/NPDs are commonly observed, sex-dependent changes in the foetal brain in response to altered foetoplacental 5-HT availability warrant further investigation.

Chapter 6:
DNA Methylation of SLC6A4 following Early Life
Stressors: Implications for Rodent MIA Models

6.1. Introduction

As mentioned in the introductory chapter to this thesis, SERT (capitalised in this chapter to refer to protein activity, the human gene, or mixed-species findings; lowercase to refer to the rodent gene) plays a vital role in the transport of 5-HT throughout the lifespan of both rodents and humans [322, 436]. During prenatal development, SERT acts as a major transporter of 5-HT in the placenta and the foetal brain. Placental SERT is likely involved in the removal of 5-HT from foetally-derived placental layers and overall foetal circulation, while brain SERT is associated with 5-HT uptake and subsequent 5-HT signalling in serotonergic neurons, mitochondria, and microglia [302, 322, 405, 437]. Changes in mRNA expression of the SERT-encoding gene *Slc6a4* (again capitalised in humans or mixed-species discussion, lowercase in rodents) may affect the transport and activity of 5-HT across these roles, leading to changes in brain development [405, 436, 438, 439].

Results in Chapter 5 suggested that *Sert*/*Slc6a4* expression in the rat E14.5 brain is altered by MIA in a sex-dependent manner, with mRNA expression demonstrating a significant sex*treatment interaction. MIA had not been previously demonstrated to affect prenatal *Sert* expression at E14.5, only at E19, with a decrease in placental mRNA (no sex effect was explored) [301]. Further postnatal changes in *Sert* expression have been demonstrated following LPS-induced MIA, including increased expression in the adult female rat hippocampus [226]. Interestingly, poly I:C-induced MIA has also been associated with postnatal changes in the epigenetic regulation of *Slc6a4*: female adult rats prenatally exposed to MIA on E12.5 demonstrated altered histone acetylation in the hippocampus, corresponding with increased accessibility to the *Slc6a4* gene and a subsequent increase in SERT protein concentrations [224]. Overall this suggests increased transcriptional activity and *Sert* mRNA expression as observed following LPS, although was not directly measured. It is currently unclear whether epigenetic modifications also influence prenatal *Sert* expression in the rat brain following MIA.

6.1.1. Human SLC6A4 Regulation

When considering how MIA may influence transcriptional regulation of *Sert* in rodents, consideration should also be given to how these regulatory mechanisms may translate to humans. The transcriptional regulation and subsequent mRNA expression of *SLC6A4* has been repeatedly investigated in human populations, particularly in relation to NPDs such as MDD where treatment options include the prescription of SSRIs [68, 146, 440]. *SLC6A4* transcriptional regulation is complex: common genotypic variation in the *SLC6A4* promoter region [441, 442], i-motif formation [443], histone acetylation [444], and DNA methylation [445, 446] have all previously been associated with mRNA expression. These mechanisms may also interact: DNA methylation for example has been shown to stabilise the formation of i-motif structures in DNA, which are four-stranded quadruplex secondary structures that form in cytosine rich DNA sequences and influence transcription factor binding [447].

6.1.1.1. DNA Methylation

DNA methylation is typically considered a repressive mechanism; DNA methyltransferases apply a methyl group to the cytosine of 'CG' DNA motifs, leading to impaired transcription factor binding and/or improved binding of repressive or maintenance machinery [448]. This occurs in regions where CG motifs are highly concentrated, labelled CpG islands. The *SLC6A4* gene (and upstream promoter region) in humans is CG rich, regulated by DNA methylation across a CpG island first published in 2007 by Philibert et al (Figure 6.1) [449]. The 779 nt CpG island in the *SLC6A4* promoter region contains 81 CpG motifs: methylation of four of these motifs (following corrections for multiple comparisons) were significantly associated with reduced transcription, as well as average methylation across all sites when accounting for underlying expression differences due to 5-HT transporter linked polymorphic region (5-HTTLPR) genotype. *SLC6A4* methylation has subsequently been associated with a range of NDD/NPD outcomes, particularly through environmental stressors such as childhood trauma or pregnancy complications such as GDM [450-453].

6.1.1.2. *SLC6A4* Genotypes

The 5-HTTLPR genotype accounted for in Philibert et al. refers to differences in the length of the variable number tandem repeat region (VNTR) found in the promoter region of *SLC6A4*, known as the 5-HTTLPR (Figure 6.1) [442, 449]. The 5-HTTLPR is found approximately 1000bp upstream of the *SLC6A4* transcription start site (TSS), and includes between 11-24 repeats of an approximately 23bp sequence. Numerous allelic variants have been identified across human populations: the first two major variants were identified in 1996, labelled as the 'short' or 'S' (14 repeats) and 'long' or 'L' (16 repeats) alleles [454]. Importantly, these were directly associated with transcriptional activity: the S allele is considered low expressing, with reduced transcriptional activity and subsequent mRNA expression compared to the high-expressing L allele. Subsequently identified L and S allele variants (containing additional repeat changes other than the original two-repeat indel) typically reflect the same positive relationship between 5-HTTLPR length and transcriptional activity [442]. The exception to this rule is the L_G variant, which contains the 16 repeats of the classical L allele as well as an A to G SNP at the rs25531 locus found within the two additional L repeats. L_G is also associated with reduced mRNA expression, and L_G carriers are often grouped with S allele carriers when investigating rs25531 concurrently with 5-HTTLPR genotype [442].

A secondary VNTR, the serotonin transporter intronic enhancer (STin2), has also been identified (Figure 6.1) [455, 456]. STin2 allelic variants include 9, 10, or 12 repeats of an approximately 17 bp sequence [456, 457]. The 12 repeat variant is associated with increased transcriptional activity compared to the shorter variants, similarly to the trend seen with the 5-HTTLPR [458]. Stin2 genotype is significantly less studied compared to 5-HTTLPR/rs25531 genotype, however the low-expressing allele variants in both VNTRs have been inconsistently associated with incidence/symptomology of NDD/NPDs as well as treatment resistance in depressive disorders [439, 441, 452, 459-463].

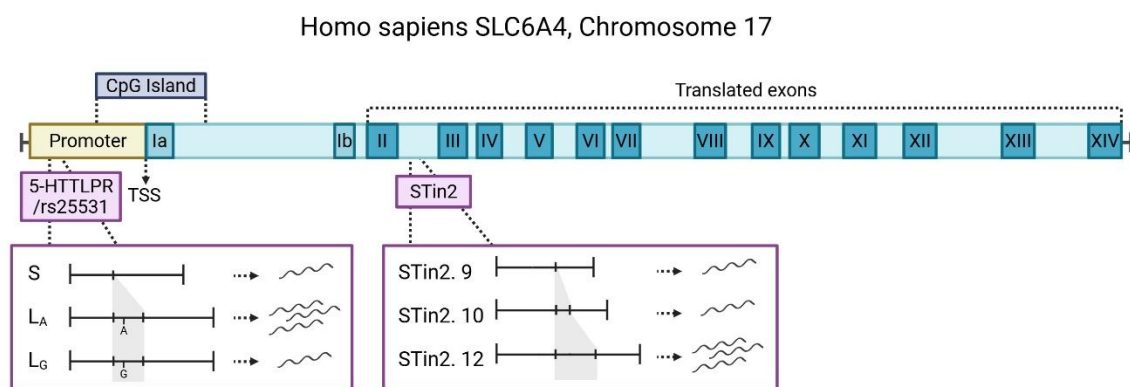


Figure 6.1: Structure of Human SLC6A4. Exons are coloured blue and labelled with roman numerals, including the non-translated segments of exon I. The location of 5-HTTLPR and STin2 VNTRs are highlighted in pink, with major allele variants presented below. The name, relative length, and associated level of mRNA transcription of each allele is presented from left to right. The 5-HTTLPR includes the rs25531 SNP, with an A to G point mutation within the 44bp insertion of the L allele associated with decreased mRNA expression. The SLC6A4 CpG island is annotated in purple, overlaying sections of the promoter, exon Ia, and first intron. TSS = transcription start site. Diagram reflective of sequence lengths but not exact to scale. Key loci using NG_011747.2 sequence alignment: 5-HTTLPR = 3488-3837, CpG island = 4769-5567, TSS = 5240, STin2 = 19347-19552. Created in <https://BioRender.com>.

6.1.2. Chapter Aims

The overall focus of this thesis is investigating the role of MIA on the serotonergic system within the foetoplacental unit, where MIA acts as an external environmental stressor during development. Environmental stressors such as MIA may induce changes in epigenetic regulation of genes, such as histone modifications and DNA methylation [464]. This is exemplified by histone acetylation changes in the adult rat hippocampus following MIA, leading to changes in SERT protein expression [224]. As SERT regulation is a popular topic of human NDD/NPD research, this chapter sought

to expand upon the sex*treatment interaction observed in *Sert* expression in Chapter 5 by investigating whether *Slc6a4* methylation was also altered in the E14.5 brain.

Initial research quickly revealed a lack of available literature on *Slc6a4* methylation in either rat or mice models, preventing quick development of an assay to explore *Slc6a4* methylation in the prenatal rat brain. This research did however identify a significant number of publications based on research in human populations. This raised questions about whether DNA methylation is present as a regulatory mechanism in *Rattus norvegicus Slc6a4* and could be meaningfully explored at all. Furthermore, it was unclear whether exploring *Slc6a4* methylation in a rodent MIA model may complement or improve our current understanding of how MIA affects *SLC6A4* transcriptional regulation in humans. As a result, the final aims of this chapter were to:

1. Outline how *SLC6A4* methylation is explored in recent literature, with a specific focus on understanding methylation changes in humans and rodents following prenatal stressors such as MIA;
2. Compile current evidence of the relationship between *SLC6A4* methylation, genotype, and mRNA expression to assess the functional relevance of *SLC6A4* methylation in models of prenatal stress;
3. Evaluate if and how DNA methylation in the *Slc6a4* promoter could be incorporated in rat MIA models.

These aims were achieved in three distinct stages. Firstly, section 6.2 presents a systematic literature review that was designed to capture all research between 2013-2023 that assessed DNA methylation of *SLC6A4/Slc6a4*. Research from predominantly human research was used to explore associations between DNA methylation and prenatal stress for aim one (section 6.2.3), then associations between DNA methylation, genotype, and mRNA expression were investigated for aim two (section 6.2.4). Section 6.3 and 6.4 address the third aim of this chapter, first by investigating whether *Rattus norvegicus Slc6a4* is likely to be regulated by DNA methylation, and then by an exploratory assay to assess DNA methylation in foetal brain tissue following MIA.

6.2. Literature Review

6.2.1. Methods

PubMed, PsychInfo, and Scopus databases were searched for papers relating to SERT methylation in humans and rodents between 1st January 2013 and 31st December 2023 (Figure 6.2). The search terms 'Slc6a4' AND 'methylation' were used for each database. Papers were then manually assessed to exclude duplicate entries, review articles, results that did not measure methylation of human or rodent *SLC6A4/Slc6a4*, or results that repeated previously published methylation data. All searches and exclusions were carried out by Alex Lister, with consideration of the 2020 PRISMA checklist and guidelines for systematic reviews [465].

Following exclusions, papers were assessed for analyses by the following predetermined information: Species, year, sample type, methylation analysis method, diagnosis/stressor of interest, and inclusion of genotype and/or expression analyses. Articles which fell into multiple subcategories (e.g. two diagnoses of interest in a single publication) were counted only once, categorised by the 'primary' variable discussed predominantly in the conclusions of the article.

Papers subcategorised as the diagnosis/stressor 'MIA', or papers investigating 'methylation/genotype/expression interactions' were predetermined as focus categories for review. As there were minimal findings classified as investigating 'MIA', this category was widened to 'prenatal stressors' during the initial categorisation process. Findings from each article in the 'prenatal stressors' and 'methylation/genotype/expression interactions' were extracted, identifying where possible the number of CpG sites analysed, the location of those sites relative to the reference gene sequence on NCBI (NG_011747.2), the significance and direction of any methylation change observed, and any additional findings associated with methylation changes. All findings discussed were directly sourced from the published articles or supplementary materials.

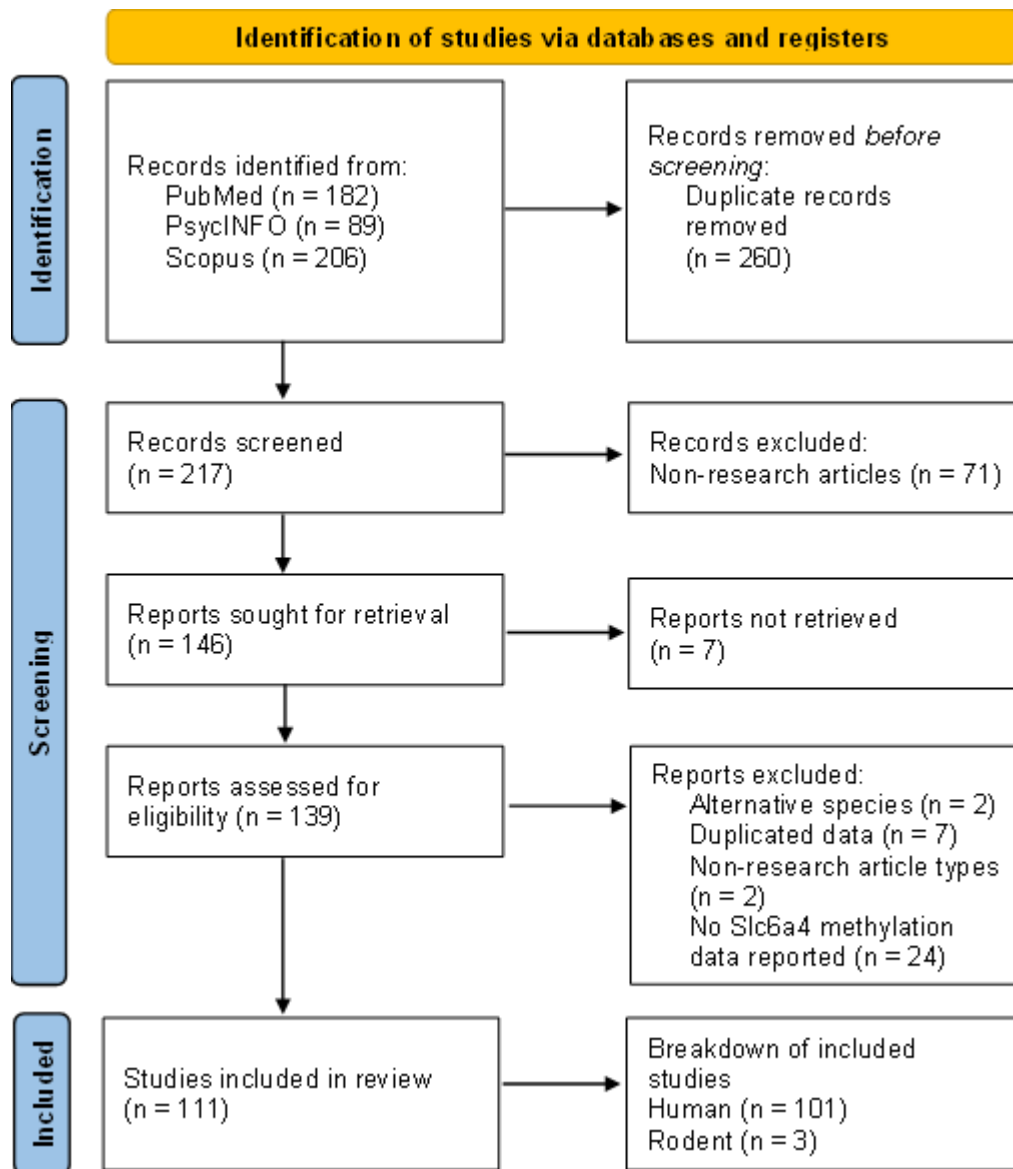


Figure 6.2: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) diagram of the selection and exclusion processes.

6.2.2. Literature Overview

Between 2013 and 2023, 104 available research articles reported findings on methylation of the *SLC6A4* gene or promoter in rodents or humans. Only 3 of these articles were in rodent models [466-468]. Two studies were performed in mice models of irritable bowel syndrome and social isolation, showing increased average methylation over a small number of CpG sites in the *Slc6a4* promoter region [466, 468]. The one additional study published by Ngai et al. used a rat model to investigate

prenatal alcohol exposure [467]. Methylation was assessed over 23 CpG sites, with average methylation increased in the hypothalamus of female ethanol-exposed offspring compared to controls. This was not seen in male ethanol-exposed offspring, however both sexes exhibited increased mRNA expression after ethanol exposure [467]. All three studies indicated that methylation is present in the *S/c6a4* promoter region and may be associated with mRNA expression in rodents, supporting further research in this area. As only one publication used a rat model, and the specific CpG sites assessed in one mouse model were not able to be identified, none of these findings could be directly compared. To expand upon this work in the context of neurodevelopment and MIA exposure in rat models, improved understanding of the *S/c6a4* promoter and its potential methylation sites is required and was explored later in this chapter (see section 6.3).

By contrast, the 101 human studies available assessed *SLC6A4* methylation in a large variety of contexts related to environmental stress, mental health disorders, and biological health and disease (Figure 6.3; full table of identified papers in Appendix H). Approximately half (51) of these studies investigated methylation in relation to a DSM-5-TR diagnosis. Half of these again focused on depression diagnoses, either in isolation or comorbid with a secondary disorder. SCZ and attention deficit hyperactivity disorder (ADHD) accounted for 5 papers overall, with no papers in this review focused on ASD. While these studies did not specifically assess MIA as a variable, MIA is associated overall with a higher risk of NDDs and NPDs in offspring – the focus on understanding *SLC6A4* methylation in these specific disorders highlights the role of environmental stressors such as MIA, and suggests understanding the direct role of MIA in *SLC6A4* methylation would be beneficial to the field.

Direct investigation of environmental stressors (both emotional and physical) accounted for approximately 33% of research available. Fifteen of these studies involved prenatal or neonatal stressors, however none directly assessed MIA. In the absence of specific MIA-based research, understanding how prenatal stress in general affects *SLC6A4* methylation may provide some useful insights for future work. As critical brain development continues shortly after birth within the neonatal period, and many stressors overlap the prenatal and neonatal periods (e.g. pre-term birth, pregnancy/birth complications and subsequent procedures), research in this area was grouped for further investigation.

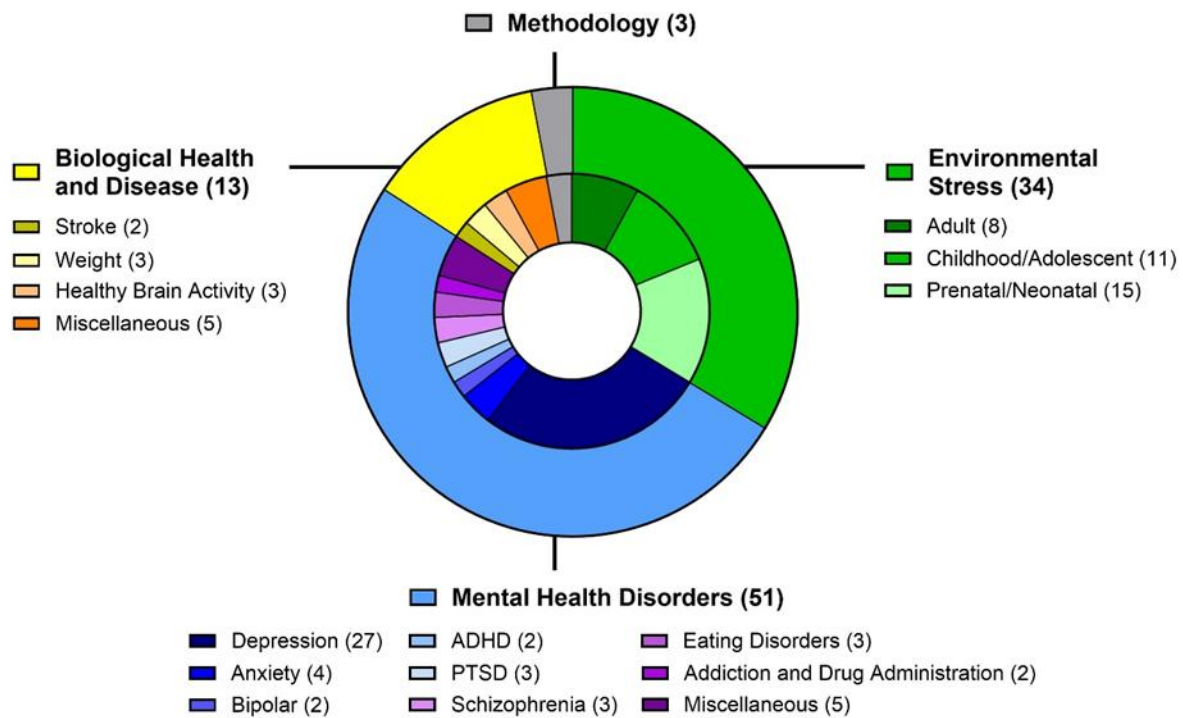


Figure 6.3: Focus of human *SLC6A4* methylation research from 2013-2023. Outer ring sections research into broad categories (bold), with inner ring displaying specific categories containing a minimum of 2 publications. Full list of publications can be found in Appendix H.

6.2.3. Prenatal and Neonatal Stress

Across the CpG sites explored, increased maternal stress (with or without accompanying air pollution exposure), pre-term birth (PTB), and neonatal intensive care unit (NICU) -related infant stress were all associated with increased *SLC6A4* methylation in offspring (Table 6.1) [238, 453, 469-475]. However, this is not consistently observed across all studies; two studies found that PTB-associated NICU stress was not significantly associated with methylation [476, 477]. Variance in the specific stressor (pain-related procedures or total chronic stress) and sampling (buccal swab, cord blood, and faecal sample) may contribute to these inconsistencies. Only maternal alcohol consumption was solely, negatively associated with average *SLC6A4* methylation, however no other study assessed methylation at the implicated sites or with parental alcohol consumption as a primary variable of interest [469]. The CpG sites of two studies were unable to be fully identified – both investigated

populations using SSRIs during pregnancy, however neither reported a specific association with *SLC6A4* methylation [478, 479].

Interestingly, the two studies investigating *SLC6A4* methylation in gestational diabetes mellitus (GDM) covered a similar selection of CpG sites with opposing findings [238, 453]. While both studies highlight a small range of sites between 4701-4865 as differentially methylated in GDM placental biopsies, as well as increased *Slc6a4* mRNA expression in GDM, opposing patterns were observed in both site-specific and average methylation (Table 6.1). Most of the sample characteristics and techniques utilised between studies appear similar, however unexplored variables alongside GDM may be contributing to this opposing effect and should be explored further. For example, 5-HTTLPR and STin2 genotype were assessed in Blazevic et al. (no association with GDM diagnosis, methylation or mRNA expression) but unexplored in Song et al [238, 453]. Maternal smoking was also reported in 33% of GDM and 22% of control pregnancies of the Blazevic cohort, and not reported in Song participants. While smoking status in isolation was not associated with *SLC6A4* methylation in Blazevic's study, nicotine is hypothesised to contribute to a reduction in SERT availability in smokers with a high-expression 5-HTTLPR genotype and has been found to be hypermethylated in subjects with alcohol and nicotine co-dependency [480, 481]. Potential interactions with GDM, genotypic variation, and/or differences in parental smoking prevalence between populations could potentially contribute to the different methylation patterns observed in offspring and warrants further exploration.

6.2.3.1 Insights for MIA and *SLC6A4* regulation

Interestingly, the CpG sites investigated in any given study of prenatal/neonatal stress tend to cluster at either the beginning (4701-4865) or in the middle (5027-5171) of the original CpG island (4769-5567) rather than covering the whole region or overlapping sections (Table 6.1). This is particularly relevant for comparisons between studies assessing a single stress such as PTB; Chau et al. finds significantly increased *SLC6A4* methylation following PTB whereas Mascheroni et al. does not find methylation changes, however there is no overlap between the CpG sites assessed in these studies [470, 476]. Multiple CpG sites in Chau et al. were found to have no association with PTB, as well as a single site with reduced methylation, which suggests site-specific changes may be associated with PTB rather than broad

hypermethylation across the *SLC6A4* promoter [470]. Further depth of research across the CpG island would strengthen the associations between *SLC6A4* methylation and PTB, as well as prenatal/neonatal stress overall.

Across the CpG sites investigated, associations identified between methylation and prenatal stress were typically positive. Implicated CpG sites were also often implicated across multiple stressors, suggesting that an understanding of sites involved in prenatal stress overall may be a useful starting point for investigating additional stressors such as MIA. A total of 12 sites (4811, 4835, 4846, 4848, 4853, 4865, 5091, 5093, 5099, 5105, 5107, 5171) were differentially methylated in at least 3 studies over 2-3 specific stressors (Table 6.1). No studies assessing CpG sites 4846, 4848, and 4853 found no association between these sites and prenatal stress, while the other sites were found to be non-significant in at least one study. Site 4853 is particularly unique, in that 2/3 studies identified negative associations between methylation and prenatal stress [453, 470]. Overall, these 12 sites provide an initial starting point for investigating *SLC6A4* methylation in MIA.

CpG Site	GDM			Maternal Alcohol Consumption			Maternal Stress ^a			PTB			PTB-associated NICU stress		
	P	N	NA	P	N	NA	P	N	NA	P	N	NA	P	N	NA
3844					[469]										
3988					[469]										
4701	[238]		[453]b												
4717	[238]		[453]b												
4728	[238]	[453]													
4769		[453]										[470]			
4780		[453]					[471]					[470]			
4811	[238]	[453]					[471]					[470]			
4816	[238]		[453]b									[470]			
4835	[238]		[453]b				[471]			[470]					
4846	[238]	[453]								[470]					
4848	[238]	[453]					[471]			[470]					
4853	[238]	[453]									[470]				
4865	[238]		[453]b				[471]			[470]					
5027												[474]	[472]		[474]
5031												[474]	[472]		[474]
5033												[474]	[472]		[474]
5040												[474]	[472]		[474]
5045												[474]	[472]		[474]

5050					[474]	[472]		[474]
5052					[474]	[472]		[474]
5066				[473, 475]		[474, 476]	[472]	[474, 477]
5070			[475]	[473]		[474, 476]	[472]	[474, 477]
5085			[473]	[475]		[474, 476]	[472]	[474, 477]
5091			[473, 475]			[474, 476]	[472]	[474, 477]
5093			[473, 475]			[474, 476]	[472]	[474, 477]
5099			[473, 475]			[474, 476]	[472]	[474, 477]
5101				[473, 475]		[474, 476]	[472]	[474, 477]
5105			[475]	[473]		[474, 476]	[472, 474]	[477]
5107			[473]	[475]		[474, 476]	[472, 474]	[477]
5128				[473, 475]		[474, 476]	[472]	[474]
5141			[473]	[475]		[474, 476]	[472]	[474]
5168			[475]	[473]		[474, 476]	[472]	[474]
5171			[473, 475]			[474, 476]	[472]	[474]

Table 6.1: CpG sites potentially associated with prenatal stress. CpG sites listed by locus as identified in NCBI SLC6A4 gene sequence (NG_011747.2). Sub-columns for each stressor represent a positive (P), negative (N), or no association (NA) with stressor and site methylation. Green, orange, and grey highlighting represents these associations respectively, with dark green/orange indicating at least one significant site-specific association and light green/orange indicating significance only when averaged across a range of sites. Only sites with a minimum of 1 significant or suggested association are reported. Highlighted sites are annotated by reference number, with blank cells indicating no data was available for this site/stress combination. a = maternal stress or maternal stress with concurrent air pollution exposure; b = sites identified as unmethylated, association not directly assessed.

6.2.4. Interactions between DNA Methylation, Genotype, and Transcriptional Regulation

Understanding the regulatory role of *SLC6A4* methylation in conjunction with genotypic variation is essential for interpreting reported changes in methylation status in humans [449]. Of the 101 human studies investigating *SLC6A4* methylation, only 17 studies investigated methylation alongside a downstream indicator of *SLC6A4* regulation. Of these, 15 measured mRNA expression [238, 439, 445, 446, 450-453, 482-486], and two selected alternative regulatory outcomes (SERT availability by PET scan, and gene promoter activity using a luciferase-based reporter gene assay) [487, 488]. Each of the 17 publications were directly compared by aligning the CpG sites studied, however two studies were unable to be included in this analysis; the first did not directly report a comparison between DNA methylation and mRNA expression [489], and the second did not report CpG locations in sufficient detail for alignment [490]. Across the 15 studies remaining, 84 sites across the promoter region were assessed. This includes all 81 sites within the original human *SLC6A4* CpG island reported in Philibert et al. (spanning 4769 – 5567, NG_011747.2) [449]. Two studies additionally reported methylation in alternative downstream sites within intron 1 and exon 9 [483, 486].

Methylation of 31 CpG sites have been associated with *SLC6A4* regulation since 2013 (Table 6.2). Eight sites exhibited significant site-specific associations between methylation and *SLC6A4* regulation, with an additional 23 additional sites implicated with weaker evidence: one study identified multiple site-specific associations that failed to meet Bonferroni post-hoc corrections, while the remaining sites were included in averaged methylation scores (not assessed individually) that significantly associated with regulation [451]. The majority of studies finding an association between CpG methylation and *SLC6A4* regulation reported a negative association (Table 6.2), supporting the common assumption that increased methylation represses transcriptional activity [448]. Seven out of eight of the significant, site-specific associations presented were negative (4811, 4835, 4846, 4848, 4853, 5052, 5107), with evidence for the majority of these sites strengthened by being implicated in more than one publication.

CpG Site	Positive Association	Negative Association	No Association	Genotype Associations
4701	[238]		[485]	
4717	[238]	[451]	[485]	
4728	[238]	[451, 453]	[485]	
4769		[451, 453]	[450, 485]	
4780		[453, 488]	[446, 450, 451, 485]	
4795		[488]	[439, 446, 450, 451, 485]	
4811	[238]	[453] , [488]	[439, 446, 450, 451, 485]	
4816	[238]	[488]	[439, 446, 450, 451, 485]	
4835	[238]	[482]a , [488]	[439, 446, 450, 451, 484, 485, 487]	
4846	[238]	[451, 453] , [488]	[439, 446, 450, 451, 484, 485, 487]	Increased methylation and L allele: [482]a
4848	[238]	[453] , [482]a , [488]	[439, 446, 450, 451, 484, 485, 487]	
4853	[238]	[453] , [488]	[439, 446, 450, 451, 485, 487]	Increased methylation and LL genotype: [484]
4865	[238]	[488]	[439, 446, 450, 451, 485, 487]	
5033		[451]	[445, 485]	
5040		[451]	[445, 485]	
5050		[451]	[445, 485]	
5052		[445] , [451]	[485]	
5093	[485]		[445, 451, 487]	
5107		[487]	[445, 451, 485]	
5229		[486]	[451, 485]	
5237		[486]	[451, 485]	

5248		[486]	[451, 485]	
5251		[486]	[451, 485]	
5254		[486]	[451, 485]	
5433		[451]	[452, 485]	
5566		[451]	[485]	
6487		[483]		
28088	[486]			
28105	[486]			
28112	[486]			
28117	[486]			

Table 6.2: CpG sites potentially associated with Slc6a4 regulation. SLC6A4 regulation includes mRNA expression, gene promoter activity, or SERT availability. CpG sites are listed by locus as identified in NCBI SLC6A4 representative gene sequence (NG_011747.2). Green, orange, or grey shading respectively indicates positive, negative, or no association, with empty cells indicating no information available. Dark green or orange shading indicates at least one site specific association was identified (reference indicated in bold), with light shading indicating a significant association over a range of averaged sites or a near-significant site-specific association. Only sites with a minimum of 1 significant or suggested association are reported. a = observed in depressed participants only.

It should also be noted that three studies found positive associations between methylation and *SLC6A4* regulation. Muller et al. identified methylation of CpG 5093 was significantly positively associated with mRNA expression, however the authors suggested interpreting this finding with caution: CpG 5093 presented relatively low methylation overall (2.93%), was relatively stable across samples (SD=0.67%), and was the only site (out of 83 investigated sites) to show any relationship with mRNA expression in their study [485]. Song et al. by contrast assessed a region of 10 sites in the promoter region, finding average methylation in foetally-derived placental tissue was positively associated with mRNA expression [238]. This contrasts with multiple studies finding a negative association with mRNA expression across these sites, including Blazevic et al. whom also investigated foetally-derived placental tissue with similar sampling parameters [453]. As discussed previously, other unexplored regulatory mechanisms may contribute to these opposed findings such as 5-HTTLPR or Stin2 genotype. Finally, Tahara et al. found a positive association between methylation and expression, where average methylation across 4 CpG sites within Exon 9 of *SLC6A4* in gastric biopsy tissue positively and significantly correlated with mRNA expression. The role of gene body DNA methylation is not well understood, however has been suggested to influence transcription through silencing of internal promoters or promoting alternative splicing [491]. Three splice variants of *SLC6A4* have been previously identified (varying in the inclusion of the first three exons), with tissue-specific expression patterns [492]. Increased methylation of Exon 9 may be involved in gut-specific expression profiles, however no other research investigating methylation at these sites was available for comparison [486].

Interestingly, only one site (4795) from the original Philibert et al. study was reinforced as functionally relevant; average or site-specific methylation of the other three sites (4938, 4952, and 5462) was not significantly associated with *SLC6A4* regulation across any of the included studies (Table 6.2) [449]. This, in combination with conflicting associations and a high number of studies finding no significant association between methylation and *SLC6A4* regulation at all, supports the idea that DNA methylation is one of many variables that contribute to complex regulation of *SLC6A4* transcription. Only 13 of the 17 studies included in Table 6.2 also reported investigating genotype as a covariate or independent variable, typically referring to the common L or S 5-HTTLPR polymorphisms. Most studies reported no significant

genotype effects, including the expected association between 5-HTTLPR genotype and mRNA expression: only Wankerl et al. reported a significant association between S allele carriers and reduced mRNA expression [451]. Three studies found a relationship between methylation and genotype: two studies found L allele carriers had significantly increased site-specific methylation (Table 6.2), while Drabe et al. identified a third site with the same trend (5128) that failed to meet significance following corrections for multiple comparisons [482, 484, 487]. As the 5-HTTLPR is located approximately 500bp upstream of the CpG sites identified in Table 6.2, increased methylation in the L allele is not a direct result of increased CpG sites being present (Figure 6.1). Both significant associations between methylation and L allele carriers were found in cohorts of depressed participants; as the low expression S allele has been inconsistently associated with depression risk, repressive methylation of the high-expression L allele may reflect a similar low-expression phenotype [482, 484].

6.2.5. Overall Findings and Implications for Future Research

Overall, while promoter DNA methylation is generally considered a repressive mechanism, recent findings are highly inconsistent. A number of methylated CpG sites have been demonstrated to be negatively associated with *SLC6A4* transcriptional regulation, notably in two clusters within the promoter region (4811-5107; Figure 6.4). These associations are confounded by further studies that report no (or occasionally a positive) association between methylation and regulatory outcomes, as well as factors such as *SLC6A4* genotype which are intermittently assessed in methylation studies and are sometimes associated with both site-specific methylation and *SLC6A4* regulation [451, 482, 484]. Understanding which CpG sites are functionally relevant, either alone or in combination, is essential for the interpretation of any methylation changes observed within NDD or NPD research. For example, methylation changes at the CpG sites 4811, 4835, 4846, 4848, 4853, 4865, 5091, 5093, 5099, 5105, 5107, and 5171 are the most commonly reported following prenatal stress: seven of these are also specifically implicated in *SLC6A4* regulation (4811, 4835, 4846, 4848, 4853, 5052, 5093, and 5107). Other sites implicated in prenatal stress models have not been demonstrated to associate with transcriptional regulation however (e.g. 3844, 3988, 5027, 5031), making it unclear what functional effects these changes might hold.

Model organisms such as *Rattus norvegicus* may be useful in further investigating mechanisms surrounding *SLC6A4/Slc6a4* methylation. Rodents do not show the same allelic variation in the 5-HTTLPR and *Stin2* regions, which have to be directly modelled using knockout rodent strains [323, 493]. This could allow for direct exploration of how methylation (either site-specific or averaged over a wider region) influences binding of transcription factors or repressive machinery, as well as the direct downstream effects on transcription, without genotype as a confounding mechanism. Methylation changes following environmental stressors such as MIA could be similarly explored, shedding further light onto the role of DNA methylation in response to prenatal stress. Current understanding of *Slc6a4* methylation in rats is poor, however: while broad assessment of the CpG island in the *SLC6A4* promoter (approximately 83 CpG sites) has been explored in human research, only one study (of 23 CpG sites) has explored *Slc6a4* methylation at all in rodents. To usefully evaluate *Slc6a4* methylation in rodents, a more detailed assessment of *Rattus norvegicus Slc6a4* is required.

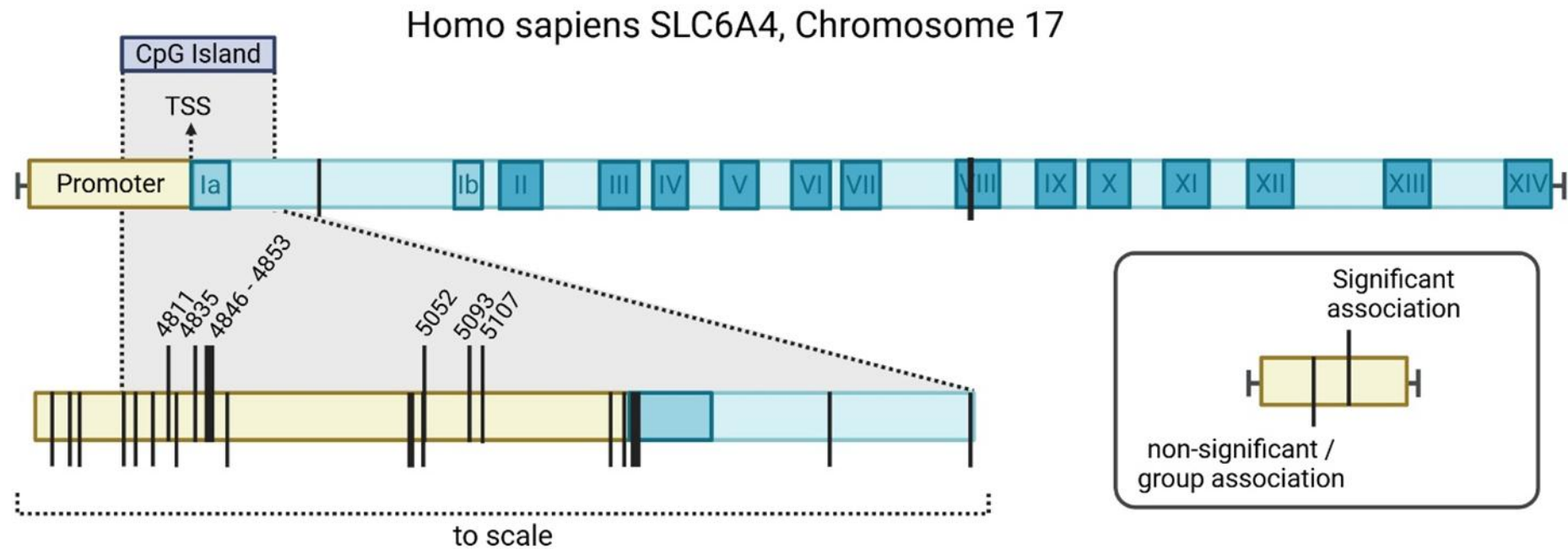


Figure 6.4: CpG sites potentially associated with *Slc6a4* regulation. *Slc6a4* regulation includes mRNA expression, gene promoter activity, or SERT availability. CpG sites identified in Table 6.2 are indicated with a black bar, with distances between CpG motifs in the promoter/CpG island presented to scale. Individual CpG sites specifically associated with mRNA expression are annotated with their locus corresponding to NG_011747.2, with bars extending upwards. All additional sites of interest are indicated with a bar extending downwards, including the 5 sites located outside of the promoter/CpG island cluster (6487, 28088, 28105, 28112, 28117). TSS = transcription start site.

6.3. Methylation of *Rattus norvegicus* *Slc6a4*

As there is minimal published literature assessing DNA methylation within *Rattus norvegicus* *Slc6a4*, the gene and promoter region were manually assessed to identify any potential CpG islands of interest. There is currently no ‘gold standard’ method to screen for CpG islands: as a result, three approaches were used sequentially to identify regions of interest [494]. First, the NCBI gene database was used to align the human *SLC6A4* CpG island to *Rattus norvegicus* *Slc6a4* and identify any homology with the regulatory region in humans (Figure 6.5). Note was also taken of any automatically annotated CpG islands. Next, *Rattus norvegicus* *Slc6a4* and the promoter region were assessed for CpG islands using two published and commonly-used identification methods: the Gardiner-Garden and Frommer criteria (Figure 6.6) and the Takai and Jones criteria (Figure 6.7) [495, 496]. The potential CpG islands identified from these approaches were then combined to a final consensus region for further investigation.

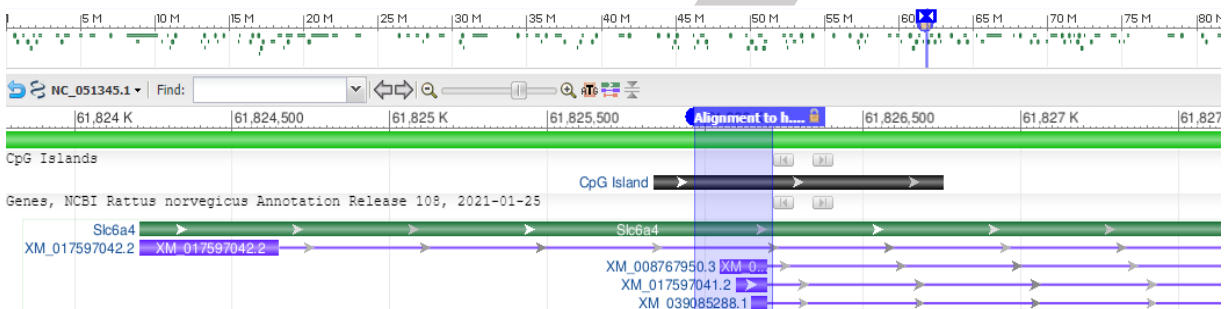
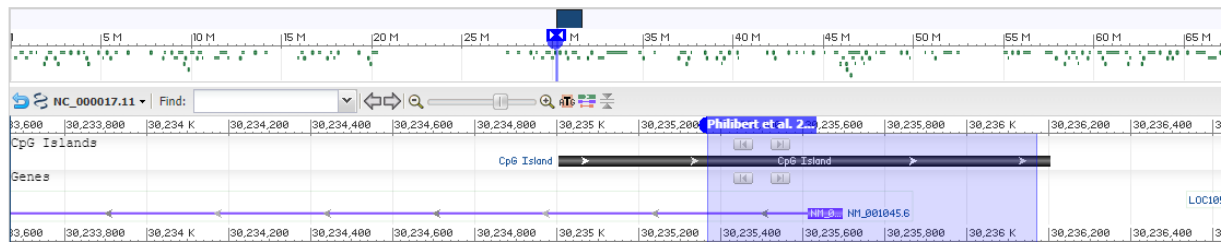
6.3.1. *Homo sapiens* to *Rattus norvegicus* alignment

Firstly, the human *SLC6A4* CpG island was identified from the NCBI annotation as a 1199nt region spanning 30235004-30236202 (*Homo sapiens* chromosome 17, GRCh38.p14 Primary Assembly NC_000017.11). This was further confirmed using the original CpG island published by Philibert et al, which identified a 779nt sequence at 30235370-30236168, falling completely within the 1199nt NCBI annotated region [449]. Nucleotide BLAST was then used to find potential homology between the 1199nt human CpG island and rat *Slc6a4* (Figure 6.5). A 256nt sequence within the 1199nt CpG island (also falling within the 779nt Philibert region) was successfully aligned to rat *Slc6a4*, with 82% identity due to the presence of 19 deletions and 29 mismatches between sequences. The homologous region in rat *Slc6a4* spans a 241nt sequence at 61825969-61826209 (*Rattus norvegicus* strain BN/NHsdMcwi chromosome 10, mRatBN7.2, whole genome shotgun sequence NC_051345.1). This overlays exon 2 of rat *Slc6a4*, as opposed to exon 1 and the promoter region of human *SLC6A4*. The NCBI annotated CpG island for rat *Slc6a4* also overlays this region, which spans a total 920nt from 61825835-61826754.

Homo sapiens chromosome 17, GRCh38.p14 Primary Assembly

NCBI Reference Sequence: NC_000017.11

[GenBank](#) [FASTA](#)



Rattus norvegicus strain BN/NHsdMcwi chromosome 10, mRatBN7.2, whole genome shotgun sequence

NCBI Reference Sequence: NC_051345.1

[GenBank](#) [FASTA](#)

Figure 6.5: Aligning the Homo sapiens CpG Island to Rattus norvegicus Slc6a4.

A 256nt sequence of the input human 1199nt CpG island, found within the highlighted originally published CpG island, aligns with a 241nt sequence in Rattus norvegicus Slc6a4. The aligned region is indicated by grey shading between human (top) and rat (bottom) sequences. The aligned sequence is additionally highlighted in the rat, shown to overlap the predicted mRNA exon two region and a small portion of the NCBI annotated CpG region. Graphic generated using NCBI Genome Data Viewer, 11th May 2022.

6.3.2. *Slc6a4* Screening Approaches

Following comparison with human *SLC6A4*, rat *Slc6a4* was screened for potential CpG islands using the original Gardiner-Garden and Frommer criteria. This approach identifies putative CpG islands of a minimum 200nt length, with a CpG (observed/expected) ratio of >0.6, and a GC content of >50% [495]. The expected number of CpG motifs is defined as the observed number of cytosines multiplied by the observed number of guanines, divided by the length of the query sequence. The entire *Rattus norvegicus Slc6a4* gene, with an additional 2000nt upstream, was screened using the Sequence Manipulation Suite (SMS) bioinformatics tool [497]. The input sequence spans a 36717nt length at 61822208 – 61858924 (*Rattus norvegicus* strain BN/NHsdMcwi chromosome 10, mRatBN7.2, whole genome shotgun sequence NC_051345.1).

Using each nucleotide sequentially as a starting point, 200nt segments that met the selection requirements were reported as predicted CpG islands alongside their CpG ratio and GC content. Overlapping/adjacent results (with start loci equal to or less than 200nt apart) were then classified into regions of interest (Table 6.3). Islands were also mapped directly to the *Slc6a4* gene to visualise regions compared to gene structure (Figure 6.6). Seven potential regions of interest were identified using this approach, the largest of these being region 1 with a length of 1,428nt. This region is located across *Slc6a4* exon 2, similarly to the human alignment shown previously (Figure 6.5).

Region	Length (nt)	Location (query sequence)	Location (NC_051345.1)	Associated Exons
1	1428	3120 - 4548	61825328 - 61826756	Exon 2
2	201	5700 - 5901	61827908 - 61828109	n/a (Intron 2)
3	692	18103 - 18795	61840311 - 61841003	Exons 6, 7
4	243	21202 - 21445	61843410 - 61843653	Exon 10
5	267	23604 - 23871	61845812 - 61846079	n/a (Intron 11)
6	255	25994 - 26249	61848202 - 61848457	Exon 14
7	218	32905 - 33123	61855113 - 61855331	n/a (Intron 16)

Table 6.3: Potential CpG islands identified using SMS identification tool.

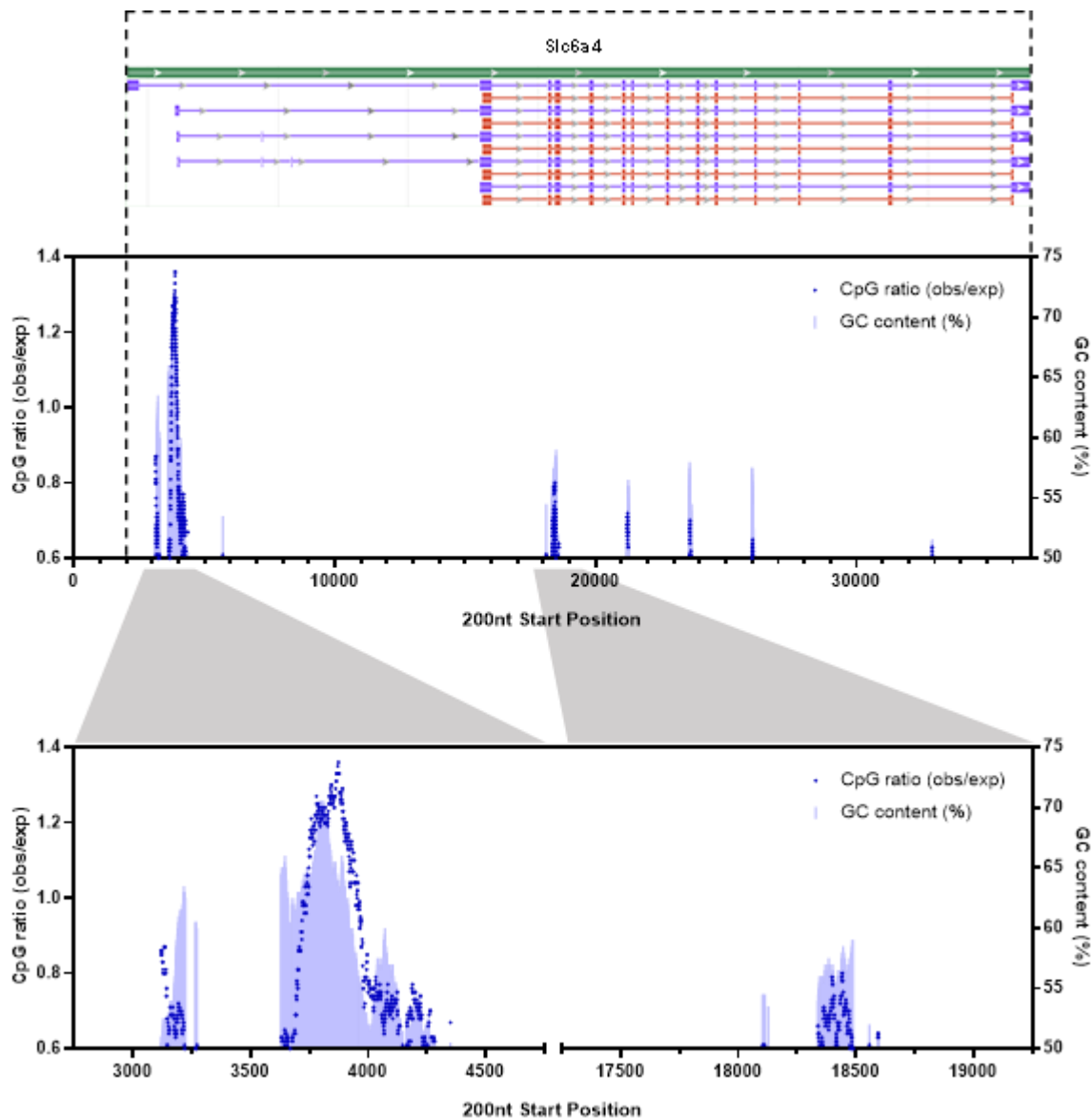


Figure 6.6: Gardiner-Garden and Frommer predicted CpG island loci. Positively identified CpG islands across the query region from the SMS CpG island bioinformatics tool are aligned by start position to the Slc6a4 gene (green bar). mRNA transcript variants and aligned protein isoforms interleaved below Slc6a4 in purple and red respectively, with the first 4 mRNA sequences representing predicted variants, and the final representing the main transcript. CpG ratio for each 200nt sequence (above 0.6) is represented by dark blue dots, overlaying GC content of the region (above 50%) in pale blue. The two regions most enriched in CpG motifs are shown in higher resolution below.

While being the classic criteria for CpG island identification and still commonly used, a critique of the Gardiner-Garden and Frommer method is its vulnerability to false positives (i.e. identification of regions which contain CpG motifs, but are not regulatory regions subject to methylation). The Takai and Jones criteria provides a more stringent version of the identification criteria, designed to avoid weaker candidate islands and focus on islands with high regulatory potential [494, 496]. Takai and Jones present an increased threshold for all criteria: a minimum length of 500nt, CpG ratio >0.65, and a GC content of >55% [496].

The seven regions identified using the Gardiner-Garden and Frommer criteria were subsequently manually screened using the Takai and Jones criteria to identify the strongest candidate CpG island(s). Six of the regions previously identified failed to meet the stringent criteria, with only a 529nt portion of Region 1 withstanding all three increased requirements (Figure 6.6). This section is found at nt 61825853 – 61826381, again aligned within the *Homo sapiens* homologous sequence. This supports the likelihood of Region 1 being a functionally relevant CpG island, with the smaller regions being less likely to hold functional significance.

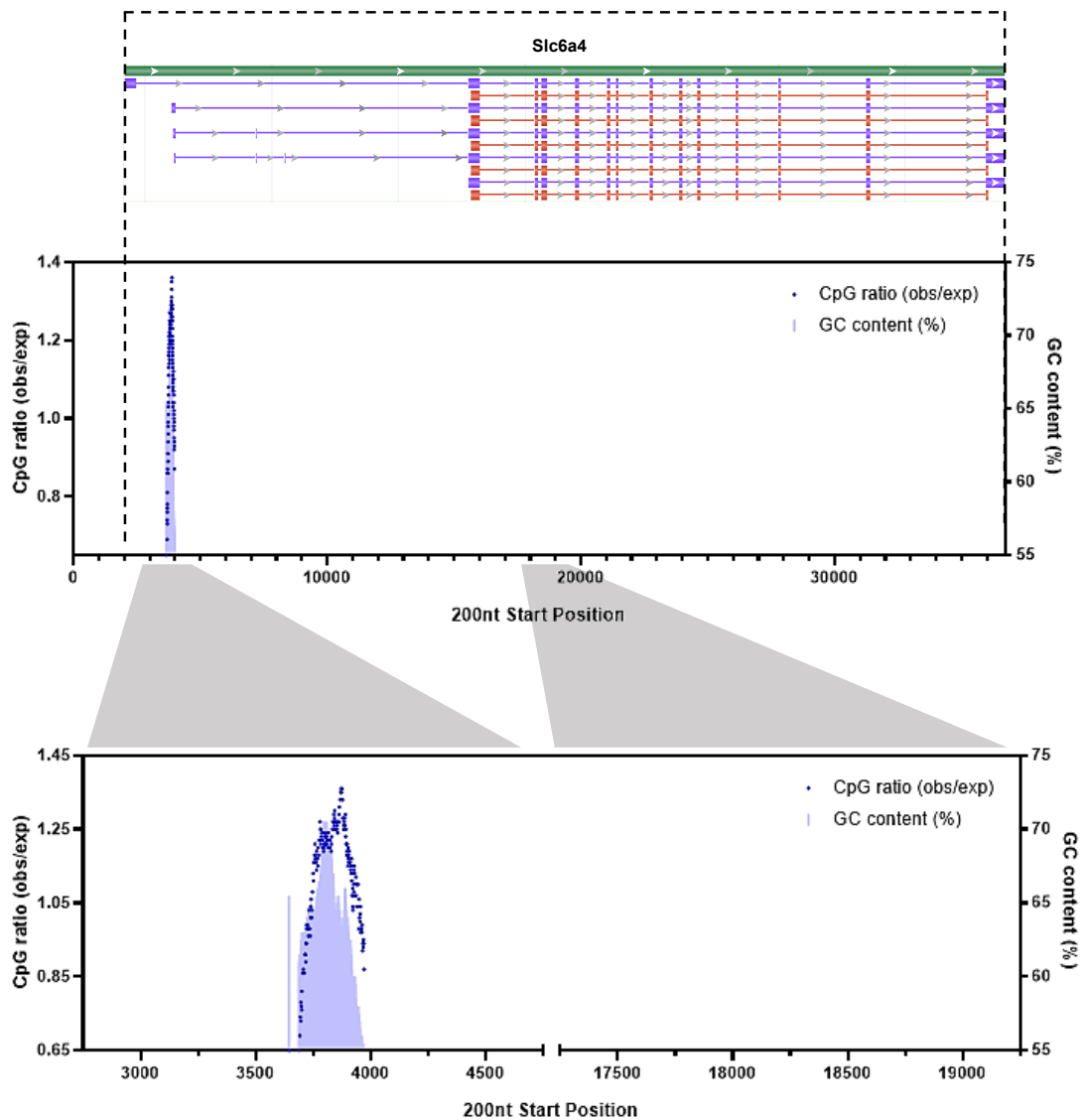


Figure 6.7: Takai and Jones predicted CpG island loci. Positively identified CpG islands (presented in Table 6.3 and Figure 6.5) following application of the stringent Takai and Jones criteria. CpG ratio for each 200nt sequence (above 0.65) is represented by dark blue dots, overlaying GC content of the region (above 55%) in pale blue. Only regions that extended a minimum of 500nt are presented. The only region to withstand the higher criteria is shown in higher resolution below the overall sequence of *Slc6a4* above.

6.3.3. Consensus CpG island

Using the aforementioned approaches, multiple putative CpG islands were identified across *Rattus norvegicus Slc6a4*. While many of the regions identified by the Gardiner-Garden and Frommer criteria only narrowly met the necessary requirements, the first CpG island listed well-exceeded these requirements (Figure 6.5). Part of this region also survived the stringent Takai and Jones criteria and shows homology with the known regulatory region of *Homo sapiens SLC6A4*, both of which suggest this CpG island may hold relevance as a regulatory region in *Rattus norvegicus*. A final consensus region was identified by selecting the overlapping sequence across all methods used (Table 6.4). This CpG island spans 241nt at 61825969 – 61826209 (*Rattus norvegicus* strain BN/NHsdMcwi chromosome 10, mRatBN7.2, whole genome shotgun sequence NC_051345.1), and contains 32 CpG sites (Figure 6.8). Although multiple approaches were used to confirm which region(s) of *Slc6a4* was the most likely to act as a functionally-relevant CpG island, this consensus region is the same region homologous with *Homo sapiens SLC6A4*.

Identification method	CpG Island loci (NC_051345.1)
NCBI annotation	61825835 - 61826754
Manual alignment with Homo sapiens SLC6A4	61825969 - 61826209
Gardiner-Garden and Frommer criteria (Region 1)	61825328 - 61826756
Takai and Jones criteria	61825853 - 61826381
Consensus region	61825969 – 61826209

Table 6.4: Summary of potential CpG islands identified in sections 6.3.1 - 6.3.2.

Following from the systematic literature review, this CpG island was compared to the single study that reported investigating *Rattus norvegicus Slc6a4* methylation. Ngai et al. investigated amplification of a 334nt region of the *Rattus norvegicus Slc6a4* promoter, which was aligned to 61825988 – 61826317 using the published primer sequences and found to partially overlap the consensus CpG island [467]. Further detail reported in Ngai et al. stated that 23 CpG sites were assessed using bisulfite pyrosequencing of the 30nt sequence following the sequencing primer. These appear

to be errors within the manuscript, with it instead being likely that an 130nt sequence was assessed including 24 potentially methylated CpG sites (Figure 6.8). Sites 15 and 23 were reported to be significantly differentially methylated, however no sequence information was available to confidently align the 23 sites from Ngai et al. with the 24 sites identified after the pyrosequencing primer site in this CpG island.

```
NC_051345.1          241nt          61825969 - 61826209

AGAGCTGAGCTGCGGCCACCGGAGAAACAAAGTTCTTTGGTCTCAAAGTGA
CGCAAAAATTCTTGCGGAGCTCTTTGGCGGCGGCTATCTAGAGATCAGAC
CATGTGAGGGGCGCGAGTACAAATAGCGCGGCGGGCGGCGCCACTCCAG
CGGCGCGCGCGGTGCCTCGAGGGCGCGAGGGTCGCGCGCCTCGCAGCC
CGGGACCGCGCGCGCGCGCGCGCGCAGGTGGGTCGTTTTCC
```

Figure 6.8: Sequence of consensus CpG island. The 32 CpG sites within this sequence are presented both bolded and underlined. Grey highlighting indicates the first 114nt of the 130nt pyrosequenced region investigated by Ngai et al, including all 24 potentially assessed CpG sites. Red text was used for the pyrosequencing primer sequence.

6.3.4. Methylation in MIA offspring

To investigate methylation of the 241nt CpG island in MIA-exposed embryonic brain tissue, three primer sets were designed to target this region using a low-cost, methylation-sensitive restriction enzyme assay (Figure 6.9A). While the Full Amplicon and Amplicon 1 failed to amplify under varied ddPCR conditions, Amplicon 2 was successful (See section 2.6.3 for detailed assay methods and primer sequences). Briefly, DNA was isolated from male E18.5 pBR tissue and split into multiple aliquots: these were treated with the methylation-sensitive restriction enzymes (msREs) HpaII and BstUI prior to ddPCR amplification alongside an undigested control at the same

volume and concentration. These msREs are unable to cleave DNA at methylated restriction sites, thus only DNA methylated at every restriction site in the amplicon will subsequently be amplified. Six HpaII restriction sites (CGCG) and two BstUI restriction sites (CCGG) are present within Amplicon 2, allowing for 8 total CpG sites to be assessed (Figure 6.9B). While each HpaII restriction site contains two CpG motifs, only methylation of the internal cytosine inhibits cleavage.

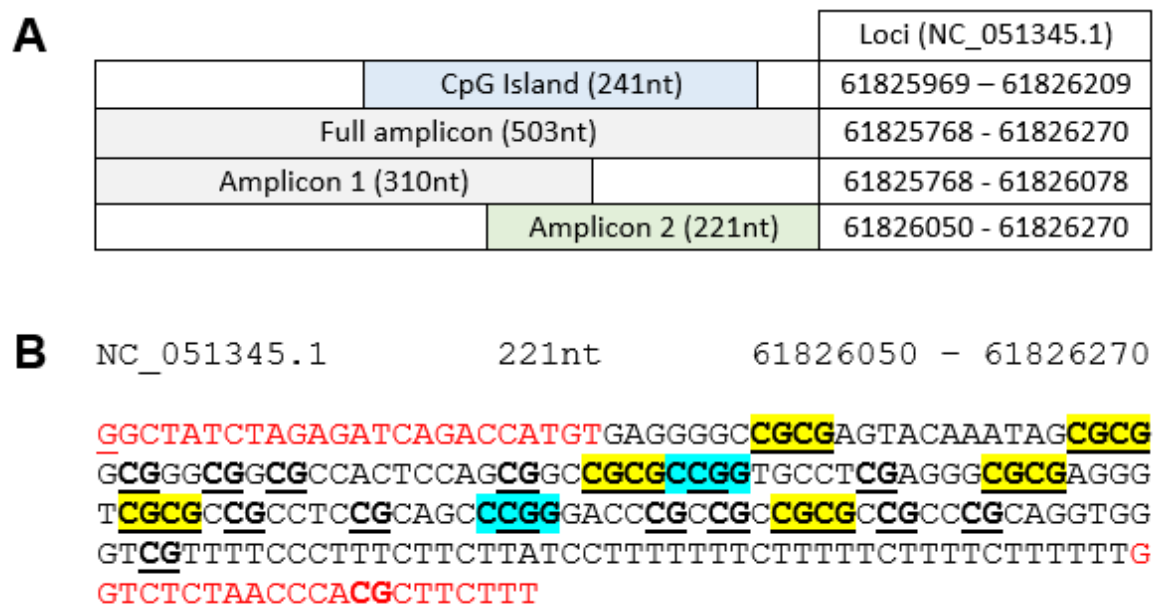


Figure 6.9: Targeting CpG island using methylation-sensitive restriction enzymes (msREs). A) Three amplicons were designed to overlay the 241nt CpG island (blue): shading differentiates successful (green) or unsuccessful (grey) amplification. B) The nt sequence of amplicon 2, with primer sequences highlighted in red and CpG sites both bolded and underlined. 26 out of 32 target CpG sites are included in this amplicon, with an additional CpG site located in the reverse primer site. HpaII restriction sites are highlighted in yellow, while BstUI sites are highlighted in blue.

Following amplification of msRE treated DNA by ddPCR, target concentration was divided by the control, undigested concentration to produce percentage amplification under both msRE treatments. All 7 samples tested in the exploratory assay resulted in low-level amplification, demonstrating that methylation is present within the rat CpG island in prenatal brain tissue (Figure 6.10). Methylation was also demonstrated by Ngai et al. in the postnatal adult brain, with site-specific increases in methylation following prenatal alcohol exposure [467]. Exploring how early methylation patterns may change in response to prenatal stressors (such as MIA and prenatal alcohol exposure), and whether they correlate to methylation patterns later in life, warrants further investigation.

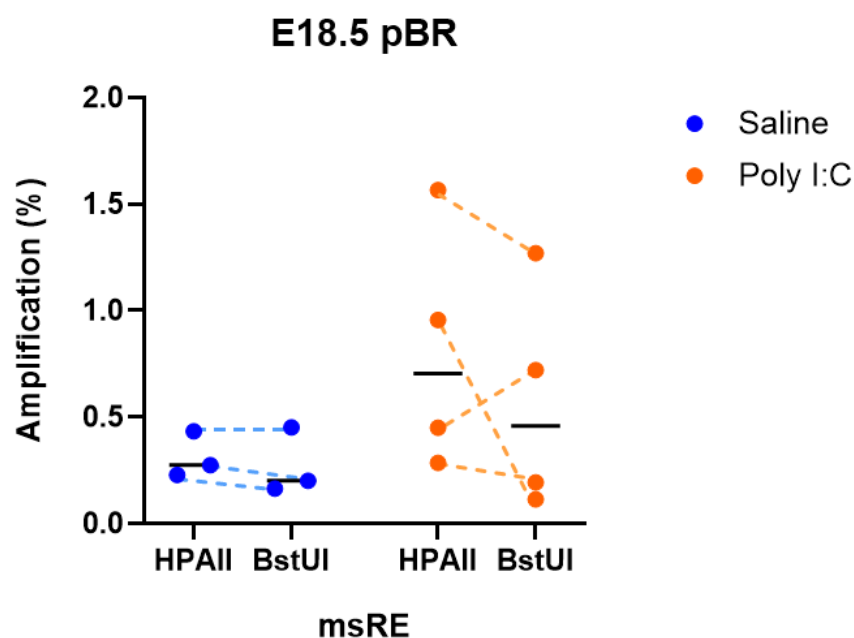


Figure 6.10: Exploratory msRE assay in embryonic frontal cortex tissue. Low amplification of methylated DNA was observed in poly I:C (orange) and saline-treated control (blue) offspring. Paired samples are indicated with dashed lines across msREs.

Although HpaII contained 4 additional CpG sites compared to BstUI, both enzymes produced comparable, low-level amplification across paired samples. This suggests that full methylation over all the HpaII sites and BstUI sites is not common within the DNA extracted from the E18.5 brain, but does occur. Furthermore, amplification of uncleaved DNA in HpaII and BstUI-treated samples was relatively stable in the saline

brain but not in the poly I:C brain, which may suggest differential methylation of the sites cleaved by these msREs (Figure 6.10). Percent amplification also tended to be lower in saline samples compared to poly I:C, although this was not significant at this small sample size by Welch's T test (HpaII: $t(3.27) = 1.71$, $p = 0.1791$; BstUI: $t(3.65) = 1.07$, $p = 0.3504$).

While this assay was able to confirm methylation exists within the CpG region identified, and whether methylation may generally vary between treatment groups, this assay lacked the specificity to explore significant methylation changes within the MIA model and was not continued further. While an increase in sample size of this assay may reveal a significant increase in poly I:C exposed offspring, the assay is limited to measuring the occurrence of DNA methylated at 4/4 or 2/2 specific sites: Any combination less than full methylation of those sites or methylation of any of the other 20 CpG sites within the amplicon would remain unexplored. As the percentage of fully methylated DNA was typically less than 1%, this provides a very limited scope of the methylation pattern that may be present in the E18.5 brain. This assay did allow a fast and cost-effective method for confirming methylation exists, however more costly methods such as bisulfite conversion and DNA sequencing would allow detailed interrogation of the CpG island and should be explored in future research.

6.4. Conclusions

While the initial goal of this chapter was to explore Slc6a4 methylation in the foetal brain following MIA, it was quickly discovered that understanding of Slc6a4 methylation in the rodent brain is poorly understood. Furthermore, studies exploring methylation of human SLC6A4 were inconsistent in the range of sites assessed and the results reported. Altogether this provided a significant barrier for exploring Slc6a4 methylation following MIA in this thesis, both in practical terms and in understanding how the overall implications of altered methylation might relate to human development.

Based upon the analyses presented in this chapter, further exploring both human and rat SLC6A4 methylation following MIA is strongly recommended for future research. In human populations, a range of prenatal stressors have been associated with altered methylation: while methylation of SLC6A4 following maternal infections has not been

directly assessed, factors such as maternal stress, GDM, and PTB, were all associated with altered SLC6A4 methylation in offspring [453, 471, 475]. These have all been further associated with an increase in maternal inflammatory cytokines, as well as increased incidence of NDD/NPDs; MIA may also therefore reflect similar changes in offspring SLC6A4 methylation [8, 94, 231, 233, 441, 498]. The one existing study in rats further suggests that prenatal stress may also effect methylation in rat Slc6a4; the analysis of Slc6a4 in this chapter provides a valuable resource for future research to thoroughly explore methylation following MIA and in other rat models [467].

This chapter also highlights key CpG sites in the human SLC6A4 associated with not only prenatal stress, but also altered SERT regulation. While it is assumed across the literature that increased methylation in the SERT promoter negatively correlates with transcriptional activity, this was not always observed in practice [451, 485]. Improved understanding of the regulatory changes resulting from differential SLC6A4 methylation therefore is necessary for future research to draw meaningful conclusions. Rodent models may be a useful tool for future research to expand on this review, further elucidating how methylation directly influences SERT regulation across tissues.

Chapter 7:

General Discussion

7.1. Thesis Aims and Rationale

This thesis aimed to explore the hypothesis that MIA leads to dysregulation of the serotonergic system within the rodent foetoplacental unit. This may have important implications for human development, as the diagnosis and symptomology of NDD/NPDs are associated with both dysregulated 5-HT signalling and exposure to maternal infections during gestation [111, 182, 220].

As of writing this thesis, only a small number of studies in rodent models have provided direct evidence to suggest that 5-HT synthesis, transport, metabolism, and signalling may be altered in the foetoplacental unit following MIA [227, 301, 319, 324]. Results from these studies are not consistent: both increases and decreases in placental 5-HT concentrations have been observed following MIA, accompanied by opposing changes in the expression of 5-HT synthesis machinery (summarised previously in Table 1.2). Alongside broader research investigating gestational 5-HT regulation, these studies have implied several suggested mechanisms by which MIA may impact 5-HT regulation within the foetoplacental unit. This includes impairment of the placental barrier and/or typical placental degradation of maternal 5-HT, which would lead to an influx of maternal 5-HT to the foetoplacental unit; altered placental synthesis and transport of 5-HT to foetal circulation; and altered synthesis and activity of foetal 5-HT in the developing brain. These suggested mechanisms may co-occur or may be specific to the gestational age when MIA is induced or the severity of the maternal immune response. While studies investigating the relationship between MIA and gestational 5-HT have increasingly been published over the last few years, further research is needed to supplement these initial findings and elaborate on the specific mechanisms that drive this relationship.

The research presented in this thesis encompasses multiple avenues of investigation, designed to supplement existing research. This includes experimental work mirroring previous publications, validating the MIA model used (Chapter 3) and exploring mRNA expression of target genes related to 5-HT regulation (Chapter 5), as well as the development of techniques to explore 5-HT concentration in relatively unexplored amniotic fluid (Chapter 4) and evaluating whether DNA methylation of *Slc6a4* should be further explored in rodent MIA models.

7.2. Key Findings

7.2.1. Model Validity and Overall Findings

This thesis utilised an MIA model established in Sprague Dawley rats, with s.c. administration of poly I:C on E12.5 and foetal sampling on E14.5 and E18.5. As with all MIA models, successful induction of the maternal immune response is a crucial assumption underlying this research [155, 156, 176]. This is acknowledged in Chapter 3, where multiple measures are used to assess the overall response to poly I:C administration. Poly I:C-treated dams exhibited a significant decrease in successful litter births, indicating an increased risk of litter resorption that is commonly reported in MIA models [139, 327, 379]. This was accompanied by a significant increase in maternal serum Crp 3-4 hours after injection, suggesting an earlier increase in IL-6, which is a common indicator of inflammation in MIA models [146, 147, 157, 158, 227].

Analysis of maternal serum in Chapter 3 was relatively limited, focused on confirming the presence of a maternal immune response through IL-6/Crp in a cost-effective manner rather than aiming for broad characterisation of MIA. Accompanied by the detailed methods reporting in Chapter 2 and Appendix C, this thesis aligns with current recommendations from MIA literature encouraging increased transparency and validation of MIA model establishment [155]. Importantly, as validation/characterisation of the immune response varies widely in published literature, future work should standardise validation measures and explore the specific immune pathway(s) associated with 5-HT dysregulation in MIA models. The four key studies investigating rodent fetoplacental 5-HT regulation following MIA vary widely in their inclusion of MIA validation/characterisation: Bondarenko et al. for example briefly reports model establishment methods and includes no assessment of immune responses, while Zengeler et al. provides detailed MIA methodology through the MIA ARRIVE guidelines (as seen in Appendix C) and a comprehensive analysis of the immune response in placental but not maternal tissues [227, 301, 319, 324]. While broader exploration of the immune response was outside the scope of this thesis, an in depth assessment of the relationships between the maternal, placental, and foetal immune response in regard to 5-HT regulation would be a valuable route of exploration.

Overall, the increased litter loss and maternal Crp demonstrated in poly I:C dams in Chapter 3 did provide evidence that poly I:C led to a maternal immune response in this thesis as anticipated. This is further supported by the treatment effects subsequently observed in foetal tissues, indicating foetal development was influenced by poly I:C administration. A summary of key results from experiments in offspring tissues can be found in Table 7.1.

Chapter	Age	Key Finding
3	E18.5	Sex difference: Total, foetal, and placental weight was reduced in female offspring (both in overall sex effect and in saline litters specifically)
3	E18.5	Treatment difference: Placental weight was correlated with foetal weight in saline litters, but not in poly I:C-exposed litters
4	E14.5, E18.5	No differences were observed in the concentration of 5-HT in AF
5	E14.5, E18.5	Sex difference or Sex*Treatment interaction: Common reference genes <i>B2m</i> , <i>Hprt</i> , <i>Ppia</i> , <i>Tpb</i> , and <i>Ywhaz</i> were differentially expressed in developing brain tissue
5	E14.5	Treatment difference: Poly I:C-exposed offspring had significantly increased placental <i>Tph1</i> and reduced brain <i>NeuroD1</i> expression compared to controls
5	E14.5	Sex difference: Placental <i>Oct3</i> and <i>Bdnf</i> expression was significantly decreased in females overall, while brain <i>MaoA</i> and <i>Ywhaz</i> expression was increased in poly I:C-exposed females compared to male littermates
5	E14.5	Sex*Treatment interaction: Brain <i>Mecp2</i> interaction driven by significantly increased expression in saline females (compared to saline males and poly I:C-exposed females); brain <i>Sert</i> interaction driven by non-significant increase in poly I:C-exposed females
5	E18.5	Sex difference: Placental <i>Bdnf</i> expression increased in saline males compared to saline females
5	E18.5	Sex*Treatment interaction: Brain <i>MaoA</i> and <i>Mecp2</i> interactions driven by non-significant increases in poly I:C male expression
6	E18.5	DNA methylation in exon 2 of <i>Rattus norvegicus Slc6a4</i> is detectable within the foetal brain and may be functionally relevant

Table 7.1: Key findings from chapters 3-6 investigating development within the fetoplacental unit following MIA.

7.2.2. Insights into Typical Gestational Development

7.2.2.1. Developmental Differences across Sexes

Within this thesis, both overall weight measurements and mRNA expression profiles highlighted sex differences in development between E14.5 and E18.5. Foetal and placental weight at E18.5 was significantly reduced in females, particularly driven by the sex difference seen in saline litters. Placental *Bdnf* expression in foetally-derived placental tissue was decreased not only in saline females compared to male littermates at E18.5 but also decreased in females overall at E14.5. This was further accompanied by sex differences in the expression of reference genes in foetal brain tissue: females overall demonstrated increased *B2m* and *Hprt* expression at E14.5, while a trend of increased *Tpb* expression in saline females appeared to drive a significant sex*treatment interaction effect at E18.5.

This study is not the first to identify differences in development between sexes, however this thesis does expand understanding of where these differences occur. Although mRNA expression changes are observed at both E14.5 and E18.5, overall weight of the foetus and placenta does not diverge at E14.5. Differential expression of placental *Bdnf* and *Oct3* at E14.5, alongside the larger placental weight in males at E18.5, likely reflects the higher metabolic function and higher energy requirement in male foetuses, with increased growth of the rapidly-developing labyrinth zone increasing surface area for higher foeto-maternal transport of nutrients to the male foetus. [188, 192, 499]. In contrast, it is less clear what role the differential expression of reference genes such as *B2m* and *Hprt* play in brain development across sexes. For example, while deficiencies in *Hprt* are associated with poorer outcomes such as cognitive impairment or Lesch-Nyhan Syndrome, baseline sex differences in *Hprt* at E14.5 have not been previously reported and may have implications for overall purine recycling and neurodevelopment [417].

Overall, these significant sex effects highlight divergent developmental trajectories between sexes at E14.5-E18.5 and underline the importance of including sex as a variable when investigating gestational models. A common approach to assessing foetal and placental tissues is to combine pups from a litter, with final sample size being the number of independent litters; without accounting for sex however, important developmental differences may be lost or confound other variables of interest.

Additionally, measures such as foetal and placental weight were heavily influenced by litter, potentially obscuring sex effects in a wider sample of offspring. The LMM approach used in this thesis enables statistical analyses to balance the importance of intra- and inter- litter variance, and is recommended for continued use in future work building upon this thesis [191, 346, 422, 500].

7.2.2.2. 5-HT Regulation during Gestation

Within existing literature, debate surrounds the role of the placenta in gestational 5-HT regulation. While most recent research suggests the placenta is a site of 5-HT production [227, 315], numerous studies have demonstrated the importance of maternal 5-HT in foetal development and suggest the placenta instead solely acts as a site of transport [270, 307]. This thesis adds evidence to support that the placenta does produce 5-HT during gestation: mRNA expression of both *Tph1* and *Ddc* was detectable and quantifiable in placental tissue at E14.5, demonstrating transcription of the machinery required to metabolise tryptophan to 5-HT within the placenta. Furthermore, expression of neither *Tph1* nor *Tph2* was quantifiable in placental tissue at E18.5. but *Tph2* mRNA was quantifiable in the foetal brain. This reflects previous work by Bonnin and Goeden in mice that proposed gestational 5-HT regulation shifts at approximately E15.5, from reliance on placental 5-HT in early/mid neurodevelopment to use of foetally-produced 5-HT in late neurodevelopment. This thesis provides initial and direct evidence that this shift may also occurs in rats between E14.5 and E18.5. While developmental translation between rats and mice suggests that E15.5 roughly aligns to E17.5 in rats, a more specific age for this transition cannot be provided by the work in this thesis.

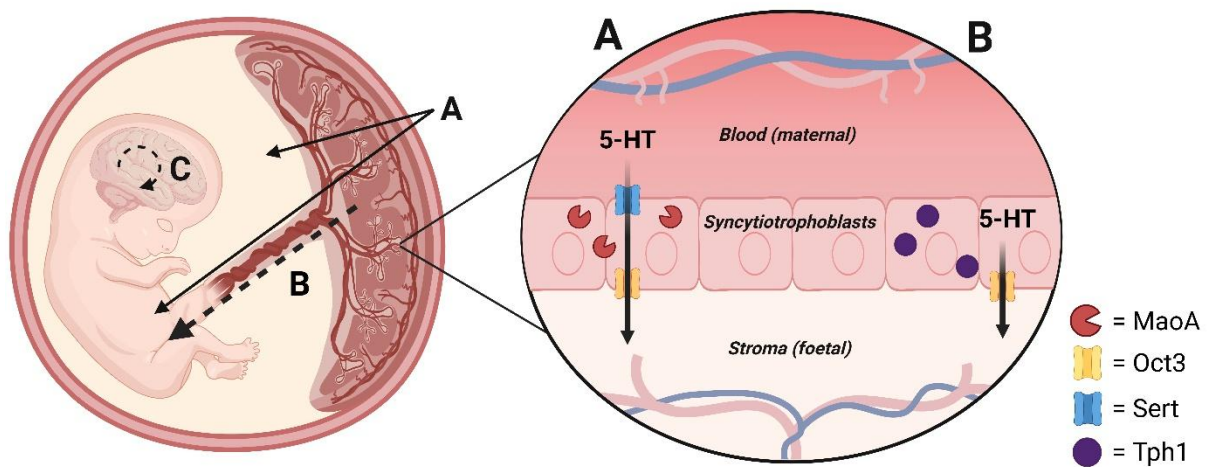
As the gestational regulation of 5-HT is important not just in the context of MIA, but in the broader contexts of gestational health, pregnancy complications, and neurodevelopment, the balance between placental and foetal 5-HT production in rat models warrants further investigation. While this thesis aligns with research demonstrating 5-HT production in the placenta in mid-gestation and in the foetal brain in late-gestation, it should not be discarded that altered maternal 5-HT regulation also has impacts on foetal development [270, 307, 501]. The mechanisms by which this occurs are currently unclear, including how maternal 5-HT may impact foetoplacental 5-HT regulation.

7.2.3. 5-HT Regulation in the Foetoplacental Unit following MIA

As previously discussed, MIA may lead to a wide range of changes in 5-HT production and transport; the mechanisms proposed in the introduction of this thesis are briefly re-depicted in Figure 7.1 (A-C). The first hypothesis was that MIA may induce an influx of maternal 5-HT, which was not directly supported by the results presented in Chapters 3 - 5. As maternal 5-HT is likely transported back into maternal circulation or metabolised within the placenta under typical conditions, an influx of maternal 5-HT would likely require altered expression of 5-HT transporters and/or *MaoA* within the placenta. In addition, impairment of typical regulation of maternal 5-HT may be detectable in the AF, which buffers a wide range of solutes for foetal development and is in direct contact with both the foetus and placenta. At E14.5 and E18.5, no change in AF 5-HT concentration was observed following MIA – this suggests that either AF 5-HT was not impacted by MIA, or that any change in 5-HT was restored to typical concentrations within 48 hours. Similarly, no treatment-associated changes were observed in placental expression of *Sert*, *Oct3*, or *MaoA* at either E14.5 or E18.5.

Overall, the suggested MIA-induced alteration of placental 5-HT production (Figure 7.1, B) is the most supported by this thesis research. Administration of poly I:C led to a number of changes within the foetoplacental unit, most notably a significant increase in placental *Tph1* expression (Figure 1.7, i). This suggests an increased capacity for 5-HT production: although *Tph1* protein expression or 5-HT concentration in placental tissue were not directly assessed in this thesis, previous work by Goeden and Abad have both demonstrated that changes in placental *Tph1* mRNA expression coincide with changes in placental 5-HT concentration [227, 301]. Interestingly, Goeden observed an increase in *Tph1* mRNA expression 24 hours after poly I:C administration that abated by 48 hours. While samples were not collected at 24 hours in this study, increased expression at 48 hours may reflect a delayed, or extended, response in the rat foetoplacental unit. As treatment effects were also observed in the foetal brain by E14.5, it is likely that placental changes began to occur prior to E14.5 – future studies assessing more acute impacts of MIA (i.e. sampling at 12 and 24 hours) would provide valuable insight to this possibility.

Potential mechanisms of 5-HT dysregulation following MIA



Mechanism supported by this thesis

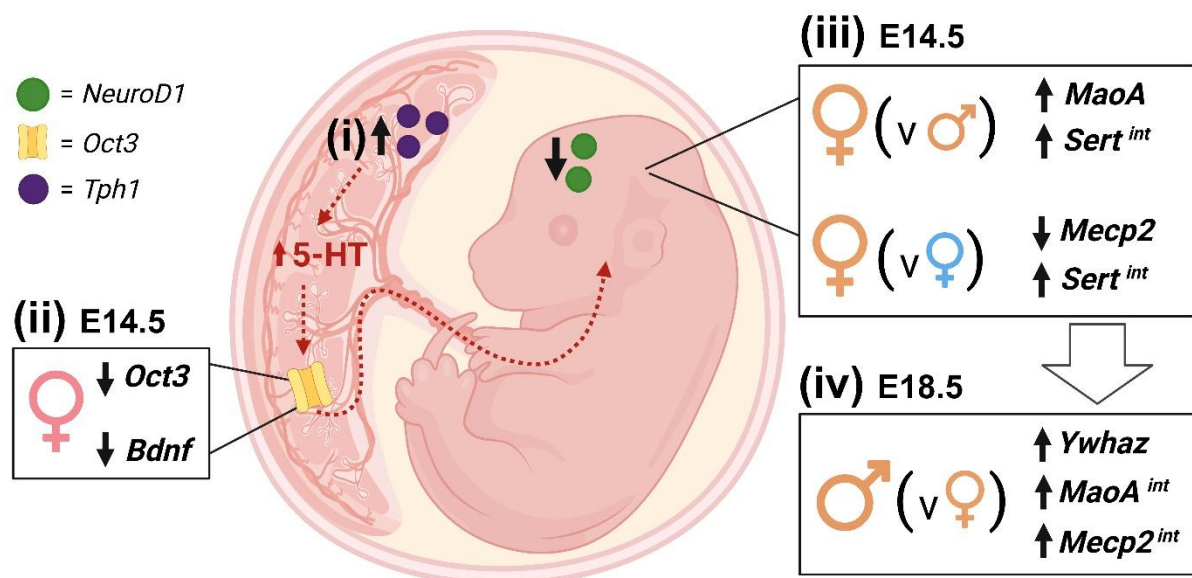


Figure 7.1: Evidence from this thesis supports the hypothesis that mid-gestational MIA induces an increase in placental 5-HT production, with the foetal response mediated by offspring sex. Top half of figure demonstrates previously hypothesised mechanisms of MIA-induced 5-HT dysregulation (adapted from Figures 1.7 and 1.8): A) MIA impairs placental regulation mechanisms and induces an influx of maternal 5-HT; B) MIA alters placental 5-HT production and/or transport to the foetus; C) MIA alters foetal 5-HT production later in development. The lower half of the figure depicts the proposed mechanism supported by findings in this thesis. Direct evidence is presented in black, while suggested changes not directly investigated are presented in red. Sex icons (♂ = male; ♀ = female) indicate observed changes that differ based on offspring sex, while colour (pink = females overall; orange = poly I:C; blue = saline) indicates whether the indicated difference relates to specific sex/treatment groups. Genes annotated with 'int' indicate that a significant sex*treatment interaction was identified without significant pairwise comparisons, with the comparison nearest to significance depicted as the direction of expression change. An increase in *Tph1* expression (i) at E14.5 may reflect an increase in *Tph1* synthesis within the placenta. Sex differences in E14.5 *Oct3* expression (ii) and overall placental growth between E14.5 and E18.5 suggest capacity for 5-HT transport to the foetal bloodstream may be reduced in females regardless of MIA exposure. In the E14.5 foetal brain (iii) *NeuroD1* is significantly reduced following MIA regardless of sex, while expression of *Sert*, *MaoA*, and *Mecp2* is differentially expressed in poly I:C females relative to male littermates and/or saline female controls. Increased *MaoA/Sert* may indicate improved uptake and metabolism of excess 5-HT in comparison to males 48 hours after MIA exposure. Sex-mediated effects extend to the E18.5 brain (iv), with reduced *MaoA*, *Mecp2*, and *Ywhaz* expression potentially indicating reduced 5-HT activity in MIA-exposed females compared to males. Created in <https://BioRender.com>.

Sampling foetal tissues earlier after MIA would also provide further context for the range of sex*treatment interactions observed here in the foetal brain at E14.5 and E18.5. It is possible that existing sex differences in placental development, such as reduced *Oct3* expression in females (Figure 1.7 ii), may have moderated the transport and/or re-uptake of excess 5-HT following MIA. Increased placental *Oct3* activity has been observed in males compared to female littermates at term, supporting the premise that *Oct3* activity differs gestationally [322]. Importantly, this requires further work to substantiate, including evaluating whether sex differences in protein or activity of the 5-HT transporters *Sert* and *Oct3* exist prior to E14.5. What is directly suggested from this thesis is that genes related to serotonergic activity differently respond to MIA across sexes (Figure 1.7, iii – iv). Sex differences in the regulation of *MaoA* (increased expression in females at E14.5, then trending higher in males at E18.5) and *Sert* (trending higher in females at E14.5) in the brain following MIA are of particular note; this may reflect different responsivity to changes in 5-HT between sexes, with increased uptake and metabolism of 5-HT in the female brain at E14.5.

While sex-dependent responses to MIA are an intriguing concept, this thesis can only provide preliminary evidence to support this due to small sample sizes across sex*treatment groups. While sex was included as a variable of interest, sampling and study design heavily prioritized treatment and accounting for potential litter effects, often leading to an unbalanced sex distribution within treatments that was unable to be remedied due to time or resource limitations. Importantly, even with limited samples and careful statistical assessment, significant sex*treatment interactions were still observed. Subsequently, future work is strongly encouraged to expand on these findings and explore sex differences in the serotonergic system following MIA. This includes both in the collection of excess tissues to determine sample sex prior to experimental work, and through increased sampling focused on assessing sex differences as a main research aim. Current understanding of how sex may influence gestational 5-HT regulation in general is poor, even though 5-HT in the postnatal human and rodent brain has previously reported to differ by sex [502, 503]. No sex differences gestational 5-HT regulation following MIA have been reported prior to this thesis, although in many cases it does not appear that offspring sex was included in analysis [319, 324].

It should also be noted that some treatment effects observed following MIA imply broader dysregulation of foeto-placental growth rather than specific 5-HT-associated changes. A decrease in *NeuroD1* expression in the E14.5 brain of poly I:C-exposed offspring for example may indicate broader changes in neurodevelopment, such as altering the epigenetic and transcriptional profile of developing neurons to influence neuronal differentiation [275, 276]. The uncoupling of placental and foetal growth at E18.5 following MIA (without overall loss in foetal weight) may also indicate a change in late-stage placental function outside of overall foetal growth restriction. For example, typical changes to hormone, steroid, and inflammatory cytokine production are reported in the rodent placenta after E18.5, and is worth investigating in future MIA models [192, 504].

7.3. Wider Implications

7.3.1. Methodological Developments for Future Research

The results presented in this thesis are underpinned by a series of methodological developments that will greatly support future research. Firstly, development of the saphenous vein blood sampling protocol (used to assess maternal serum IL-6 and Crp concentrations) allows future researchers in the VUW Small Animal Facility to rapidly sample small volumes of rat blood with minimal handling stress or animal recovery time required. This is beneficial not just for MIA research that may follow this project, but also for studies in other developmental models or experiments where animal behaviour measures may be impacted by sedation or other blood sampling methods [325, 326, 368].

In addition to blood sampling, numerous experimental assays were optimised and/or evaluated in chapters 4-6. Chapter 4 details the optimisation and successful employment of an LC-MS method to detect 5-HT in amniotic fluid, adapted from previously published protocols assessing 5-HT in dialysate samples [333, 393, 394]. Derivatization and quantification of 5-HT in rat amniotic fluid by LC-MS has not previously been published. Amniocytes are a known site of 5-HT metabolism within the foetoplacental unit, however amniocytes and AF overall has previously been overlooked in research assessing 5-HT regulation following MIA [306]. Development of this LC-MS technique will hopefully encourage further studies investigating foetoplacental 5-HT regulation to include AF as a medium of interest and potentially develop this approach further to include other target metabolites of interest such as 5-HIAA. Furthermore, one of the advantages of using LC-MS over other HPLC methods for 5-HT quantification is the depth of the data collected for each sample, which can be used for wider metabolomics analyses [395, 505]. Although outside the scope of this thesis, this dataset may be used by future students to explore further AF changes within the foetoplacental unit following MIA.

In chapter 5, two ddPCR approaches were tested and evaluated prior to the final quantification of target mRNA expression. Established selfie-ddPCR methods were found to produce inconsistent results in foetal brain tissue, while common reference genes were found to vary in expression across sex and treatment groups. This

provides useful insight for the selection of ddPCR assays and reference genes in future work, avoiding extensive and/or costly method development processes. Along with the tissue library generated by the breeding program in this thesis, this optimisation will be particularly beneficial for future MIA projects, which may investigate mRNA expression of further targets of the 5-HT system (e.g. 5-HT receptor expression), 5-HT-adjacent systems (e.g. mitochondrial copy number and mRNA expression), or broader developmental outcomes [63].

Chapter 6 outlines the identification of a potentially functionally-relevant CpG island in rat *Slc6a4*, with the recommendation that this should be investigated further in MIA models and rodent research generally. Although literature assessing *SLC6A4* methylation in humans is highly variable, a small number of sites were differentially methylated in models of prenatal stress and also associated with changes in transcriptional activity or protein expression of SERT (see Figure 6.4). Of note, it is unclear whether the homologous region of rat *Slc6a4* identified holds the same functional significance, although methylation was observed in a region of rodent *Slc6a4* homologous with the human *SLC6A4* CpG island. Future work should expand on this chapter by exploring methylation of this region following MIA, with particular interest in whether methylation following MIA reflects the sex*treatment interaction in *Sert* mRNA expression identified in the foetal brain at E14.5. Expanding beyond this model, this chapter has wider implications for understanding the role of SERT directly in human research. Ensuring that future methylation research includes a validity measure such as mRNA expression or ensuring that the methylation sites selected have previously been associated with changes in *SLC6A4* regulation, would improve validity and consistency across all relevant research fields.

7.3.2. Human Development and Disorders

Maternal infections during pregnancy lead to an increased risk of NDD/NPDs in offspring, although the mechanisms underlying this increased risk are not well understood [12]. Along with recently growing literature, this thesis supports the hypothesis that MIA may induce changes in gestational 5-HT, contributing to increased risk of 5-HT-associated disorders later in life [227]. Although the ability to directly explore these mechanisms in humans is limited, rat models have demonstrated

reasonable translatability for early human development and MIA models show strong face, predictive and construct validity to human NDD/NPD outcomes [143, 182, 192, 193, 351, 506].

Along with other research focusing on early/mid-gestational MIA, this thesis suggests that MIA is associated with an increase in placental 5-HT production in offspring. Increased 5-HT exposure may significantly influence neuronal connectivity and lead to long-lasting differences in rodent behaviour [227, 438, 507, 508]. These outcomes reflect common findings in NDDs and NPDs, where altered neuronal connectivity and 5-HT signalling are associated with disorder symptomology. Furthermore, increased placental serotonin is consistent with human pregnancy complications such as preeclampsia and GDM, both of which are also associated with NDD/NPD risk [234, 235, 240, 242].

This thesis also suggests that offspring sex may play a role in the foetal response to increased 5-HT following MIA. Although not commonly reported, some studies find MIA risk may be sex dependent; Debost et al. for example report maternal infection alone only resulted in an increased risk of SCZ in females, while Brown et al found an increased risk of ASD in males only following maternal infection [102, 107]. More commonly reported are sex differences in NDD/NPDs. This includes not only incidence and common trends in symptomology, but in treatment efficacy; common medications such as haloperidol (in treating SCZ) and fluoxetine (in treating MDD) for example exhibit sex differences in their effectiveness and severity of side effects [440, 509-511]. Sex differences in NDD/NPDs overall are likely driven by a complex interplay of social and biological factors, which may include sex differences in the prenatal response to risk factors such as prenatal SSRI exposure or MIA [39, 41, 42, 80, 512, 513]. Further research exploring sex-specific differences in the gestational serotonergic system may help explain sex differences in NDD/NPDs, leading to earlier diagnoses and improved treatment outcomes.

7.4. Conclusions

Overall, this thesis provides further evidence that mid-gestational MIA alters offspring development through changes to the serotonergic system. Furthermore, MIA-induced alterations in the serotonergic system may be mediated by offspring sex and require further exploration. The complex interplay between MIA, 5-HT signalling, and sex is not thoroughly understood. The experimental findings and methodological developments produced in this thesis encourage further exploration of 5-HT regulation in rodent MIA models, leading to improved understanding of human development and the developmental processes that lead to NDD/NPD outcomes in later life.

Appendix A

This Appendix includes a draft publication, “Genetic Reduction in the Serotonin Transporter Influences Behavioural and Molecular Outcomes in Rodent Offspring Following Maternal Immune Activation”, including ddPCR experiments performed during this PhD alongside behavioural research reported in a previously completed Masters Thesis [221]. This manuscript is not currently under review (pending feedback from the corresponding author, who is currently unavailable), however includes experimental work relevant to this thesis and is cited accordingly.

Genetic Reduction in the Serotonin Transporter Influences Behavioural and Molecular Outcomes in Rodent Offspring Following Maternal Immune Activation

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Abstract

Background: Maternal Immune Activation (MIA) during early pregnancy is an established risk factor for the occurrence of neurodevelopmental disorders (NDDs) such as Autism Spectrum Disorder (ASD) and schizophrenia (SCZ) in offspring. NDDs are widely understood to have both genetic and environmental components, however risk factors need to be studied in combination to understand potential interactions during development. Genetic variation in the human serotonin transporter (SERT) gene SLC6A4 has also been linked to NDD pathology, however the interaction between MIA and SERT genotype remains poorly understood. **Methods:** Using a haploinsufficient rat model to reflect low expression human SERT genotypes, we investigate the impact of differing SERT expression levels in combination with mid-gestational MIA exposure on offspring development. Ultrasonic vocalisations (USVs) during maternal separation were used as a measure of offspring communication at postnatal day (PND)7, while mRNA expression of the *Rac1/Kal7/Disc1* signalosome was measured by selfie-digital droplet PCR (ddPCR) and used as an indicator of hippocampal synaptic plasticity at PND21. Offspring were assessed by treatment, genotype (wildtype/full expression or heterozygous/halved expression), and sex using a linear mixed modelling approach to account for litter associations between offspring. **Results:** USV analysis reveals highly increased call complexity in male SERT heterozygous offspring compared to wildtypes, which significantly interacts with treatment to provide initial evidence of male SERT heterozygotes having a higher susceptibility to MIA challenge in utero. A weaker yet similar trend is observed in females SERT heterozygotes, suggesting sex may mediate this relationship. Hippocampal expression of *Rac1* and *Kal7* is significantly altered following MIA exposure in SERT wildtype offspring, with *Rac1* expression again revealing a significant treatment*genotype interaction indicating changes in synaptic plasticity. **Conclusions:** Overall, these results highlight important behavioural and molecular alterations in MIA exposed pups that are modulated by SERT genotype and offspring sex. USV and selfie-ddPCR findings together indicate reduced SERT expression may not simply amplify MIA effects, but instead alter offspring responses to MIA in a complex manner that warrants further research.

Keywords

Maternal immune activation; poly I:C; serotonin; serotonin transporter; neurodevelopmental disorders; autism spectrum disorder

1. Introduction

Neurodevelopmental disorders (NDDs) such as autism spectrum disorder (ASD) and schizophrenia (SCZ) are heterogeneous, complex disorders associated with altered brain development in utero. Lifelong changes in neural connectivity, signalling, and plasticity are observed in NDDs, and underlie a wide range of symptoms where social communication, mood, and learning may be affected. Symptomology and symptom severity within each disorder is highly variable, making causative mechanisms difficult to identify; this is further confounded by overlapping symptomology between NDDs, as well as high comorbidity rates with mood disorders such as major depressive disorder (MDD) [1]. Instead of single driving causes, an array of genetic and environmental risk factors have been identified as possible contributors to NDD incidence. These risk factors interact in a 'multiple hit' model to induce disorder phenotypes: genetic pre-dispositions and environmental insults combine to disrupt connectivity, signalling, and plasticity mechanisms in the brain [2]. With the large number of risk factors and potential mechanisms altering



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41 typical brain development, identifying whether and how specific risk factors interact remains a crucial field of
42 investigation.

43 1.1 Maternal Immune Activation

44 One strongly implicated risk factor for NDD incidence is maternal immune activation (MIA) during early
45 pregnancy [3, 4]. While some pathogens such as Zika virus are known to directly enter and influence the foetal
46 compartment, MIA describes a generalised response to infection irrespective of the specific pathogen involved [2].
47 Briefly, both viral and bacterial agents display pathogen-associated molecular patterns (PAMPs) that are recognised
48 by toll-like receptors 3 and 4 respectively within innate immune cells [5]. This triggers the upregulation of
49 proinflammatory cytokines in maternal circulation, promoting the adaptive immune response to defend against
50 infection. The maternal bloodstream also carries these inflammatory markers to the foeto-placental unit however,
51 exposing both the foetus and placenta to inflammation during critical development of the foetal brain [6].

52 The viral mimic polyinosinic:polycytidylic acid (poly I:C) is commonly used in rodent research to induce MIA,
53 by activating toll-like receptor 3 to trigger an acute immune response in the absence of any secondary pathogen-
54 specific effects [5]. When administered to pregnant dams during mid-pregnancy at approximately gestational day
55 (GD)12.5, offspring display numerous behavioural and neurological changes reflective of NDDs. For example, MIA-
56 exposed pups typically emit fewer (and less complex) ultrasonic vocalisations upon maternal separation, indicative
57 of altered social communication [7]. Offspring also show altered repetitive and anxiety-like behaviours, as well as
58 deficits in learning and memory, which reflects symptomology of both ASD and SCZ [8]. Dysregulation of
59 hippocampal connectivity and synaptic plasticity are thought to underlie many of these processes [9], and have been
60 further implicated in the poly I:C MIA model. For example, both increased and decreased hippocampal long term
61 potentiation (LTP) and neurogenesis have been observed in poly I:C-exposed offspring alongside poorer cognitive
62 performance [10, 11]. NDD-associated genes such as Disrupted-in-schizophrenia 1 (DISC1) are known to mediate
63 LTP in the hippocampus, with rodent studies demonstrating that expression of NDD-associated Disc1 variants
64 interacts with both prenatal and postnatal immune stimulation [12-14]. Interestingly, the transgenic DISC1 mouse
65 also shows a significant reduction in brain serotonin (5-HT), with a further (yet non-significant) reduction following
66 juvenile immune stimulation [12].

67 1.2 The Serotonergic and Immune Systems

68 Dysregulation of the serotonergic system is a risk factor for NDDs in its own right, as well as in combination
69 with the immune system. 5-HT signalling is a well established and key mechanism in early neural development,
70 regulating processes including neural patterning, interneuron growth and synaptogenesis [15]. Many associations
71 between 5-HT and NDDs have been identified previously, such as blood hyperserotonemia in ASD and changes in
72 5-HT receptor expression in the hippocampus of SCZ individuals [16, 17]. These associations further extend to
73 include immune dysregulation; for example, altered synaptic pruning in both ASD and SCZ is associated with 5-HT
74 mediated changes in microglial activation [18, 19]. Specifically during gestational development, maternal 5-HT
75 influences immune activity in conditions such as preeclampsia and gestational diabetes mellitus (both of which are
76 also associated with increased NDDs in offspring) [20, 21]. While evidence demonstrating 5-HT/MIA interactions
77 remain limited, Goeden *et al.* reported an increase in 5-HT synthesis in the rodent placenta following MIA leading to
78 subsequent axonal changes in the foetal brain [22]. MacDowell *et al.* further extend on this relationship by providing
79 evidence of long-term dysregulation of 5-HT and kynurenine metabolism in offspring exposed to MIA [23]. This
80 dysregulation was partially rescued through administration of the SCZ medication paliperidone, which is known to
81 be an inverse agonist of the 5-HT_{2A} receptor. Overall, there is a strong argument suggesting MIA may influence the
82 serotonergic system (and vice versa), and that these interactions may be key mechanisms underlying NDD
83 incidence/symptomology.

84 1.3 The Serotonin Transporter

The serotonin transporter (SERT) facilitates 5-HT transport throughout the brain and body, supporting the regulatory roles of 5-HT as well as its function as a neurotransmitter. Interestingly, the promoter region (5-HTTLPR) of the human SERT-encoding gene SLC6A4 is highly polymorphic and associated with changes in mRNA transcript expression. Following the identification of the two original 'long' (L) and 'short' (S) variants in 1996, many allelic variants have been identified and discussed extensively elsewhere [24, 25]. Briefly explained, alleles are typically categorised as either L or S variants based on the number of poly-C repeats in the variable number tandem repeat region (VNTR) [25, 26]. S alleles are associated with reduced SERT expression compared to L variants, leading to reduced serotonin reuptake activity [26]. SERT genotype is commonly discussed in relation to mood disorders, where S alleles are inconsistently associated with increased risk of depressive and anxiety disorders [27-29]. SERT genotype has also been inconsistently implicated in both ASD and SCZ [30, 31]. One current hypothesis suggests the contradictory findings across NDDs and neuropsychiatric comorbidities are driven by a combination of environmental variation and cognitive function, leading to genotype-moderated differences in responding to the environment [32]. S alleles are suggested to confer a 'reactive' phenotype, where both positive or negative stimuli will have an amplified effect, driving mood disorder incidence only in the presence of sufficient additional risk factors.

The identification of SERT genotype as an environmentally-responsive risk factor in both neuropsychiatric and neurodevelopmental disorders requires further investigation. Rodent research has shown that during early development, offspring of mothers with the ASD-associated Ala56 SERT genotype have reduced placental and embryonic 5-HT [33]. This normalises to wild-type levels after the foetus starts to produce endogenous 5-HT mid-gestation, suggesting impaired transport of external 5-HT to the foetus during early development is associated with ASD-like outcomes. Rats with a genetic reduction in SERT expression have further shown NDD-like outcomes including altered sociality [34], memory deficits [35], and responsivity to antipsychotic medication [36]. The Wistar SERT knockout rat model produces animals with full, approximately 50-60%, or negligible SERT protein in wildtype, heterozygous (HETs), and homozygous (HOM) offspring respectively [37]. SERT HETs, similarly to MIA-exposed offspring, show altered hippocampal LTP and cognitive flexibility compared to wildtype (WT) controls [38]. While much of this evidence suggests MIA exposure and reduced SERT expression influence similar processes within the brain, few studies have investigated direct interactions. One notable study by Reisinger *et al.* did observe a direct interaction between SERT and MIA by demonstrating hippocampal epigenetic down-regulation of SERT in MIA-exposed rodents, suggesting decreased SERT expression is associated with MIA outcomes [39].

1.4 Aims and Study Design

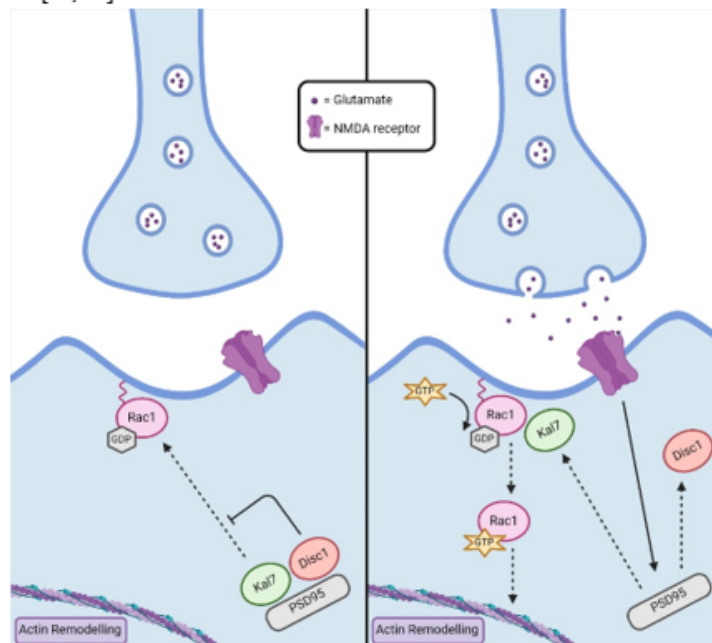
To further explore the relationship between the serotonergic system and MIA, we utilised the Wistar SERT knockout rat model as a genetic background for poly I:C -induced MIA to assess whether reduced SERT expression influences behavioural and molecular changes driven by MIA during gestation. As there is no precedent for a complete genetic SERT knockout in humans, only WT and HET pups were produced in order to represent the natural variation of the human population [37, 40].

Following administration of poly I:C at GD12.5, dams were assessed for poly I:C response using non-invasive methods of temperature and weight measurement. Offspring were also weighed to assess potential delays to overall growth and development, with the addition of two experimental approaches to reflect and measure NDD-associated outcomes. Behavioural changes at postnatal day (PND)7 were assessed using pup ultrasonic vocalisations (USVs) following maternal separation. This paradigm produces an acute stress, where increased and more complex calling is considered indicative of improved communication for maternal attention. As mentioned previously, MIA-exposed rodent offspring have shown an altered response upon maternal separation and typically emit fewer calls with reduced complexity [7]. To our knowledge no ultrasonic vocalisation differences have been observed between Wistar SERT WTs and HET pups, however male adult HOM rodents have shown altered call patterns during social interaction compared to WT counterparts [34].

As a molecular marker, mRNA expression of the Rac1/Kal7/Disc1 signalosome was assessed in hippocampal tissue by selfie-digital droplet PCR (ddPCR) [41]. This signalosome is a regulator of LTP and is strongly implicated

131 in altered learning and memory processes (Fig. 1) [13, 42, 43]. As mentioned previously, DISC1 variants are strongly
 132 implicated in NDDs and show potential interactions with both immune activation and the serotonergic system [12].
 133 Kalirin-7 (Kal7) and Rac1 are both further associated with serotonergic signalling in animal models, through the 5-
 134 HT2A receptor; Kal7 as a crucial factor in serotonin-regulated dendritic spine plasticity, while Rac1 may be
 135 constitutively activated following 5-HT2A signalling [44, 45].

136 Through these measures we sought to determine whether treatment, genotype, or interaction effects were
 137 observed in both male and female offspring following MIA exposure. We hypothesised a synergistic effect between
 138 treatment and genotype across both tests, whereby an initial treatment effect would be observed in WT offspring and
 139 be exacerbated in HETs. Considering that many NDDs show diagnostic and symptomatic differences based on sex,
 140 we also hypothesised that male offspring may show an increased sensitivity to MIA and genotype effects in
 141 comparison to females [46, 47].



142 **Fig. 1. The Rac1/Kal7/Disc1 signalosome.** Activation of Rho-GTPase Rac1 in the established synapse leads to remodelling of the actin
 143 cytoskeleton during long term potentiation and long-term depression (LTP/LTD) [46]. Rac1 activation occurs through the guanine
 144 exchange factor Kalirin-7 (Kal7), which is sequestered by Disrupted in Schizophrenia 1 (Disc1) to scaffolding protein PSD95 under
 145 baseline conditions. Upon the increased NMDA receptor activation through glutamatergic signalling that triggers LTP, the
 146 Disc1/Kal7/PSD95 complex dissociates and free Kal7 promotes activation of Rac1 [40]. Figure made using BioRender.
 147

148 2. Materials and Methods

149 2.1 Animal Breeding

150 All animal work was carried out in accordance with animal ethics guidelines and approved by the Victoria
 151 University of Wellington Animal Ethics Committee (Approval #25848). Prior to breeding, animals were socially
 152 housed in individually ventilated cage (IVC) systems. Naive female Wistar SERT HETs were crossed with male
 153 wildtype Wistars in a timed breeding protocol. Breeding occurred in IVCs with metal grate flooring until at least one
 154 vaginal sperm plug was identified, up to a maximum period of 5 days, before being returned to individual housing.
 155 Day of plug identification was designated GD0.5. Animals at all times had access to food and water ad libitum. Adults
 156 post-breeding were assigned to other projects where possible to improve animal use, as per general animal ethics
 157 recommendations.

158 2.2 Maternal Treatment and Dam Exclusion

159 Paired females had temperature and weight recorded on GD12.5 prior to injection (poly I:C n=6, saline n=4).
 160 Control dams were given one subcutaneous injection between the shoulder blades using 0.9% saline vehicle at a
 161 1ml/kg volume. Treatment group dams were also given one subcutaneous injection, with a dosage of 5mg/kg Poly
 162 I:C (Sigma-Aldrich P1530, Lot #116M4039V) suspended in 0.9% saline vehicle at a volume of 1ml/kg. Temperature
 163 was then re-recorded at 3, 6, and 24 hours post injection with an infrared thermometer at the base of the tail, with
 164 weight measured again at 24 hours. Three females (1 saline, 2 poly I:C) did not produce a litter and thus excluded
 165 from further study. An additional dam received poly I:C treatment and was excluded from further study as there was
 166 no observed fever response. Fever in MIA-exposed rodents is typically observed at approximately 1°C above baseline,
 167 with our threshold for exclusion set at less than 0.5°C [49].

168 2.3 Litter Assessment

169 Litters produced were counted, weighed and visually sexed on PND7 following maternal separations, and given
 170 identifying marks in non-toxic marker for tracking individual pups. Genotyping was carried out using 1mm tail tip
 171 samples taken at PND11/12. Litters with a skewed sex ratio over 80% (n=2, both Poly I:C group) were excluded from
 172 this study, due to evidence suggesting this may affect both maternal and litter behaviour [50, 51]. Additionally, litters
 173 with less than 10 pups (n=1, Saline group) were excluded due to both nutritional and behavioural effects of variable
 174 litter size that may contribute to development [52, 53]. This results in a final n=3 for each treatment group. Pups were
 175 finally weighed again at PND21 prior to sacrifice.

176 2.4 Ultrasonic Vocalisations

177 Individual pups were removed from the home cage at PND7 and placed inside a shallow transport tray containing
 178 fresh bedding, which was then placed inside a custom-built 24cm x 24cm x 26cm Styrofoam chamber with a
 179 microphone suspended through a hole in the chamber lid. Pups were recorded for 5 min (UltraMic 250K, 16 bit)
 180 using Audacity software (Audacity team, version 2.4.2) before labelling with a non-toxic marker for later
 181 identification and return to the home cage. Researcher was blind to offspring sex and genotype at time of recording.

182 Recordings were edited where necessary to remove background noise (Audacity, Audacity team version 2.4.2)
 183 before initial generation of call detection files in machine learning MATLAB program DeepSqueak (MATLAB,
 184 version 2018a; Coffey and Marx, version 2.6.1). Detection files were manually corrected in MATLAB program fork
 185 DeepSqueak Screener (MATLAB, version 2019b; UEFepilepsyAIVI, version 0.9) due to a high false negative rate
 186 in call identification. Calls were required to have a minimum duration of 10 milliseconds, minimum recorded
 187 amplitude of 0.15 decibels, and have the selection area correctly altered to fully include each call contour and exclude
 188 noise. Call statistics were generated from manually corrected detection files in the original DeepSqueak program
 189 (Marx and Coffey, version 2.6.1). Metrics were calculated for each 5-minute offspring recording as described in the
 190 table below:

191 **Table 1. Call metrics extracted from DeepSqueak.**

Metric	Calculation
Number of Calls	Total number of calls in 5 minute recording
Average Call Length	Sum length of each call (ms) / total number of calls
Total Time Calling	Sum of call lengths (ms) in recording
Average Call Frequency	Sum of median call frequency (kHz) per call / total number of calls
Average Change in Frequency	Sum of highest frequency (kHz) – lowest frequency (kHz) per call / total number of calls
Average Call Sinuosity	Sum sinuosity index (Length of call curvature / linear length of call (ms)) per call / total number of calls

2.5 Selfie Digital Droplet PCR

Pups were euthanised at PND21 with carbon dioxide gas immediately prior to decapitation and brain extraction. Whole hippocampi were dissected from coronal brain slices cut in 3D-printed brain blocks and immediately snap frozen on dry ice before storage at -80°C.

Samples were later homogenised and a subsample removed for cell lysis, to extract both DNA and RNA based on the methods described in Thorne *et al* [54]. Gene-specific reverse primers (as shown in the below table) were pre-annealed to RNA at a concentration of 4µM in a 70°C-4°C slow ramp incubation (0.2°C/sec). Lysates were then split into paired samples for reverse transcription, where both samples were treated identically with 0.5µL 0.1M DTT (ThermoFisher Scientific, 18080-044; USA), 0.5µL RNaseOUT (ThermoFisher Scientific, 10777019; USA), 1µM dNTPs, and 2µL 5x reverse transcriptase buffer. One sample was then treated with 0.5µL Maxima H- reverse transcriptase (ThermoFisher Scientific, EP0751; USA) while the other received equal volume reverse transcriptase buffer. Following reverse transcription incubation at 58°C for 30 min and 90°C for 3 min, samples were restriction digested with 1µL of 1U/µL EcoRI (ThermoFisher Scientific, ER0271; USA) diluted in specific EcoRI buffer at 37°C for 1 hour and 65°C for 20 min before dilution to 100µL working volume.

Table 2. Primer pairs designed and used for ddPCR gene expression analysis.

Target	Forward primer (5' > 3')	Reverse primer (5' > 3')	Amplicon length (bp)
Rac1	TGCTGTCAAATACCTGGAGTG	ACAGAGAACGGCTCGGATAG	85
Kal7	TGTATCCAACAACAAGACGCC	TCTGTACCTGAGCATCGTCC	86
Disc1	CTGGACCTTCAGTCTTCACG	GAGGAGACCATGCCACATTC	93

All ddPCR reactions were prepared with 11µL EvaGreen Supermix (Bio-Rad, 186-4034; USA), 0.25µM gene-specific primers, and sample volumes equivalent between paired samples (dependent on template abundance) and nuclease free water to a volume of 23.5µL. 20µL of this mixture was used alongside 70µL droplet generation oil (Bio-Rad, 186-4005; USA) to generate droplets from PCR at a maximum volume of 40µL. Thermocycling conditions were as follows: 95°C for 5 min (initiation only), 95°C for 30s, 64°C for 1 min (for 35 cycles), 4°C for 5min, 90°C for 5 min. All amplification was carried out with a slow ramp rate of 2°C/sec. Droplet fluorescence was then assessed using the QX200 Droplet Reader (Bio-Rad, 1864003; USA) and QuantaSoft Analysis Pro software (Version 1.0.596, Bio-Rad; USA), using a Poisson distribution analysis to quantify template concentration within each sample as copies/µL. Samples containing DNA + cDNA were normalised to their paired DNA-only reference sample using the 2ddCt method. As ddPCR provides an absolute measure of quantification and was normalised to DNA copies of template, mRNA expression for this study is provided as copies/cell [41, 54].

2.6 Statistical Analyses

Changes in maternal temperature and weight were analysed using GraphPad Prism 10 (GraphPad, Boston, MA, USA) by repeated measures 2-way ANOVA, with Bonferroni multiple comparisons and post-hoc correction applied where appropriate [55]. Initial pup weight analyses were also carried out in GraphPad Prism by repeated measures 2-way ANOVA, Welch's t test, or Mann-Whitney U test as appropriate based on Levene's and Shapiro-Wilk tests for unequal variances and normality.

All other offspring data analysis was conducted in RStudio, using the lme4, lmerTest, car, and afex packages for Linear Mixed Modelling (LMM) and Type 1 ANOVA [56-61]. Sequential analysis of variables was carried out as follows: Genotype, Sex, Treatment, Genotype*Sex, Genotype*Treatment, Sex*Treatment, Genotype*Sex*Treatment. To compensate for the limited independence between offspring in the MIA model, LMM was used as a data frame for ANOVA comparisons with litter designated as a fixed random effect. All other ANOVA assumptions were validated. In cases where homoskedasticity and/or normality requirements failed to be met, the robust LMM (rLMM) package robustlmm was utilised as an alternative to differentially weight data and improve

these criteria [62]. ANOVA tables are incompatible with this modelling approach, therefore experiments where one or more targets are assessed by rLMM present main findings as comparable rLMM or LMM summary tables. LMM-based ANOVA results for variables in these data sets that do not require the robust modelling approach are presented alongside full LMM tables in supplementary material, providing additional context for comparison with previous literature. All LMM, rLMM, and ANOVA analyses are reported to 3dp.

Direct comparisons between relevant groups were carried out using the *emmeans* package in RStudio, with pairwise comparisons determining significance markers for all figures presenting offspring data shown in this paper [63]. Given the potential role of offspring sex as a main or interaction effect in this study, all pairwise analyses were initially carried out separated by sex. Sexes were grouped for analysis where no differences were observed by either initial modelling or pairwise analysis. Bonferroni post-hoc adjustments were used for multiple comparisons, accounting for all relevant pairwise comparisons of offspring genotype, sex, and treatment.

3. Results

3.1 Maternal Immune Response

Maternal weight and temperature were assessed as non-invasive indicators of maternal immune activation (Fig. 2A). Across all 10 dams included in this study, no maternal weight loss was observed 24hrs after injection in either treatment group. A significant weight gain was observed in poly I:C treated mothers ($p = 0.0008$) compared to baseline, with saline dams also showing a weight increase that did not reach significance.

As with weight, saline-treated dams do not show significant changes in body temperature (Fig. 2B). Saline dam temperature at 24 hours appears more variable, although this is not significant by Levene's test for unequal variance across groups ($p = 0.11$) and was potentially driven by unanticipated additional stress from handling and cage changing shortly before follow-up temperature measurement [64]. Maternal temperature in the poly I:C-treated dams rose significantly following poly I:C injection at 3-6 hours ($p = 0.0004$) and returned to near-baseline measurement at 24 hrs, suggestive of a short-term immune-driven fever that was absent in saline treated dams during the same 3-6 hour period.

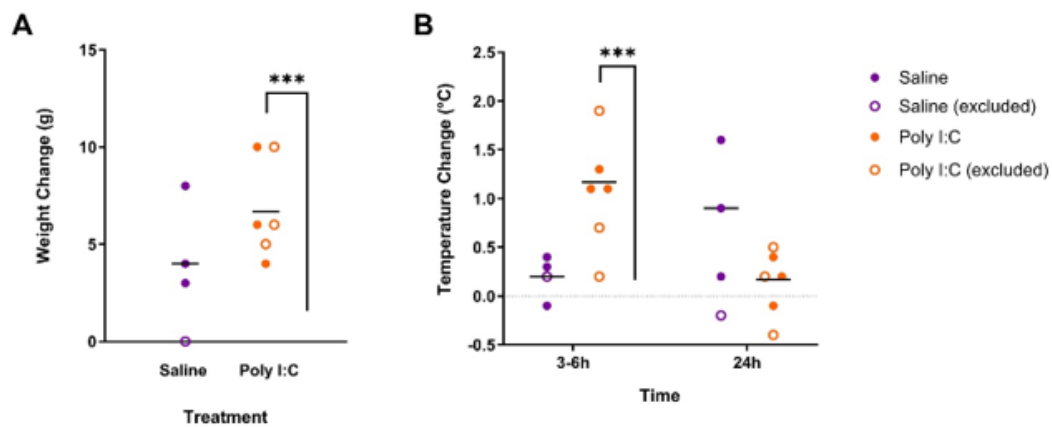


Fig. 2. Individual dam weight and temperature change following injection. (A) Dams administered poly I:C show a significant weight increase 24 hours after injection ($p = 0.0008$), while saline weight gain did not reach significance in this time period ($p = 0.0614$). (B) Dams administered poly I:C also display a significant temperature increase 3-6 hours after injection ($p = 0.0004$) that is not observed at 24 hours ($p > 0.9999$). Saline controls do not display a significant temperature difference at either 3-6 or 24 hours ($p = 0.9221$, $p = 0.0626$ respectively). All dams treated in this study are presented, with excluded dams presented as hollow data points. All p -values are adjusted for Bonferroni's multiple comparisons following repeated measures 2-way ANOVA. *** = $p < 0.001$.

3.2 Offspring Weight

All offspring were weighed on PND7 and PND21, as a commonly utilised and non-invasive measure of physical development following MIA exposure (Fig. 3). Initial analyses of pup weight across treatment by two-way repeated measures ANOVA and multiple comparisons with Bonferroni corrections resulted in a significant reduction in poly I:C-exposed pup weight at PND7 ($t(152) = 2.6814, p = 0.0163$) and PND21 ($t(152) = 6.9516, p < 0.0001$) compared to saline controls. This further extended to overall weight gain (Mann-Whitney U test, $U = 351, p < 0.0001$) but not percentage weight gained by each pup (Welch's t test, $t(64.66) = 0.8663, p = 0.3895$), indicating the respective size of each pup relative to the total cohort was conserved over this time period. This initial result reflects previous MIA studies, with poly I:C exposed offspring body weight being reduced compared to controls [65, 66]. A clear bimodal distribution of weights emerged in the poly I:C-exposed offspring post-PND7 however (Figure 3B-C), with further investigation revealing a significant negative correlation (Fig. 3D) between weight and litter size at weaning. Across both treatment groups, increased litter size corresponded to decreased overall weight gain in pups ($r = -0.9361, r^2 = 0.8762, p = 0.0060$). Percentage weight gain shows a similar negative correlation to litter size, although this does not reach significance ($r = -0.5918, r^2 = 0.3502, p = 0.2159$). To address this significant litter bias, linear mixed modelling (LMM) was used to account for litter as a random fixed effect [67]. Re-assessment of the offspring weight data following this adjustment produced no significant differences in offspring weight or weight gain between treatment groups (Fig. 3). In addition, no significant genotype or sex effects were observed on offspring weight (full LMM analyses are available in supplementary material).

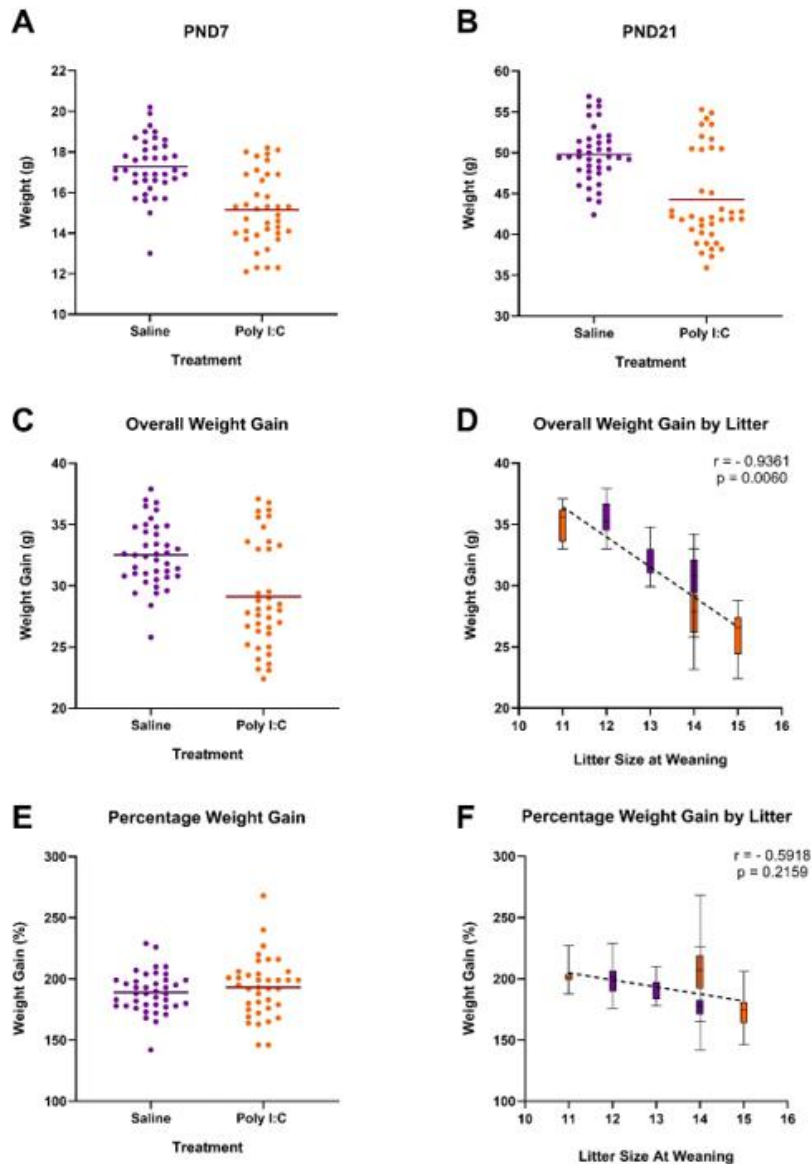


Fig. 3. Offspring weight is primarily driven by litter size. Analysis by linear mixed modelling methods reveals no significant treatment effect on pup weight at PND7 (A, $p = 0.095$), PND21 (B, $p = 0.304$), overall weight gain (C, $p = 0.406$), or percentage weight gain (E, $p = 0.671$). Presentation of weight gain and percentage weight gain (C, E) differentiated by litter (D, F) presents a negative relationship between litter size and pup weight. Overall weight gain is significantly affected by this relationship ($r = -0.9361$, $r^2 = 0.8762$, $p = 0.0060$), with percentage weight gain showing a weaker trend ($r = -0.5918$, $r^2 = 0.3502$, $p = 0.2159$). All offspring and litters are presented in either purple (saline) or orange (poly I:C).

3.3 Ultrasonic Vocalisations following Maternal Separation

As an indication of pup communication during early development, maternal separation tests were used to induce offspring calling. Calls were measured over a 5-minute period and assessed over a range of metrics targeting either

call volume (number, length, and total time calling) or call complexity (frequency, change in frequency, and sinuosity). Analysis of ultrasonic vocalisations emitted over the 5-minute test period by LMM and Type I ANOVA revealed a significant genotype main effect across all tested metrics, except for Average Call Length (Table 1). The strongest genotype effect was observed in Average Call Sinuosity, which also exhibited a significant genotype*treatment interaction ($p = 0.041$). No significant sex or treatment main effects were observed across any metric.

Table 3. Ultrasonic Vocalisations: Summary of Type I ANOVA p -values

	Number of Calls	Average Call Length	Total Time Calling	Average Call Frequency	Average Change in Frequency	Average Call Sinuosity
Genotype (G)	0.009	0.972	0.032	0.003	<0.001	<0.001
Sex (S)	0.612	0.333	0.211	0.874	0.463	0.513
Treatment (T)	0.249	0.705	0.362	0.830	0.148	0.120
G*S	0.154	0.239	0.741	0.576	0.266	0.101
G*T	0.436	0.361	0.824	0.956	0.303	0.043
S*T	0.588	0.457	0.758	0.963	0.822	0.518
G*S*T	0.111	0.662	0.452	0.497	0.669	0.256

Table 3. SERT genotype has a strong effect on USV communication during maternal separation. All p -values in table are presented to 3 decimal places, with significant findings highlighted in green. Full summary statistics are available in supplementary material. LMM with Type I ANOVA of 69 offspring reveals a significant Genotype main effect across all metrics excluding Average Call Length (Number of Calls $F(1,58.75) = 7.249, p = 0.009$; Total Time Calling $F(1,59.03) = 4.831, p = 0.032$; Average Call Frequency $F(1,59.98) = 9.901, p = 0.003$; Average Change in Frequency $F(1, 60.15) = 14.20, p < 0.001$; Average Call Sinuosity $F(1,58.80) = 14.79, p < 0.001$). Average Call Sinuosity also displays a significant genotype*treatment interaction ($F(1,58.89) = 4.298, p = 0.043$). No other main or interaction effects reach statistical significance.

To further explore the relationship between treatment and genotype, pairwise comparisons were carried out for all characteristics of interest. Number of Calls, Average Change in Frequency, and Average Call Sinuosity (Fig. 4A, E, F) all display a significant increase in saline male HET offspring compared to WT controls ($t(59.40) = 3.213, p = 0.026$; $t(60.46) = 3.042, p = 0.042$; $t(59.51) = 4.284, p < 0.001$ respectively), reflecting the overall genotype effect observed previously. Saline male HETs also show a visual increase compared to poly I:C male HET counterparts in these metrics. This shift fails to meet statistical significance after post-hoc corrections, however Average Call Sinuosity in particular falls close to this threshold ($t(16.10) = -3.045, p = 0.092$). Female HET offspring also show a similar trend to male littermates across metrics, although this again does not reach significance.

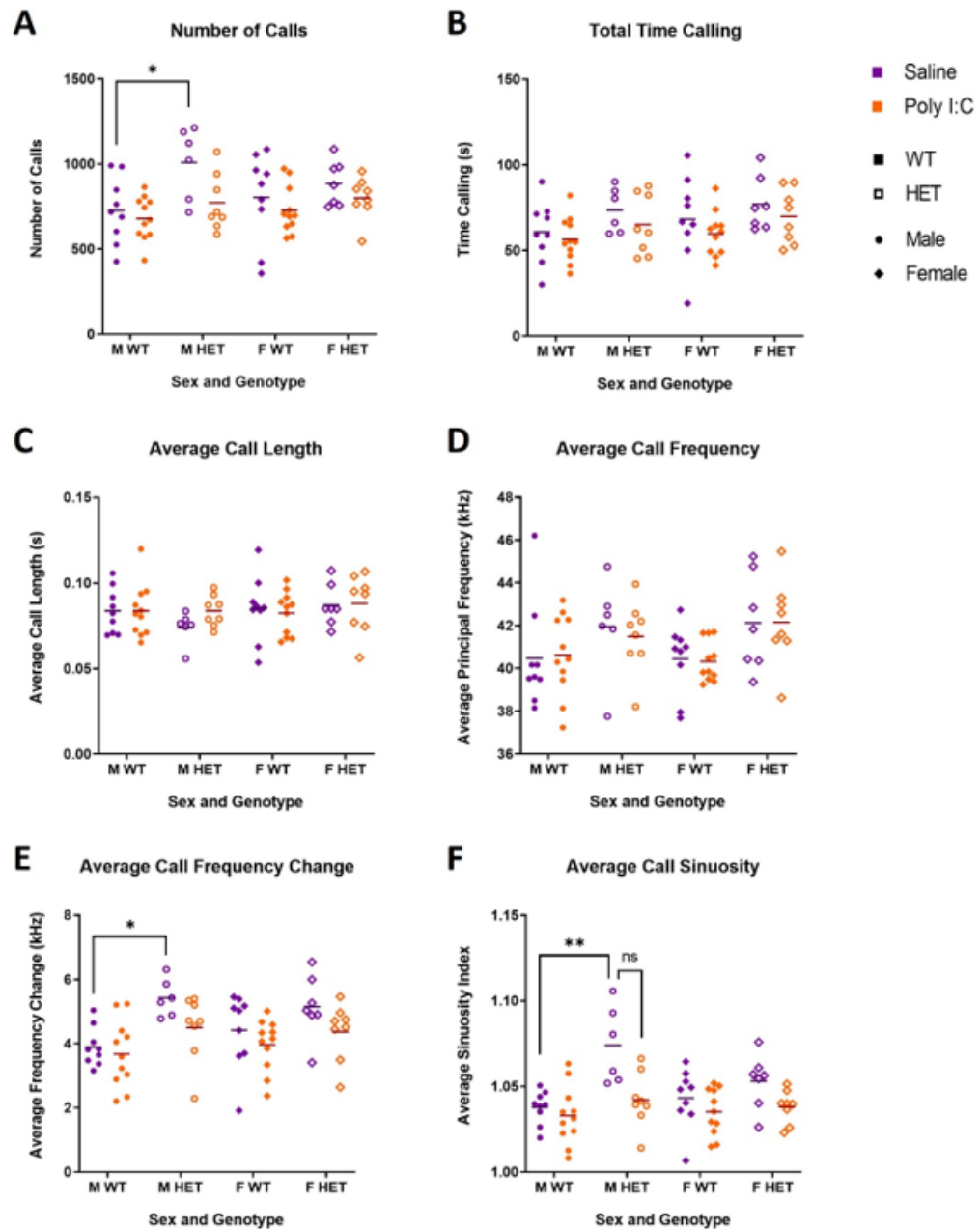


Fig. 4. Offspring USV characteristics presented across genotype, treatment, and sex. (A) Pairwise comparisons by LMM show a significant increase in number of calls from saline male HET compared to WT littermates ($t(59.40) = 3.213, p = 0.025$). (B-D) No significant pairwise differences were observed in time calling, call length, or call frequency following Bonferroni post-hoc corrections. (E-F) Saline male HET offspring also show a significant increase compared to WT counterparts in average frequency change and average call sinuosity ($t(60.46) = 3.042, p = 0.042$; $t(59.51) = 4.284, p < 0.001$ respectively). Comparison of average call sinuosity between saline male HET and poly I:C male HET offspring nears but does not reach significance ($t(16.10) = -3.045, p = 0.092$). Groups are differentiated by point colour (treatment), shading (genotype), and shape (sex). * = $p < 0.05$, ** = $p < 0.005$.

3.4 *Rac1/Kal7/Disc1* Signaling Expression

mRNA expression of *Rac1*, *Kal7*, and *Disc1* was quantified by selfie-ddPCR on PND21 hippocampal tissue as a measure for the potential for synaptic plasticity across this region. Statistical analysis was performed by LMM for *Rac1* and *Kal7*, and rLMM for *Disc1* due to homoskedasticity limitations (full data tables are presented in supplementary material). Overall LMM analysis of *Rac1* and *Kal7* mRNA expression shows a significant treatment effect (*Rac1* $t(23) = -2.886$, $p = 0.008$; *Kal7* $t(23) = -3.119$, $p = 0.005$). *Rac1* also shows a significant genotype*treatment interaction ($t(23) = 2.537$, $p = 0.018$), whereas the genotype*treatment interaction for *Kal7* mRNA expression nears but does not reach significance ($t(23) = 1.879$, $p = 0.073$). This difference may be driven by a stronger genotype main effect in *Kal7*, which also nears but does not reach significance ($t(23) = -1.862$, $p = 0.076$). *Disc1* by comparison did not present treatment or genotype main or interaction effects, however, was the only target to exhibit a main sex effect ($t = -2.379$, $p = 0.017$). No significant sex interaction effects were observed, nor any significant pairwise comparisons in *Disc1* expression when comparing across sex. As a result, sex is grouped for pairwise comparisons to improve statistical power for treatment and genotype effects, however still visually identifiable by symbol (Fig. 5).

Table 4. Signalingosome mRNA Expression: Summary of LMM/rLMM p -values

	Rac1	Kal7	Disc1
Genotype (G)	0.547	0.076	0.096
Sex (S)	0.979	0.995	0.017
Treatment (T)	0.008	0.005	0.325
G*S	0.377	0.230	0.174
G*T	0.018	0.073	0.152
S*T	0.463	0.815	0.821
G*S*T	0.341	0.224	0.836

Table 4. Treatment and genotype play a role in *Rac1/Kal7/Disc1* signalingosome mRNA expression. Modelling of *Rac1* mRNA expression by LMM resulted in a significant treatment effect ($t(23) = -2.886$, $p = 0.008$) and genotype*treatment interaction ($t(23) = 2.537$, $p = 0.018$). LMM for *Kal7* also shows a significant treatment effect ($t(23) = -3.119$, $p = 0.005$), while *Disc1* modelling by rLMM presents only a significant sex effect ($t = -2.379$, $p = 0.017$). A representative subsample of 31 offspring across all litters was used. As previously, all p -values in table are presented to 3 decimal places with significant findings highlighted in green. Full summary statistics (including additional ANOVA modelling for *Rac1/Kal7*) can be found in supplementary material.

Further comparisons show both *Rac1* and *Kal7* mRNA expression was significantly reduced in WT offspring exposed to poly I:C compared to saline controls (*Rac1* $t(23) = -3.914$, $p = 0.003$; *Kal7* $t(23) = -3.607$, $p = 0.006$). This pairwise effect was not reflected in HET offspring, which did not differ significantly by treatment nor compared to respective WT controls. Interestingly, no significant pairwise comparisons were observed for *Disc1*, irrespective of dividing offspring by sex.

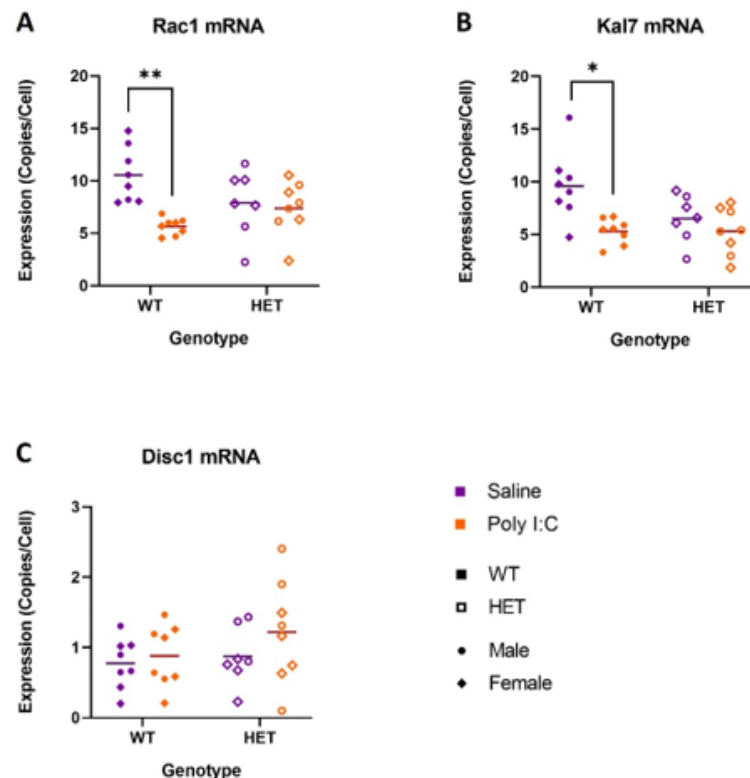


Fig. 5. Rac1/Kal7/Disc1 signalosome mRNA expression assessed by selfie-ddPCR. (A) Rac1 mRNA expression in WT offspring is significantly reduced in poly I:C exposed pups following Bonferroni post-hoc corrections ($t(23) = -3.914$, $p = 0.003$), as is Kal7 (B) ($t(23) = -3.607$, $p = 0.006$). (C) No significant pairwise differences were observed in Disc1 mRNA expression. Offspring are again differentiated by point colour (treatment), shading (genotype), and shape (sex), with sex grouped in columns for pairwise comparisons. * = $p < 0.05$, ** = $p < 0.005$.

4. Discussion

4.1 MIA Validation and Reproducibility

Poly I:C administration is a well-established model for investigating the role of MIA in NDDs, however there are large methodological variations between studies. This variation includes the dosage, reagent batch, timing, and number of immune challenges; the strain of rat and characterisation (or lack thereof) of the maternal immune response; and statistical analyses varying in their independence assumptions of littermates [67-69]. These areas of variation are considered a strong source of confounding error in the current literature. Importantly, studies that initially seem equivalent but with conflicting findings may be subject to methodological differences not distinguishable to the reader by the published information alone [69].

To investigate immune response using a non-invasive and low-cost approach in the present study, maternal weight and temperature were used as markers of immune activation following poly I:C or vehicle administration. Previous literature indicates poly I:C induces a stalling or reduction in dam weight; however, this was not seen in the present study [66]. Differences in rat strain, dam age, and variation in poly I:C molecular weight/dosage may all contribute to this effect [68]. Our study did however demonstrate an expected short-lived increase in dam temperature following poly I:C administration (Fig. 2), suggestive of a maternal fever accompanying an induced immune response.

369 While temperature is a surrogate measure of immune response and could be affected by other factors such as stress,
370 more invasive measures such as blood sampling for cytokine assessment may in turn cause additional stress and affect
371 development in both groups [69].

372 An additional concern of the poly I:C driven MIA model is exemplified in the evident litter bias revealed during
373 pup weight analysis (Fig. 3). Offspring produced in a single litter are exposed to the same maternal bloodstream and
374 overall uterine environment, which additionally prevents them from being assigned independently to treatment
375 conditions. Each pup, however, differs in sex, genotype, and placement in the uterine horn, all of which may influence
376 neurodevelopment in unique ways and thus prevent offspring from being regarded as simple technical replicates.
377 Approaches to address the limitation of litter-relatedness include accounting for litter statistically (by using models
378 which allow for limited independence) or developing experimental designs to mitigate the effects of litter bias (such
379 as controlling the number of animals used per litter, culling to normalise litter size, or cross-fostering offspring) [67,
380 69]. In this study, preliminary analysis of offspring weight by two-way repeated measures ANOVA indicated a
381 significant treatment effect in almost all weight comparisons (See 3.2). This reflects findings observed in MIA
382 literature, with poly I:C resulting in reduced offspring weights both prenatally and postnatally [5, 70]. Further analysis,
383 however, revealed litter size had a highly significant correlation with offspring weight ($p = 0.0060$), leading to the
384 decision to account for litter statistically as a random effect within linear mixed effects modelling. Analysis with this
385 model resulted in no significant treatment effect on pup weight, which is less common yet has been observed
386 elsewhere where litter effects are investigated [7].

387 4.2 Behavioural Assessment

388 To assess communication alterations in this study, ultrasonic vocalisations from PND7 pups were recorded for
389 5-minute maternal separation periods. Ultrasonic vocalisations emitted by offspring in this study first revealed a large
390 genotype-associated difference across metrics, including those associated with the volume of calling (number of calls,
391 time calling) and the complexity of calls (change in frequency, sinuosity) (Table 3). Both an increased number of
392 calls and increased call complexity is considered to be more urgent and attention-demanding, producing a more
393 successful outcome with increased maternal responsiveness to offspring. As the maternal separation paradigm uses acute
394 separation stress to induce calling behaviour in pups, the strong genotype effect seen here could reflect an increased
395 stress response from HET offspring to an acute environmental stressor such as isolation. This would support the
396 'reactive' model of SERT genotype, with a stronger response to an environmental trigger observed in offspring with
397 a reduction in serotonin transporter expression [32].

398 Using pairwise comparisons to explore this genotype effect further, it becomes evident that male offspring
399 display this difference to a greater extent than females. While female offspring reflect similar trends to males across
400 call metrics, it is only male saline-exposed HET offspring which show significantly higher call number, call frequency
401 change, and call sinuosity compared to WT counterparts (Fig. 4). Sex differences in diagnostic rates and
402 symptomology are well-described in NDD research, particularly cases such as ASD where males are approximately
403 four times as likely to be diagnosed [46]. Males displaying more extreme calling patterns in this study than females
404 reflect this trend, albeit limited to the relative simplicity of animal model compared to human behaviour.

405 Interestingly, poly I:C exposure in isolation did not produce a significant main treatment effect on USVs in any
406 metric. Treatment effect may have been weakened in the LMM analysis by low statistical power, given the limited
407 number of independent litters available within each treatment group ($n = 3$). Previous literature has also shown
408 variables such as rat strain and testing laboratory may impact MIA outcomes and have reduced treatment effect here
409 [71, 72]. It is also possible that communication differences were not captured in this snapshot of development, but
410 may be observed with a wider developmental window or by encompassing additional tests. For example, a paradigm
411 that does not rely on stress for USV generation such as social play USVs during the adolescent period would be
412 particularly interesting, considering the above hypothesis that genotype effect is driven by separation stress rather
413 than inherent communication differences.

414 While no treatment main effects were observed, a significant genotype*treatment interaction was observed in

415 Average Sinuosity. This is again exemplified by an increase in saline-treated male HET offspring: pairwise
 416 comparisons show call sinuosity is significantly decreased in male HETs following poly I:C exposure in comparison
 417 to saline controls, but this relationship is not seen in WT littermates (Fig. 4). This suggests an interaction between
 418 MIA and SERT as NDD risk factors as initially hypothesised, however the nature of this interaction is perplexing.
 419 Rather than a synergistic interaction, the relationship seen in male HET offspring indicates that poly I:C ameliorates
 420 the effect of genotype, reducing call sinuosity to a WT equivalent. The lack of treatment response seen in WT call
 421 sinuosity may be a floor effect, as a sinuosity index of 1 would indicate a completely flat call, so individual values
 422 less than 1 are not possible. Given that average sinuosity is calculated over 500-1000 calls however, and call sinuosity
 423 reflects the visual genotype and treatment trends seen across multiple metrics, a floor effect is unlikely to be the main
 424 driver of this USV pattern. If HET offspring are more sensitive to the isolation stress of maternal separation, this
 425 interaction may instead suggest a situational effect where poly I:C exposure reduces communication behaviour under
 426 intense but not mild stress.

427 4.3 Signalosome Expression

428 Expression of the Rac1/Kal7/Disc1 signalosome by selfie-ddPCR was utilised as a molecular metric to
 429 investigate the effects of poly I:C and SERT genotype on synaptic plasticity, in relation to neurodevelopmental
 430 disorders. Total Disc1 mRNA expression in this model however did not display any treatment or genotype-associated
 431 changes (Fig. 5), although an overall sex effect was observed. Disc1 was typically increased in male offspring
 432 compared to female, however no pairwise comparisons of relevant groups resulted in significance alone. This may
 433 be due to low statistical power, with more complex interactions across sex/treatment/genotype groups being limited
 434 by sample size. Interestingly, sex-specific outcomes are often associated with Disc1 splice variants and disorder-
 435 associated alleles; expression in this study captures overall expression by targeting a conserved exon, however does
 436 not assess the underlying ratios of splice variants that may vary in their contribution to the overall level of expression
 437 [12, 73, 74].

438 Both Rac1 and Kal7 show a significant decrease in mRNA expression in the poly I:C exposed cohort (Fig. 5),
 439 with significant treatment effects observed by LMM (Table 4). Additionally, Rac1 mRNA expression exhibits a
 440 significant interaction between Genotype and Treatment, with Kal7 presenting a similar but non-significant result.
 441 Pairwise comparisons support this relationship by highlighting a treatment effect limited to WT offspring only:
 442 heterozygotes do not exhibit any treatment effect however show a visual trend toward lower expression overall
 443 compared to WT counterparts.

444 The Rac1/Kal7 expression decrease exemplified in WT offspring reflects previous research indicating activity
 445 of Rac1 and Kal7 is an important factor underlying NDD-associated changes in neuroplasticity [43, 75]. Our initial
 446 hypothesis of a synergistic effect between MIA and SERT HET genotype was again unsupported by our findings;
 447 expression in heterozygote offspring appears uninfluenced by MIA exposure, and only WT offspring show a clear
 448 treatment effect. This contrast in treatment outcome between genotypes, alongside the visual yet non-significant
 449 decrease in HET expression overall compared to WT, rather suggests another complex interaction. A floor effect may
 450 explain this relationship, where both genotype and treatment tend to reduce mRNA expression but only to a required
 451 baseline capacity. As 5-HT signalling is known to regulate plasticity through many mechanisms, SERT HET offspring
 452 may also be utilising a compensatory pathway that results in reduced Rac1/Kal7 expression irrespective of MIA
 453 exposure [76, 77]. It should be noted that decreased mRNA expression in MIA-exposed WT offspring does not
 454 necessitate reduced activity however does suggest a reduced capacity for Rac1-dependent plasticity within the
 455 hippocampus.

456 5. Conclusions

457 This study overall supports both SERT genotype and MIA exposure in-utero have an important influence on
 458 neurodevelopment, with initial evidence to indicate these factors may interact to contribute to dysregulation in NDD
 459 associated markers. It was initially hypothesised that these NDD risk factors may produce a synergistic effect, with

poly I:C-exposed HET offspring displaying the strongest behavioural and molecular disturbances compared to controls. Furthermore, males were predicted to display stronger effects than females, given the discrepancy between sexes in ASD and other NDDs. Examples in both behavioural and molecular analysis do indicate an interaction between genotype and treatment, however neither show a synergistic relationship. Instead, SERT genotypes appear to respond to MIA differentially. Male SERT HET offspring show strong communication differences compared to WT in saline offspring which are ameliorated following poly I:C exposure, however appear to show resilience to any treatment effect in *Rac1/Kal7* mRNA expression. WT offspring in comparison show no altered USV communication between treatments, however significantly lowered *Rac1/Kal7* expression following MIA.

Overall, while this research suggests interactions between these risk factors do occur, the relationship between these factors appears to be more nuanced than overall sensitivity or NDD incidence. Given that USV and expression analyses present treatment*genotype interactions with vastly different patterns, we hypothesise MIA exposure in individuals with a low-expression SERT genotype may influence the presence or severity of NDD-associated symptomology. As mentioned previously, individuals diagnosed with ASD and other NDDs display unique combinations of symptoms including both communicative and cognitive differences. This study suggests that the effect of SERT genotype (and likely the activity of other 5-HT-associated genes) may be influential in NDD symptomology, as opposed to simply NDD incidence, and may contribute to symptom heterogeneity.

Abbreviations

5-HT, serotonin; ASD, autism spectrum disorder; ddPCR, digital droplet polymerase chain reaction; GD, gestational day; HET, heterozygous; HOM, homozygous knockout; LMM, linear mixed model; LTP, long term potentiation; MIA, maternal immune activation; NDD, neurodevelopmental disorder; PND, postnatal day; rLMM, robust linear mixed model; SCZ, schizophrenia; SERT, serotonin transporter; USV, ultrasonic vocalization; WT, wildtype.

Supplementary Material

[supplementary material attached as separate file. Contains full statistical tables for all LMM/rLMM analyses of results sections 3.2-3.4.]

Availability of Data and Materials

All data is available upon request to the corresponding author.

Author Contributions

AL, BE, and DD conceptualized research. AL performed all research and analyzed data. AL wrote manuscript, with editorial advice from BE and DD. All authors read and approved the final manuscript.

Ethics Approval and Consent to Participate

This research protocol was approved by the Animal Ethics Committee of Te Herenga Waka - Victoria University of Wellington (Ethics Approval Number: 25848).

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500 **Conflict of Interest**

501 The author(s) declare no conflict of interest.

502 **References and Notes**

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Appendix B

This Appendix includes the LC-MS/MS Acquisition Methods Report, detailing the settings used for data collected in Chapter 4.

Acquisition Method Report



Acquisition Method Info

Method Name 5-HT v3 msms.m

Method Path D:\MassHunter\Methods\Alex L\5-HT v3 msms.m

Method Description

Device List

Sampler
Binary Pump
Column Comp.
VWD
Q-TOF

TOF/Q-TOF Mass Spectrometer

Component Name	MS Q-TOF	Component Model	G6530A
Ion Source	AJS ESI	Stop Time (min)	No Limit/As Pump
Can wait for temp.	Enable	Fast Polarity	False
MS Abs. threshold	200	MS Rel. threshold(%)	0.010
MS/MS Abs. threshold	5	MS/MS Rel. threshold(%)	0.010

Time Segments

Time Segment #	Start Time (min)	Diverter Valve State	Storage Mode	Ion Mode
1	0	Waste	Both	AJS ESI
2	0.5	MS	Both	AJS ESI
3	11	Waste	Both	AJS ESI

Time Segment 1

Acquisition Mode MS1

Min Range (m/z)	100
Max Range (m/z)	1000
Scan Rate (spectra/sec)	1.00

Instrument Parameters

Parameter	Value
Gas Temp (°C)	275
Gas Flow (l/min)	5
Nebulizer (psig)	30
SheathGasTemp	275
SheathGasFlow	8

Scan Segments

Scan Seg #	Ion Polarity	Collision Energy
1	Positive	0

Scan Segment 1

Scan Source Parameters

Parameter	Value
VCap	2750
Nozzle Voltage (V)	0
Fragmentor	130
Skimmer1	65
OctopoleRFPeak	750

Acquisition Method Report



Agilent Technologies

Time Segment 2

Acquisition Mode AutoMS2

MS Min Range (m/z) 100
MS Max Range (m/z) 1000
MS Scan Rate (spectra/sec) 2.00
MS/MS Min Range (m/z) 100
MS/MS Max Range (m/z) 1000
MS/MS Scan Rate (spectra/sec) 2.00
Isolation Width MS/MS Medium (~4 amu)
Decision Engine Adv
Use Fixed Collision Energies 10.00,20.00,40.00

Auto MS/MS Preferred/Exclude Table

Mass	Delta Mass (ppm)	Charge	Type	Retention Time (min)	Delta Ret. Time (min)	Isolation Width	Collision Energy
385.15	100	1	Preferred	5	9.9	Medium (~4 amu)	
922.004	100	1	Exclude	5	9.9	Medium (~4 amu)	

Precursor Selection

Max Precursors Per Cycle 2
Threshold (Abs) 10000
Threshold (Rel)(%) 0.010
Precursor abundance based scan speed No
Purity Stringency (%) 100.000
Purity Cutoff (%) 30.000
Isotope Model Common
Active exclusion enabled No
Sort precursors By abundance only

Charge State Preference

Selected
Charges
1
2
Unk

Instrument Parameters

Parameter	Value
Gas Temp (°C)	275
Gas Flow (l/min)	5
Nebulizer (psig)	30
SheathGasTemp	275
SheathGasFlow	8

Scan Segments

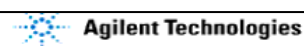
Scan Seg # Ion Polarity
1 Positive

Scan Segment 1

Scan Source Parameters

Parameter	Value
VCap	2750
Nozzle Voltage (V)	0
Fragmentor	130
Skimmer1	65
OctopoleRFPeak	750

Acquisition Method Report



Time Segment 3

Acquisition Mode MS1

Min Range (m/z)	100
Max Range (m/z)	1000
Scan Rate (spectra/sec)	2.00

Instrument Parameters

Parameter	Value
Gas Temp (°C)	275
Gas Flow (l/min)	5
Nebulizer (psig)	30
SheathGasTemp	275
SheathGasFlow	8

Scan Segments

Scan Seg #	Ion Polarity	Collision Energy
1	Positive	0

Scan Segment 1

Scan Source Parameters

Parameter	Value
VCap	2750
Nozzle Voltage (V)	0
Fragmentor	130
Skimmer1	65
OctopoleRFPeak	750

ReferenceMasses

Ref Mass Enabled	Enabled
Use Bottle A RefNebulizer	True
Ref Nebulizer (psig)	

AutoRecalibration

Average Scans	1
Detection Window (ppm)	100
Min Height (counts)	1000

Reference Masses

<Positive>
121.05087300
922.00979800

Chromatograms

Chrom Type	Label	Offset	Y-Range
TIC	TIC	15	300000000

Name: **Sampler**

Model: **G1329B**

Auxiliary

Draw Speed	200 µL/min
Eject Speed	200 µL/min
Draw Position Offset	2.0 mm

Injection

Injection Mode	Injection with needle wash
Injection Volume	10.00 µL
Needle Wash	

Needle Wash Location	Wash Vial
Wash Location	Vial 100

High throughput

Overlapped Injection

Enable Overlapped Injection	No
-----------------------------	----

Stop Time

Stoptime Mode	As pump/No limit
---------------	------------------

Post Time

Posttime Mode	Off
---------------	-----

Report generation date: 12/16/2025 10:42:21 AM

Page 3 of 5

Acquisition Method Report



Name: Binary Pump

Model: G1312C

Flow 0.450 mL/min
Low Pressure Limit 0.00 bar
High Pressure Limit 400.00 bar
Maximum Flow Gradient 100.000 mL/min²

Stroke A

Automatic Stroke Calculation A Yes

Stroke B

Automatic Stroke Calculation B Yes

Compress A

Compressibility Mode A Compressibility Value Set
Compressibility A 50 10e-6/bar

Compress B

Compressibility Mode B Compressibility Value Set
Compressibility B 115 10e-6/bar

Stop Time

Stoptime Mode Time set
Stoptime 11.70 min

Post Time

Posttime Mode Time set
Posttime 3.50 min

Solvent Composition

	Channel	Name 1	Used	Percent
1	A		Yes	70.0 %
2	B		Yes	30.0 %

Timetable

	Time	A	B	Flow	Pressure
1	0.50 min	70.0 %	30.0 %	0.450 mL/min	400.00 bar
2	9.00 min	2.5 %	97.5 %	0.450 mL/min	400.00 bar
3	10.00 min	2.5 %	97.5 %	0.450 mL/min	400.00 bar
4	10.50 min	50.0 %	50.0 %	0.450 mL/min	400.00 bar
5	10.70 min	2.5 %	97.5 %	0.500 mL/min	400.00 bar
6	11.50 min	2.5 %	97.5 %	0.500 mL/min	400.00 bar
7	11.70 min	70.0 %	30.0 %	0.450 mL/min	400.00 bar

Name: Column Comp.

Model: G1316A

Left Temperature Control

Temperature Control Mode Temperature Set
Temperature 36.0 °C

Enable Analysis Left Temperature

Enable Analysis Left Temperature On No

Right Temperature Control

Right temperature Control Mode Temperature Set
Right temperature 36.0 °C

Enable Analysis Right Temperature

Enable Analysis Right Temperature On No

Stop Time

Stoptime Mode As pump/injector

Post Time

Posttime Mode Off

Timetable

Acquisition Method Report



Agilent Technologies

Name: VWD

Model: G1314B

Signal Peakwidth	> 0.1 min (2 s resp. time) (3.43 Hz)
Analog Output Source Channel	1
Signal Polarity	Positive (+)
Lamp on required for acquisition	No
Acquire Signal without Reference	No
Acquire Reference only	No

Analog Output

Analog Zero Offset	5 %
Analog Attenuation	1000 mAU

Signals

Signal

Acquire Signal	No
Signal ID	Signal A
Signal Wavelength	250 nm

Prepare Mode

Margin for negative Absorbance	100 mAU
--------------------------------	---------

Autobalance

Autobalance Prerun	Yes
Autobalance Postrun	No

Scan Variables

Scan Range From	190 nm
Scan Range To	400 nm
Scan Range Step	2 nm

Stoptime

Stoptime Mode	As pump/injector
---------------	------------------

Posttime

Posttime Mode	Off
---------------	-----

Timetable

Appendix C

This Appendix includes the MIA ARRIVE methods reporting table fully completed for the MIA model generated for this thesis [155].

Maternal Immune Activation Model Reporting Guidelines Checklist

ARRIVE Reporting Guideline & Recommendation	Arrive Item	MIA Model Specific Reporting Recommendation <i>Please complete this chart for each point outlined below. If not applicable, write N/A</i>
Study design ➤ Overview of immune activation issues For each experiment, give brief details of the study design including: a. The number of experimental and control groups. b. Any steps taken to minimize the effects of subjective bias when allocating animals to treatment (e.g. randomization procedure) and when assessing results (e.g. if done, describe who was blinded and when). c. The experimental unit (e.g. a single animal, group or cage of animals). A time-line diagram or flow chart can be useful to illustrate how complex study designs were carried out.	6	MIA Specific Reporting: a. General need for improved reporting in MIA model methods + reporting pilot data Details on pilot data: No specific pilot data for this study is available. Initial dosage of 5mg/kg based on experience/previous studies in laboratory, with first animals treated providing positive results by validation metrics. To minimize animal use and reagent variability, continued with same treatment batch and dosage for full study.
Experimental procedures ➤ Compounds ➤ Validation measures For each experiment and each experimental group, including controls, provide precise details of all procedures carried out. For example: a. How (e.g. drug formulation and dose, site and route of administration, anaesthesia and analgesia used [including monitoring], surgical procedure, method of euthanasia). Provide details of any specialist equipment used, including supplier(s). b. When (e.g. time of day). c. Where (e.g. home cage, laboratory, water maze). d. Why (e.g. rationale for choice of specific anaesthetic, route of administration, drug dose used).	7	Provide details of: a. Compounds – source, vehicle, preparation/storage, administration route, volume administered, whether anesthetics were used at time of immune challenge. - Name of compound: Polyinosinic:polycytidylic acid - Catalogue number: P1530-100MG (Sigma Aldrich) - Lot number: 12190304, Batch #0000067639 - Vehicle control used: 0.9% saline solution (one batch for all treatments) - Route of administration: Subcutaneous injection - Volume administered: 200-500µL per animal (5mg/mL, 1mL/kg) - Storage conditions: Fresh dilution per cohort, stored <5 days at -4°C - Anesthetic (type, dose, duration) used: not applicable (N/A) b. Housing variables at injection - temperature of room at injection time, cage change at time of injection or not - Light cycle of animal housing room: Reverse cycle (dark from 7am-7pm) - Time of day of injection: approx. 10am - Room temperature at injection time: 21°C ± 2°C - Did a cage change occur at time of injection: NO ☐

Kentner AC, Bilbo AD, Brown AS, Hsiao EY, McAllister AK, Meyer U, Pearce BD, Pletnikov MV, Yonken RH, Bauman MD. (2018). Maternal immune activation: reporting guidelines to improve the rigor, reproducibility, and transparency of the model. *Neuropsychopharmacology*, <https://doi.org/10.1038/s41386-018-0185-7>.

		c. Validation of immune activation – behavior, physiological indices and/or cytokine data, including pilot dosing data - Method used to verify immune activation: Cytokine measurement by maternal saphenous vein blood sampling (every dam in study tested, by same handler, to minimize variability in stress) d. Validation of gestational timing – vaginal plug, estrous cycle, weight gain - Method of validating gestational timing: Timed mating protocol (naïve female, paired <5 days or until vaginal plug identified). Identification of vaginal plug designated as gestational day (GD)0.5. Additional comments: N/A
Experimental animals ➤Species/strain/vendor a. Provide details of the animals used, including species, strain, sex, developmental stage (e.g. mean or median age plus age range) and weight (e.g. mean or median weight plus weight range). b. Provide further relevant information such as the source of animals, international strain nomenclature, genetic modification status (e.g. knock-out or transgenic), genotype, health/immune status, drug or test naïve, previous procedures, etc.	8	Provide details of: a. Species – considerations for appropriate species (mouse, rat, non human primate, other) - Species: Rat (Rattus norvegicus) b. Strain – variability in strain can influence model - Strain: Sprague-Dawley Rat c. Maternal/Offspring Physiological Variables at time of immune challenge – age, body weight - Maternal Age at challenge: Postnatal day (PD)65-100 - Maternal Body weight: 200-500 grams - Offspring Age at challenge: GD12.5 - Offspring Sex: Both males and females tested - Offspring Body weight: N/A d. Vendor – even within the same strain, vendor can influence endpoints - Vendor: Victoria University of Wellington - Location of Vendor: Wellington, New Zealand - Room/area where animals originated from: VUW Small Animal Facility

		Additional Comments: N/A
Housing and husbandry ➤Cage, ventilation, bedding, enrichment Provide details of: a. Housing (type of facility e.g. specific pathogen free [SPF]; type of cage or housing; bedding material; number of cage companions; tank shape and material etc. for fish). b. Husbandry conditions (e.g. breeding program, light/dark cycle, temperature, quality of water etc for fish, type of food, access to food and water, environmental enrichment). c. Welfare-related assessments and interventions that were carried out prior to, during, or after the experiment.	9	Provide details of: a. Caging systems - At breeding - Material of cage: Polysulfone (PSU) - Cage dimensions: 38.9 cm L x 56.9 cm W (front) x 26.2 cm H. Floor area: 1040 cm² - After parturition - Material of cage: polysulfone (PSU) - Cage dimensions: 38.9 cm L x 56.9 cm W (front) x 26.2 cm H. Floor area: 1040 cm² - At weaning - Material of cage: N/A - Cage dimensions: N/A b. Animal Holding room - Temperature in room: 21°C ± 2°C - Humidity in room: 55-60% - Ventilation system: Individually ventilated caging (IVC) system - Specific pathogen free [SPF]: NO - Are males & females housed in the same or separate rooms: Housed in same room after weaning c. Bedding exchanges/bedding type - Type of cage bedding used: SAFE Select Rodent Bedding (aspen woodflake) - Frequency of cage changes per week - during gestation: Once per week - during neonatal period: N/A - following weaning: N/A d. Breeding - bred on site or timed pregnant, how many different sires (are the same fathers breeding with both experimental and control dams) Breeding location: On site, VUW SAF

		○ Gestational age at shipping: N/A ○ Biological age of dams (if not listed in Section 8c): - ○ Number of Dams bred: 38 ○ How many times have dams been mated previously: None ○ How many times did the dams mate and not become pregnant: None ○ Are the dams primiparous or multiparous? All dams are primiparous v ○ What was the frequency of maternal handling during the gestational/neonatal period (e.g. cage cleanings, weighing, blood collection manipulations): Cage change once per week, maternal blood ○ Biological age of sires: PD80-150 ○ Number of sires bred: 16 ○ How many times have sires been mated previously: >1 (taken from ○ How many times did the sires mate successfully (e.g. mating resulted in pregnancy, full term birth): 1-3 successful pregnancies (in this project) ○ If bred previously, what was the interval between mating times: ○ Are sires matched to experimental and control dams: YES v ○ Describe the mating design (1:1, 1:2 etc): 1:2 target (one sire 1:3) e. Social enrichment – number of cage companions ○ Number of cage companions prior to breeding: 2-3 ○ Gestational age when dam separated for parturition: GD0 (seperated) ○ Number of cage companions at weaning: N/A f. Physical enrichment – describe enrichment devices, and when enrichment is in the cage (removed when pups born? Or present throughout study), does the enrichment type change? How frequently? ○ Describe what type of enrichment devices (and how many) are included in cage/housing room: From GD0.5 (post timed mating protocol), cages always contained: 1 red/blue transparent plastic tunnel (allowing shelter for animals but maintaining visibility for researchers) 1 wooden block (for chewing etc) 1 clear mezzanine platform (provides climbing platform and additional spage in cage).
--	--	--

		- Does enrichment type/access change across study? NO - If so, when does enrichment type/access change (e.g. enrichment removed prior to parturition and replaced in late neonatal period): N/A Additional Comments: Breeding: mating protocol was designed for 2:1 ratio for mothers: fathers (with mothers split across treatments), however due to a number of failed pairings this is not always the case (some fathers sired one litter, some 3). Enrichment: During timed mating protocol, plastic tunnel and mezzanine are removed (these may prevent plugs over the 24 hour breeding period falling through metal grate and preserved for identification). Consistent access to this enrichment is available for animals at all other stages, as standard practice. As this is before treatment and pregnancy, is not considered applicable to above question (but potentially still of note to other researchers).
Sample size ➤ Litter versus offspring a. Specify the total number of animals used in each experiment, and the number of animals in each experimental group. b. Explain how the number of animals was arrived at. Provide details of any sample size calculation used. c. Indicate the number of independent replications of each experiment, if relevant.	10	Provide details of: a. Maternal N vs offspring N - What is the total number of dams/litters included in the study: 27 - What is the total number of offspring per litter included the study: 215 b. Litter size and sex distribution - What size was each litter maintained at: N/A - What age did culling take place at: GD14.5 or GD18.5 - How many males and females were maintained in each litter: N/A c. Cross fostering - Did cross fostering occur: NO - If so, at what age did cross fostering occur: N/A Additional Comments: N/A

Allocating animals to experimental groups a. Give full details of how animals were allocated to experimental groups, including randomization or matching if done. b. Describe the order in which the animals in the different experimental groups were treated and assessed.	11	a. How many offspring per litter were used in each measure: approx. 2 per assay (1 b. Randomization/Matching procedures - What procedures were used to assign animals to groups: 1 male and 1 female per litter were used where possible. Litters were selected in random order, matched by treatment at every step (e.g. during sample preparation, both treatment types were included to minimise technical variation between days/batches of reagent). c. Sex as a biological variable (behavioral and physiological outcomes) - Were both males and females evaluated in each behavioral and physiological outcome: YES v Additional Comments: N/A
Experimental outcomes ➤ Behavioral testing ➤ Physiological endpoints Clearly define the primary and secondary experimental outcomes assessed (e.g. cell death, molecular markers, behavioral changes).	12	a. Maternal behavior and pup interactions - If maternal care was evaluated, were there differences following immunogen challenge (if so, please briefly describe): N/A b. Age(s) of offspring at behavioral testing/physiological evaluation endpoints: GD14.5 and GD18.5 c. Order of testing (e.g. behavioral test order) - Were animals evaluated in a counter-balanced order in terms of: <i>presentation of tests to each animal:</i> NO v <i>order of experimental/control groups run through each test:</i> NO v - What was the inter-test interval if a single animal underwent a battery of tests: N/A - no behavioural tests were used

		Additional Comments: N/A
Statistical methods a. Provide details of the statistical methods used for each analysis. b. Specify the unit of analysis for each dataset (e.g. single animal, group of animals, single neuron). c. Describe any methods used to assess whether the data met the assumptions of the statistical approach.	13	a. Unit of analysis for each data set Is the unit (n) of each analysis based on number of litters, or number of animals used per group: Validation data (e.g. maternal responses, whole litter analyses) will utilise Litter as the primary 'n', with linear mixed modelling to account for limited independence between pups in a litter (but still using each data point). All analyses following this using pup samples (e.g. mRNA expression, methylation analyses) consider each animal as independent, given that litter effect is being managed by limiting pups per litter.
Other Disclosures		Please make note of any other extraneous variables that you would like to report (e.g. fire alarms, construction, temporary relocations, other variables that you think we should be considering in our studies etc.): Construction on campus throughout this breeding program may have impacted animal stress in this study. While this was a mostly consistent issue here (and both treatments were applied in any given week to minimise timing effect as much as possible), this may have increased stress in both groups.

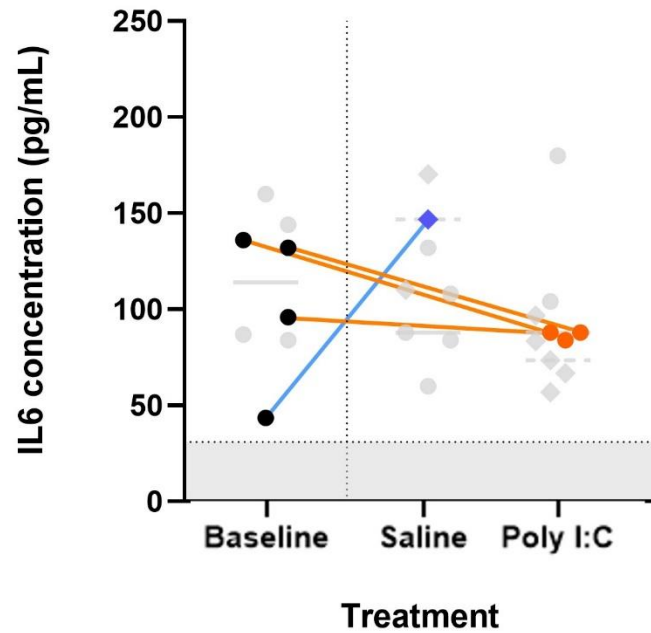
The recommended use of this reporting form is to fill it out and include it as supplemental material for each of your laboratory's research publications. If there are difficulties utilizing/adapting this fillable form, please contact one of the corresponding authors to request a copy. The authors give permission for this table to be edited for use in reporting on other animal models (e.g. postnatal immune challenge models, early life stress models) as appropriate.

Kentner AC, Bilbo AD, Brown AS, Hsiao EY, McAllister AK, Meyer U, Pearce BD, Pletnikov MV, Yolken RH, Bauman MD. (2018). Maternal immune activation: reporting guidelines to improve the rigor, reproducibility, and transparency of the model. *Neuropsychopharmacology*, <https://doi.org/10.1038/s41386-018-0185-7>.

Appendix D

This Appendix includes all linear mixed modelling performed in Chapter 3, along with additional figures providing further detail on maternal cytokine analysis and litter characteristics.

A) IL-6 concentration: 3-4hrs post-injection



B) CRP concentration: 3hrs post-injection

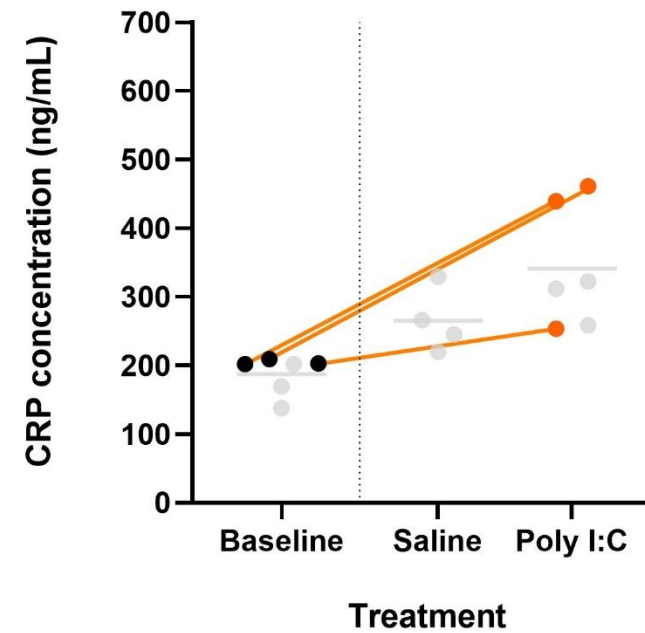


Figure D1: Repeated presentation of Figure 3.1 with only paired samples remaining. Unpaired samples remain translucent in background for reference. Lines connect paired samples, with blue and orange indicating saline and poly I:C respectively.

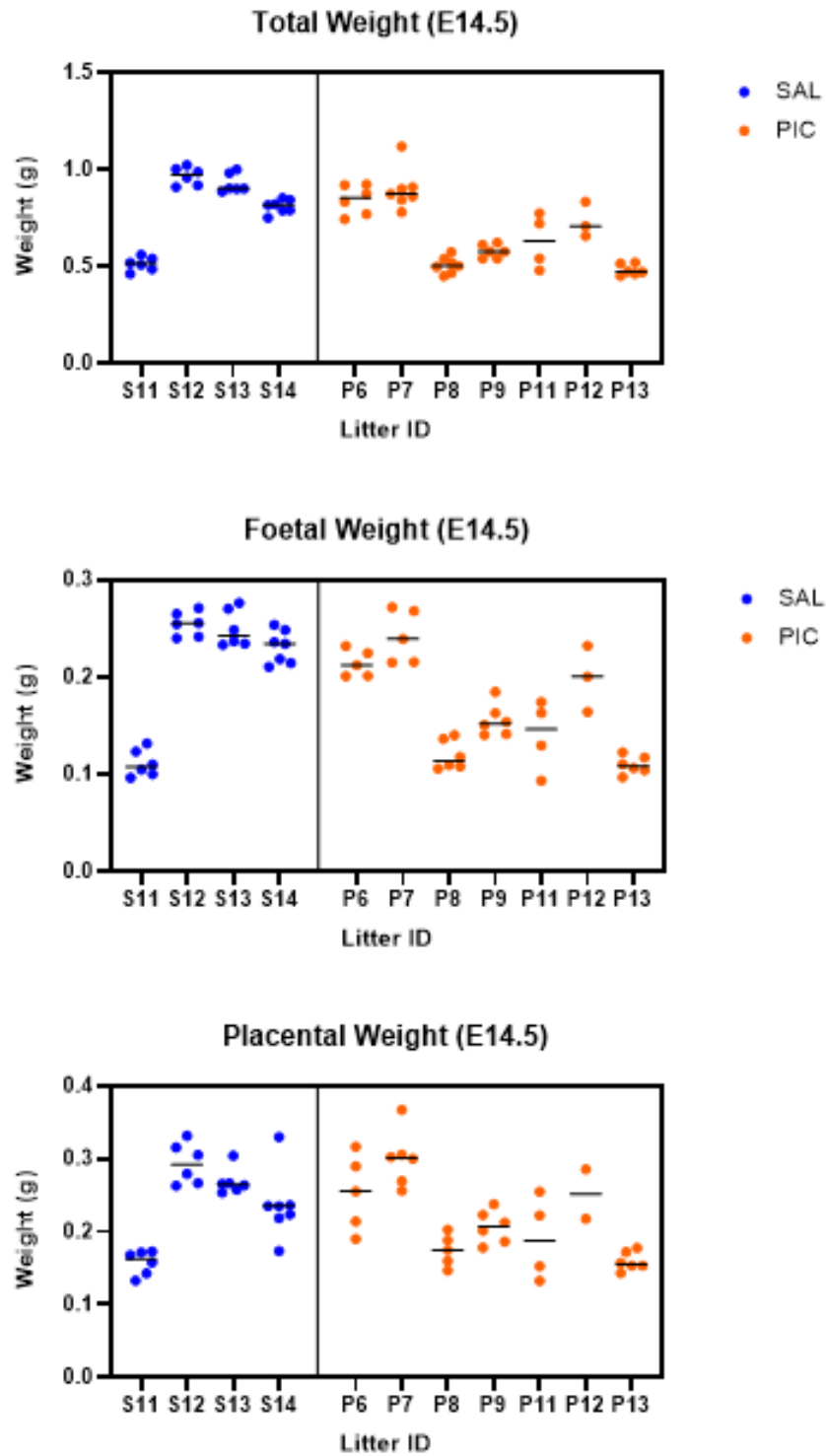


Figure D2: Litter variance is observed across all E14.5 weights assessed in chapter 3.

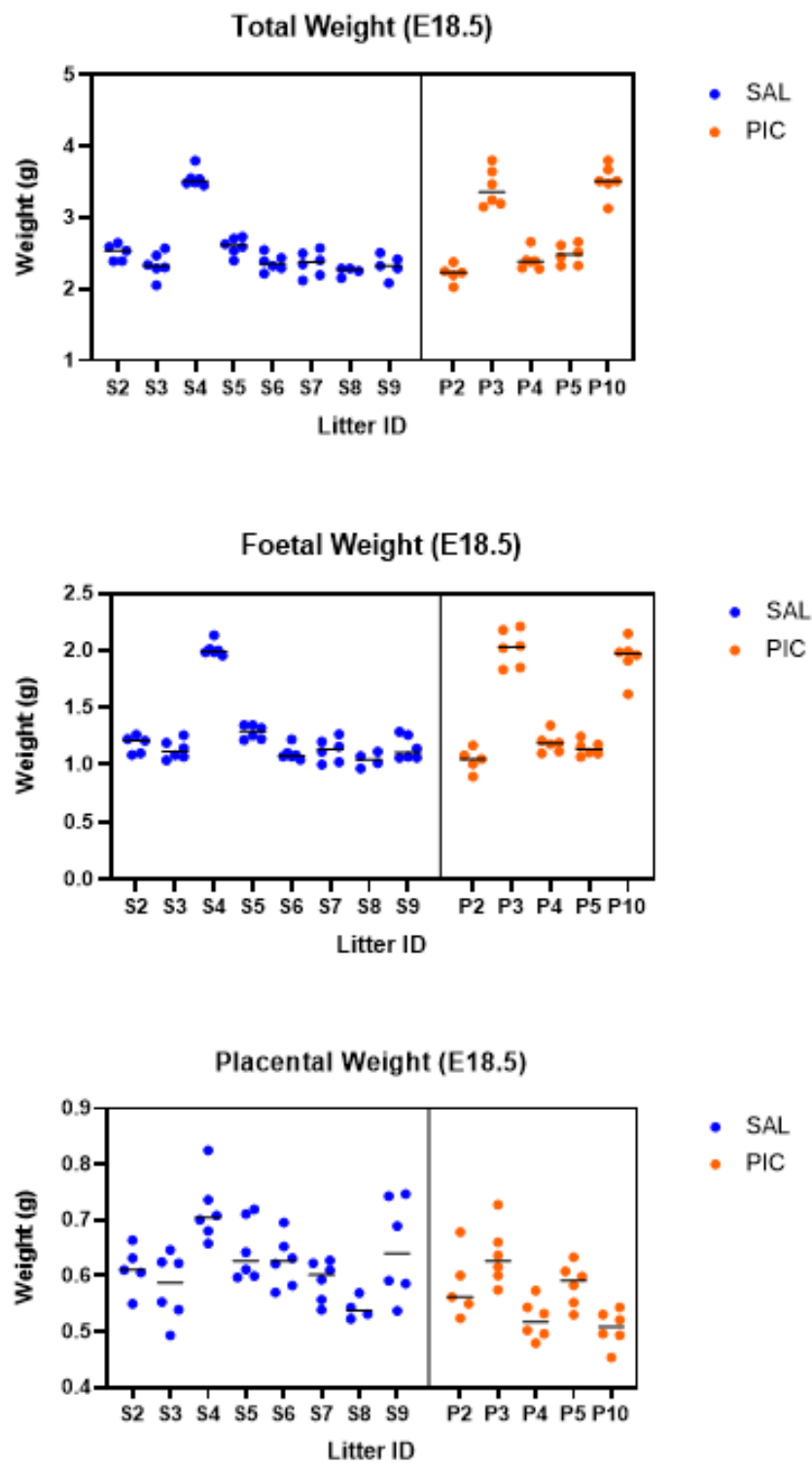


Figure D3: Litter variance is observed across all E18.5 weights assessed in chapter 3.

E14.5	Number of obs: 56, groups: Litter, 11							
Total Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-93.7	
Intercept	0.804736	0.091339	8.918522	8.8105	1.08E-05			
Sex1	0.003965	0.013933	43.539062	0.2846	0.7773		Variance	Std Dev
Treatment(PIC)	-0.14471	0.114915	9.043537	-1.2593	0.2395	Litter(Intercept)	0.032717	0.18088
Sex1*Treatment(PIC)	0.014636	0.021489	44.75387	0.6811	0.4993	Residual	0.004065	0.06376
Foetal Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-219.6	
Intercept	0.2114649	0.0285862	8.8834963	7.3974	4.41E-05			
Sex1	0.0030074	0.0041078	43.43768	0.7321	0.4680		Variance	Std Dev
Treatment(PIC)	-0.0433138	0.0359508	8.9949843	-1.2048	0.2590	Litter(Intercept)	0.003212	0.05667
Sex1*Treatment(PIC)	0.0007033	0.0063406	44.538171	0.1109	0.9122	Residual	0.000353	0.01879
Placental Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-170.7	
Intercept	0.239197	0.02693	8.715933	8.8823	1.17E-05			
Sex1	-0.002895	0.007288	44.55305	-0.3972	0.6931		Variance	Std Dev
Treatment(PIC)	-0.023312	0.034118	9.054577	-0.6833	0.5115	Litter(Intercept)	0.00272	0.05216
Sex1*Treatment(PIC)	0.006349	0.011084	47.409785	0.5728	0.5695	Residual	0.001122	0.0335
Foeto-placental Ratio								
	Estimate	Std. Error	df	t value	p value	REML criterion	-56.2	
Intercept	0.87994	0.04575	7.81965	19.2332	7.26E-08			
Sex1	0.02459	0.02402	47.75178	1.0238	0.3111		Variance	Std Dev
Treatment(PIC)	-0.11446	0.05928	8.67682	-1.9310	0.0867	Litter(Intercept)	0.006344	0.07965
Sex1*Treatment(PIC)	-0.02716	0.03515	51.6555	-0.7728	0.4431	Residual	0.012617	0.11232

Figure D4: Full linear mixed model tables for E14.5 weight data. Green shading indicates a significant value ($p < 0.05$), yellow indicates near-significance ($p \leq 0.10$).

E18.5	Number of obs: 73, groups: Litter, 13							
Total Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-11.7	
Intercept	2.53847	0.17834	11.01374	14.2336	1.95E-08			
Sex1	0.07618	0.02445	58.45321	3.1162	0.00284		Variance	Std Dev
Treatment(PIC)	0.26078	0.28751	11.00524	0.9070	0.38383	Litter(Intercept)	0.25063	0.5006
Sex1*Treatment(PIC)	-0.01143	0.03694	58.25492	-0.3094	0.75809	Residual	0.02056	0.1434
Foetal Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-70.2	
Intercept	1.25186	0.13414	11.01094	9.3326	1.46E-06			
Sex1	0.04138	0.01559	58.32696	2.6545	0.0102		Variance	Std Dev
Treatment(PIC)	0.20828	0.21626	11.00481	0.9631	0.3562	Litter(Intercept)	0.14239	0.37735
Sex1*Treatment(PIC)	0.00087	0.02355	58.1843	0.0370	0.9706	Residual	0.00835	0.09138
Placental Weight								
	Estimate	Std. Error	df	t value	p value	REML criterion	-188.8	
Intercept	0.619119	0.019843	10.917151	31.2003	5.05E-12			
Sex1	0.022107	0.007934	61.61078	2.7862	0.00708		Variance	Std Dev
Treatment(PIC)	-0.05342	0.031945	10.85189	-1.6722	0.12303	Litter(Intercept)	0.002737	0.05231
Sex1*Treatment(PIC)	-0.019476	0.012068	59.97852	-1.6138	0.11181	Residual	0.00223	0.04723
Foeto-placental Ratio								
	Estimate	Std. Error	df	t value	p value	REML criterion	40.3	
Intercept	2.01374	0.20596	11.02685	9.7773	9.08E-07			
Sex1	-0.01083	0.03719	58.78767	-0.2912	0.7719		Variance	Std Dev
Treatment(PIC)	0.59486	0.33199	11.01222	1.7918	0.1007	Litter(Intercept)	0.3305	0.5749
Sex1*Treatment(PIC)	0.0693	0.05622	58.44482	1.2328	0.2226	Residual	0.0477	0.2184

Figure D5: Full linear mixed model tables for E18.5 weight data. Green shading indicates a significant value ($p < 0.05$), yellow indicates near-significance ($p \leq 0.10$).

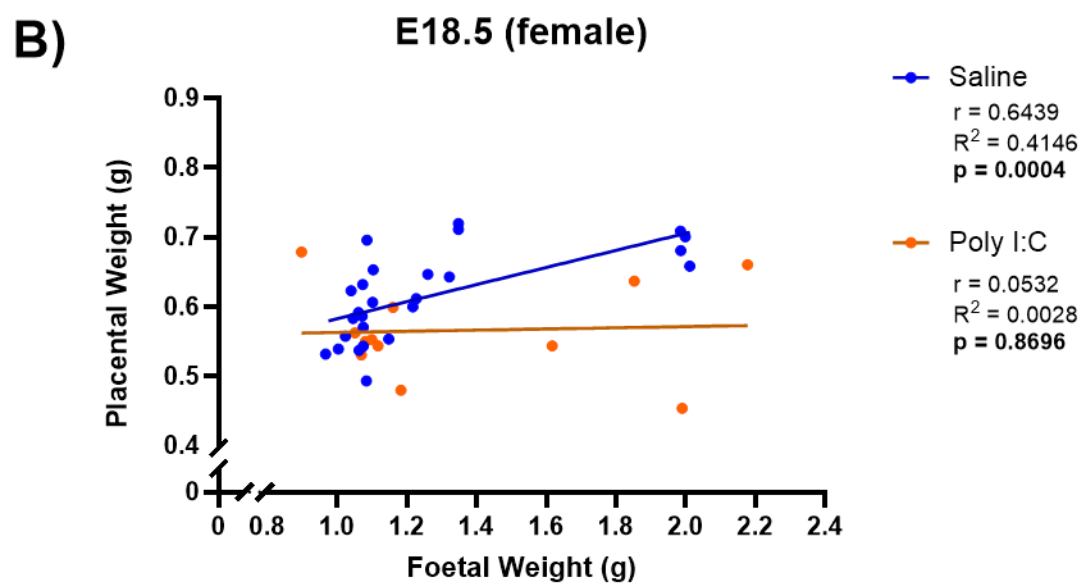
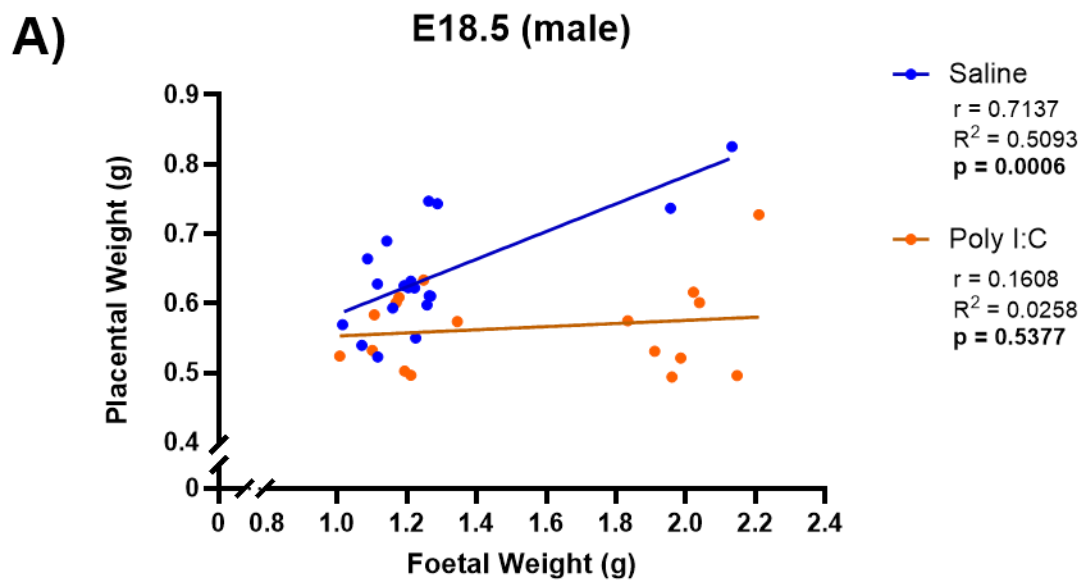


Figure D6: Sex differences at E18.5 do not significantly affect foetal placental weight correlations.

Appendix E

This Appendix includes statistics tables for the LMM models presented in Chapter 4.

E14.5

Linear Mixed Model

	Estimate	Std. Error	df	t value	p value
Intercept	216.894	45.005	5.228	4.819	0.00427
Sex	44.921	39.187	11	1.146	0.27598
Treatment	15.415	45.005	5.228	0.343	0.7453
Sex*Treatment	29.115	39.187	11	0.743	0.4731

Type III ANOVA with Satterthwaite's method: E14.5

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	21672.8	21672.8	1	10.9999	1.3141	0.276
Treatment	1934.8	1934.8	1	5.2276	0.1173	0.7453
Sex:Treatment	9104.3	9104.3	1	10.9999	0.552	0.4731

REML criterion

151.6

Linear Mixed Model Random Effects:

	Variance	Std Dev
Litter(Intercept)	6763	82.24
Residual	16493	128.42

Estimated litter variance (%)

29.08357

E18.5

Linear Mixed Model

	Estimate	Std. Error	df	t value	p value
Intercept	288.156	48.942	1.604	5.888	0.045
Sex	-39.181	46.027	11.499	-0.851	0.412
Treatment(Poly I:C)	-17.059	48.942	1.604	-0.349	0.768
Sex*Treatment(Poly I:C)	-14.18	46.027	11.499	-0.308	0.764

Type III ANOVA with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	17738.3	17738.3	1	11.4992	0.7246	0.412
Treatment	2974	2974	1	1.6041	0.1215	0.7677
Sex:Treatment	2323.3	2323.3	1	11.4992	0.0949	0.7635

REML criterion

167.5

Linear Mixed Model Random Effects:

	Variance	Std Dev
Litter(Intercept)	4192	64.75
Residual	24479	156.46

Estimated litter variance (%)

14.62235

Figure E1: Full linear mixed model and 2-way ANOVA tables for data presented in Figure 4.8. Green highlighting indicates significant result ($p < 0.05$).

Appendix F

This Appendix includes an additional analysis of the RT-ddPCR data presented in Chapter 5, normalised to reference gene expression. The results (particularly from E14.5 brain tissue) may be biased by unstable reference tissue.

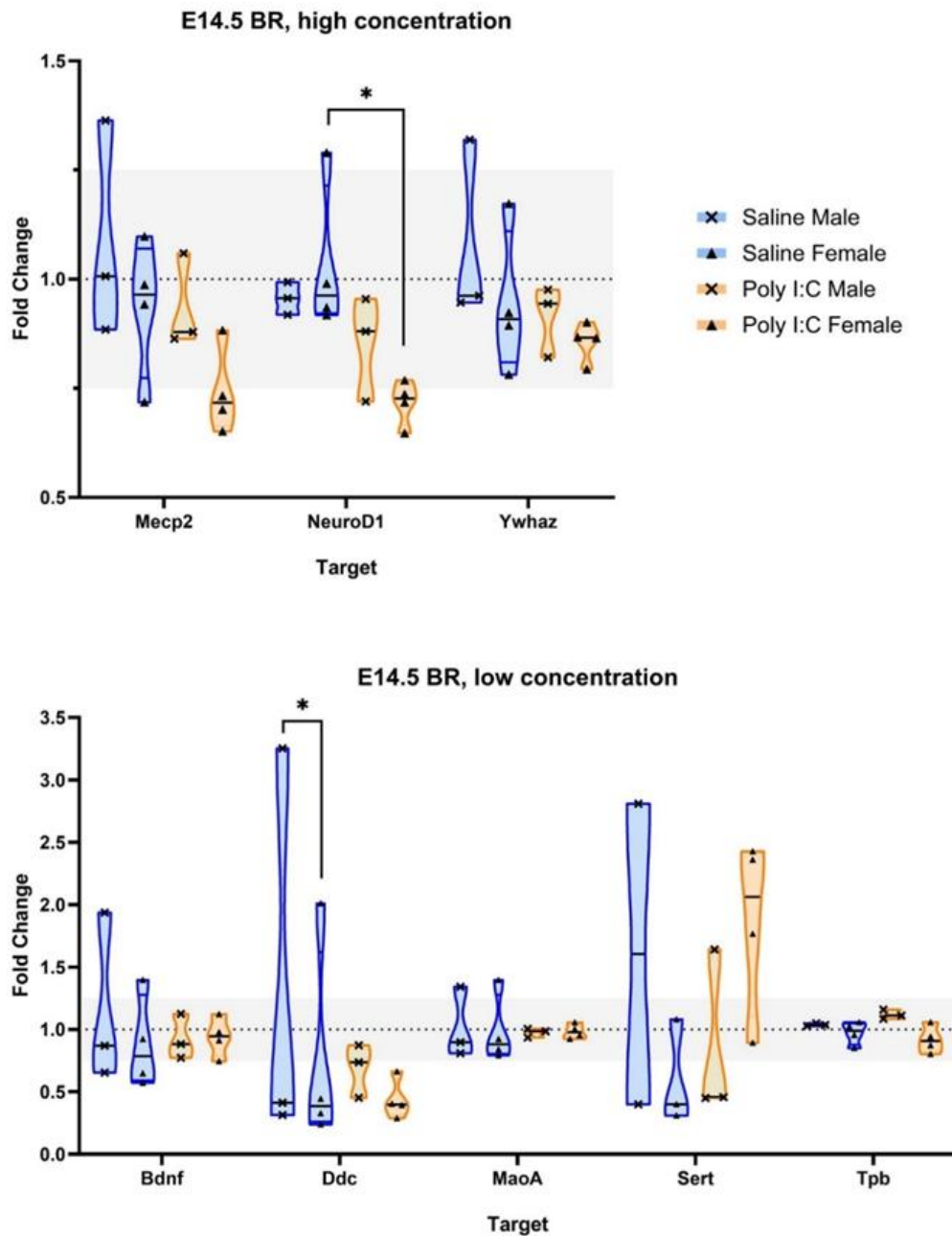


Figure F1: Target expression in the E14.5 brain. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. * = $p < 0.05$.

BDNF											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	1.0036	0.1317	5.9893	7.6230	0.0003		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	0.1175	0.0464	5.0668	2.5340	0.0516	Sex	0.0809	0.0809	1	5.0668	6.4226 0.0516
Treatment	0.0477	0.1317	5.9893	0.3620	0.7295	Treatment	0.0017	0.0017	1	5.9893	0.1313 0.7295
Sex*Treatment	0.1317	0.0464	5.0668	2.8390	0.0357	Sex:Treatment	0.1015	0.1015	1	5.0668	8.0598 0.0358
Ddc											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	0.8169	0.3001	6.0216	2.7220	0.0344		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	0.3638	0.0790	4.5977	4.6040	0.0072	Sex	0.7431	0.7431	1	4.5977	##### 0.0072
Treatment	0.2487	0.3001	6.0216	0.8290	0.4390	Treatment	0.0241	0.0241	1	6.0216	0.6865 0.4390
Sex*Treatment	0.2316	0.0790	4.5977	2.9300	0.0361	Sex:Treatment	0.3009	0.3009	1	4.5977	8.5831 0.0361
MaoA											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	0.9791	0.0682	6.0188	14.3670	6.94E-06		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	-0.0025	0.0241	5.1093	-0.1050	0.9200	Sex	0.0000	0.0000	1	5.1093	0.0110 0.9205
Treatment	-0.0019	0.0682	6.0188	-0.0280	0.9790	Treatment	0.0000	0.0000	1	6.0188	0.0008 0.9786
Sex*Treatment	-0.0161	0.0241	5.1093	-0.6660	0.5340	Sex:Treatment	0.0015	0.0015	1	5.1093	0.4434 0.5344
Mecp2											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	0.9255	0.0516	5.6262	17.9210	3.53E-06		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	0.1067	0.0390	9.1396	2.7390	0.0226	Sex	0.0986	0.0986	1	9.1396	7.5023 0.0226
Treatment	0.0897	0.0516	5.6262	1.7360	0.1365	Treatment	0.0396	0.0396	1	5.6262	3.0135 0.1365
Sex*Treatment	-0.0116	0.0390	9.1396	-0.2970	0.7732	Sex:Treatment	0.0012	0.0012	1	9.1396	0.0881 0.7732
NeuroD1											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	0.8670	0.0480	4.5087	18.0600	2.23E-05		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	-0.0148	0.0217	4.3783	-0.6810	0.5302	Sex	0.0014	0.0014	1	4.3783	0.4637 0.5302
Treatment	0.0962	0.0480	4.5087	2.0040	0.1077	Treatment	0.0118	0.0118	1	4.5087	4.0148 0.1077
Sex*Treatment	-0.1037	0.0217	4.3783	-4.7830	0.0070	Sex:Treatment	0.0670	0.0670	1	4.3783	##### 0.0070
Sert (weighted model, unequal sex variance)											
Number of obs: 12. Groups: Litter, 7											
	Estimate	Std. Error	df	t value	p value						
Intercept	1.2101	0.3248	5	3.7262	0.0136						
Sex	0.0951	0.2036	3	0.4672	0.6722						
Treatment	-0.2047	0.3248	5	-0.6304	0.5561						
Sex*Treatment	0.4533	0.2036	3	2.2262	0.1124						
Tpb (weighted model, unequal sex variance)											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value						
Intercept	1.0112	0.0203	6	49.8638	0.0000						
Sex	0.0696	0.0183	4	3.7941	0.0192						
Treatment	-0.0067	0.0203	6	-0.3290	0.7533						
Sex*Treatment	-0.0362	0.0183	4	-1.9705	0.1201						
Ywhaz											
Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method					
Intercept	0.941992	0.047185	5.998115	19.964	1.03E-06		Sum Sq	Mean Sq	NumDF	DenDF	F value Pr(>F)
Sex	0.065976	0.019468	5.527998	3.389	0.0167	Sex	0.0264	0.0264	1	5.528	11.485 0.0167
Treatment	0.066745	0.047185	5.998115	1.415	0.207	Treatment	0.0046	0.0046	1	5.9981	2.0009 0.207
Sex*Treatment	0.006464	0.019468	5.527998	0.332	0.7521	Sex:Treatment	0.0003	0.0003	1	5.528	0.1103 0.7521

Figure F2: LMM analyses of target expression in the E14.5 brain. Green shading indicates significant p values < 0.05 , with yellow for near-significance ($p < 0.10$).

BDNF						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.4984373	0.1495703	5.32	3.332	0.0189	0.0755658
Saline Male vs Poly I:C Male	0.3587698	0.2841007	7.57	1.263	0.2441	0.9765788
Saline Female vs Poly I:C Female	-0.1679312	0.2741771	6.86	-0.612	0.5599	1
Poly I:C Male vs Poly I:C Female	-0.0282638	0.1097641	4.63	-0.257	0.8078	1
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	1.1907708	0.2563403	4.72	4.645	0.0065	0.0258494
Saline Male vs Poly I:C Male	0.9604378	0.6272868	7.01	1.531	0.1696	0.6782718
Saline Female vs Poly I:C Female	0.0342572	0.6141268	6.54	0.056	0.9572	1
Poly I:C Male vs Poly I:C Female	0.2645902	0.1850177	4.37	1.43	0.2202	0.8806768
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.0371881	0.077793	5.36	-0.478	0.6515	1
Saline Male vs Poly I:C Male	-0.0359358	0.1471593	7.61	-0.244	0.8135	1
Saline Female vs Poly I:C Female	0.02832319	0.1419764	6.9	0.199	0.8476	1
Poly I:C Male vs Poly I:C Female	0.02707082	0.0571109	4.67	0.474	0.6568	1
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.1902969	0.1451356	9.85	1.311	0.2195	0.8781731
Saline Male vs Poly I:C Male	0.1561741	0.1480983	8.24	1.055	0.3216	1
Saline Female vs Poly I:C Female	0.2024291	0.1308249	7.21	1.547	0.1645	0.6579303
Poly I:C Male vs Poly I:C Female	0.2365519	0.1148519	6.74	2.06	0.08	0.3198412
NeuroD1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2368557	0.0692812	4.86	-3.419	0.0197	0.0789621
Saline Male vs Poly I:C Male	-0.0149579	0.1081152	7.11	-0.138	0.8938	1
Saline Female vs Poly I:C Female	0.3997149	0.1024983	5.97	3.9	0.0081	0.0323026
Poly I:C Male vs Poly I:C Female	0.1778171	0.0521226	3.66	3.412	0.0311	0.1242531
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	1.0969043	0.6470674	3	1.695	0.1886	0.7544395
Saline Male vs Poly I:C Male	0.4971803	0.8989337	3	0.553	0.6188	1
Saline Female vs Poly I:C Female	-1.3160756	0.6060835	3	-2.171	0.1183	0.4731824
Poly I:C Male vs Poly I:C Female	-0.7163516	0.4946754	3	-1.448	0.2434	0.9736465
Tpb						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.06692067	0.0530692	4	1.261	0.2758	1
Saline Male vs Poly I:C Male	-0.0856514	0.028302	4	-3.026	0.0389	0.155681
Saline Female vs Poly I:C Female	0.05896386	0.0719886	4	0.819	0.4587	1
Poly I:C Male vs Poly I:C Female	0.21153588	0.0506952	4	4.173	0.014	0.0560024
Ywhaz						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.144881	0.0624636	5.9	2.319	0.0602	0.2409957
Saline Male vs Poly I:C Male	0.1464196	0.1044193	7.96	1.402	0.1986	0.7944531
Saline Female vs Poly I:C Female	0.120562	0.0996994	7.13	1.209	0.2651	1
Poly I:C Male vs Poly I:C Female	0.1190234	0.0464991	4.92	2.56	0.0515	0.2058809

Figure F3: Pairwise comparisons based on LMM analyses in the E14.5 brain.
Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

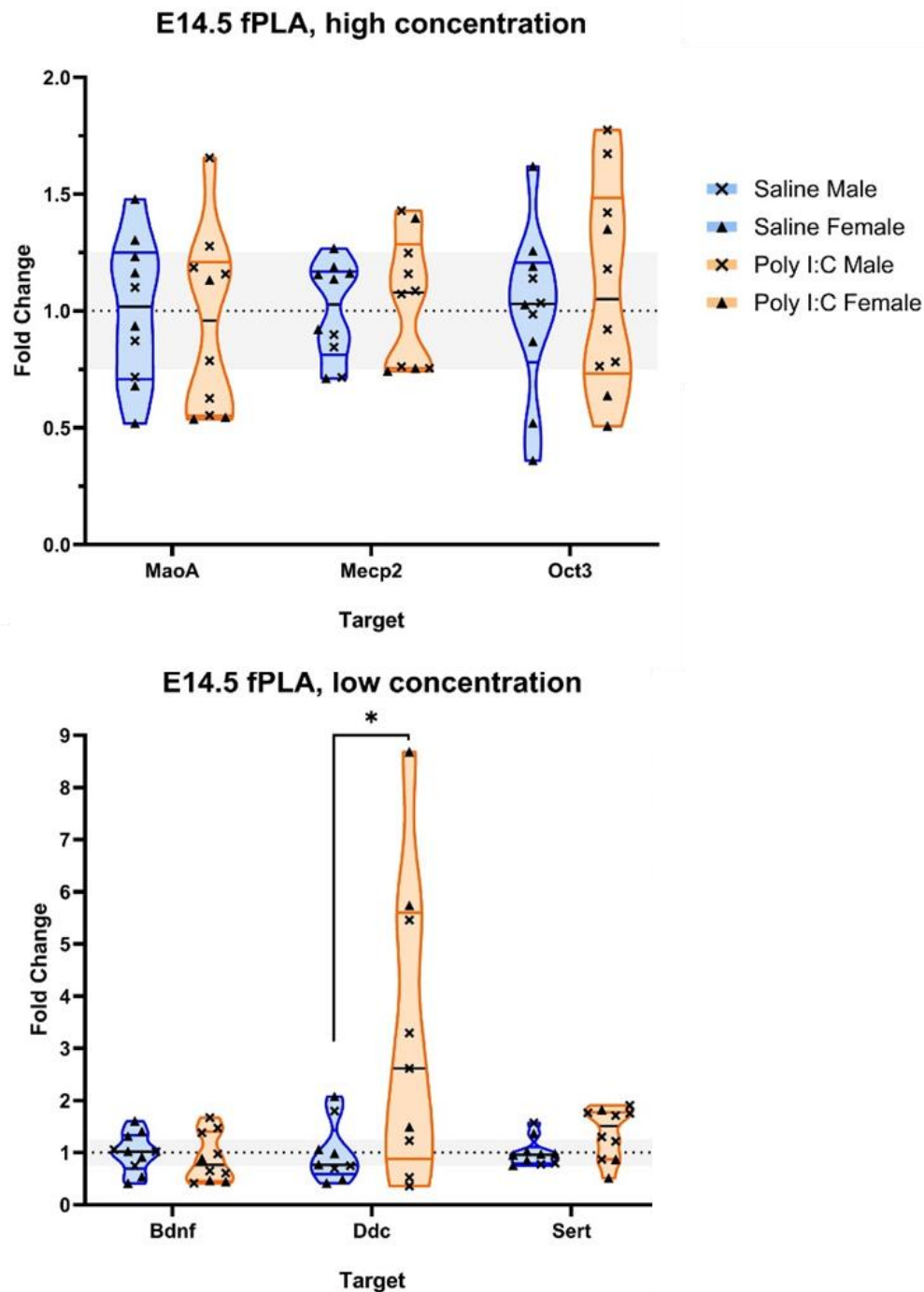


Figure F4: Target expression in the E14.5 placenta. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. * = $p < 0.05$.

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.2	0.232	4.58	0.862	0.4315	
Saline Male vs Saline Female	-0.0447045	0.2703739	15.97	-0.165	0.8707	1
Saline Male vs Poly I:C Male	-0.0306238	0.2846165	9.03	-0.108	0.9167	1
Saline Female vs Poly I:C Female	0.430996	0.3235607	9.02	1.332	0.2155	0.8620691
Poly I:C Male vs Poly I:C Female	0.4169153	0.2872516	15.72	1.451	0.1663	0.6653403
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-2.66	0.96	5	-2.771	0.0393	
Saline Male vs Saline Female	0.098116	0.3668516	9	0.267	0.7951	1
Saline Male vs Poly I:C Male	-1.138685	1.1093084	5	-1.026	0.3517	1
Saline Female vs Poly I:C Female	-4.183733	1.5168472	5	-2.758	0.0399	0.1596836
Poly I:C Male vs Poly I:C Female	-2.946932	1.799831	9	-1.637	0.136	0.5439428
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.113	0.168	16	0.675	0.5093	
Saline Male vs Saline Female	-0.0021188	0.2295071	16	-0.009	0.9927	1
Saline Male vs Poly I:C Male	-0.0361763	0.2228533	16	-0.162	0.8731	1
Saline Female vs Poly I:C Female	0.26296841	0.2514124	16	1.046	0.3111	1
Poly I:C Male vs Poly I:C Female	0.29702595	0.2453534	16	1.211	0.2436	0.9745089
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.0348	0.116	16	-0.3	0.7679	
Saline Male vs Saline Female	-0.1600828	0.1583289	16	-1.011	0.327	1
Saline Male vs Poly I:C Male	-0.1693239	0.1537387	16	-1.101	0.287	1
Saline Female vs Poly I:C Female	0.09975885	0.1734407	16	0.575	0.5732	1
Poly I:C Male vs Poly I:C Female	0.10900004	0.1692607	16	0.644	0.5287	1
Oct3						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.00963	0.189	16	-0.051	0.9601	
Saline Male vs Saline Female	0.146653	0.2586533	16	0.567	0.5786	1
Saline Male vs Poly I:C Male	-0.1288349	0.2511545	16	-0.513	0.615	1
Saline Female vs Poly I:C Female	0.109576	0.2833405	16	0.387	0.7041	1
Poly I:C Male vs Poly I:C Female	0.3850639	0.276512	16	1.393	0.1828	1
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.283	0.186	6	-1.52	0.1792	
Saline Male vs Saline Female	-0.0057827	0.1817038	10	-0.032	0.9752	1
Saline Male vs Poly I:C Male	-0.5059384	0.2267812	6	-2.231	0.0672	0.2687107
Saline Female vs Poly I:C Female	-0.0596796	0.2949344	6	-0.202	0.8463	1
Poly I:C Male vs Poly I:C Female	0.4404761	0.3246533	10	1.357	0.2047	0.8187746
Tph1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-1.62	0.968	4	-1.674	0.1695	
Saline Male vs Saline Female	0.2485693	0.5349496	4	0.465	0.6663	1
Saline Male vs Poly I:C Male	-0.6387896	0.9033183	4	-0.707	0.5185	1
Saline Female vs Poly I:C Female	-2.6027724	1.7132529	4	-1.519	0.2033	0.8133424
Poly I:C Male vs Poly I:C Female	-1.7154136	1.861464	4	-0.922	0.4089	1
Tph2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.0408	1.12	8	-0.037	0.9717	
Saline Male vs Saline Female	-0.800979	1.579482	8	-0.507	0.6258	1
Saline Male vs Poly I:C Male	-0.844061	1.579482	8	-0.534	0.6076	1
Saline Female vs Poly I:C Female	0.7623716	1.579482	8	0.483	0.6423	1
Poly I:C Male vs Poly I:C Female	0.8054536	1.579482	8	0.51	0.6239	1

Figure F6: Pairwise comparisons from LMM analyses in the E14.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

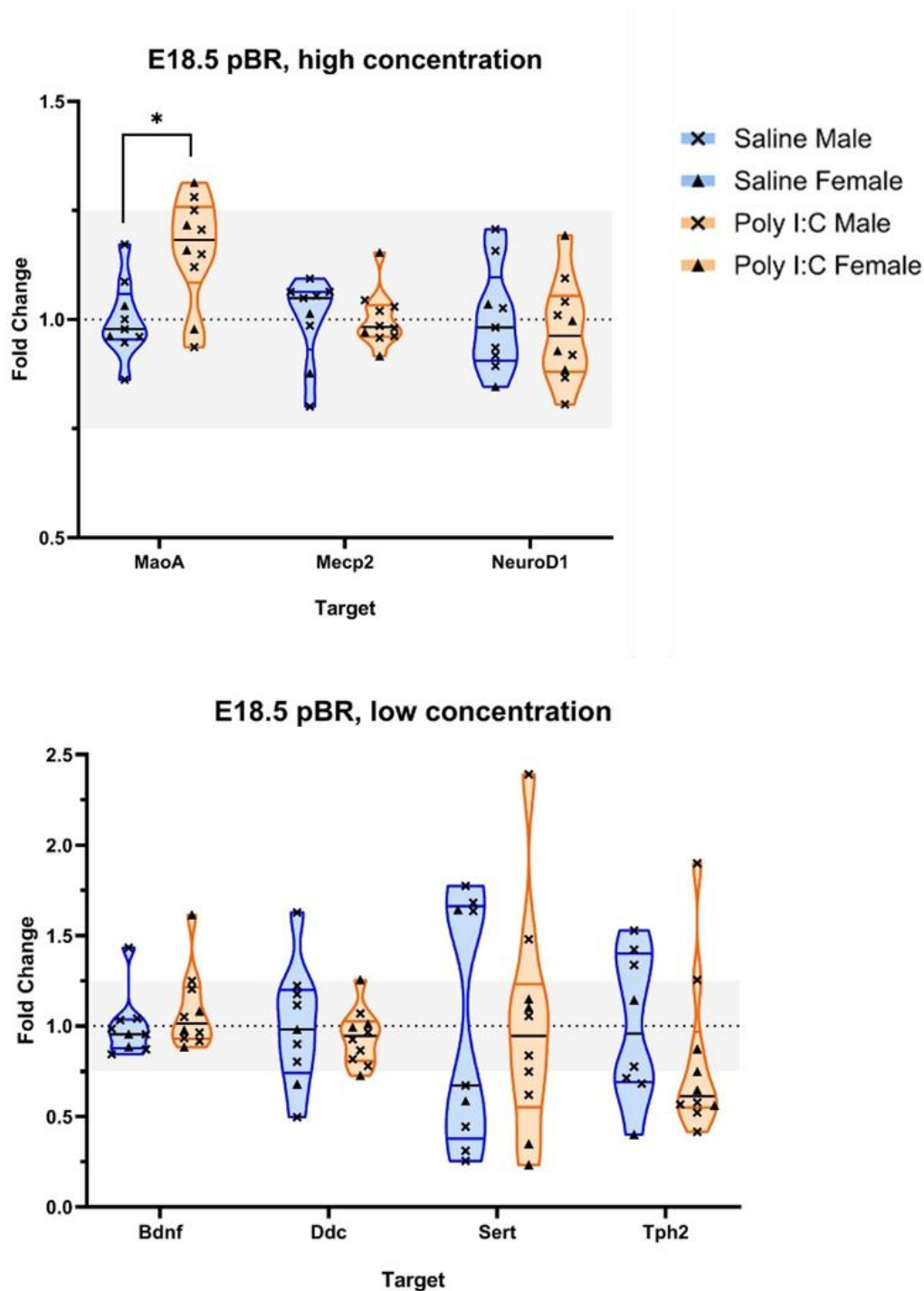


Figure F7: Target expression in the E18.5 posterior brain. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. * = $p < 0.05$.

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.127	0.115	7	-1.104	0.3062	
Saline Male vs Saline Female	0.0970829	0.105995	8	0.916	0.3865	1
Saline Male vs Poly I:C Male	-0.056207	0.116842	7	-0.481	0.6452	1
Saline Female vs Poly I:C Female	-0.198573	0.166463	7	-1.193	0.2718	1
Poly I:C Male vs Poly I:C Female	-0.045283	0.134945	8	-0.336	0.7458	1
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.0232	0.144	7.41	-0.161	0.8765	
Saline Male vs Saline Female	0.2502721	0.204748	14.03	1.222	0.2417	0.9668362
Saline Male vs Poly I:C Male	0.1521328	0.153258	8.93	0.993	0.347	1
Saline Female vs Poly I:C Female	-0.198443	0.228403	14.26	-0.869	0.3993	1
Poly I:C Male vs Poly I:C Female	-0.100303	0.162911	12.98	-0.616	0.5487	1
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.163	0.0594	15	-2.751	0.0148	
Saline Male vs Saline Female	0.0047415	0.092479	15	0.051	0.9598	1
Saline Male vs Poly I:C Male	-0.15636	0.06417	15	-2.437	0.0278	0.1110394
Saline Female vs Poly I:C Female	-0.170307	0.099889	15	-1.705	0.1088	0.4352766
Poly I:C Male vs Poly I:C Female	-0.009205	0.074453	15	-0.124	0.9032	1
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.0223	0.0437	15	-0.511	0.617	
Saline Male vs Saline Female	0.0708504	0.068142	15	1.04	0.3149	1
Saline Male vs Poly I:C Male	0.0166854	0.047283	15	0.353	0.7291	1
Saline Female vs Poly I:C Female	-0.061356	0.073602	15	-0.834	0.4176	1
Poly I:C Male vs Poly I:C Female	-0.00719	0.054859	15	-0.131	0.8975	1
NeuroD1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.000256	0.0629	15	0.004	0.9968	
Saline Male vs Saline Female	0.0765758	0.097936	15	0.782	0.4464	1
Saline Male vs Poly I:C Male	0.0607677	0.067956	15	0.894	0.3853	1
Saline Female vs Poly I:C Female	-0.060257	0.105783	15	-0.57	0.5774	1
Poly I:C Male vs Poly I:C Female	-0.044448	0.078846	15	-0.564	0.5813	1
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.0926	0.335	15	0.276	0.7861	
Saline Male vs Saline Female	-0.146783	0.522221	15	-0.281	0.7825	1
Saline Male vs Poly I:C Male	-0.221247	0.362363	15	-0.611	0.5506	1
Saline Female vs Poly I:C Female	0.4064913	0.564063	15	0.721	0.4822	1
Poly I:C Male vs Poly I:C Female	0.4809549	0.420428	15	1.144	0.2706	1
Tph2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.0574	0.259	15	0.222	0.8277	
Saline Male vs Saline Female	0.1509422	0.403804	15	0.374	0.7138	1
Saline Male vs Poly I:C Male	0.0499186	0.280195	15	0.178	0.861	1
Saline Female vs Poly I:C Female	0.0649191	0.436158	15	0.149	0.8837	1
Poly I:C Male vs Poly I:C Female	0.1659427	0.325093	15	0.51	0.6172	1

Figure F9: Pairwise comparisons from LMM analyses in the E18.5 brain. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

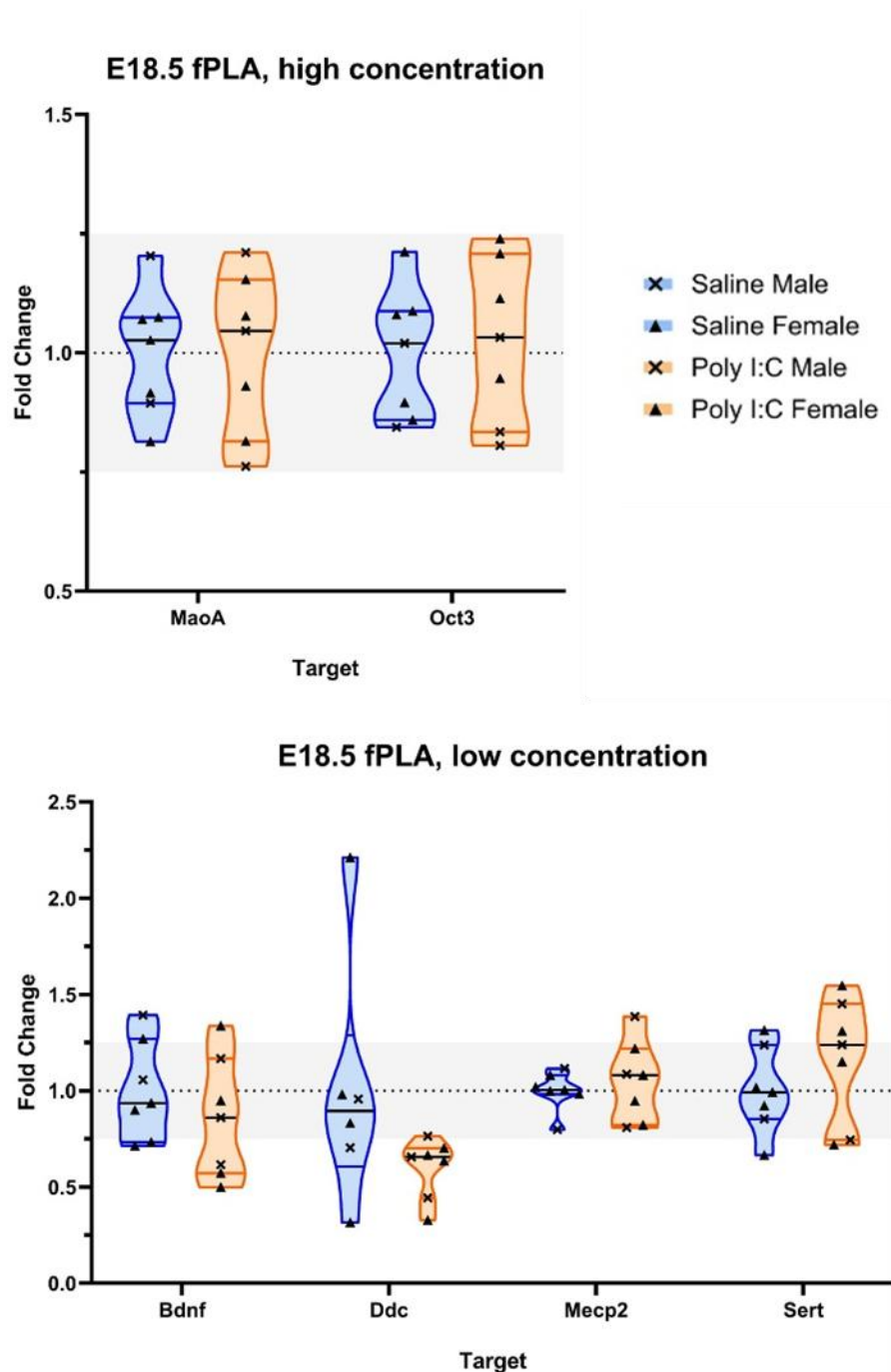


Figure F10: Target expression in the E18.5 placenta. Targets detected at a high concentration during ddPCR (A) are presented separately from targets detected at a lower concentration (B). Colour of violin plot denotes treatment group, while symbol denotes offspring sex. Mean for each violin is represented by black bar. Grey background shading indicates 0.75 – 1.25 fold change range for improved comparison between graph scales. * = $p < 0.05$.

Bdnf	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method						
Intercept	0.9746	0.0946	4.7656	10.3050	0.0002		Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	0.1057	0.0709	4.2380	1.4910	0.2063	Sex	0.1240	0.1240	1	4.2380	2.2238	0.2063
Treatment	0.0838	0.0946	4.7656	0.8860	0.4180	Treatment	0.0438	0.0438	1	4.7656	0.7853	0.4180
Sex* Treatment	0.0536	0.0709	4.2380	0.7560	0.4895	Sex: Treatment	0.0319	0.0319	1	4.2380	0.5715	0.4895

Ddc (weighted for treatment vari	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value							
Intercept	0.7797	0.1552	6	5.0227	0.0024							
Sex	-0.0537	0.1552	3	-0.3461	0.7521							
Treatment	0.1781	0.1552	6	1.1470	0.2950							
Sex* Treatment	-0.0732	0.1552	3	-0.4713	0.6696							

MaoA	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method						
Intercept	1.0075	0.0466	10	21.6350	9.94E-10		Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	0.0203	0.0466	10	0.4360	0.6720	Sex	0.0052	0.0052	1	10	0.1905	0.6718
Treatment	0.0073	0.0466	10	0.1570	0.8780	Treatment	0.0007	0.0007	1	10	0.0247	0.8782
Sex* Treatment	0.0142	0.0466	10	0.3040	0.7670	Sex: Treatment	0.0025	0.0025	1	10	0.0926	0.7672

Mecp2 (weighted for sex)treatment	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value							
Intercept	1.0212	0.0615	6	16.6063	<0.0001							
Sex	0.0041	0.0615	4	0.0666	0.9501							
Treatment	-0.0342	0.0615	6	-0.5559	0.5984							
Sex* Treatment	-0.0345	0.0615	4	-0.5603	0.6052							

Oct3	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method						
Intercept	0.9943	0.0384	10	25.8830	1.70E-10		Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	-0.0827	0.0384	10	-2.1520	0.0569	Sex	0.0852	0.0852	1	10	4.6318	0.0569
Treatment	-0.0146	0.0384	10	-0.3810	0.7112	Treatment	0.0027	0.0027	1	10	0.1451	0.7112
Sex* Treatment	0.0353	0.0384	10	0.9180	0.3801	Sex: Treatment	0.0155	0.0155	1	10	0.8431	0.3801

Sert	Number of obs: 14. Groups: Litter, 8											
	Estimate	Std. Error	df	t value	p value	Type III ANOVA with Satterthwaite's method						
Intercept	1.0864	0.0915	6.2759	11.8750	1.56E-05		Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
Sex	0.0096	0.0799	6.6004	0.1200	0.9080	Sex	0.0011	0.0011	1	6.6004	0.0143	0.9083
Treatment	-0.0802	0.0915	6.2759	-0.8770	0.4130	Treatment	0.0575	0.0575	1	6.2759	0.7693	0.4128
Sex* Treatment	0.0240	0.0799	6.6004	0.3000	0.7730	Sex: Treatment	0.0067	0.0067	1	6.6004	0.0900	0.7734

Figure F11: LMM analyses of target expression in the E18.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.168	0.189	4.77	0.886	0.418	
Saline Male vs Saline Female	0.318	0.215	4.8	1.483	0.201	0.801996
Saline Male vs Poly I:C Male	0.275	0.264	9.98	1.041	0.322	1
Saline Female vs Poly I:C Female	0.060	0.205	7.41	0.295	0.776	1
Poly I:C Male vs Poly I:C Female	0.104	0.185	3.6	0.563	0.607	1
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.356	0.310	6	1.147	0.295	
Saline Male vs Saline Female	-0.254	0.607	3	-0.418	0.704	1
Saline Male vs Poly I:C Male	0.210	0.505	3	0.415	0.706	1
Saline Female vs Poly I:C Female	0.502	0.361	3	1.393	0.258	1
Poly I:C Male vs Poly I:C Female	0.039	0.129	3	0.301	0.783	1
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	0.015	0.093	10	0.157	0.878	
Saline Male vs Saline Female	0.069	0.138	10	0.501	0.627	1
Saline Male vs Poly I:C Male	0.043	0.150	10	0.286	0.781	1
Saline Female vs Poly I:C Female	-0.014	0.110	10	-0.124	0.904	1
Poly I:C Male vs Poly I:C Female	0.012	0.126	10	0.098	0.924	1
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.068	0.123	6	-0.556	0.598	
Saline Male vs Saline Female	-0.061	0.160	4	-0.381	0.723	1
Saline Male vs Poly I:C Male	-0.137	0.230	4	-0.597	0.583	1
Saline Female vs Poly I:C Female	0.001	0.087	4	0.006	0.995	1
Poly I:C Male vs Poly I:C Female	0.077	0.187	4	0.412	0.702	1
Oct3						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.029	0.077	10	-0.381	0.711	
Saline Male vs Saline Female	-0.095	0.113	10	-0.835	0.423	1
Saline Male vs Poly I:C Male	0.041	0.124	10	0.333	0.746	1
Saline Female vs Poly I:C Female	-0.100	0.091	10	-1.097	0.298	1
Poly I:C Male vs Poly I:C Female	-0.236	0.104	10	-2.277	0.046	0.1840663
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline vs Poly I:C (sexes grouped)	-0.160	0.183	6.28	-0.877	0.413	
Saline Male vs Saline Female	0.067	0.239	7.43	0.280	0.787	1
Saline Male vs Poly I:C Male	-0.113	0.275	10	-0.410	0.691	1
Saline Female vs Poly I:C Female	-0.208	0.206	9.09	-1.010	0.339	1
Poly I:C Male vs Poly I:C Female	-0.029	0.212	5.61	-0.136	0.897	1

Figure F12: Pairwise comparisons from LMM analyses in the E18.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Appendix G

This Appendix includes full linear mixed model tables for the final ddPCR analyses presented in Chapter 5.

B2m (weighted for sex variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	9. 84981	
Intercept	0. 988837	0. 0397693	6	24. 86435	<0. 0001	Linear Mixed Model Random Effects:		
Sex	-0. 1549902	0. 0331156	4	-4. 680283	0. 0094		Std Dev	
Treatment	-0. 0140641	0. 0397693	6	-0. 353642	0. 7357	Litter(Intercept) 0. 077123		
Sex*Treatment	-0. 0344786	0. 0331156	4	-1. 04116	0. 3566	Residual 0. 156425		
Bdnf			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	REML criterion	8. 2	
Intercept	0. 987797	0. 110752	6. 023274	8. 919	0. 000108	Linear Mixed Model Random Effects:		
Sex	-0. 097586	0. 042072	5. 297235	-2. 32	0. 065184		Variance	Std Dev
Treatment	-0. 004705	0. 110752	6. 023274	-0. 042	0. 967489	Litter(Intercept) 0. 09107 0. 3018		
Sex*Treatment	0. 014209	0. 042072	5. 297235	0. 338	0. 748534	Residual 0. 01053 0. 1026		
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0. 056627	0. 056627	1	5. 2972	5. 3801	0. 06518		
Treatment	0. 000019	0. 000019	1	6. 0233	0. 0018	0. 96749		
Sex:Treatment	0. 001201	0. 001201	1	5. 2972	0. 1141	0. 74853		
Ddc (weighted for treatment variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	33. 9973	
Intercept	0. 7730361	0. 2747573	6	2. 8135237	0. 0306	Linear Mixed Model Random Effects:		
Sex	0. 0580739	0. 0600222	4	0. 9675406	0. 3881		Std Dev	
Treatment	0. 1594291	0. 2747573	6	0. 5802541	0. 5829	Litter(Intercept) 0. 765647		
Sex*Treatment	-0. 0233675	0. 0600222	4	-0. 389314	0. 7169	Residual 0. 059258		
Hprt			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	REML criterion	-1. 5	
Intercept	0. 971172	0. 035811	10	27. 119	1. 08E-10	Linear Mixed Model Random Effects:		
Sex	-0. 081572	0. 035811	10	-2. 278	0. 046		Variance	Std Dev
Treatment	0. 016123	0. 035811	10	0. 45	0. 662	Litter(Intercept) 0 0		
Sex*Treatment	-0. 007368	0. 035811	10	-0. 206	0. 841	Residual 0. 01759 0. 1326		
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0. 091254	0. 091254	1	10	5. 1884	0. 04596		
Treatment	0. 003565	0. 003565	1	10	0. 2027	0. 66216		
Sex:Treatment	0. 000744	0. 000744	1	10	0. 0423	0. 84112		
MaoA (weighted for treatment variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	7. 866723	
Intercept	0. 9824625	0. 0526649	6	18. 654972	<0. 0001	Linear Mixed Model Random Effects:		
Sex	-0. 1059854	0. 0501656	4	-2. 112712	0. 1022		Std Dev	
Treatment	-0. 0011065	0. 0526649	6	-0. 02101	0. 9839	Litter(Intercept) 0. 056082		
Sex*Treatment	-0. 0153113	0. 0501656	4	-0. 305215	0. 7754	Residual 0. 253625		
Mecp2			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	REML criterion	-12. 2	
Intercept	0. 90599	0. 03751	5. 385	24. 151	1. 07E-06	Linear Mixed Model Random Effects:		
Sex	-0. 03336	0. 01633	5. 10346	-2. 043	0. 0954		Variance	Std Dev
Treatment	0. 06994	0. 03751	5. 385	1. 865	0. 1171	Litter(Intercept) 0. 010164 0. 10082		
Sex*Treatment	-0. 08049	0. 01633	5. 10346	-4. 928	0. 00413	Residual 0. 001645 0. 04056		
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0. 006863	0. 006863	1	5. 1035	4. 1723	0. 0954		
Treatment	0. 005718	0. 005718	1	5. 385	3. 4764	0. 1171		
Sex:Treatment	0. 039943	0. 039943	1	5. 1035	24. 2833	0. 00413		

NeuroD1 (weighted for treatment variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	11. 21078	
Intercept	0. 8698902	0. 0406851	6	21. 381029	<0. 0001	Linear Mixed Model Random Effects:		
Sex	-0. 0800227	0. 0406851	4	-1. 966877	0. 1206		Std Dev	
Treatment	0. 1076027	0. 0406851	6	2. 644767	0. 0383	Litter(Intercept)	1. 52E-06	
Sex*Treatment	-0. 0775266	0. 0406851	4	-1. 905526	0. 1294	Residual	0. 198259	
Ppia (weighted for treatment variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	8. 090859	
Intercept	0. 9928007	0. 050107	6	19. 813603	<0. 0001	Linear Mixed Model Random Effects:		
Sex	-0. 0940826	0. 0412481	4	-2. 280897	0. 0847		Std Dev	
Treatment	-0. 012646	0. 050107	6	-0. 25238	0. 8092	Litter(Intercept)	0. 095973	
Sex*Treatment	-0. 0232515	0. 0412481	4	-0. 5637	0. 6031	Residual	0. 18871	
Sert			Number of obs: 12. Groups: Litter, 7					
	Estimate	Std. Error	df	t value	p value	REML criterion	26. 2	
Intercept	1. 3084	0. 3213	4. 7724	4. 072	0. 0106	Linear Mixed Model Random Effects:		
Sex	-0. 1201	0. 1521	4. 3139	-0. 79	0. 4708		Variance	Std Dev
Treatment	-0. 3795	0. 3213	4. 7724	-1. 181	0. 2931	Litter(Intercept)	0. 6163	0. 785
Sex*Treatment	0. 4933	0. 1521	4. 3139	3. 244	0. 0283	Residual	0. 1435	0. 3788
Tpb			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	REML criterion	-7. 1	
Intercept	0. 993764	0. 029203	3. 342156	34. 03	2. 30E-05	Linear Mixed Model Random Effects:		
Sex	-0. 03832	0. 027221	9. 989075	-1. 408	0. 1896		Variance	Std Dev
Treatment	-0. 008517	0. 029203	3. 342156	-0. 292	0. 7877	Litter(Intercept)	0. 001606	0. 04008
Sex*Treatment	-0. 056075	0. 027221	9. 989075	-2. 06	0. 0664	Residual	0. 008775	0. 09368
Ywhaz (weighted for treatment variance)			Number of obs: 14. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	-1. 72222	
Intercept	0. 9363616	0. 0315133	6	29. 71324	<0. 0001	Linear Mixed Model Random Effects:		
Sex	-0. 055471	0. 0315133	4	-1. 760241	0. 1532		Std Dev	
Treatment	0. 0546246	0. 0315133	6	1. 733384	0. 1337	Litter(Intercept)	1. 15E-06	
Sex*Treatment	-0. 0076252	0. 0315133	4	-0. 241968	0. 8207	Residual	0. 162972	

Figure G1: LMM analyses of target and reference gene expression in the E14.5 brain. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

B2m						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.3789377	0.09769	4	-3.879	0.0179	0.0714268
Saline Male vs Poly I:C Male	-0.0970854	0.0730082	4	-1.33	0.2544	1
Saline Female vs Poly I:C Female	0.0408291	0.1268688	4	0.322	0.7637	1
Poly I:C Male vs Poly I:C Female	-0.2410232	0.0894591	4	-2.694	0.0544	0.2176751
BDNF						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1667529	0.1353768	5.6	-1.232	0.2673	1
Saline Male vs Poly I:C Male	0.01900924	0.2416831	7.78	0.079	0.9393	1
Saline Female vs Poly I:C Female	-0.0378282	0.2321168	7.01	-0.163	0.8751	1
Poly I:C Male vs Poly I:C Female	-0.2235904	0.0999683	4.79	-2.237	0.0779	0.3116796
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.0694128	0.0834623	4	0.832	0.4524	1
Saline Male vs Poly I:C Male	0.2721231	0.5634805	4	0.483	0.6544	1
Saline Female vs Poly I:C Female	0.3655931	0.5614656	4	0.651	0.5505	1
Poly I:C Male vs Poly I:C Female	0.1628828	0.2251147	4	0.724	0.5094	1
Hprt						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1778796	0.1012902	10	-1.756	0.1096	0.4383475
Saline Male vs Poly I:C Male	0.01750977	0.1082838	10	0.162	0.8748	1
Saline Female vs Poly I:C Female	0.0469814	0.0937765	10	0.501	0.6272	1
Poly I:C Male vs Poly I:C Female	-0.148408	0.1012902	10	-1.465	0.1736	0.6943759
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2425933	0.1996448	4	-1.215	0.2911	1
Saline Male vs Poly I:C Male	-0.0328356	0.1553185	4	-0.211	0.8429	1
Saline Female vs Poly I:C Female	0.02840956	0.1348984	4	0.211	0.8435	1
Poly I:C Male vs Poly I:C Female	-0.1813482	0.020181	4	-8.986	0.0008	0.0033954
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2277157	0.0522963	5.54	-4.354	0.0058	0.0231601
Saline Male vs Poly I:C Male	-0.0210986	0.083866	7.68	-0.252	0.808	1
Saline Female vs Poly I:C Female	0.30087316	0.079743	6.7	3.773	0.0075	0.0301606
Poly I:C Male vs Poly I:C Female	0.09425608	0.0391677	4.41	2.406	0.0679	0.2714544
NeuroD1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.3150986	0.1514229	4	-2.081	0.1059	0.4236517
Saline Male vs Poly I:C Male	0.0601522	0.1230203	4	0.489	0.6505	1
Saline Female vs Poly I:C Female	0.3702587	0.1065387	4	3.475	0.0255	0.1018381
Poly I:C Male vs Poly I:C Female	-0.0049921	0.0596289	4	-0.084	0.9373	1
Ppia						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2346683	0.1639741	4	-1.431	0.2256	0.9025649
Saline Male vs Poly I:C Male	-0.0717951	0.1381969	4	-0.52	0.6308	1
Saline Female vs Poly I:C Female	0.02121102	0.1208244	4	0.176	0.8692	1
Poly I:C Male vs Poly I:C Female	-0.1416622	0.0183014	4	-7.741	0.0015	0.0060019
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.7464122	0.4893636	4.54	1.525	0.1934	0.7737384
Saline Male vs Poly I:C Male	0.2275659	0.7148588	6.57	0.318	0.7601	1
Saline Female vs Poly I:C Female	-1.7457044	0.7071661	6.2	-2.469	0.0473	0.1891536
Poly I:C Male vs Poly I:C Female	-1.226858	0.3613018	3.92	-3.396	0.0283	0.1131121
Tpb						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1887908	0.0789112	8.21	-2.392	0.0429	0.171677
Saline Male vs Poly I:C Male	-0.1291854	0.0849121	6.85	-1.521	0.1729	0.6916427
Saline Female vs Poly I:C Female	0.09511642	0.0744318	5.58	1.278	0.2519	1
Poly I:C Male vs Poly I:C Female	0.03551093	0.0750241	8.34	0.473	0.6481	1
Ywhaz						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1261924	0.124472	4	-1.014	0.368	1
Saline Male vs Poly I:C Male	0.09399884	0.0952872	4	0.986	0.3797	1
Saline Female vs Poly I:C Female	0.12449965	0.0825211	4	1.509	0.2059	0.8234919
Poly I:C Male vs Poly I:C Female	-0.0956916	0.0199023	4	-4.808	0.0086	0.0343896

Figure G2: Pairwise comparisons for the E14.5 brain. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf (weighted for treatment and sex variance)						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	AIC	31.92236	
Intercept	0.884337	0.07307694	10	12.101451	<0.0001	Linear Mixed Model Random Effects:		
Sex	0.1669623	0.07128269	10	2.342256	0.0412		Std Dev	
Treatment	0.1404387	0.07307694	6	1.921793	0.103	Litter(Intercept)	0.051924	
Sex*Treatment	-0.0289173	0.07128269	10	-0.40567	0.6935	Residual	0.249975	
Ddc (weighted for treatment and sex variance)						Number of obs: 18. Groups: Litter, 7		
	Estimate	Std. Error	df	t value	p value	AIC	65.74765	
Intercept	2.62533	0.856717	9	3.064408	0.0135	Linear Mixed Model Random Effects:		
Sex	-0.8767339	0.8352661	9	-1.049646	0.3212		Std Dev	
Treatment	-1.442772	0.856717	5	-1.684077	0.153	Litter(Intercept)	0.54623	
Sex*Treatment	1.0453903	0.8352661	9	1.251566	0.2423	Residual	0.170362	
Gapdh						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion	4.3	
Intercept	0.988702	0.045214	16	21.867	2.41E-13	Linear Mixed Model Random Effects:		
Sex	-0.008848	0.045214	16	-0.196	0.847		Variance	Std Dev
Treatment	0.007307	0.045214	16	0.162	0.874	Litter(Intercept)	0	0
Sex*Treatment	-0.011109	0.045214	16	-0.246	0.809	Residual	0.03663	0.1914
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.00140276	0.00140276	1	16	0.0383	0.8473		
Treatment	0.00095676	0.00095676	1	16	0.0261	0.8736		
Sex:Treatment	0.00221162	0.00221162	1	16	0.0604	0.809		
Hprt						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion	10.6	
Intercept	0.985624	0.055056	16	17.902	5.24E-12	Linear Mixed Model Random Effects:		
Sex	0.014374	0.055056	16	0.261	0.797		Variance	Std Dev
Treatment	0.019078	0.055056	16	0.347	0.733	Litter(Intercept)	0	0
Sex*Treatment	0.009138	0.055056	16	0.166	0.87	Residual	0.05432	0.2331
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.0037027	0.0037027	1	16	0.0682	0.7974		
Treatment	0.0065225	0.0065225	1	16	0.1201	0.7335		
Sex:Treatment	0.0014962	0.0014962	1	16	0.0275	0.8703		
MaoA						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion	21.8	
Intercept	0.92598	0.07809	16	11.858	2.45E-09	Linear Mixed Model Random Effects:		
Sex	0.12145	0.07809	16	1.555	0.139		Variance	Std Dev
Treatment	0.09615	0.07809	16	1.231	0.236	Litter(Intercept)	0	0
Sex*Treatment	-0.01081	0.07809	16	-0.138	0.892	Residual	0.1093	0.3306
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.264323	0.264323	1	16	2.4188	0.1394		
Treatment	0.165677	0.165677	1	16	1.5161	0.236		
Sex:Treatment	0.002093	0.002093	1	16	0.0192	0.8916		
Mecp2						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion	1.3	
Intercept	1.003031	0.041186	16	24.354	4.50E-14	Linear Mixed Model Random Effects:		
Sex	0.042489	0.041186	16	1.032	0.318		Variance	Std Dev
Treatment	0.005701	0.041186	16	0.138	0.892	Litter(Intercept)	0	0
Sex*Treatment	0.001173	0.041186	16	0.028	0.978	Residual	0.0304	0.1743
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.032351	0.032351	1	16	1.0643	0.3176		
Treatment	0.000582	0.000582	1	16	0.0192	0.8916		
Sex:Treatment	0.000025	0.000025	1	16	0.0008	0.9776		
Oct3 (weighted for treatment and sex variance)						Number of obs: 20. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	AIC	34.0467	
Intercept	0.9931818	0.06888819	10	14.417301	<0.0001	Linear Mixed Model Random Effects:		
Sex	0.1812867	0.06888819	10	2.631607	0.0251		Std Dev	
Treatment	0.0418518	0.06888819	6	0.607532	0.5658	Litter(Intercept)	2.35E-06	
Sex*Treatment	0.006119	0.06888819	10	-0.088825	0.931	Residual	0.062488	

Ppia			Number of obs: 20. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	REML criterion	34.2	
Intercept	0.98127	0.13517	14	7.259	4.16E-06	Linear Mixed Model Random Effects:		
Sex	0.04279	0.13517	14	0.317	0.756		Variance	Std Dev
Treatment	0.01873	0.13517	14	0.139	0.892	Litter(Intercept)	0	0
Sex*Treatment	0.15806	0.13517	14	1.169	0.262	Residual	0.2995	0.5473
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.03001	0.03001	1	14	0.1002	0.7562		
Treatment	0.00575	0.00575	1	14	0.0192	0.8918		
Sex:Treatment	0.40948	0.40948	1	14	1.3673	0.2618		
Sert (weighted for treatment variance)			Number of obs: 20. Groups: Litter, 8					
	Estimate	Std. Error	df	t value	p value	AIC	34.15476	
Intercept	1.1610879	0.1328871	10	8.737401	<0.0001	Linear Mixed Model Random Effects:		
Sex	0.1587333	0.08918932	10	1.779734	0.1055		Std Dev	
Treatment	-0.0494373	0.1328871	6	-0.372025	0.7227	Litter(Intercept)	0.289637	
Sex*Treatment	-0.0829048	0.08918932	10	-0.929538	0.3745	Residual	0.128359	
Tph1 (weighted for treatment variance)			Number of obs: 13. Groups: Litter, 6					
	Estimate	Std. Error	df	t value	p value	AIC	40.94317	
Intercept	1.9526468	0.261506	5	7.466928	0.0007	Linear Mixed Model Random Effects:		
Sex	-0.3244774	0.261506	5	-1.240803	0.2697		Std Dev	
Treatment	-0.7279754	0.261506	4	-2.783781	0.0496	Litter(Intercept)	1.47E-05	
Sex*Treatment	0.6614845	0.261506	5	2.529519	0.0526	Residual	0.294132	
Tph2			Number of obs: 12. Groups: Litter, 6					
	Estimate	Std. Error	df	t value	p value	REML criterion	34.5	
Intercept	0.80523	0.46914	8	1.716	0.124	Linear Mixed Model Random Effects:		
Sex	0.08156	0.46914	8	0.174	0.866		Variance	Std Dev
Treatment	0.0034	0.46914	8	0.007	0.994	Litter(Intercept)	0	0
Sex*Treatment	-0.36861	0.46914	8	-0.786	0.455	Residual	1.467	1.211
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.04435	0.04435	1	8	0.0302	0.8663		
Treatment	0.00008	0.00008	1	8	0.0001	0.9944		
Sex:Treatment	0.90582	0.90582	1	8	0.6173	0.4547		

Figure G3: LMM analyses of target and reference gene expression in the E14.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.2760901	0.2344689	10	1.178	0.2663	1
Saline Male vs Poly I:C Male	0.2230429	0.2068201	6	1.078	0.3223	1
Saline Female vs Poly I:C Female	0.338712	0.2014874	6	1.681	0.1437	0.5749937
Poly I:C Male vs Poly I:C Female	0.3917592	0.1622464	10	2.415	0.0364	0.1455857
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.337313	0.168725	9	1.999	0.0767	0.3066142
Saline Male vs Poly I:C Male	-0.794774	0.738787	5	-1.076	0.3312	1
Saline Female vs Poly I:C Female	-4.976335	3.302614	5	-1.507	0.1922	0.7688827
Poly I:C Male vs Poly I:C Female	-3.844248	3.336801	9	-1.152	0.279	1
Gapdh						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.0399136	0.1235479	16	-0.323	0.7508	1
Saline Male vs Poly I:C Male	-0.0076048	0.119966	16	-0.063	0.9502	1
Saline Female vs Poly I:C Female	0.0368323	0.1353399	16	0.272	0.789	1
Poly I:C Male vs Poly I:C Female	0.00452349	0.1320783	16	0.034	0.9731	1
Hprt						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.04702394	0.1504424	16	0.313	0.7586	1
Saline Male vs Poly I:C Male	0.05643162	0.1460808	16	0.386	0.7044	1
Saline Female vs Poly I:C Female	0.01988123	0.1648014	16	0.121	0.9055	1
Poly I:C Male vs Poly I:C Female	0.01047356	0.1608296	16	0.065	0.9489	1
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.2212843	0.213384	16	1.037	0.3151	1
Saline Male vs Poly I:C Male	0.1706898	0.2071977	16	0.824	0.4222	1
Saline Female vs Poly I:C Female	0.2139219	0.2337505	16	0.915	0.3737	1
Poly I:C Male vs Poly I:C Female	0.2645164	0.2281171	16	1.16	0.2632	1
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.08732317	0.1125408	16	0.776	0.4491	1
Saline Male vs Poly I:C Male	0.01374825	0.109278	16	0.126	0.9014	1
Saline Female vs Poly I:C Female	0.00905626	0.1232822	16	0.073	0.9424	1
Poly I:C Male vs Poly I:C Female	0.08263118	0.1203111	16	0.687	0.502	1
Oct3						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.3503354	0.1992113	10	1.759	0.1091	0.4365952
Saline Male vs Poly I:C Male	0.0714656	0.1639604	6	0.436	0.6782	1
Saline Female vs Poly I:C Female	0.0959415	0.221464	6	0.433	0.68	1
Poly I:C Male vs Poly I:C Female	0.3748113	0.190379	10	1.969	0.0773	0.3092124
Ppia						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.401703	0.386966	14	1.038	0.3168	1
Saline Male vs Poly I:C Male	0.353573	0.3430087	14	1.031	0.3201	1
Saline Female vs Poly I:C Female	-0.2786701	0.417971	14	-0.667	0.5158	1
Poly I:C Male vs Poly I:C Female	-0.2305402	0.3776401	14	-0.61	0.5513	1
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.1516569	0.1073184	10	1.413	0.188	0.7518961
Saline Male vs Poly I:C Male	-0.2646843	0.2783873	6	-0.951	0.3784	1
Saline Female vs Poly I:C Female	0.0669351	0.3569457	6	0.188	0.8574	1
Poly I:C Male vs Poly I:C Female	0.4832763	0.340233	10	1.42	0.1859	0.7436121
Tph1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.6740141	0.3222053	5	2.092	0.0907	0.3627336
Saline Male vs Poly I:C Male	-0.1329819	0.6078411	4	-0.219	0.8375	1
Saline Female vs Poly I:C Female	-2.7789197	0.8512906	4	-3.264	0.031	0.1238129
Poly I:C Male vs Poly I:C Female	-1.9719237	0.9951634	5	-1.982	0.1044	0.417507
Tph2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.5741029	1.326927	8	-0.433	0.6767	1
Saline Male vs Poly I:C Male	-0.7304197	1.326927	8	-0.55	0.597	1
Saline Female vs Poly I:C Female	0.7440208	1.326927	8	0.561	0.5903	1
Poly I:C Male vs Poly I:C Female	0.9003377	1.326927	8	0.679	0.5166	1

Figure G4: Pairwise comparisons for the E14.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	REML criterion	17.7		
Intercept	1.0720	0.0935	8.247351	11.465	2.36E-06	Linear Mixed Model Random Effects:			
Sex	-0.0091	0.0721	12.2476	-0.126	0.902		Variance	Std Dev	
Treatment	-0.0522	0.0935	8.247351	-0.558	0.591	Litter(Intercept)	0.03771	0.1942	
Sex*Treatment	-0.0581	0.0721	12.2476	-0.806	0.436	Residual	0.06362	0.2522	
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0010	0.0010	1	12.2476	0.0159	0.9016			
Treatment	0.0198	0.0198	1	8.2474	0.3117	0.5915			
Sex:Treatment	0.0413	0.0413	1	12.2476	0.6493	0.4357			
Ddc	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	REML criterion	22		
Intercept	0.9729	0.1047	7.720923	9.291	1.85E-05	Linear Mixed Model Random Effects:			
Sex	0.0186	0.0842	12.33053	0.22	0.829		Variance	Std Dev	
Treatment	-0.0036	0.1047	7.720923	-0.034	0.974	Litter(Intercept)	0.04207	0.2051	
Sex*Treatment	0.0128	0.0842	12.33053	0.151	0.882	Residual	0.08882	0.298	
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0043	0.0043	1	12.3305	0.0485	0.8292			
Treatment	0.0001	0.0001	1	7.7209	0.0012	0.9736			
Sex:Treatment	0.0020	0.0020	1	12.3305	0.023	0.882			
Gapdh	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	REML criterion	5.3		
Intercept	1.0539	0.0524	8.02133	20.101	3.79E-08	Linear Mixed Model Random Effects:			
Sex	-0.0169	0.0507	14.51553	-0.333	0.744		Variance	Std Dev	
Treatment	-0.0268	0.0524	8.02133	-0.51	0.624	Litter(Intercept)	0.00223	0.04722	
Sex*Treatment	-0.0350	0.0507	14.51553	-0.691	0.5	Residual	0.03729	0.1931	
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0041	0.0041	1	14.5155	0.111	0.7438			
Treatment	0.0097	0.0097	1	8.0213	0.2604	0.6236			
Sex:Treatment	0.0178	0.0178	1	14.5155	0.4777	0.5004			
MaoA	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	REML criterion	11.9		
Intercept	1.1149	0.0996	6.87436	11.193	1.16E-05	Linear Mixed Model Random Effects:			
Sex	-0.0156	0.0499	8.88798	-0.313	0.7619		Variance	Std Dev	
Treatment	-0.0540	0.0996	6.87436	-0.542	0.6049	Litter(Intercept)	0.07046	0.2654	
Sex*Treatment	-0.1341	0.0499	8.88798	-2.687	0.0252	Residual	0.02685	0.1639	
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0026	0.0026	1	8.888	0.0977	0.76186			
Treatment	0.0079	0.0079	1	6.8744	0.2938	0.60494			
Sex:Treatment	0.1939	0.1939	1	8.888	7.2203	0.02519			
Mecp2	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	REML criterion	-1.7		
Intercept	1.0382	0.0643	7.10252	16.155	7.34E-07	Linear Mixed Model Random Effects:			
Sex	0.0040	0.0315	9.049248	0.127	0.902		Variance	Std Dev	
Treatment	-0.0019	0.0643	7.10252	-0.03	0.977	Litter(Intercept)	0.02968	0.1723	
Sex*Treatment	-0.0956	0.0315	9.049248	-3.038	0.014	Residual	0.01063	0.1031	
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0002	0.0002	1	9.0492	0.0161	0.90169			
Treatment	0.0000	0.0000	1	7.1025	0.0009	0.97717			
Sex:Treatment	0.0981	0.0981	1	9.0492	9.2306	0.01397			
NeuroD1 (weighted for sex variance)	Number of obs: 19. Groups: Litter, 9								
	Estimate	Std. Error	df	t value	p value	AIC	12.50735		
Intercept	1.0279	0.0367	8	27.993066	<0.0001	Linear Mixed Model Random Effects:			
Sex	-0.0077	0.0327	8	-0.235952	0.8194		Std Dev		
Treatment	-0.0114	0.0367	7	-0.311415	0.7646	Litter(Intercept)	0.058374		
Sex*Treatment	-0.0281	0.0327	8	-0.85982	0.4149	Residual	0.050933		

Sert		Number of obs: 19. Groups: Litter, 9							
	Estimate	Std. Error	df	t value	p value		REML criterion	40.2	
Intercept	1.0263	0.1713	7.42675	5.993	0.000435		Linear Mixed Model Random Effects:		
Sex	0.0196	0.1612	14.11112	0.121	0.905112			Variance	Std Dev
Treatment	0.0384	0.1713	7.42675	0.224	0.828749		Litter(Intercept)	0.04063	0.2016
Sex*Treatment	-0.1601	0.1612	14.11112	-0.994	0.33718		Residual	0.36651	0.6054
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0054	0.0054	1	14.1111	0.0147	0.9051			
Treatment	0.0184	0.0184	1	7.4268	0.0502	0.8287			
Sex:Treatment	0.3618	0.3618	1	14.1111	0.9871	0.3372			
Tpb									
		Number of obs: 19. Groups: Litter, 9							
	Estimate	Std. Error	df	t value	p value		REML criterion	17	
Intercept	1.0364	0.1055	6.74918	9.825	3.05E-05		Linear Mixed Model Random Effects:		
Sex	-0.0544	0.0643	9.48493	-0.845	0.419			Variance	Std Dev
Treatment	0.0528	0.1055	6.74918	0.501	0.6323		Litter(Intercept)	0.06871	0.2621
Sex*Treatment	-0.1544	0.0643	9.48493	-2.4	0.0386		Residual	0.04649	0.2156
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0332	0.0332	1	9.4849	0.7137	0.41902			
Treatment	0.0117	0.0117	1	6.7492	0.2509	0.63235			
Sex:Treatment	0.2678	0.2678	1	9.4849	5.7591	0.03859			
Tph2									
		Number of obs: 19. Groups: Litter, 9							
	Estimate	Std. Error	df	t value	p value		REML criterion	33.6	
Intercept	0.8853	0.1401	14	6.318	1.90E-05		Linear Mixed Model Random Effects:		
Sex	0.0958	0.1401	14	0.683	0.505			Variance	Std Dev
Treatment	0.0731	0.1401	14	0.521	0.61		Litter(Intercept)	0	0
Sex*Treatment	-0.0126	0.1401	14	-0.09	0.93		Residual	0.29	0.5385
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.1355	0.1355	1	14	0.4671	0.5055			
Treatment	0.0789	0.0789	1	14	0.2719	0.6102			
Sex:Treatment	0.0023	0.0023	1	14	0.008	0.9298			
Ywhaz									
		Number of obs: 19. Groups: Litter, 9							
	Estimate	Std. Error	df	t value	p value		REML criterion	0.6	
Intercept	1.0117	0.0789	7.15488	12.821	3.39E-06		Linear Mixed Model Random Effects:		
Sex	0.0109	0.0307	8.69039	0.356	0.73027			Variance	Std Dev
Treatment	0.0274	0.0789	7.15488	0.347	0.73874		Litter(Intercept)	0.048762	0.22082
Sex*Treatment	-0.1253	0.0307	8.69039	-4.082	0.00296		Residual	0.009839	0.09919
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.0012	0.0012	1	8.6904	0.1268	0.73027			
Treatment	0.0012	0.0012	1	7.1549	0.1203	0.73874			
Sex:Treatment	0.1639	0.1639	1	8.6904	16.6615	0.00296			

Figure G5: LMM analyses of target and reference gene expression in the E18.5 brain. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1343528	0.2266315	12.5	-0.593	0.5639	1
Saline Male vs Poly I:C Male	-0.220554	0.1952866	9.41	-1.129	0.2867	1
Saline Female vs Poly I:C Female	0.01174569	0.2708468	14.46	0.043	0.966	1
Poly I:C Male vs Poly I:C Female	0.09794691	0.1781808	11.83	0.55	0.5927	1
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.06261022	0.2645141	12.64	0.237	0.8167	1
Saline Male vs Poly I:C Male	0.01836213	0.2196566	9	0.084	0.9352	1
Saline Female vs Poly I:C Female	-0.0326583	0.310139	14.41	-0.105	0.9176	1
Poly I:C Male vs Poly I:C Female	0.01158983	0.2084535	11.82	0.056	0.9566	1
Gapdh						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1038458	0.1582294	14.77	-0.656	0.5217	1
Saline Male vs Poly I:C Male	-0.1235714	0.1127105	9.58	-1.096	0.2997	1
Saline Female vs Poly I:C Female	0.01656691	0.1727468	14.18	0.096	0.9249	1
Poly I:C Male vs Poly I:C Female	0.03629256	0.1267868	13.97	0.286	0.7789	1
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2994168	0.1575998	8.96	-1.9	0.0901	0.3602536
Saline Male vs Poly I:C Male	-0.3761943	0.2026068	7.37	-1.857	0.1036	0.4144669
Saline Female vs Poly I:C Female	0.1602505	0.2413535	12.19	0.664	0.5191	1
Poly I:C Male vs Poly I:C Female	0.237028	0.1225506	8.77	1.934	0.0859	0.3437675
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.1831592	0.0993437	9.11	-1.844	0.0979	0.3916914
Saline Male vs Poly I:C Male	-0.1949615	0.1306127	7.57	-1.493	0.176	0.7039095
Saline Female vs Poly I:C Female	0.1873425	0.1545892	12.17	1.212	0.2486	0.9942491
Poly I:C Male vs Poly I:C Female	0.1991448	0.0772312	8.94	2.579	0.0299	0.11964
NeuroD1						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.0717266	0.0939423	8	-0.764	0.4671	1
Saline Male vs Poly I:C Male	-0.0791518	0.1210586	7	-0.654	0.5341	1
Saline Female vs Poly I:C Female	0.03341167	0.0685627	7	0.487	0.6409	1
Poly I:C Male vs Poly I:C Female	0.04083691	0.0911791	8	0.448	0.6661	1
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2811311	0.5037363	14.45	-0.558	0.5853	1
Saline Male vs Poly I:C Male	-0.2435085	0.3666057	9	-0.664	0.5232	1
Saline Female vs Poly I:C Female	0.396995	0.5549989	14.16	0.715	0.486	1
Poly I:C Male vs Poly I:C Female	0.3593723	0.4023026	13.45	0.893	0.3874	1
Tpb						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.4174726	0.2029084	9.62	-2.057	0.0678	0.2710364
Saline Male vs Poly I:C Male	-0.2030985	0.21648	7.5	-0.938	0.3774	1
Saline Female vs Poly I:C Female	0.4144494	0.274333	13.37	1.511	0.1541	0.6164897
Poly I:C Male vs Poly I:C Female	0.2000753	0.1582666	9.26	1.264	0.2371	0.9482661
Tph2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.1664147	0.4396893	14	0.378	0.7107	1
Saline Male vs Poly I:C Male	0.1210163	0.3109073	14	0.389	0.703	1
Saline Female vs Poly I:C Female	0.1712709	0.4663609	14	0.367	0.7189	1
Poly I:C Male vs Poly I:C Female	0.2166694	0.3476049	14	0.623	0.5431	1
Ywhaz						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.2286832	0.096977	8.72	-2.358	0.0436	0.1743499
Saline Male vs Poly I:C Male	-0.1958131	0.1593842	7.44	-1.229	0.2567	1
Saline Female vs Poly I:C Female	0.3052647	0.1787246	10.73	1.708	0.1163	0.4653913
Poly I:C Male vs Poly I:C Female	0.2723946	0.0752652	8.64	3.619	0.006	0.0238833

Figure G6: Pairwise comparisons for the E18.5 brain. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf (weighted for sex variance)						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	AIC		19.3065	
Intercept	1.0038831	0.06458838	6	15.54278	<0.0001	Linear Mixed Model Random Effects:			
Sex	0.1353525	0.06458838	4	2.095617	0.1042	Std Dev			
Treatment	0.0968063	0.06458838	6	1.498819	0.1846	Litter(Intercept)		#####	
Sex*Treatment	0.0995894	0.06458838	4	1.541909	0.198	Residual		0.31505	
Ddc (weighted for sex and treatment variance)						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	AIC		33.3764	
Intercept	0.8588987	0.1127408	6	7.618348	0.0003	Linear Mixed Model Random Effects:			
Sex	0.0287987	0.1127408	3	0.255441	0.8149	Std Dev			
Treatment	0.1324305	0.1127408	6	1.174645	0.2846	Litter(Intercept)		#####	
Sex*Treatment	-0.054811	0.1127408	3	-0.48617	0.6601	Residual		0.23364	
MaoA						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	REML criterion		6.1	
Intercept	1.05511	0.05538	10	19.054	#####	Linear Mixed Model Random Effects:			
Sex	0.06685	0.05538	10	1.207	0.255			Variance	Std Dev
Treatment	-0.01411	0.05538	10	-0.255	0.804	Litter(Intercept)		0	0
Sex*Treatment	0.0288	0.05538	10	0.52	0.614	Residual		0.03823	0.1955
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.055717	0.055717	1	10	1.4574	0.2551			
Treatment	0.002484	0.002484	1	10	0.065	0.804			
Sex:Treatment	0.010341	0.010341	1	10	0.2705	0.6143			
Mecp2						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	REML criterion		4.5	
Intercept	1.06218	0.05129	10	20.711	#####	Linear Mixed Model Random Effects:			
Sex	0.04332	0.05129	10	0.845	0.418			Variance	Std Dev
Treatment	-0.05267	0.05129	10	-1.027	0.329	Litter(Intercept)		0	0
Sex*Treatment	-0.02114	0.05129	10	-0.412	0.689	Residual		0.03279	0.1811
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.023401	0.023401	1	10	0.7136	0.418			
Treatment	0.03459	0.03459	1	10	1.0548	0.3286			
Sex:Treatment	0.005574	0.005574	1	10	0.17	0.6888			
Oct3 (weighted for treatment variance)						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	AIC		21.9653	
Intercept	1.0570304	0.07496446	6	14.10042	<0.0001	Linear Mixed Model Random Effects:			
Sex	-0.0395306	0.07496446	4	-0.52732	0.6259	Std Dev			
Treatment	-0.0494314	0.07496446	6	-0.6594	0.5341	Litter(Intercept)		#####	
Sex*Treatment	0.0572616	0.07496446	4	0.76385	0.4875	Residual		0.36768	
Ppia						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	REML criterion		1.6	
Intercept	1.038938	0.050308	5.658616	20.652	#####	Linear Mixed Model Random Effects:			
Sex	0.057564	0.039501	5.361702	1.457	0.201			Variance	Std Dev
Treatment	-0.006566	0.050308	5.658616	-0.131	0.901	Litter(Intercept)		0.00824	0.0908
Sex*Treatment	-0.000658	0.039501	5.361702	-0.017	0.987	Residual		0.01758	0.1326
Type III ANOVA with Satterthwaite's method									
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)			
Sex	0.037331	0.037331	1	5.3617	2.1236	0.201			
Treatment	0.000299	0.000299	1	5.6586	0.017	0.9007			
Sex:Treatment	0.000005	0.000005	1	5.3617	0.0003	0.9873			
Sert (weighted for sex variance)						Number of obs: 14. Groups: Litter, 8			
	Estimate	Std. Error	df	t value	p value	AIC		25.2909	
Intercept	1.1437944	0.07498726	6	15.25318	<0.0001	Linear Mixed Model Random Effects:			
Sex	0.0342012	0.07498726	4	0.456093	0.672	Std Dev			
Treatment	-0.1032291	0.07498726	6	-1.37662	0.2178	Litter(Intercept)		#####	
Sex*Treatment	0.0604511	0.07498726	4	0.806152	0.4654	Residual		0.43636	

Tpb						Number of obs: 14. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion		27.9
Intercept	1.079321	0.16466	10	6.555	#####	Linear Mixed Model Random Effects:		
Sex	0.029627	0.16466	10	0.18	0.861		Variance	Std Dev
Treatment	-0.063197	0.16466	10	-0.384	0.709	Litter(Intercept)	0	0
Sex*Treatment	0.007996	0.16466	10	0.049	0.962	Residual	0.338	0.5814
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.010943	0.010943	1	10	0.0324	0.8608		
Treatment	0.049794	0.049794	1	10	0.1473	0.7092		
Sex:Treatment	0.000797	0.000797	1	10	0.0024	0.9622		
Ywhaz						Number of obs: 14. Groups: Litter, 8		
	Estimate	Std. Error	df	t value	p value	REML criterion		15.9
Intercept	1.04076	0.09072	10	11.472	#####	Linear Mixed Model Random Effects:		
Sex	0.05326	0.09072	10	0.587	0.57		Variance	Std Dev
Treatment	-0.01595	0.09072	10	-0.176	0.864	Litter(Intercept)	0	0
Sex*Treatment	0.00463	0.09072	10	0.051	0.96	Residual	0.1026	0.3203
Type III ANOVA with Satterthwaite's method								
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)		
Sex	0.035364	0.035364	1	10	0.3446	0.5702		
Treatment	0.003172	0.003172	1	10	0.0309	0.864		
Sex:Treatment	0.000267	0.000267	1	10	0.0026	0.9603		

Figure G7: LMM analyses of target and reference gene expression in the E18.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Bdnf						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.4698839	0.094062	4	4.995	0.0075	0.030058
Saline Male vs Poly I:C Male	0.3927914	0.1985065	4	1.979	0.119	0.4758958
Saline Female vs Poly I:C Female	-0.0055662	0.1653533	4	-0.034	0.9748	1
Poly I:C Male vs Poly I:C Female	0.0715263	0.2406219	4	0.297	0.7811	1
Ddc						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	-0.0520247	0.3116732	3	-0.167	0.878	1
Saline Male vs Poly I:C Male	0.1552389	0.3060757	3	0.507	0.647	1
Saline Female vs Poly I:C Female	0.374483	0.331188	3	1.131	0.3404	1
Poly I:C Male vs Poly I:C Female	0.1672194	0.3259259	3	0.513	0.6433	1
MaoA						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.19129913	0.1635905	10	1.169	0.2694	1
Saline Male vs Poly I:C Male	0.02936983	0.1784919	10	0.165	0.8726	1
Saline Female vs Poly I:C Female	-0.0858279	0.1311642	10	-0.654	0.5276	1
Poly I:C Male vs Poly I:C Female	0.0761014	0.149337	10	0.51	0.6214	1
Mecp2						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.04435914	0.1515073	10	0.293	0.7757	1
Saline Male vs Poly I:C Male	-0.1476341	0.165308	10	-0.893	0.3928	1
Saline Female vs Poly I:C Female	-0.0630558	0.1214761	10	-0.519	0.615	1
Poly I:C Male vs Poly I:C Female	0.12893746	0.1383066	10	0.932	0.3732	1
Oct3						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.03546208	0.1051467	4	0.337	0.7529	1
Saline Male vs Poly I:C Male	0.01566041	0.2301288	4	0.068	0.949	1
Saline Female vs Poly I:C Female	-0.213386	0.1922381	4	-1.11	0.3292	1
Poly I:C Male vs Poly I:C Female	-0.1935843	0.2808183	4	-0.689	0.5285	1
Ppia						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.11381123	0.1193484	6.01	0.954	0.377	1
Saline Male vs Poly I:C Male	-0.0144483	0.1433899	9.99	-0.101	0.9217	1
Saline Female vs Poly I:C Female	-0.0118164	0.1103136	8.28	-0.107	0.9172	1
Poly I:C Male vs Poly I:C Female	0.11644307	0.1035457	4.6	1.125	0.316	1
Sert						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.1893046	0.201626	4	0.939	0.4009	1
Saline Male vs Poly I:C Male	-0.085556	0.0654647	4	-1.307	0.2613	1
Saline Female vs Poly I:C Female	-0.3273604	0.292718	4	-1.118	0.326	1
Poly I:C Male vs Poly I:C Female	-0.0524998	0.2220729	4	-0.236	0.8247	1
Tpb						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.0752455	0.4864379	10	0.155	0.8801	1
Saline Male vs Poly I:C Male	-0.1104023	0.5307472	10	-0.208	0.8394	1
Saline Female vs Poly I:C Female	-0.1423864	0.390018	10	-0.365	0.7227	1
Poly I:C Male vs Poly I:C Female	0.04326143	0.444055	10	0.097	0.9243	1
Ywhaz						
	Estimate	Std. Error	df	t value	p value	adjusted p value
Saline Male vs Saline Female	0.11577824	0.2680122	10	0.432	0.6749	1
Saline Male vs Poly I:C Male	-0.0226398	0.2924253	10	-0.077	0.9398	1
Saline Female vs Poly I:C Female	-0.0411606	0.2148878	10	-0.192	0.8519	1
Poly I:C Male vs Poly I:C Female	0.09725739	0.2446606	10	0.398	0.6993	1

Figure G8: Pairwise comparisons for the E18.5 placenta. Shading indicates significance (green; $p < 0.05$) or near-significance (yellow; $p < 0.10$).

Appendix H

This Appendix contains a table of all 101 human studies identified in the SLC6A4 methylation literature review presented in Chapter 6.

Number	First Author	Title	Year
1	Alexander	Serotonin transporter gene methylation predicts long-term cortisol concentrations in hair	2019
2	Alexander	DNA methylation profiles within the serotonin transporter gene moderate the association of 5-HTTLPR and cortisol stress reactivity	2014
3	Asai	Effect of mood stabilizers on DNA methylation in human neuroblastoma cells	2013
4	Bakusic	Role of NR3C1 and SLC6A4 methylation in the HPA axis regulation in burnout	2021
5	Bakusic	Increased methylation of NR3C1 and SLC6A4 is associated with blunted cortisol reactivity to stress in major depression	2020
6	Bandeira	Methylation of BDNF and SLC6A4 Gene Promoters in Brazilian Patients With Temporal Lobe Epilepsy Presenting or Not Psychiatric Comorbidities	2021
7	Beach	Differential impact of cumulative SES risk on methylation of protein-protein interaction pathways as a function of SLC6A4 genetic variation in African American young adults	2015
8	Bishop	Methylation of FKBP5 and SLC6A4 in Relation to Treatment Response to Mindfulness Based Stress Reduction for Posttraumatic Stress Disorder	2018
9	Blazevic	Epigenetic adaptation of the placental serotonin transporter gene (SLC6A4) to gestational diabetes mellitus	2017
10	Boehm	Peripheral serotonin transporter DNA methylation is linked to increased salience network connectivity in females with anorexia nervosa	2020
11	Booij	DNA methylation of the serotonin transporter gene in peripheral cells and stress-related changes in hippocampal volume: a study in depressed patients and healthy controls	2015
12	Chagnon	DNA methylation and single nucleotide variants in the brain-derived neurotrophic factor (BDNF) and oxytocin receptor (OXTR) genes are associated with anxiety/depression in older women	2015
13	Chau	Neonatal pain and COMT Val158Met genotype in relation to serotonin transporter (SLC6A4) promoter methylation in very preterm children at school age	2014
14	Chiarella	DNA methylation differences in stress-related genes, functional connectivity and gray matter volume in depressed and healthy adolescents	2020
15	Clausing	Epigenetics as a Mechanism of Developmental Embodiment of Stress, Resilience, and Cardiometabolic Risk Across Generations of Latinx Immigrant Families	2021
16	Comtois-Cabana	Association between child maltreatment and depressive symptoms in emerging adulthood: The mediating and moderating roles of DNA methylation	2023
17	Craig	Exploring the Contribution of Proximal Family Risk Factors on SLC6A4 DNA Methylation in Children with a History of Maltreatment: A Preliminary Study	2021
18	DeLano	Associations Between Maternal Community Deprivation and Infant DNA Methylation of the SLC6A4 Gene	2020
19	Domschke	Serotonin transporter gene hypomethylation predicts impaired antidepressant treatment response	2014
20	Drabe	Serotonin transporter gene promoter methylation status correlates with in vivo prefrontal 5-HTT availability and reward function in human obesity	2017
21	Dannlowski	Serotonin transporter gene methylation is associated with hippocampal gray matter volume	2014
22	Dukal	New-born females show higher stress- and genotype-independent methylation of SLC6A4 than males	2015
23	Duman	Influence of life stress, 5-HTTLPR genotype, and SLC6A4 methylation on gene expression and stress response in healthy Caucasian males	2015
24	Franzago	SLC6A4 DNA Methylation Levels and Serum Kynurenine/Tryptophan Ratio in Eating Disorders: A Possible Link with Psychopathological Traits?	2023
25	Frodl	DNA methylation of the serotonin transporter gene (SLC6A4) is associated with brain function involved in processing emotional stimuli	2015
26	Gartstein	Sculpting infant soothability: the role of prenatal SSRI antidepressant exposure and neonatal SLC6A4 methylation status	2016
27	Grunblatt	Combining genetic and epigenetic parameters of the serotonin transporter gene in obsessive-compulsive disorder	2017
28	He	Hypermethylation of the serotonin transporter gene and paternal parenting styles in untreated anorexia nervosa patients: A pilot study	2023

29	Hildonen	Elevated Expression of SLC6A4 Encoding the Serotonin Transporter (SERT) in Gilles de la Tourette Syndrome	2021
30	Hossack	Adverse Childhood Experience, Genes, and PTSD Risk in Soldiers: A Methylation Study	2020
31	Hummel	No evidence for intervention-associated DNA methylation changes in monocytes of patients with posttraumatic stress disorder	2017
32	Iga	Association study of polymorphism in the serotonin transporter gene promoter, methylation profiles, and expression in patients with major depressive disorder	2016
33	Ikegame	Promoter Activity-Based Case-Control Association Study on SLC6A4 Highlighting Hypermethylation and Altered Amygdala Volume in Male Patients With Schizophrenia	2020
34	Ismaylova	Serotonin transporter promoter methylation in peripheral cells and neural responses to negative stimuli: A study of adolescent monozygotic twins	2018
35	Ismaylova	Serotonin transporter gene promoter methylation in peripheral cells in healthy adults: Neural correlates and tissue specificity	2017
36	Jahn	Serotonin system-associated genetic and epigenetic changes in pedophilia and child sexual offending	2022
37	Kang	Association of SLC6A4 methylation with early adversity, characteristics and outcomes in depression	2013
38	Kang	Association of SLC6A4 methylation with long-term outcomes after stroke: focus on the interaction with suicidal ideation	2021
39	Kim	A longitudinal study of SLC6A4 DNA promoter methylation and poststroke depression	2013
40	Kouter	Targeted sequencing approach: Comprehensive analysis of DNA methylation and gene expression across blood and brain regions in suicide victims	2023
41	Lam	Genotype-dependent associations between serotonin transporter gene (SLC6A4) DNA methylation and late-life depression	2018
42	Lee	Changes in the methylation status of DAT, SERT, and MeCP2 gene promoters in the blood cell in families exposed to alcohol during the periconceptional period	2015
43	Lei	Neighborhood crime and depressive symptoms among African American women: Genetic moderation and epigenetic mediation of effects	2015
44	Leibold	DNA methylation in the 5-HTT regulatory region is associated with CO2-induced fear in panic disorder patients	2020
45	Lillycrop	Differential SLC6A4 methylation: a predictive epigenetic marker of adiposity from birth to adulthood	2019
46	Liu	Sex-dependent DNA hypermethylation of SLC6A4 in patients with schizophrenia	2022
47	Longtain	Methylation of the serotonin transporter gene moderates the depressive subjective effect of cocaine	2022
48	Marzi	Analysis of DNA Methylation in Young People: Limited Evidence for an Association Between Victimization Stress and Epigenetic Variation in Blood	2018
49	Mascheroni	Serotonin transporter gene methylation and emotional regulation in preschool children born preterm: A longitudinal evaluation of the role of negative emotionality in infancy	2022
50	Mason	Epigenetic marks as the link between environment and development: examination of the associations between attachment, socioeconomic status, and methylation of the SLC6A4 gene	2016
51	Mendonca	Epigenetic variation at the SLC6A4 gene promoter in mother-child pairs with major depressive disorder	2019
52	Moon	Influence of antidepressant treatment on SLC6A4 methylation in Korean patients with major depression	2022
53	Moser	Targeted bisulfite sequencing: A novel tool for the assessment of DNA methylation with high sensitivity and increased coverage	2020
54	Muehlhan	Epigenetic variation in the serotonin transporter gene predicts resting state functional connectivity strength within the salience-network	2015
55	Muench	Lack of Association Between Serotonin Transporter Gene (SLC6A4) Promoter Methylation and Amygdala Response During Negative Emotion Processing in Individuals With Alcohol Dependence	2019
56	Mulder	Epigenomics of being bullied: changes in DNA methylation following bullying exposure	2020
57	Muller	No long-term effects of antenatal synthetic glucocorticoid exposure on epigenetic regulation of stress-related genes	2022
58	Nikolova	Beyond genotype: serotonin transporter epigenetic modification predicts human brain function	2015
59	Non	DNA methylation at stress-related genes is associated with exposure to early life institutionalization	2017
60	Okada	The potential of SLC6A4 gene methylation analysis for the diagnosis and treatment of major depression	2014

60	Okada	The potential of SLC6A4 gene methylation analysis for the diagnosis and treatment of major depression	2014
61	Ouellet-Morin	Increased serotonin transporter gene (SERT) DNA methylation is associated with bullying victimization and blunted cortisol response to stress in childhood: a longitudinal study of discordant monozygotic twins	2013
62	Palma-Gudiel	Epigenetics-by-sex interaction for somatization conferred by methylation at the promoter region of SLC6A4 gene	2018
63	Park	Associations between serotonin transporter gene (SLC6A4) methylation and clinical characteristics and cortical thickness in children with ADHD	2015
64	Peng	Childhood Trauma, DNA Methylation of Stress-Related Genes, and Depression: Findings From Two Monozygotic Twin Studies	2018
65	Pfisterer	Sudden infant death syndrome revisited: serotonin transporter gene, polymorphisms and promoter methylation	2022
66	Provenzi	Hidden pandemic: COVID-19-related stress, SLC6A4 methylation, and infants' temperament at 3 months	2021
67	Provenzi	Pain-related stress during the Neonatal Intensive Care Unit stay and SLC6A4 methylation in very preterm infants	2015
68	Rhein	DNA methylation results depend on DNA integrity-role of post mortem interval	2015
69	Santoro	Association Study of SLC6A4 (5-HTTLPR) Polymorphism and Its Promoter Methylation with Rehabilitation Outcome in Patients with Subacute Stroke	2021
70	Sanwald	Factors related to age at depression onset: the role of SLC6A4 methylation, sex, exposure to stressful life events and personality in a sample of inpatients suffering from major depression	2021
71	Sbarra	DNA Methylation Across the Serotonin Transporter Gene Following Marital Separation: A Pilot Study	2019
72	Schiele	Serotonin transporter gene promoter hypomethylation in obsessive-compulsive disorder - Predictor of impaired response to exposure treatment?	2021
73	Schiele	Hypermethylation of the serotonin transporter gene promoter in panic disorder-Epigenetic imprint of comorbid depression?	2019
74	Schiele	Serotonin Transporter Gene Promoter Hypomethylation as a Predictor of Antidepressant Treatment Response in Major Depression: A Replication Study	2021
75	Schneider	Association of Serotonin Transporter Gene Alu/b Methylation with Major Depression, Amygdala Responsiveness, 5-HTTLPR/rs25531 Polymorphism, and Stress	2018
76	Smith	Neighborhood characteristics influence DNA methylation of genes involved in stress response and inflammation: The Multi-Ethnic Study of Atherosclerosis	2017
77	Song	Maternal Gestational Diabetes Influences DNA Methylation in the Serotonin System in the Human Placenta	2022
78	Stoffel	Associations of SLC6A4 methylation with salivary cortisol, salivary alpha-amylase, and subjective stress in everyday life	2023
79	Stonawski	Associations of prenatal depressive symptoms with DNA methylation of HPA axis-related genes and diurnal cortisol profiles in primary school-aged children	2018
80	Sugawara	Effects of quetiapine on DNA methylation in neuroblastoma cells	2015
81	Swartz	An epigenetic mechanism links socioeconomic status to changes in depression-related brain function in high-risk adolescents	2016
82	Tabano	Mental health and the effects on methylation of stress-related genes in front-line versus other health care professionals during the second wave of COVID-19 pandemic: an Italian pilot study	2022
83	Tahara	Change in DNA methylation patterns of SLC6A4 gene in the gastric mucosa in functional dyspepsia	2014
84	Timmers	Methylation of the serotonin reuptake transporter gene and non-motor symptoms in dystonia patients	2022
85	Timothy	Influence of early adversity on cortisol reactivity, SLC6A4 methylation and externalizing behavior in children of alcoholics	2019
86	van der Knaap	Methylation of NR3C1 and SLC6A4 and internalizing problems. The TRAILS study	2015
87	van Mil	DNA methylation profiles at birth and child ADHD symptoms	2014
88	Wan	Early-Onset Schizophrenia: A Special Phenotype of the Disease Characterized by Increased MTHFR Polymorphisms and Aggravating Symptoms	2021
89	Wankerl	Effects of genetic and early environmental risk factors for depression on serotonin transporter expression and methylation profiles	2014
90	Won	Association between reduced white matter integrity in the corpus callosum and serotonin transporter gene DNA methylation in medication-naïve patients with major depressive disorder	2016
91	Yamazaki	Association Study and Meta-Analysis of Polymorphisms, Methylation Profiles, and Peripheral mRNA Expression of the Serotonin Transporter Gene in Patients with Alzheimer's Disease	2016

92	Zhao	Association between promoter methylation of serotonin transporter gene and depressive symptoms: a monozygotic twin study	2013
93	Zhao	Promoter methylation of serotonin transporter gene is associated with obesity measures: a monozygotic twin study	2013
94	Kallak	DNA methylation in cord blood in association with prenatal depressive symptoms	2023
95	Malin	Feasibility of Stress Research in Premature Infant-Maternal Dyads During and After Neonatal Intensive Care Unit Hospitalization	2023
96	Moore	Variability in DNA methylation at the serotonin transporter gene promoter: epigenetic mechanism or cell-type artifact?	2023
97	Nazzari	Prenatal exposure to environmental air pollution and psychosocial stress jointly contribute to the epigenetic regulation of the serotonin transporter gene in newborns	2023
98	van Dokkum	Neonatal stress exposure and DNA methylation of stress-related and neurodevelopmentally relevant genes: An exploratory study	2023
99	Vidovic	DNA Methylation Patterns in Relation to Acute Severity and Duration of Anxiety and Depression	2023
100	Wannemuller	DNA methylation levels of the serotonin transporter gene are not associated with the outcome of highly standardized one-session exposure-based fear treatment	2023
101	Xiong	Exploring mental health from the perspective of DNA methylation: evidence from the hospitality employees	2023

Table H: Extended table of all 101 studies included in preliminary analysis of human SLC6A4 methylation. First author, title, and year of publication are provided. Key studies used for CpG site analysis can also be found cited in analysis tables within Chapter 6 and within the full references list for this thesis.

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